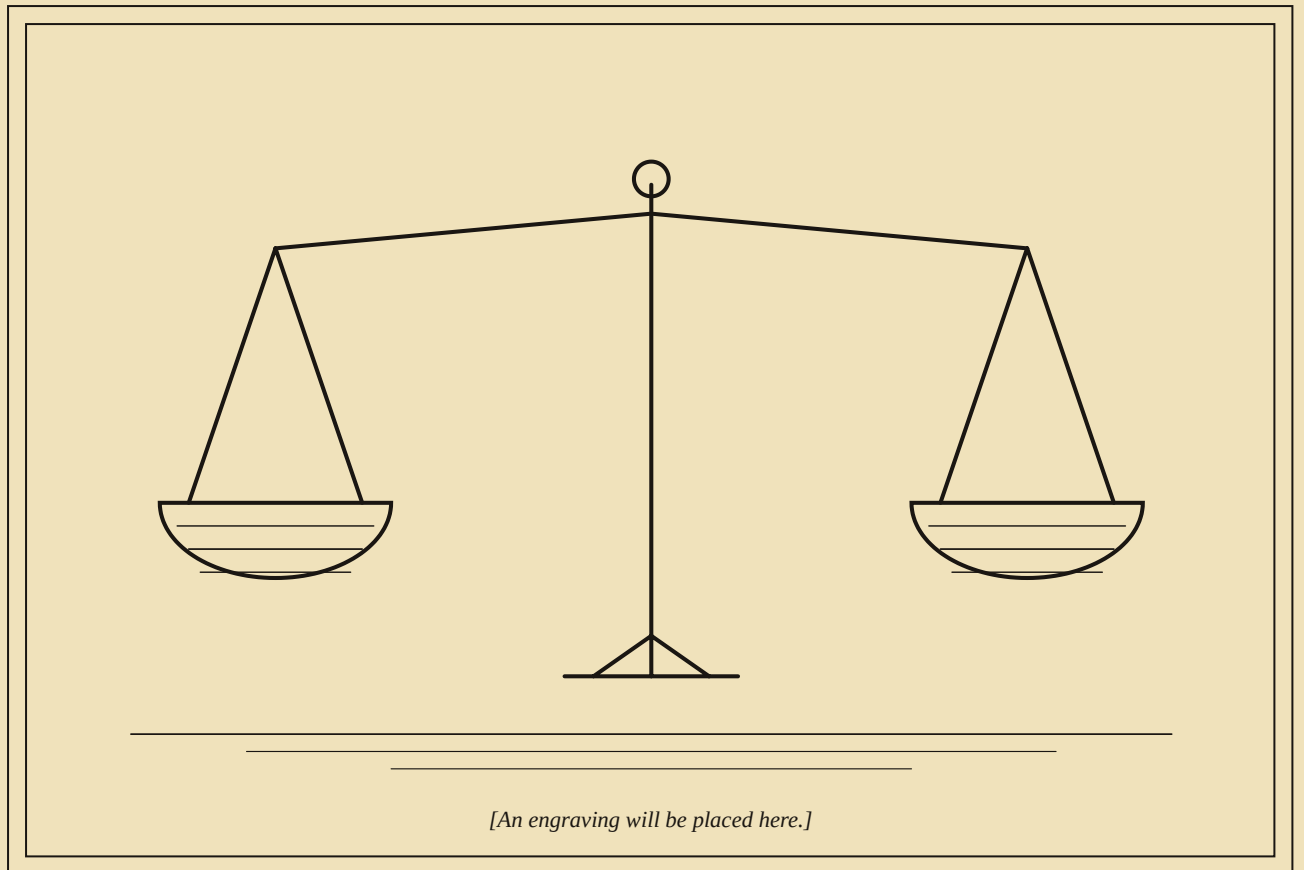




CHAPTER I

Organisations and Managers

Who answers for the morning bread?



A group of people, a result they want, and someone who answers for the work: these three things appear in every shop, school and hospital. This chapter shows what they are called, what the work of guiding a group means, and who does it.

A QUESTION BEFORE WE BEGIN

One woman, ten loaves, and no questions to ask.

Two years later, the bread sells so well that she needs help.

The recipe has not changed; the work of arranging the people has.

The chapter asks four plain questions, one after another.

This chapter needs nothing from earlier pages.

Some tasks cannot be done alone, however hard one tries.

GO SLOWLY HERE.

A family is the first group most people belong to.

Goal. A result someone wants to reach. Example: the rent paid by the first of the month.

A goal is a result that people want to reach.

Resource. Anything a group uses to do its work. Example: money, tools, flour, time, skill.

A resource is anything a group uses to do its work.

Organisation. Word in use. In daily life, "she is well organised" means tidy. Here, a group working for shared goals.

A group with a shared goal is an organisation.

Imagine a woman in a quiet street in Dhaka who bakes ten loaves each morning in her own kitchen. She buys the flour, mixes the dough, bakes the bread and sells it to her neighbours at the door. Nobody asks who decides what is baked, or when, or at what price. She does. Every decision is hers, and every task is hers, so the question never arises.

Two years later her bread is known across the district. Each morning she must now produce two hundred loaves. One oven and one pair of hands cannot do it, so she hires four helpers. At once, new questions appear. Who mixes the dough, and who bakes? Who stands at the counter? Who buys the flour, and who pays for it? What happens when the oven fails at five o'clock in the morning?

The loaves are the same, and so is the recipe. What has changed cannot be baked or sold. It is the work of deciding, arranging and checking, so that five people reach one result. This work has a name, and the people who do it have a name. Both are the subject of this chapter.

We ask four plain questions. Why do people join together to work? What is the work of guiding such a group? Why does that work matter? And who does it, using what? By the end, you will be able to answer each one in your own words.

You will need: nothing. This is the first chapter, and it starts from everyday life. Each idea is built from things you have already seen: a household, a school club, a market stall.

1.1 WHY PEOPLE FORM ORGANISATIONS

Start with a heavy wooden table in one room that must move to another. Alone, you can push one end a few centimetres. With a friend, you lift it together and carry it through the door. Two people have done what neither could do alone. This small fact explains why groups of people work together, and it explains the whole of this chapter.

Now think of a household. A family has one shared aim: to eat well, to pay the rent and to send the children to school. One person earns, another shops, a third cooks. Each month they talk about money and agree what to spend. Nobody calls this a business. Yet it already contains the parts we will meet again and again: an aim that everyone shares, several people, a part for each person, and some agreement about who does what.

We give the shared aim a name. A **goal** means a result that someone wants to reach. The family's goal may be to pay the rent by the first of the month. A baker's goal may be two hundred loaves by seven o'clock. A goal always points to something that does not exist yet and that people wish to bring about.

People also pool what they have. The family's wages, the kitchen, the food in the cupboard and the cook's knowledge are all used to reach the goal. Each such thing is a **resource**. A resource means anything that a group uses to do its work. Money is a resource. So are time, tools, materials and skill. Section 1.4 shows the main kinds in more detail.

A household is a small group, but the idea reaches much further. An **organisation** means a group of people working together for shared goals. Our family is one example. A school football club is another. Suppose fifteen students form a club. Their goal is to play matches and win some. Their resources are a ball, a field, an hour after lessons and the permission

of a teacher. Each student plays a position, and the club agrees who plays where.¹

How can you tell an organisation from other groups? Three tests help. First, the people share a goal. Second, they work together, and not only side by side. Third, there is some agreement about who does which part. Consider a crowd waiting at a bus stop. The people stand in the same place, but each has a separate aim. They do not work together, and nobody has agreed a part for anyone. A crowd is not an organisation. A football team is.

Why do people form organisations? Look back at the heavy table. The first reason is strength: a group can do what one person cannot. The second reason is skill. In a group, each person can do the part that suits them best, and the work improves. The third reason is the pooling of resources. One baker cannot buy a large oven, but five people who share the cost can. In each case, the group reaches a goal that the individual cannot reach alone.

Suppose five neighbours agree to cook for a wedding feast. Each is willing, and each has a pot. None of them agrees who cooks what. By evening, there is too much rice, there is no water, and nobody has bought onions. The goal was shared and the people were willing. The third test, an agreement on who does which part, was missing. The group wasted much of its effort. This is why a group of any size soon needs someone to settle who does what.

People in an organisation also keep goals of their own. A baker wishes to earn wages and to learn the trade. The owner wishes to earn a living and to build the shop. A customer wishes for fresh bread at a fair price. These goals are not the same as the shop's goal, and they may not agree with each other. Part of the work of guiding a group is to make the shop's goal and the people's own goals meet, so that people are glad to work toward it.

The edge of an organisation is not always sharp. A baker who works every day plainly belongs to the bakery. A supplier who delivers flour three times a week works closely with it but belongs to another organisation. Customers deal with the bakery daily, yet they are outside it. For this chapter, the organisation means the people who work together inside it for its goals. Chapter IV returns to those outside.

Organisations are everywhere. Families, school clubs, mosque committees, hospitals, shops, factories and government offices are all organisations. Some, like a shop, work to earn money. Others, like a school or a charity, work for different goals. This difference matters little for our purpose. Wherever a group of people works for a shared goal with shared resources, the same questions arise. The ideas in this book apply to all of them.

Return to the bakery from the opening page. The goal is two hundred loaves each morning. The people are the owner and four helpers. The resources are money, flour, an oven and the owner's recipe. Each person has a part. The bakery passes all three tests, so it is an organisation. It also faces a new problem: the group is large enough that nobody can carry everything in one head.

Before moving on. A goal is a result people want. A resource is anything they use to reach it. An organisation is a group that works together for a shared goal. A crowd at a bus stop shares a place but not a goal, so it is not an organisation.

You should now be able to say: "An organisation is a group of people who work together, with agreed parts, for goals they share."

GO SLOWLY HERE.

Three tests tell an organisation from a mere crowd.

People gain three things by joining a group.

A group without agreed parts can fail even when everyone is willing.

Each person in a group also has a private goal.

Who belongs to an organisation can sometimes be uncertain.

Organisations are everywhere, and not only in business.

The bakery from the opening page fits all three tests.

Before moving on: a group with a shared goal is an organisation.

You should now be able to say what an organisation is.

¹ The word *organisation* is also used for the act of arranging things, as in "the organisation of the event". This book uses it only for the group.

Test yourself before the next section.

At five o'clock, two bakers, one oven, and one question.

Willing people and resources are not enough on their own.

GO SLOWLY HERE.

Management. Getting things done through people and resources to reach goals.

Management is getting things done through people and resources.

The first part: management is about results, not about effort alone.

The second part: managers work through other people.

The third part: people are not the only thing a manager uses.

The fourth part: all of this is done for goals, not for its own sake.

Manager. A person responsible for the work of others.

A manager is the person who answers for others' work.

Three ideas often get mixed: doing, owning, managing.

Pause and check. A group of passengers sits on the same bus. Do they form an organisation? Then consider the bus crew: a driver and a conductor who share the aim of reaching the terminus on time. Does the crew pass the three tests? Give a reason for each answer.

1.2 WHAT MANAGEMENT IS

Picture the bakery at five in the morning. Two bakers have arrived. There is one oven, and each baker wants to use it first. One has the largest order. The other needs the oven for a short bake and has come early. The goal is clear, the people are willing and the flour is ready. Still, the bread will not be baked until someone decides who goes first.

This small scene shows a large truth. A goal, people and resources do not by themselves produce results. Someone must decide, arrange and check. If nobody does, willing people may waste time, resources may be wasted and the goal may be missed. The work of deciding, arranging and checking is the work we are about to name.

THE MEANING OF MANAGEMENT

The name for this work is **management**. Management means getting things done through people and resources to reach goals. This one sentence has four parts, and each part matters. We take them one at a time.

Getting things done. Management is about results. A manager who works very hard but reaches no goal has not managed well. The test is what was done, not how busy people were. At the bakery, the test is whether the loaves are baked, sold and paid for.

Through people. A manager does not do everything alone. A manager gets results by working with and through others. The bakery owner who bakes every loaf personally is working as a baker. The owner who arranges the work of five people so that two hundred loaves are ready is managing.

And resources. People need money, flour, an oven and a recipe. A manager must also see that these are bought, kept safe and used well. Managing people while wasting materials still fails.

To reach goals. The goal gives the work its direction. Without a goal, there is nothing to decide, because every choice would be as good as any other. Management always points toward a result that people have agreed to seek.

We now name the people who do this work. A **manager** means a person who is responsible for the work of others. Responsible here means that the person must see that the work is done and must answer if it is not. A manager of a shop answers for the shop's workers and its results. A school head answers for the school.²

Three ideas are often mixed, so we keep them apart. A person may do the work, own the business or manage the work. The same person can do all three, as in the one-woman kitchen. In a larger firm, the three are often separate. The table sets out the difference, starting with the question each role answers.

² In this book, a *worker* means a person who does the work and directs no one. An *employee* means any person paid by an organisation, whether worker or manager.

THREE ROLES THAT ARE OFTEN CONFUSED

ROLE	THE QUESTION IT ANSWERS	BAKERY EXAMPLE
Worker	What do I make or do today?	A baker shapes the loaves
Owner	Whose money started the business, and who receives its gains?	The person whose savings paid for the oven
Manager	Who sees that the work reaches its goal?	The person who decides tomorrow's order and the work of each helper

Read the table row by row. A worker answers a daily question about tasks. An owner answers a question about money and gains. A manager answers a question about results through others. An owner who hires a manager and goes home has stopped managing but still owns. A manager who owns nothing, such as an employee hired to run a shop, still manages. The roles overlap, but they are not the same.

The table shows that one person may play one, two or all three roles.



FIG. 1. Goal, organisation and manager: each calls for the next.

Figure 1 shows the order in which the three ideas arise. A goal calls for an organisation, because one person cannot reach it alone. An organisation calls for a manager, because a group of people needs someone who answers for the work. Read it from left to right, and notice that the bakery example appears under each box.

The figure shows the order in which these three ideas arise.

Return to the bakery at five in the morning. The owner sees that only one of the two bakers has arrived. The owner decides that the counter worker will help with shaping the dough until seven. This is *getting things done*. The owner explains the change to each person, so the work goes *through people*. The owner checks that there is enough flour for two hundred loaves, which is care of *resources*. And the owner keeps one result in mind: bread on the shelf when the doors open at seven, which is the *goal*. In a few minutes, all four parts of the definition were at work.

A morning at the bakery shows the four parts together.

The word management is used in three ways. It can name the activity of getting things done. It can name the group of people who do that work, as in the phrase "the management has decided". It can name a field of study, as in "she studies management". To keep one word to one meaning, this book uses management only for the activity and for the field of study. For the people, it says *managers*. Some books use the word *administration* for the same activity. This book uses management throughout.

The word has three uses, but this book keeps to one.

MANAGEMENT IN EVERYDAY LIFE

Management is not found only in business. Suppose a school plans a fair to raise money for a new library. The goal is Tk 100,000 in one day. The people are thirty students and five teachers. The resources are the school hall, some tables, a small amount of money and the time of everyone involved. One teacher

A school fair shows management at work outside business.

arranges all of this. She decides which class runs the food stalls and which runs the games. She sees that the tables arrive before the doors open. At the end of the day, she asks how much has been raised. She sells nothing herself, yet she is a manager.

Each part of the definition can be found in this example.

Look at the four parts of the definition in turn. She *gets things done*: the tables, stalls and games are ready by nine. She works *through people*: she does not run every stall herself. She watches the *resources*: the hall, the tables and the money. And she works for a *goal*: the library fund. If one part is missing, the work suffers. Without a goal, the day has no purpose. Without people, nothing happens. Without resources, the plans stay on paper.

The word "through" separates managing from doing the work oneself.

A note on the word *through*. It does not mean *over*. A manager does not stand above others and push them. A manager works through others in the sense that results arrive by means of their efforts. The teacher at the fair succeeds when the students succeed. Her own hands may sell nothing. This is why a good manager spends more time arranging, explaining and checking than doing the tasks that workers do.

Management looks different from four different seats.

Management also looks different from different seats. The *owner* sees management as the protector of the money. The *manager* sees it as the daily work of making things happen. The *worker* sees it as the way that tasks, hours and pay are fixed. The *customer* does not see it at all, and sees only its results: bread that is fresh and a counter that is open. All four views are partly correct. A manager who understands all four is better placed to decide well.

GO SLOWLY HERE.

THE NATURE OF MANAGEMENT

Nature means the features that all management shares.

What is management like? Writers describe its **nature**, which means the main features that management shows wherever it is found. Six features are enough for a beginner. Each can be checked against the bakery.

Management is always aimed at a goal.

Management is *goal-directed*. It exists to reach goals, and its quality is judged by whether they are reached. A manager who cannot say what the goal is cannot say whether the work is going well. In the bakery, the goal is bread that is baked on time, sold and paid for.

Management is found in every kind of organisation.

Management is *universal*. A school, a hospital, a cricket club, a charity and a government office all need people who answer for the work of others. The details differ, but the need does not. A shop with one worker and a hospital with one thousand both need it, though in different amounts.

Management never finishes; one task leads to the next.

Management is *continuing*. It is not done once and then put aside. Each day brings new decisions: what to order, who does what, whether yesterday's results were good. The work is made of several kinds of activity that repeat in a cycle. Chapter II names them: planning, organising, leading and controlling.

Management always involves dealing with people.

Management is *people-centred*. Machines and money matter, but the manager's central task is to work with people who have their own needs, fears and abilities. Two bakers with the same oven may produce different results because of how they feel about their work. A manager who forgets this will meet surprises.

Management is shared by a group of managers.

Management is a *group activity*. In a large firm, many managers do the work together, each in a different place and on a different part of the task. No single person manages a whole hospital. Section 1.5 shows how these managers are arranged.

Management cannot be seen directly, but its results can.

Management is *intangible*. You cannot hold management in your hand, as you can hold a loaf. You see it in what it produces: bread that is ready, workers who know their tasks, a shop that runs without

confusion. When management is poor, you also see the results: delays, waste and disorder.

Writers also ask whether management is a science, an art or a profession. That question is taken up in Chapter III, after you have seen what managers do.

THE PURPOSE OF MANAGEMENT

Why does management exist? Its **purpose** means the reason it is done, the result it is for. Four purposes can be named. They overlap, but each adds something.

The first purpose is to *reach the goals of the organisation*. This is the main one. All other purposes serve it. A bakery that does not bake its bread has failed in the one thing it was formed to do.

The second is to *use resources well*. Money, time and flour are limited. A manager must see that they are used with little waste. A bakery that throws away one loaf in ten wastes money, flour and the baker's work. Chapter II gives precise measures for this.

The third is to *keep people working together*. People can pull in different directions, and good intentions do not prevent it. If the counter seller promises delivery by noon and the baker bakes by two, both worked hard and the customer is still disappointed. Management links the parts so that they fit.

The fourth is to *help the group meet change*. Prices move, customers change their tastes, and rules are altered. An organisation that cannot adjust will lose ground. Chapter IV shows how outside forces reach inside an organisation and how managers respond.

Three cautions end this section. Management is not *ownership*: the table showed that a person can own without managing, and manage without owning. Management is not only *giving orders*: orders are one tool among many. And management is not the same as *leadership*, which means influencing people toward a goal; Chapter XII compares the two carefully. Keeping these ideas apart will save you from many confusions later.

Before moving on. Management means getting things done through people and resources to reach goals. A manager is the person responsible for the work of others. Management is goal-directed, universal, continuing, people-centred, a group activity and intangible. Its purposes are to reach goals, use resources well, keep people working together and meet change.

You should now be able to say: "Management is the work of getting results through other people and the resources they use, so that the group reaches its goals."

Pause and check. Look at the three roles in the table. A woman owns a shop, hires a cousin to run it and lives abroad. Which role does each person play? Who is the manager?

Chapter III asks whether management is a science or an art.

GO SLOWLY HERE.

Purpose means why management exists at all.

The first purpose: to reach the group's goals.

The second purpose: to use resources well and avoid waste.

The third purpose: to keep people working together.

The fourth purpose: to help the group adapt to change.

Management is not the same as owning, ordering or leading.

Before moving on: management is getting results through people and resources.

You should now be able to define management in your own words.

Test yourself before the next section.

1.3 WHY MANAGEMENT MATTERS

Consider two bakeries on opposite sides of the same road. Each has one oven, the same flour supplier and the same price for bread. One makes a steady living and its customers return. The other is often late, has no flour on Fridays and loses its helpers within months. The ovens are equal and the flour is equal. What differs is the way the work is guided. This is the first reason management matters.

Management matters first because it decides whether goals are reached. Good intentions, hard work and good equipment are all needed, but none is enough

Two bakeries with the same oven can have very different results.

Reason one: groups with good management reach their goals more often.

Reason two: resources are limited, and management reduces waste.

Reason three: management makes the parts of the group fit together.

Reason four: management helps a group meet change.

Reason five: society benefits when organisations work well.

Poor management shows itself in everyday signs.

Management matters differently in small and large organisations.

Management matters in hospitals and schools as much as in shops.

GO SLOWLY HERE.

A manager is answerable to several groups at once.

Accountability. Being answerable for results.

Accountability means being answerable for results.

Managers owe different things to different groups.

alone. Management joins them. Where it is weak, the other three are often wasted. Where it is strong, even modest resources can produce good results.

It matters second because resources are limited. Money runs out, time passes and flour spoils. Every organisation must choose what to do with what it has. A manager who spends carefully can do more with the same resources than one who does not. Over a year, a small saving each day becomes a large one.

It matters third because people and tasks must fit. A baker, a seller and a rider each do a different part. If the parts do not meet at the right time, the customer sees only the failure. Management sets who does what and when, so that the parts join into one result.

It matters fourth because the world changes. A new competitor opens nearby. The price of flour rises. A rule is altered. Someone must notice these changes and decide how the organisation will respond. Without management, an organisation tends to continue as before until the change harms it.

Last, it matters beyond the organisation. A well-managed firm gives its workers steady pay, gives its customers goods they can trust and pays what it owes to the government. A poorly managed firm may fail, leaving workers without jobs and suppliers unpaid. A country in which many organisations are well managed is usually a country in which many people live better.

Poor management is easy to recognise by its signs. Customers wait too long. Workers are unsure what is asked of them and ask the same question several times. Materials are bought twice or not at all. The same mistake repeats month after month. Good management shows itself in the opposite way: tasks are clear, work flows and problems are met early and not late. Few of these signs need a report. A visitor to a shop often feels them within ten minutes.

Management matters in small and in large organisations, but it shows itself differently. In a small shop, the owner sees everything and may repair a mistake within the hour. In a large hospital, no single person sees everything. Many managers must pass information to one another, or an error in one ward may go unnoticed for days. The larger the organisation, the more its results depend on the quality of its management.

The same is true outside business. In a hospital, poor management can mean that a patient waits hours for a bed or that a medicine is not in the ward when it is needed. In a school, it can mean that a class has no teacher for a week. The cost is then not lost money but lost health or lost learning. Wherever people depend on an organisation, its management matters to them.

A manager's work also carries duties. Managers act for other people. They spend money that others have given, they direct the work of employees, they serve customers and they operate within the law. For this reason, a manager is answerable to several groups. We now give a name to this idea.

Accountability means being answerable for results. A person who is accountable must be able to explain what was done and what happened, and must accept the consequences. If a bakery's loaves are always late, the manager cannot say, "The bakers did that." The manager is accountable for the work of the bakers. Chapter X explains how this idea works when authority is passed down to others.

Which groups does a manager answer to? Four are enough to begin with. To the *owners*, a manager owes careful use of their money and honest reports on the results. To the *employees*, a manager owes fair treatment, safe conditions and clear instructions. To the *customers*, a manager owes goods of the quality that was promised and at an honest price. To the *public*, a

manager owes obedience to the law and care for the people and places the work touches.

These duties do not always agree. A lower price pleases customers but may reduce the owner's gains. A higher wage helps employees but raises costs. A manager's responsibility is to weigh these claims fairly and to explain the choice. Chapter V returns to this question and treats the firm's duties to society in full. The groups that depend on an organisation, and are affected by it, also receive a precise name in Chapter IV.

A caution is needed. Good management does not guarantee success. A well-managed bakery may still fail if a flood closes the road, or if a rival opens next door with lower prices. Good management also cannot create resources that do not exist: it cannot bake without flour. What management does is improve the chances. Managers who think carefully, arrange the work well and check the results will, over time, do better than those who do not.

Before moving on. Management matters because it brings goals within reach, reduces waste, makes the parts fit together, helps a group meet change and benefits society. A manager is accountable, which means answerable, to owners, employees, customers and the public.

You should now be able to say: "Management matters because good intentions and good equipment are wasted without someone who joins them toward a goal, and because managers answer to many groups."

These duties can pull against each other; managers must weigh them.

Management cannot guarantee success; it only improves the chances.

Before moving on: management matters for results, resources, fit, change and society.

You should now be able to give reasons why management is important.

1.4 THE RESOURCES MANAGERS WORK WITH

A manager does not work in empty space. Every day, a manager uses things: people to do the work, money to pay for it, flour and tools to make the product, and knowledge to guide the whole. We have already met the word resource in section 1.1. It means anything an organisation uses to do its work. Now we divide resources into five kinds.

People. The first resource is the people who work in the organisation: their time, their skill, their strength and their ideas. People are different from the other four kinds, because they have wishes of their own. Money does not leave the bank because it is tired, but a worker may. At the bakery, the people are the owner, the bakers, the counter worker and the rider.

Money. The second resource is money. It buys materials, pays wages, covers rent and keeps the organisation running while it waits for its own income. Money may come from the owner's savings, from loans or from sales. Without enough of it, the other resources cannot be obtained or kept.

Materials. The third is the raw stuff that the organisation turns into something else. For a bakery, these are flour, yeast, salt, sugar and the bags in which bread is sold. Materials are used up in the work, so they must be bought again and again. A shortage stops the work at once.

Equipment. The fourth is the tools, machines and buildings that serve the work over a long period, not a single day. The bakery's equipment includes the oven, the mixer and the shop premises. Unlike flour, equipment is not used up in a single batch, though it wears and needs repair. It costs money to buy, and it often limits how much an organisation can produce.

Information. The fifth is what people know and can find out: orders received, prices, recipes, the customer's address, yesterday's sales. A manager decides on the basis of information, so poor information leads to poor decisions. At the bakery, the order sheet and the price list

Everything a manager uses falls into five kinds.

People are the first resource and the most important.

Money is the second resource: it buys the others.

Materials are the third resource: what the product is made of.

Equipment is the fourth resource: what is used again and again.

Information is the fifth resource: knowledge needed to decide.

are information. So is the owner's knowledge of which loaves sell best on Friday.

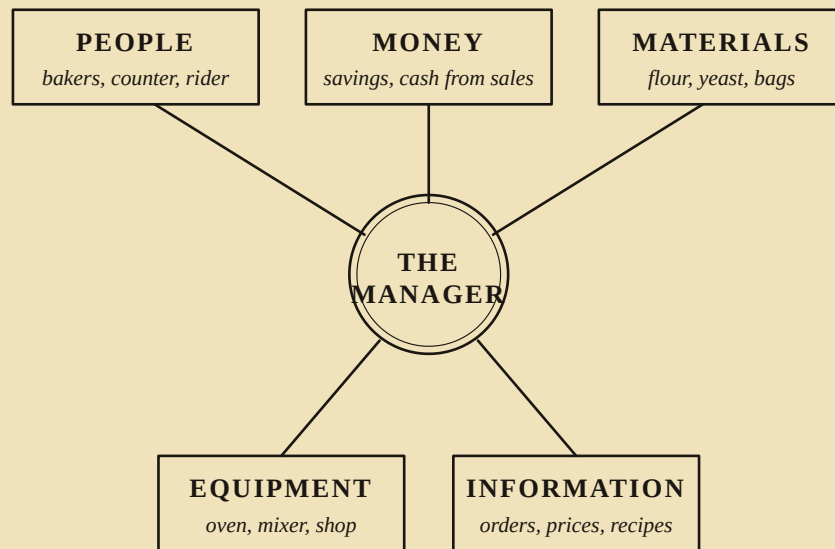


FIG. 2. The five resources a manager works with, with a bakery example for each.

The figure places the manager among the five resources.

The five kinds are separate, but they depend on each other.

One resource can sometimes replace another.

The mix of resources changes as an organisation grows.

People are different from the other resources in one important way.

Figure 2 places the manager in the centre. Each line shows one resource flowing to the manager's hands. The manager's work is to bring these five together in the right amounts and at the right time. The bakery examples in each box show that every organisation uses all five, though in different amounts.

The five kinds depend on each other. Money buys materials and equipment, and pays people. People use equipment and materials, and produce goods that bring money. Information tells the manager which of the other four is short. A shortage of any one can stop the work, which is why a manager watches all five. Now we can say the definition of management with the word *resources* in its place: management is getting things done through people and other resources to reach goals.

Resources can sometimes replace one another. Money can buy time, as when a bakery pays for a faster oven. A machine can replace the work of several people. A better price list, which is information, can save materials by showing which loaves waste the most. A manager's skill lies partly in seeing which exchanges are worthwhile. Each exchange has a cost, and the cost must be weighed against the gain.

The mix of resources changes as an organisation grows. A one-woman kitchen depends mostly on one person's skill and a small amount of money. A shop with five workers needs much more information, because several people must know what is ordered and what is done. A large factory depends heavily on equipment and on money, and the handling of information becomes the task of a whole department. A manager must therefore ask, from time to time, which resource is now the scarcest.

Return to the first resource for a moment. Flour obeys the manager's decisions, and so does the oven. People respond with their own thoughts. A baker who feels treated unfairly may work slowly, and a baker who feels trusted may suggest a better method. This fact is the reason that Chapters XI and XII study motivation and leadership. A manager who treats people as if they were flour will lose the benefits that only people can give.

Resources are always limited. A bakery cannot buy unlimited flour, hire unlimited bakers or run an oven twenty-five hours a day. A manager must decide which use of each resource brings the best result. This is one reason why Chapter II measures how well resources are used.

Before moving on. The five resources are people, money, materials, equipment and information. They depend on each other, and they are limited.

You should now be able to say: "A manager works with people, money, materials, equipment and information, and must use them well because they are limited."

Pause and check. A school canteen serves lunch to four hundred students. Name one example of each of the five resources in the canteen.

Resources are limited, so a manager must choose how to use them.

Before moving on: five resources, each needed, none unlimited.

You should now be able to name and describe the five resources.

Test yourself before the next section.

1.5 LEVELS AND KINDS OF MANAGERS

In a large organisation, many people carry the title of manager. A bank may have a managing director, a branch manager and a counter supervisor. They are all managers, but their work is not the same. To see how they differ, we sort managers in two ways: by *level*, which means how far they are from the daily work, and by *kind*, which means what they direct.

Start with level. Imagine the organisation as a stack, with the people who do the daily work at the base. Managers stand above them in steps. Those on the lowest step are close to the daily work. Those on the highest step are far from it, and look at the organisation as a whole. Most organisations have three steps.

At the highest step are the **top managers**. A top manager is a manager who directs the whole organisation. Typical titles are chief executive, managing director and chairman. Top managers decide what the organisation will try to do over the coming years, which markets to enter and how much money to spend. They also speak for the organisation to the public and to the government. They see little of the daily work, but their decisions shape it.

Below them are the **middle managers**. A middle manager is a manager who runs a department or a branch and reports upward to a top manager. A bank's branch manager and a factory's production manager are typical examples. Middle managers take the direction given by the top and convert it into work for their departments. They also pass information back up. They stand between those who set the direction and those who do the daily work.

At the lowest step are the **first-line managers**. A first-line manager is a manager who guides the daily work of workers who manage no one. Common titles are supervisor, team leader and head baker. They explain tasks, check quality, solve small problems on the spot and report to the middle manager. They are closest to the work and to the people who do it.³

Some managers decide for the whole, others for a small team.

GO SLOWLY HERE.

Level means a manager's distance from the daily work.

Top manager. A manager who directs the whole organisation.

Top managers direct the whole organisation.

Middle manager. A manager who runs a department or branch and answers to top managers.

Middle managers convert direction into department work.

First-line manager. A manager who directly guides the daily work of workers.

First-line managers guide the daily work of workers.

³ Some books call first-line managers *supervisors* or *operating managers*. This book keeps to the term first-line manager.

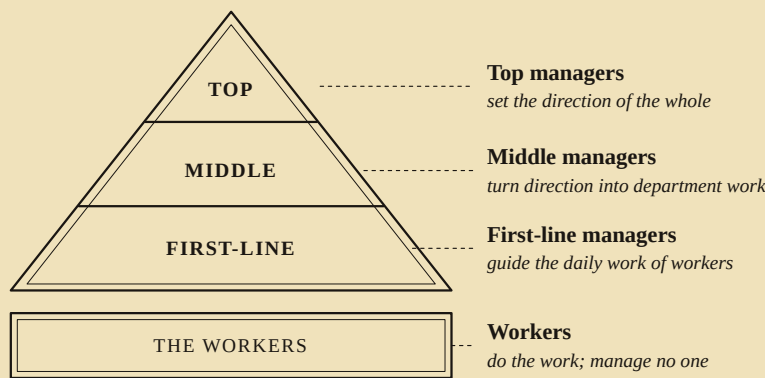


FIG. 3. Three levels of managers, above the workers who do the work.

Figure 3 shows the three steps above the workers.

The table compares the three levels by their main concern.

Figure 3 shows the three steps above the workers. Notice that the workers themselves do not appear as a manager level. They do the work and manage no one. Notice too that the stack narrows toward the top. A large organisation has many first-line managers, fewer middle managers and fewer still at the top. Small organisations may have one person standing on all three steps.

The table compares the three levels. Read it across one row at a time: each row describes one level, its main concern, how far ahead it looks and the titles it commonly carries. The titles are examples; organisations use many different names for the same work.

THE THREE LEVELS OF MANAGERS COMPARED

LEVEL	MAIN CONCERN	LOOKS AHEAD	COMMON TITLES
Top	Direction of the whole organisation	Years	Managing director, chief executive
Middle	Turning direction into work for a department	Months	Branch manager, department head
First-line	Daily work of the workers	Days and weeks	Supervisor, team leader

The levels differ in concern and time, not in importance.

Each level makes different kinds of decisions.

Managers in hospitals and schools follow the same pattern.

The levels differ in concern and in how far ahead they look. They do not differ in importance. An organisation with excellent top managers and poor first-line managers will fail at the counter, where customers meet it. An organisation with excellent first-line managers and no clear direction will work hard and achieve little. All three levels are needed, and they depend on each other.

An example of decisions makes the levels concrete. In a chain of bakeries, a top manager decides whether to open ten new shops next year and how to find the money. A middle manager decides how many loaves each shop will bake next month and which shop needs a new oven. A first-line manager decides who works at the counter on Friday and what to do when the mixer stops. The three kinds of decision differ in size and in time. A mistake at the top can affect every shop for years. A mistake at the first-line level is usually small and quick to repair, yet repeated small mistakes can damage the whole.

The three levels are not found only in business. A hospital has a chief executive at the top. Heads of departments, such as surgery or nursing, stand in the middle. Ward supervisors guide the daily work. A school has a head teacher at the top, heads of departments in the middle, and senior teachers of each year group nearest to the daily

work. The titles change from place to place, but the pattern of three steps is common.

Now consider kind. Two managers on the same level may be in charge of very different things. A manager's *kind* depends on the area of work, not on the level. Two kinds are met most often.

A **functional manager** is a manager who runs one specialist area of an organisation. Examples are a production manager, who runs the making of goods; a sales manager, who runs the selling; and a finance manager, who runs the handling of money. Each is expert in one area. A functional manager may stand at any level. The sales manager of a small firm and the head of sales of a large one are both functional managers.

A **general manager** is a manager who runs a whole unit and all of its specialist areas. The manager of a hotel, for example, answers for rooms, food, accounts and employees together. The unit may be a whole firm or one branch of it. A general manager must see how the areas fit, so this kind of manager needs knowledge of many fields and not only one.

The two kinds work side by side. In a factory, a general manager runs the whole site, and the production, sales and finance managers each run one area under that manager. The general manager joins what the functional managers do. The functional managers bring the deep knowledge of each area that the general manager may lack.

Titles can mislead. A person called "manager" in a small shop may be a first-line manager, and a person called "supervisor" in another firm may hold a larger role. To decide the level, ignore the title and ask three questions. Whom does this person direct? How far ahead must the person look? Does the person answer for a department or for the whole organisation? The answers show the level.

Level and kind are separate ways of sorting. The same manager can be described by both. The table shows four examples from a chain of bakeries. In each row, the third column gives the level, and the fourth gives the kind.

SORTING FOUR MANAGERS BY LEVEL AND BY KIND

JOB	WHAT THE PERSON DIRECTS	LEVEL	KIND
Managing director of a bakery chain	All the shops	Top	General
Head of production at the chain	The making of bread in all shops	Middle	Functional
Manager of one shop	That shop and all its sections	Middle	General
Head baker of one shop	The bakers on one shift	First-line	Functional

Read the table row by row. The managing director directs the whole chain, so the level is top and the kind is general. The head of production directs only the making of bread, so the kind is functional. In a chain with many shops, we place this job at the middle level. The shop manager directs one whole unit, so the kind is general, yet in the chain this is a middle role. The head baker directs a daily task, so the level is first-line and the kind is functional. The same word, manager, covers all four jobs.

Most managers begin at the lowest level and move up. A skilled baker may become head baker, then manager of a shop, then head of production. At each step, the work changes. The head baker spends most of the day with workers. The shop manager spends more time on orders, money and the results of the

GO SLOWLY HERE.

Kind means what a manager directs.

Functional manager. A manager who runs one specialist area of an organisation.

A functional manager runs one specialist area.

General manager. A manager who runs a whole unit, with all its specialist areas.

A general manager runs a whole unit and all its areas.

The two kinds are not opposites; they depend on each other.

A title does not always show the level.

Level and kind are separate ways of sorting the same manager.

Read the table row by row to see why each job is placed as it is.

Most managers begin at the first-line level and move upward.

In a small firm, one person may be all of these at once.

Before moving on: managers differ by level and by kind.

You should now be able to place a manager by level and kind.

Test yourself before the next section.

The case file records the bakery's first month in plain facts.

We ask which resources the shop uses and at which levels the owner manages.

Step one: find each resource in the given facts.

Step two: find the levels at which the owner works.

whole shop. The head of production spends more time with other managers. A person who rises must learn to let others do the work that the person once did well. Chapter II shows which skills each level needs most.

A small organisation may have only one manager. The owner of a small bakery sets the direction, arranges the work of the departments and guides the daily work of the workers. That one person stands on all three levels at once. As the bakery grows, the owner will pass some of these duties to others. Later chapters follow that change.

Before moving on. Managers are sorted by level into top, middle and first-line, and by kind into functional and general. Level shows distance from the daily work. Kind shows what the manager directs.

You should now be able to say: "A top manager directs the whole organisation, a middle manager runs a department, and a first-line manager guides daily work; a functional manager runs one area, and a general manager runs a whole unit."

Pause and check. A hospital has a chief executive, a head of surgery and a ward supervisor. Place each one by level. Which of them is most likely a functional manager?

WORKED CASE: MORNING LOAF BAKERY

Case file, illustrative. Month 1. Morning Loaf Bakery is a single shop in Dhaka. Its owner opens it as the only owner, with start capital of Tk 600,000 from personal savings. The first oven cost Tk 150,000. Rent is Tk 40,000 a month. The shop trades 26 days a month. A loaf sells at Tk 70. Ingredients cost about Tk 24 a loaf, of which flour is 0.24 kg at Tk 60 a kilogram. Four workers are employed: two bakers, one counter worker and one rider. Output is 200 loaves a day, and the four workers together work 32 hours a day. The owner is the only manager.

QUESTION

The case asks us two things. First, which of the five resources does the shop use, and what is the evidence for each? Second, at which of the three levels of management does the owner work?

REASONING

Step one: resources. We take the five kinds in turn and look for a fact that matches each. *People:* the owner and four workers. *Money:* the start capital of Tk 600,000, of which Tk 150,000 is now held in the oven, and the rent of Tk 40,000 a month. *Materials:* flour and other ingredients. At 200 loaves a day and 0.24 kg a loaf, the shop uses $200 \times 0.24 = 48$ kg of flour a day. At Tk 60 a kilogram, that costs $48 \times 60 = \text{Tk } 2,880$ a day. *Equipment:* the oven and the shop itself. *Information:* the price of Tk 70 a loaf, the daily order list and the owner's knowledge of how much to bake.

Step two: levels. We ask what the owner does and match each task to a level. When the owner decides to buy a second oven or to change the price, the owner is directing the whole shop. That is the top level. When the owner decides that the bakers start at four and the rider leaves at six, the owner is arranging the parts of the shop to fit. That is the middle level. When the owner shows a new baker how to shape loaves, the owner is guiding daily work. That is the first-line level.

CONCLUSION

The shop uses all five kinds of resources. The owner, as the only manager, works at all three levels at once. This is common in a small organisation. As the shop grows, the owner will pass some of these levels to other people. Later chapters follow that growth month by month.

The shop uses all five resources, and one person works at all three levels.

CHECK

To check the resources, we remove each one in turn and ask whether the shop could still make its bread. Without flour, no loaves are made. Without the oven, none is baked. Without workers, nobody makes or sells. Without money, none of these can be bought. Without information about orders and prices, the work has no direction. Each resource is therefore needed. To check the levels, we confirm that each of the three tasks above was a task done by the same person: the owner. The count of managers is one, and the count of levels is three.

A quick test confirms each answer against the facts.

SUMMARY

People form organisations because a group can do what one person cannot. An organisation is a group of people working together, with agreed parts, for shared goals. Management is getting things done through people and resources to reach those goals. A manager is the person responsible for the work of others. Management is goal-directed, universal, continuing, people-centred, a group activity and intangible. Its purposes are to reach goals, use resources well, keep people working together and meet change.

The chapter's main ideas, in short.

Management matters because it brings goals within reach, reduces waste, makes the parts fit, helps an organisation meet change and benefits society. Managers are accountable, which means answerable, to owners, employees, customers and the public. They work with five resources: people, money, materials, equipment and information. They stand on three levels, top, middle and first-line, and they are of two main kinds, functional and general.

Management matters, and managers are accountable.

THINK IT THROUGH

Rina runs a roadside stall in Dhaka that sells mobile phone credit and snacks. She works alone. She buys stock each morning, sets the prices, serves customers and keeps the cash in a tin. One person claims that her stall needs no management, because there is nobody for her to manage. Test this claim in three steps. First, state the goal of the stall. Second, find an example of each of the five resources at the stall. Third, describe one decision Rina makes in a day and say which level of management it belongs to. Then decide whether the claim holds, and say what Rina is when she is a worker, an owner and a manager at the same time.

A problem to reason through, with no single fixed answer.

WHERE PEOPLE GO WRONG

"Management means giving orders." Orders are one tool. Management is the whole work of getting results through people and resources, which includes deciding, arranging, explaining, listening and checking. A manager who only gives orders, and never asks whether the work is going well, is managing poorly.

Mistake one: thinking that management means giving orders.

"Only businesses and large firms need management." Every organisation has goals, people and resources, so every organisation needs management. A school, a hospital, a mosque committee and a football

Mistake two: thinking that only businesses need management.

Mistake three: thinking the owner is always the manager.

Mistake four: thinking only top managers are managers.

Mistake five: judging a manager by how hard the manager works.

A shipyard in Venice shows goals, people and resources organised on a large scale.

The Arsenal divided the building of a ship into stages and used standard parts.

Historians disagree on how modern the Arsenal's methods were.

The Arsenal can be read with this chapter's ideas.

A short list of where this chapter's central ideas first appeared.

club all have managers. Size changes the amount of management, not the need for it.

"The owner is the manager." Not always. Ownership concerns whose money is in the business. Management concerns who answers for the work. They may be the same person, as in a small shop, but they may also be different, as in a firm that employs a manager to run its branch.

"Only the people at the top are managers." A first-line manager who guides ten workers is as much a manager as a chief executive. The levels differ in concern and in how far ahead they look. They do not differ in whether the person is a manager.

"The hardest-working manager is the best." A manager is judged by results, not by effort. A manager who works fourteen hours a day but whose bakery is always late has not managed well. A manager who arranges the work so that the bread is ready on time, with no one overworked, has.

IN HISTORY

Long before modern firms, an organisation in Venice showed what managing at scale could mean. The Arsenal was the state shipyard of the city. Its beginnings are traditionally dated to the year 1104.⁴ Its goal was plain: to build and keep ready the ships on which Venice depended for trade and for defence. By the sixteenth century, many historians regard it as one of the largest industrial sites in Europe before the Industrial Revolution.

The Arsenal's resources included timber, rope, iron and the money of the state. Its people were thousands of skilled workers, known as the Arsenalotti. Some sources give a figure as high as sixteen thousand at its peak, around 1600. Ships were built in stages, with hulls moved from one work area to the next, and with standard parts made ahead of time and kept in store. When war threatened, an officer of the Arsenal could order more of these parts to be made. Each of these arrangements shows decisions made by people who answered for the result.

Historians disagree on how closely the Arsenal resembled a modern factory. They also record that it had trouble with its workers, including a dispute over working hours in 1569. These facts do not weaken the lesson. They support it. A large goal, many people and valuable resources required people to decide and to answer for the outcome. Venice had managers long before the word management was in use.

Read the Arsenal with the ideas of this chapter. It had a goal: ships. It was an organisation, with thousands of people working together. Its resources were of all five kinds: workers, money from the state, timber and iron, docks and tools, and knowledge of how to build a galley. Some officials directed the whole yard, others directed sections, and others guided small crews. The names differ from ours, but the three levels are recognisable. The idea of management is older than the word.

SOURCES

The idea that managers perform a continuing set of connected activities was described by the French mining engineer and company director Henri Fayol (1916), whose work is the subject of Chapter III. The idea that management is a

⁴ Sources differ on the founding date and on the number of workers, and the date of 1104 is traditional rather than certain. The workforce figure is also uncertain.

distinct task, and that results through people are its measure, was developed by the writer Peter Drucker in *The Practice of Management* (1954). A fuller reading list appears at the back of the book.

REVIEW QUESTIONS

Short answers to these questions are collected at the back of the book. The first set tests whether you have understood the ideas. The second set asks you to apply them to new cases.

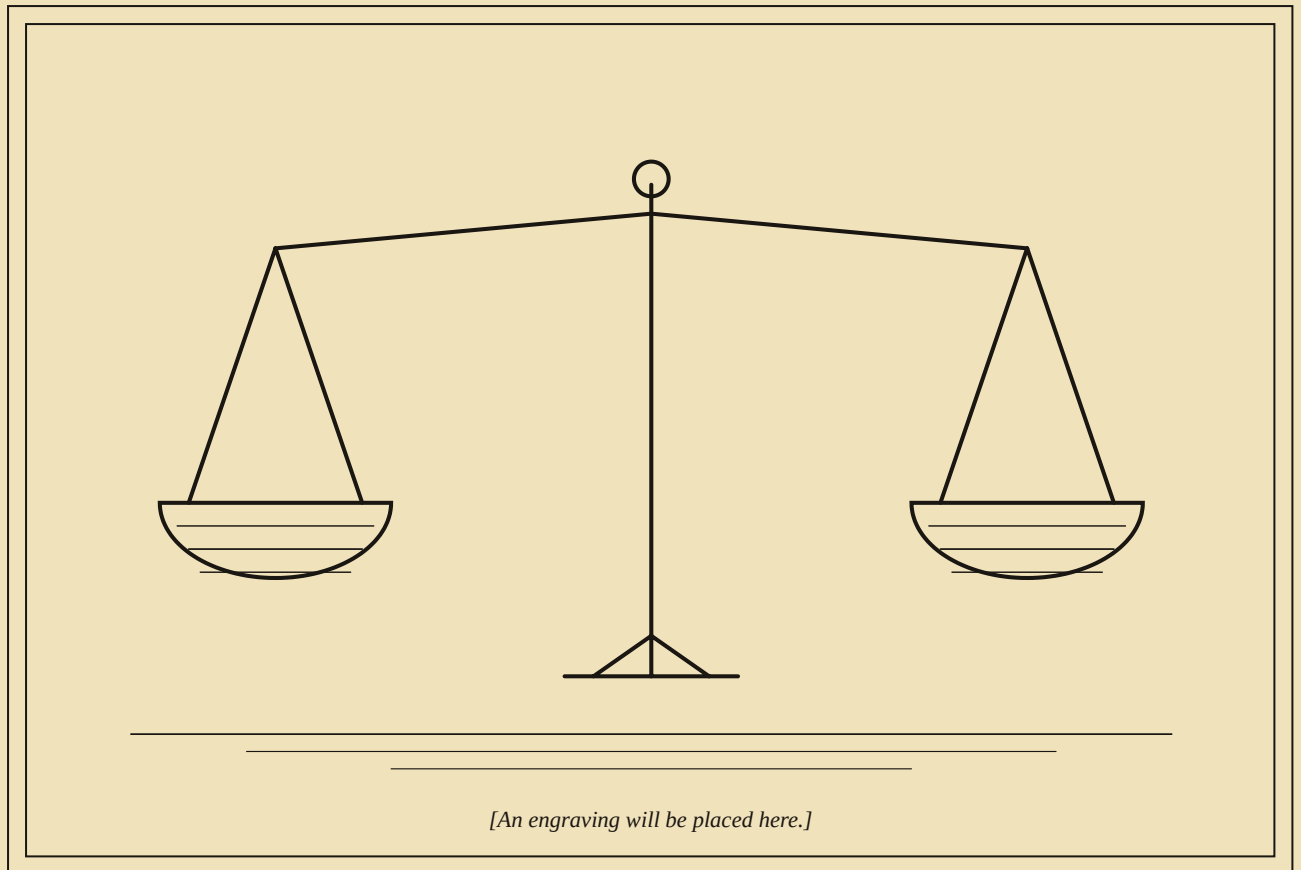
Short answers to these questions are collected at the back of the book.

REVIEW QUESTIONS: UNDERSTANDING

NO.	QUESTION
1	Define organisation in your own words, and give one example that is not a business.
2	State what management means, and explain each of the four parts of the definition.
3	Describe three features of the nature of management, giving a short example of each.
4	State four purposes of management.
5	Give four reasons why management is important to an organisation.
6	Explain accountability, and name the groups to which a manager answers.
7	Name the five resources, and show how each depends on at least one of the others.
8	Distinguish top, middle and first-line managers by their main concern.
9	Explain the difference between a functional manager and a general manager.

REVIEW QUESTIONS: APPLYING

NO.	QUESTION
10	Sort these jobs into levels: head of a bank's regional office; supervisor of a garment-factory floor; chief executive of a telephone company; manager of one bank branch; team leader in a call centre.
11	Three friends start a tea stall with a kettle, a table and Tk 20,000. List the resources they have under each of the five kinds, and say which one is most limited.
12	In a school, name a person who plays the role of worker, one who plays the role of owner and one who plays the role of manager. Explain your choices.
13	A factory has a production manager, a sales manager, a finance manager and one person above all three. State the kind of each manager, and explain the work of the person at the top.
14	Two clothing shops sell the same goods at the same prices, yet one earns steady profits and the other keeps losing workers. Using this chapter, suggest three questions you would ask to find the cause.
15	A friend says that a one-person business needs no management. Reply to the friend with two reasons.



CHAPTER II

What Managers Do

If the manager never touches the dough, what is the day spent on?



Managers do not bake the bread, yet their days are full. This chapter sorts managerial work into four kinds, shows the skills each level needs, and teaches the first measure of how well an organisation uses what it has.

A QUESTION BEFORE WE BEGIN

*One morning in the bakery,
written down entry by entry.*

*The owner never touched the
dough, yet every entry was work.*

*The chapter answers both
questions in turn.*

*This chapter builds on the
definition, resources and levels of
Chapter I.*

*Four words cover everything in
the owner's diary.*

Function. A main kind of work
that managers do. Example:
checking results.

*A function is a main kind of work
that managers do.*

GO SLOWLY HERE.

Planning. Choosing goals and
deciding how to reach them.

*Planning means choosing goals
and the way to reach them.*

Organising. Arranging people,
tasks and resources so that the
work can be done.

*Organising means arranging
people and resources for the work.*

Leading. Guiding and influencing
people so that they work toward
the goals.

*Leading means guiding people
toward the goals.*

Suppose the owner of a small bakery keeps a diary of one morning. The entries read like this. At five o'clock, checks the flour in the store. At ten past five, decides that the counter worker will help with the dough. At twenty past, explains the change to both workers. At six, notes that the oven has been late for three days. At seven, writes down how many loaves are ready. At eight, thinks about next month's orders and decides to ask a supplier for a lower price.

Not one entry says that the owner shaped a loaf or served a customer. Yet nobody would say that the owner did nothing. The morning was full, and each entry was a real piece of work. This raises two questions. Can the many things a manager does be sorted into a few kinds? And, once we know what a manager does, how can we tell whether it was done well?

The first question takes up the first two sections. We name four kinds of managerial work, and then we ask which skills a manager needs for them. The second question takes up the last two sections. We learn a ratio that measures how well resources are used, and we separate it from two ideas that are often confused with it.

You will need: the meaning of management, the five resources, and the three levels of managers, all from Chapter I. If any of these is unclear, read the summary at the end of that chapter first.

2.1 THE MANAGEMENT PROCESS

Return to the owner's diary. Read the entries again and ask what each is *for*. Checking the flour prepares for the day. Moving the counter worker arranges people. Explaining the change guides them. Noting the late oven checks a result. Asking the supplier for a lower price looks ahead. These five purposes, and a few more like them, can be gathered under four headings. Each heading is a function.

A **function** means a main kind of work that managers do. The word does not mean a party or a duty of the day. It names a group of activities that belong together because they have the same purpose. Management has four functions. We take them one at a time, and we go slowly, because they are the frame for the rest of the book.

The first function is **planning**. Planning means choosing goals and deciding how to reach them. It looks ahead. Before the bakery can bake, someone must decide how many loaves are wanted, when, and how they will be made. The owner who asks a supplier for a lower price next month is planning. A plan is the result: a chosen route from where the group is to where it wishes to be.

The second function is **organising**. Organising means arranging people, tasks and resources so that the work can be done. A goal and a plan are not enough if nobody knows who does what. The owner who moves the counter worker to the dough is organising. So is the owner who decides that bakers start at four and the rider leaves at six. The result is a pattern of tasks, people and tools that fits the plan.

The third function is **leading**. Leading means guiding and influencing people so that they work toward the goals. People are not machines that start at a signal. They need to understand what is asked, to be encouraged and sometimes to be corrected. The owner who explains the change to two workers is leading. Chapter XII studies this function in depth. For now, notice that it is the function most closely tied to people.

The fourth function is **controlling**. Controlling means checking results against the goals and correcting what is wrong. Without it, a manager would not know whether the plan was working. The owner who notes that the oven has been late three days running is controlling. So is the owner who counts the loaves at seven. Controlling asks the same question every time: are we where we meant to be?

These four functions are not special to bakeries. A hospital head plans next year's beds, organises the wards, leads the nurses and controls the cost of medicines. A school head does the same for classrooms and teachers. The four words describe what managers do in every organisation, at every level. Only the details change.

The functions do not stand apart. Together they form a cycle. The **management process** means the continuing cycle of the four functions. Planning leads to organising, organising leads to leading, leading leads to controlling, and controlling leads back to planning, because what the manager learns from checking changes the next plan. Figure 4 shows this.

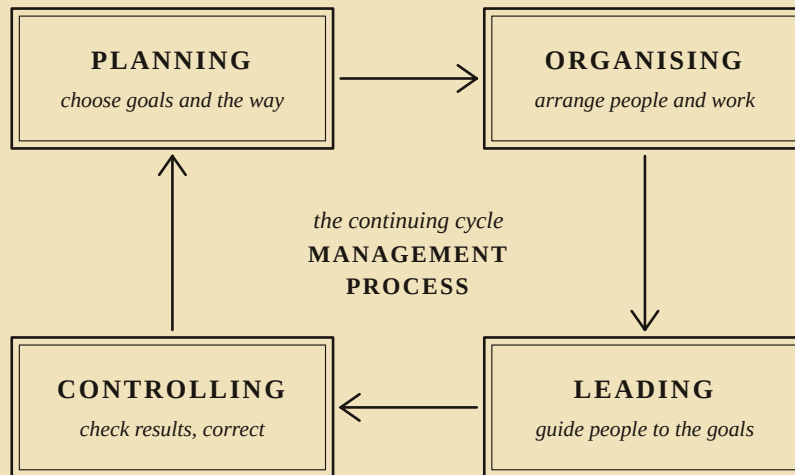


FIG. 4. The four functions form a continuing cycle: the management process.

Look at Figure 4 and follow the arrows. They run clockwise from planning. The arrow from controlling back to planning is the most important one. It shows that management does not finish. A manager who checks results and finds a gap does not stop. The finding becomes the starting point of the next plan, and the cycle turns again.

Why do we place planning first? Because the other functions depend on it. You cannot arrange people for work until you know what the work is. You cannot guide people toward a goal until the goal is chosen. You cannot check results without something to check them against. In this sense the order is logical. Each function needs the one before it.

In real life, the order is not strict. The owner does not plan for an hour, then organise for an hour, and so on. In one morning the owner may plan, organise, lead and control within ten minutes, and then repeat the sequence on a different subject. The four functions overlap. A single conversation, such as the owner's talk with the two workers, may contain organising, leading and controlling together.

Now we can answer the puzzle. The table sorts the owner's morning. Read each row from left to right: the time, what the owner did, and the function it belongs to.

Controlling. Checking results against the goals and correcting what is wrong.

Controlling means checking results and correcting them.

The four functions are the same wherever management is found.

Management process. The continuing cycle of the four functions.

The functions form a continuing cycle called the management process.

Figure 4 shows each function leading to the next, and the last leading back.

The order is a teaching order, because later functions depend on earlier ones.

In real life, a manager moves between the functions many times a day.

Sorting the owner's morning shows each entry has a home.

THE OWNER'S MORNING, SORTED BY FUNCTION

TIME	WHAT THE OWNER DID	FUNCTION
5:00	Checked the flour in the store	Controlling
5:10	Decided that the counter worker will help with the dough	Organising
5:20	Explained the change to both workers	Leading
6:00	Noted that the oven has been late three days	Controlling
7:00	Wrote down how many loaves were ready	Controlling
8:00	Decided to ask a supplier for a lower price next month	Planning

Controlling appears most often here, but this is only one morning.

Do not confuse the work of managing with the work of the bakery.

Levels differ in how much time they give to each function.

A small organisation may have all four functions done by one person.

Before moving on: four functions make a continuing cycle.

You should now be able to name and define the four functions.

Test yourself before the next section.

The table shows controlling three times and planning once. That does not mean that controlling is more important. It shows that one early morning, with the shop already open, is mostly about checking. A different day, such as the first of the month, would show more planning. The sorting works because every entry has a home, and only four homes are needed.

One caution. Baking, selling and delivering are not functions of management. They are the operating work of the bakery. A manager's functions are about arranging, guiding and checking that work. When the owner personally bakes a loaf, the owner is working as a baker. The same person may switch between the two kinds of work many times a day, and the switch is easy to miss.

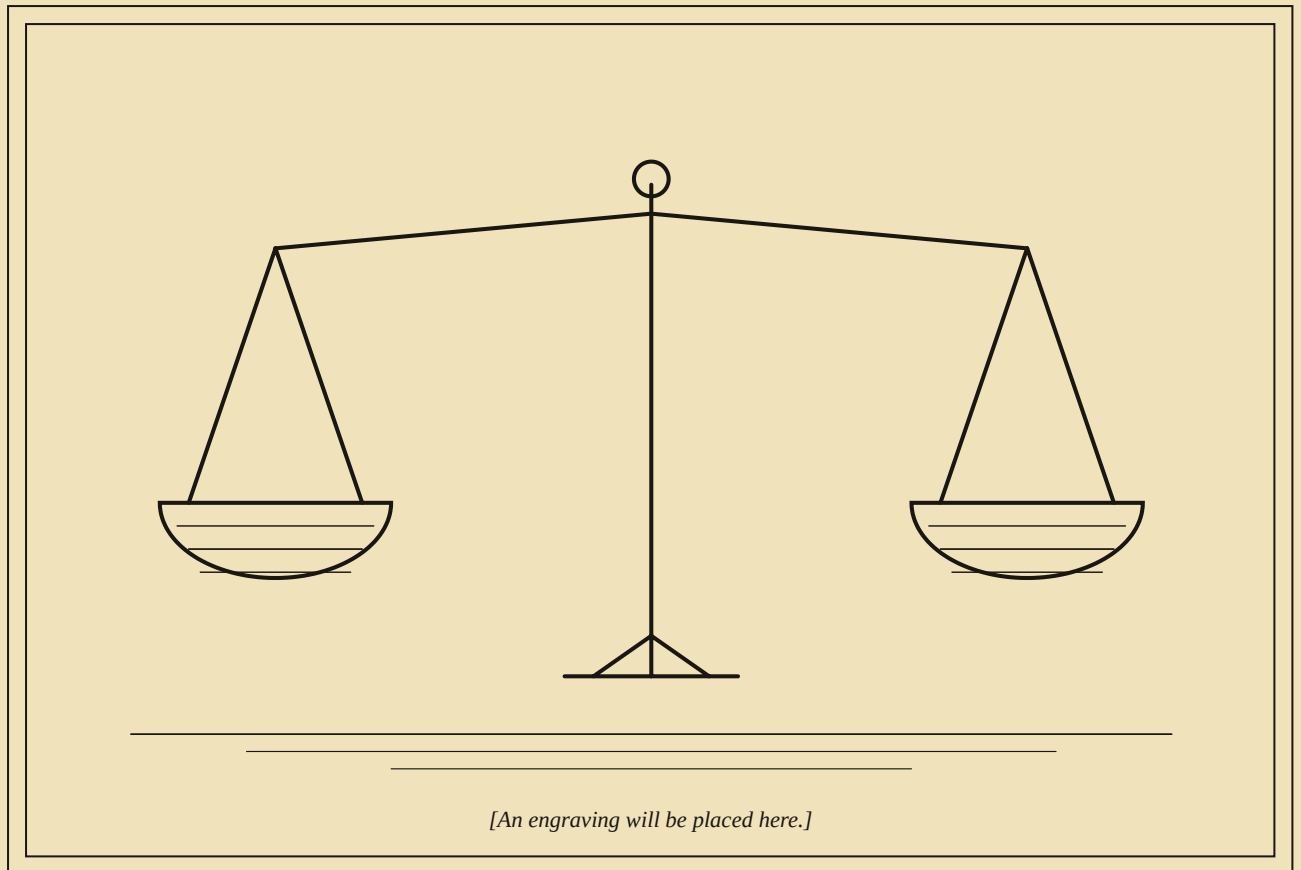
Do all managers spend the same time on each function? No. The mix changes with level. Top managers give more of their time to planning, because they look furthest ahead. First-line managers give more to leading and controlling, because they stand beside the daily work. Middle managers divide their time more evenly, and organising is the heart of their role. In every case, however, all four functions appear at every level.

In the small bakery, one person performs all four functions at all three levels. As the shop grows, the owner will pass some of the work to others, and the pattern of who does what will change. Later chapters study each function in turn: planning in Chapter VI, organising in Chapters IX and X, leading in Chapters XI and XII, and controlling in Chapter XIII.

Before moving on. A function is a main kind of work that managers do. The four functions are planning, organising, leading and controlling. Together they form the management process, a continuing cycle in which controlling leads back to planning. All four appear at every level, though in different amounts.

You should now be able to say: "Managers plan, organise, lead and control, and the four functions form a cycle that never finishes."

Pause and check. A hospital head decides that the surgery wing needs two more nurses at night, asks the nurses' supervisor to adjust the rota, tells the nurses why the change is needed and later counts the number of patient complaints. Which function does each of these four actions belong to?



CHAPTER III

How Management Thinking Developed

*People managed long before they studied management.
Then they began to ask which methods work, and why.*



This chapter follows the main schools of thought about management, from the study of factory work to the study of whole organisations. It looks closely at Taylor's scientific management and at Fayol's fourteen principles. It ends by asking whether management is a science, an art or a profession.

A QUESTION THAT SOMEONE HAS ALREADY ASKED

A bakery owner meets two problems and wonders whether anyone has solved them before.

The owner suspects that these troubles are common, not special to one shop.

This chapter tells that story and tests each idea against the bakery.

You need two earlier ideas: the four functions, and productivity.

Return to the Morning Loaf Bakery in its third month. The owner has learned to see the four functions at work, and has measured productivity. Now two small troubles appear at the same time. Packing bread at the counter bench is slow, and the person packing walks back and forth for bags, labels and string. At the same moment, two people give orders to the counter assistant, and each order is different.

The owner could solve each trouble by trial and error. A better way is to ask a question first. Has anyone studied these troubles carefully before? The answer is yes. For more than a century, careful people have studied how work is arranged and how orders flow. Some of them watched workers with a clock. Some of them wrote rules about command. Some of them asked why workers behave as they do. Their ideas did not agree, and the disagreements teach us something.

In this chapter we follow those ideas in the order in which they appeared. We ask what each school of thought believed, what it got right, and what it missed. We give most space to two landmark writers, Frederick Taylor and Henri Fayol. At the end we ask a harder question: is management a science, an art, or a profession?

You will need two ideas from Chapter II. The first is the four functions of management: planning, organising, leading and controlling. The second is productivity, which is output divided by input. If either idea feels uncertain, read the matching part of that chapter again before going on.



WHY IDEAS ABOUT MANAGEMENT CHANGED

People managed for thousands of years, but they began to study management only about a century ago.

Large factories created problems that old habits could not solve.

GO SLOWLY HERE. THIS IS A NEW KIND OF IDEA: A WAY OF GROUPING IDEAS.

School of thought. A group of ideas about management that share one main belief.

A school of thought is a group of writers who share one main belief about how to manage.

A new school usually appears because the old one left a problem unsolved.

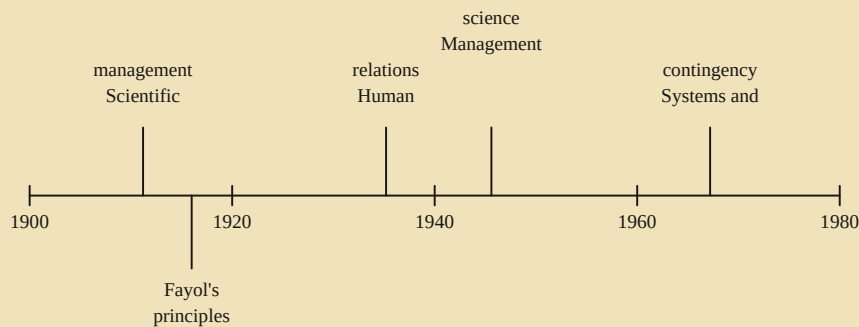
Kings built pyramids. Merchants ran caravans across deserts. Armies marched in order, and cities collected taxes. All of this needed managers, and the managers learned their work by doing it. For most of history, a young manager learned from an older one, and nobody wrote down a general theory.

Around the year 1800, factories began to replace small workshops in Britain, and the change spread to other countries. A workshop might hold six people who knew each other. A factory held hundreds, working with machines, in rows, for long hours. Suddenly an owner had to ask new questions. How should hundreds of tasks be divided? Who gives orders to whom? How much should each worker produce, and how should the owner know?

Writers tried to answer these questions, and their answers fell into groups. We call each group a **school of thought**. A school of thought is a set of ideas about management that share one main belief. For example, one school believed that the best way to raise output is to study the work. Another believed that the best way is to understand the worker.

Think of a school timetable as a bridge to this idea. A school may first give every teacher the same number of lessons, because that is easy to arrange. Soon teachers complain that some subjects are harder to teach than others. The school then adjusts the timetable. Later, it finds that students need breaks, and it adjusts again. Each change answers a weakness in the earlier one. The history of management ideas moves in the same way.

Keep two cautions in mind. First, the schools overlap in time, and their dates are approximate. Second, an old school is not wrong only because a new one follows it. Managers still use time study today, and they still use Fayol's rules. Each school left something of lasting value. The figure below shows the order in which the main schools appeared.¹



Each approach answered a weakness in the one before it.

FIG. 5. *Main schools of thought on a timeline (dates approximate)*

Why should a student of management study history at all? There is a practical reason. Every manager meets the same old problems: slow work, unclear orders, unhappy staff, changing markets. Earlier writers have already tried answers to each of them. When you know the history, you do not need to invent a solution from the beginning. You can ask which school met your kind of problem, what it advised, and where it failed. The story that follows is therefore a store of tried methods, and not only a record of the past.

You should now be able to say this: a school of thought is a group of ideas that share one main belief, and a new school usually appears to answer a weakness in an earlier one.

Two cautions: schools overlap, and old ideas do not disappear.

Studying the history of ideas helps a manager choose between them.

You should now be able to say why a new school of thought appears.

SCIENTIFIC MANAGEMENT

THE MAN AND THE PROBLEM

Frederick Winslow Taylor (1856–1915) was an American engineer. He began as a worker in a steel company and rose to be its chief engineer. Because he had worked on the shop floor, he knew what workers did when nobody watched. What he saw troubled him.

Taylor saw that many workers worked well below their ability. He called this **soldiering**. Soldiering means working at a slow pace on purpose, so that nobody expects more. The workers had reasons. They feared that if they produced more, the owner would cut the pay for each piece, or that fewer workers would be needed. Managers, for their part, did not know how long a task should take. Neither side had facts.

Managers of that time relied on a **rule of thumb**. A rule of thumb is a method based on habit and guesswork, not on measurement. Each worker used a personal method. The manager could not say which method was best, because nobody had compared them. Taylor wanted to replace guesses with facts.

Taylor believed that the way work is done can be studied and improved, like any other problem.

Frederick Taylor was an American engineer who watched factory workers closely.

Soldiering. Working slower than one could, on purpose, so as not to do more than others.

Taylor found that workers often worked slowly on purpose.

Rule of thumb. A method based on habit and guesswork, not on measurement.

Rule of thumb means a method learned from habit, not from measurement.

¹ The dates are broad guides. Scientific management and Fayol's principles come from the early twentieth century; the other approaches grew over the following decades.

Scientific management. Studying work by measurement to find the best method, then teaching it.

Taylor's book of 1911 named four principles for managers.

The first principle replaces rule of thumb with a method found by study.

The second principle is to select and train workers for the method.

The third principle joins the method to the worker through cooperation and pay.

Piece rate. Pay that depends on the number of pieces a worker produces.

A piece rate pays by the number of units a worker makes.

The fourth principle divides the work fairly between managers and workers.

Time-and-motion study. Timing a task and examining each movement to remove waste.

Taylor's main tool was the clock, used with careful method.

A time study measures how long each movement takes.

The packing bench shows how a time-and-motion study works.

THE FOUR PRINCIPLES

In 1911 Taylor published *The Principles of Scientific Management*. In it he named an approach that he called science, not guesswork. We call it **scientific management**. Scientific management means studying work with measurement in order to find the one best method, and then teaching that method to every worker.

Taylor set out four principles, which we state in plain words. The first principle is to develop a science of each task. A manager studies the task, tests different methods, and chooses the best one. The chosen method replaces the rule of thumb.

The second principle is to choose workers with care and to train them. Different tasks need different strengths. A manager should match each worker to a task, and then teach the best method. Before Taylor, workers often trained themselves, and trained each other badly.

The third principle is cooperation. The manager works with the workers so that the task is done in the way the study found. Taylor wanted workers to follow the method because they gained from it. He therefore linked pay to output. His best known plan paid a higher rate for each piece to the worker who reached the standard, and a lower rate to the worker who did not.

The word **piece rate** needs a short explanation. A piece rate is a payment for each unit that a worker produces. A baker who is paid Tk 2 for each loaf packed is on a piece rate. The more loaves the baker packs, the more the baker earns.

The fourth principle is a division of duty. Managers take over the planning of work, which workers had done by habit. Workers carry out the work according to the plan. Taylor believed that each side would do what it did best, and that the two sides would share the gain.

TIME AND MOTION STUDY

To find the best method, Taylor needed facts. His tool was the **time-and-motion study**. A time-and-motion study means timing a task and examining each movement in it, in order to remove movements that waste time.

The study has two parts. In the first part, the observer breaks a task into small movements and times each one with a stopwatch. In the second part, the observer asks whether each movement is needed. Could the worker reach less far? Could a tool be kept closer? Could two movements become one? Frank and Lillian Gilbreth, an American husband and wife, developed this second part and used it to study how workers lay bricks and perform other tasks.²

Apply this to the bakery's packing bench. The owner watches the person who packs for an afternoon. Each loaf takes 40 seconds to pack. Most of the time is not spent on packing. The packer walks to a shelf for a bag, walks back for a label, and walks again for string. The owner moves bags, labels and string to one place within the packer's reach. The task now takes 30 seconds. We shall return to these figures in the worked case at the end of this chapter.

² Frank Gilbreth (1868–1924) and Lillian Gilbreth (1878–1972) worked on motion study and on fatigue. Their work added the study of movement to Taylor's study of time.

Notice what happened. The person packing did not work harder. That person worked in a better arrangement. This is the central idea of scientific management: much waste hides in the way work is arranged, and a careful study can find it.

Scientific management is not only for factories. Consider a hospital ward where nurses give medicine to patients each morning. A manager times the round and finds that the nurse walks to the store room many times, because each trip brings the medicine for one patient only. The manager then arranges the medicine for all the patients in a tray before the round begins. The nurse makes one trip and spends the saved time with the patients. The method is the same as at the bakery's bench: time the task, remove the movements that waste time, and teach the better arrangement. The setting differs, but the idea carries over.

A caution belongs here. In a hospital, a faster round that causes a mistake is worse than a slow one. The goal of the study is therefore not speed alone. The study must keep the task safe, and any new arrangement must be checked for that before it is adopted. This shows that a method found by study still needs judgement.

WHAT SCIENTIFIC MANAGEMENT GOT RIGHT

Scientific management gave management three lasting gifts. The first gift is measurement: a manager can ask how long a task takes, and can answer with a number. The second gift is method: tasks can be improved by study, and not only by effort. The third gift is training: a worker should learn the best method and not be left alone to find one.

These gifts raised productivity, the ratio of output to input that we met in Chapter II. When a task takes less time for the same output, the same workers produce more. Later in the twentieth century, factories around the world used time study, standard methods and piece rates. Some still do.

One caution is needed. Taylor described spectacular results at a steel works in Pennsylvania, where he said the output of workers who handled heavy iron bars rose several times. Later writers have questioned the figures and the way he reached them.³ We therefore treat these stories as claims, not as proven facts. The idea of studying work does not depend on them.

WHAT SCIENTIFIC MANAGEMENT MISSED

Critics point to three weaknesses. The first is that Taylor treated workers mainly as parts of a machine. He assumed that money was the chief reason people work. He paid little attention to pride, friendship, boredom or fear. As later chapters show, people work for many reasons, and money is only one of them.

The second weakness is that the method works best for tasks that are repeated many times. A packing bench is a good example. The work of a manager who deals with new problems each day is harder to time and divide. The third weakness is that workers and unions often saw the method as a way of forcing a faster pace. Where trust was low, workers resisted the stopwatch, and the gains were small.

Taylor looked closely at the task, at the level of the workshop. A French engineer asked a different question. What does a manager do, and by what rules should the whole firm be run? We turn to him next.

GO SLOWLY HERE. NOTICE THAT THE METHOD CHANGED, NOT THE EFFORT.

The gain came from a better arrangement, not from harder work.

A hospital ward shows that the same method can serve a very different task.

In a ward, the method must protect safety as well as save time.

Scientific management brought measurement, method and training into management.

Taylor's ideas raised productivity in many factories.

Taylor's claims about particular results need careful handling.

The first weakness is that it treated workers as parts of a machine.

The second weakness is that it suited repeated tasks better than varied ones.

Taylor looked at the task; another writer looked at the whole firm.

³ Historians disagree about how accurate Taylor's reports were. Treat any single figure from his own accounts with caution.



FAYOL AND THE FOURTEEN PRINCIPLES

Fayol studied the work of the manager, and Taylor had studied the work of the worker.

Henri Fayol was a French mining engineer who became the head of a large company.

Fayol asked what managers do and what rules guide them.

Fayol divided the work of a firm into six groups, and one of them was management.

Fayol listed five elements of management, close to our four functions.

The table lines up Fayol's five elements with the four functions of Chapter II.

THE MAN AND HIS QUESTION

Henri Fayol (1841–1925) was a French mining engineer. He worked for a company that mined coal and made iron and steel, and in 1888 he became its managing director. The company was then in serious difficulty. Under his direction it recovered, and he later gave the credit to the way it was managed.⁴

Fayol wrote down what he had learned. He published *Administration industrielle et générale* in 1916. It reached English readers much later, in a translation of 1949. Taylor started with the worker and worked upward. Fayol started with the whole firm and worked downward. He wanted every manager, in any firm, to have a clear set of rules.

Fayol saw six groups of activity in any firm. They are technical work (making), commercial work (buying and selling), financial work (finding and using money), security work (protecting people and property), accounting work (recording), and managerial work (planning, organising and the rest). He believed that managerial work was the least understood of the six, and he gave it most of his attention.

Fayol named five elements of management: to plan, to organise, to command, to coordinate and to control. You will recognise them. They are close to the four functions of Chapter II. To command, which means to give orders and direct people, belongs to what we call leading. To coordinate, which means to fit the parts of the firm together, we place within organising. The idea that management is a process of connected activities comes largely from Fayol.

The match between Fayol's five elements and our four functions is easy to state in a table. Read each row as one pair.

FAYOL'S FIVE ELEMENTS AND THE FOUR FUNCTIONS

FAYOL'S ELEMENT	MEANING IN PLAIN WORDS	OUR FUNCTION
To plan	Look ahead and prepare a course of action	Planning
To organise	Build the structure of people and materials	Organising
To command	Give orders and direct people	Leading
To coordinate	Fit the parts of the firm together	Organising
To control	Check that results match the plan	Controlling

WHAT A PRINCIPLE IS

Principle. A general rule that guides action in many different situations.

A principle is a general rule, not a law of nature.

Fayol offered fourteen **principles** of management. A principle is a general rule that guides action in many situations. A principle is not a law of nature. Water always boils at the same temperature at sea level, but a principle of management holds with exceptions. Fayol himself said that management is a matter of proportion, and that no principle should be applied as an unbreakable rule.

⁴ The company is usually named as Commentry-Fourchambault. Accounts of its recovery come mainly from Fayol's own writing and should be read with care.

We call these rules **Fayol's principles**. Fourteen separate rules are hard to remember, so we number them in our own order and sort them into three groups: principles about work and order, principles about authority and command, and principles about people and spirit. The figure below shows the groups, and the sections that follow explain each rule in turn.

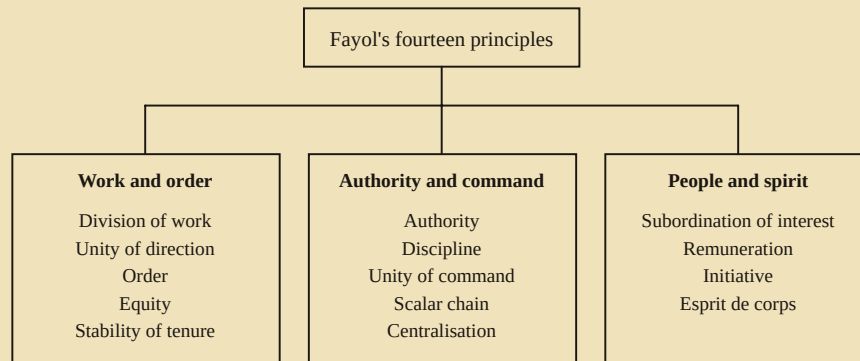


FIG. 6. Fayol's fourteen principles in three groups

PRINCIPLES ABOUT WORK AND ORDER

The first principle is **division of work**. Division of work means splitting a job into parts, so that each person does one kind of task. Return to the school timetable. If one teacher taught every subject to every class, that teacher would know each subject only slightly. When each teacher teaches one subject, each becomes more expert, and the school teaches better. In a bakery, one person may bake while another packs and sells. Each person becomes quick at one task.

The principle has a limit. If a job becomes very narrow, the worker may become bored, and the worker may not see how the task joins the others. Fayol saw the gain from division of work as real, but he did not ask for more division without end.

The second principle is **unity of direction**. It means that a group of activities with the same aim should have one head and one plan. Suppose a firm's production team has two plans at once: one to bake more bread and one to bake new kinds of cake. The team would pull in two directions. One head and one plan keep the effort together.

The third principle is **order**. Order means a proper place for each thing and each person. Fayol spoke of material order, which concerns tools and goods, and of social order, which concerns people and jobs. A bench where bags, labels and string lie in fixed places has material order. A chart that shows who holds each job has social order.

The fourth principle is **equity**. Equity means treating staff with fairness and kindness. Fayol argued that staff who feel fairly treated give loyalty and effort. Equity is not the same as equal treatment. A fair manager may treat two workers differently if their tasks and their needs are different, and can explain why.

The fifth principle is **stability of tenure**. Tenure means the length of time a person holds a job. Fayol advised that staff should not be moved or dismissed without good reason, because new staff need time to learn. A baker who leaves after four months takes the skill that the bakery has recently paid to teach.

Fayol's principles. Fourteen general rules of management set out by Henri Fayol.

The fourteen principles fall into three groups for easier learning.

Division of work. Splitting a job into parts so that each person does one kind of task.

The first principle is that dividing work raises output.

Division of work has a limit, because a very narrow job becomes dull.

Unity of direction. One head and one plan for each group of activities with the same aim.

Unity of direction means one plan and one head for each group of related activities.

Order. A proper, known place for each thing and each person.

Order means that each person and each thing has a proper place.

Equity. Fair and kind treatment of staff.

Equity means fair and kind treatment of staff.

Stability of tenure. Keeping staff in their posts long enough to learn and give good service.

Stability of tenure means keeping good staff and avoiding needless turnover.

Authority. The right to give orders and to use the firm's resources to get results.

Authority is the right to give orders, and it must come with responsibility.

Discipline. Obedience to agreed rules and respect for the agreements between firm and staff.

Discipline means obedience and respect for agreed rules, kept by good leaders.

Unity of command. One boss for each person: orders come from one superior only.

Unity of command means that each person receives orders from one superior only.

GO SLOWLY HERE. UNITY OF COMMAND AND UNITY OF DIRECTION SOUND ALIKE.

The two unities differ: one concerns a person's boss, the other a team's plan.

A hospital ward shows how hard unity of command can be to keep.

Scalar chain. The line of authority from the top of the firm to the lowest level.

The scalar chain is the line of authority from top to bottom.

Fayol allowed a direct link between equals, with their superiors' knowledge.

Centralisation. The degree to which authority is held at the top of the firm.

Centralisation concerns how much authority is held at the top.

The eleventh principle asks workers to put the firm's interest first.

PRINCIPLES ABOUT AUTHORITY AND COMMAND

The sixth principle in our list is **authority**, the right to give orders and to expect obedience. Fayol joined authority to responsibility, which is the duty to answer for results. Whoever receives authority also receives responsibility. A manager who has the right to order but no duty to answer for results may use power carelessly. A manager who answers for results but has no authority cannot act. Chapter X studies this pair in detail.

The seventh principle is **discipline**. Discipline means obedience to agreed rules and respect for the agreements between the firm and its staff. Fayol said that discipline depends on good leaders at every level, on clear agreements, and on sensible penalties used with fairness. Discipline that rests only on fear does not last.

The eighth principle is **unity of command**, and it is the one that the bakery's second trouble breaks. Unity of command means that each person receives orders from one superior only. When two people give different orders, the worker does not know whom to obey. One order is ignored, or the worker tries both and does neither well. Quarrels follow, because each superior believes the worker belongs to him or her.

Keep the two unities apart. Unity of command is about one person: who is that person's boss? Unity of direction is about a group: do all its members follow the same plan? A team can have unity of command, with every member holding a single boss, and still lack unity of direction if the boss gives the team two conflicting plans.

Consider a hospital ward, a setting where this principle is tested. A nurse may receive instructions from the ward's senior nurse about bed care and from a doctor about treatment. Both people have a right to speak, and the nurse serves two bosses in part. Hospitals handle this by dividing the subjects: the doctor decides treatment, and the senior nurse decides ward routine. The principle is not abandoned. It is applied to each kind of order.

The ninth principle is the **scalar chain**. The scalar chain is the line of authority that runs from the top of the firm to the lowest level. Orders travel down it, and reports travel up. In a firm with an owner, a supervisor and two workers, the chain has three levels. A message should normally pass through each level.

Fayol saw that a long chain can be slow. Suppose a baker needs help from a rider, and both sit on the same level but in different parts of the firm. To pass the message up and down the chain would waste days. Fayol therefore allowed a bridge, which he called a gang plank. The two may speak directly, provided their superiors know of it.

The tenth principle is **centralisation**. It concerns how far authority is held at the top and how far it is shared with lower levels. Fayol did not say that more is always better, or that less is. He said that the right amount depends on the firm, its size and its people. In a very small firm, the owner holds nearly all authority, and that suits the firm. Chapter X returns to this principle.

PRINCIPLES ABOUT PEOPLE AND SPIRIT

The eleventh principle is **subordination of individual interest to general interest**. Here to subordinate means to place second. When a worker's wish and the firm's good conflict, the firm's good should usually come first. Fayol knew that people forget this, from ignorance or ambition or weakness. He asked

managers to set an example and to keep agreements fair, so that sacrifice is not demanded from one side only.

The twelfth principle is **remuneration**. Remuneration means payment for work. It should be fair to the worker and to the firm, and it should reward effort that deserves reward. Pay that is too low drives good workers away. Pay that is too high may put the firm in danger.

The thirteenth principle is **initiative**. Initiative means the freedom to think out a plan and to carry it through. Fayol counted it among the keenest satisfactions of a person at work. A manager must sometimes give up personal vanity in order to let others act. In a bakery, a helper who suggests a new bread, and is allowed to try it, is using initiative.

The fourteenth principle is **esprit de corps**, a French phrase for team spirit.⁵ It means harmony and unity among the members of a firm. Fayol warned managers against dividing a team, and against the habit of sending written notes when a short talk would do. Where staff feel that they belong, they work better together.

HOW THE PRINCIPLES ARE USED

A manager can use the principles as a checklist. When a firm is in trouble, the manager runs through the fourteen and asks which one is being broken. Many problems of small firms have simple causes. Two bosses break unity of command. Unclear jobs break order. A worker with no freedom to try new ideas breaks initiative.

Critics have named three limits. The first is that the principles are general, and in a real case two of them may pull in opposite directions. Unity of command may conflict with the need to use expert advice. The second is that the principles were written from the experience of one man in large industrial firms, and he did not test them in other settings. The third is that they say little about how people feel. Even so, they remain a useful starting point, and many later writers built on them.

Fayol would have agreed with these cautions. He wrote that the principles are flexible and must be used with judgement. The skill of the manager lies in knowing which principle matters most in a given case, and in what proportion.

TAYLOR AND FAYOL COMPARED

Taylor and Fayol worked in the same years, but in different countries, and they did not copy each other. Taylor began at the level of the worker and the task. Fayol began at the level of the whole firm and the manager. Taylor's question was how a task should be done. Fayol's question was how a firm should be run. Their answers fit together like two halves of one picture, and a student gains most by seeing both.

On one point they agreed strongly. Both believed that management is not a gift that a few people are born with. Both believed that careful study can produce rules that anyone can learn. This belief, more than any single rule, is their shared gift to the subject. On other points they differed, and one difference deserves a closer look.

Remuneration means pay that is fair to workers and to the firm.

Initiative. The freedom to think out a plan and to carry it through.

Initiative means freedom to think out a plan and carry it through.

esprit de corps. Team spirit: harmony and unity among the members of a firm.

Esprit de corps means unity and good feeling among the members of a firm.

The principles form a checklist for diagnosing a troubled firm.

The principles have limits, and critics have named three.

The principles are best used as guides to be weighed, not rules to be applied blindly.

Taylor and Fayol looked at management from opposite ends.

The two men agreed that management can be taught, and disagreed on where to begin.

⁵ The phrase is French. It is pronounced roughly "es-PREE duh KOR". Use it with that meaning only: the spirit of a united group.

Functional foremanship. Taylor's plan to replace one foreman with several specialist foremen, each directing one part of a worker's task.

One of Taylor's own plans seems to break Fayol's rule of unity of command.

The table sets out the main differences between the two writers.

Taylor proposed a plan that he called **functional foremanship**. Functional foremanship means replacing a single foreman with several specialist foremen, each of whom directs one part of a worker's task. One foreman might direct speed, another the care of the machine, and another the order of the work.⁶ The idea has a logic: each foreman is an expert. But the worker now has several bosses. Fayol's rule of unity of command says that this causes confusion. Here two great writers disagree, and a manager must decide which risk is greater.

We now list the differences in a table. The first column names the feature, and the next two columns describe each writer.

TAYLOR AND FAYOL COMPARED

FEATURE	TAYLOR	FAYOL
Starting point	The worker and the task	The whole firm and the manager
Main tool	Time-and-motion study	Fourteen principles
Main concern	Output per worker	Sound running of the firm
View of orders	Several specialist foremen	One boss for each person
Main weakness	Treats workers as machines	Principles are general and can conflict

The comparison shows two views that a manager needs together.

A manager of a bakery can use both writers on the same day. In the morning the manager studies the packing bench, as Taylor would. In the afternoon the manager checks that each person has one boss, as Fayol would. The two views do not compete. They answer different questions.



Four later approaches added people, mathematics, whole systems and situations.

LATER APPROACHES TO MANAGEMENT

Taylor and Fayol gave management its first body of ideas. Their ideas were useful, but each had a gap. Taylor said little about what workers feel. Fayol said little about how a firm reacts to the world outside it. Later writers filled the gaps, and four approaches stand out. We take them in the order of their rise: the human relations approach, management science, the systems approach and the contingency approach.

THE HUMAN RELATIONS APPROACH

In the 1920s, researchers began a series of studies at the Hawthorne Works, a factory near Chicago owned by the Western Electric Company.⁷ They wanted to learn how conditions such as lighting and rest breaks affect output. The results surprised them. Output rose when the lighting improved. It also rose when the lighting was made worse. Something other than lighting was at work.

The researchers reached a new idea. The workers knew that they were being watched, and they felt that managers cared about them. They also formed a close group among themselves, with its own unwritten rules about how much to produce. From these findings grew the **human relations approach**. The human relations approach stresses that workers are people with feelings, and that attention, respect and group membership strongly affect how well they work.

The human relations approach began when a study of lighting found that people, not lights, changed output.

Human relations approach. An approach that stresses people's feelings, attention and group membership.

The researchers concluded that attention and group belonging affect how people work.

⁶ Taylor's plan is usually described as using several specialist foremen, some in a planning room and some on the shop floor. Check the number and the roles before citing them.

⁷ The studies ran from the middle 1920s into the early 1930s. Elton Mayo and colleagues from Harvard joined the work. Dates and details vary between accounts, and later critics have questioned the methods.

In one study, a small group of women who assembled telephone relays were moved to a separate room and watched, while the researchers changed their hours, breaks and pay in turn. Output rose again and again, whatever the researchers changed. The researchers decided that the workers' pride in being chosen, and their friendly relations with each other and with the observers, had raised their effort.

In another study, a group of men who wired telephone equipment was observed in a room of their own. Here the researchers found that the group had an unwritten rule about how much each member should produce. A member who produced too much, or too little, was teased or pushed back into line. The group, and not the manager, set the limit. This finding showed that an informal group can hold great power over how people work, a point we take up in Chapter XI.

You can see how this approach answers Taylor. Taylor believed that pay and a good method drive effort. The Hawthorne researchers found that feelings and groups matter as well. A worker who feels respected may work harder than one who is only well paid. We study this idea more fully in Chapter XI, where we ask what motivates people.

The approach also has limits. Critics say that the Hawthorne studies were not carefully controlled, and that their findings were stated too strongly. Critics also say that a contented worker is not always a productive one. Still, the approach changed management for good. After it, managers could no longer ignore how people feel.

MANAGEMENT SCIENCE

A different line of thought grew from the Second World War. Teams of scientists were asked to solve practical problems, such as how to place ships and aircraft for the best protection. They built mathematical models of the problem, tested different choices on paper, and recommended the best. After the war, firms borrowed the method. The result is called **management science**. Management science means using mathematical models and calculation to help managers solve problems.

The word **model** needs a short explanation. A model is a simplified picture of a real problem, drawn with numbers and rules. A bakery might build a model that shows how many loaves to bake each day. The model uses the average number of customers, the cost of ingredients, and the loss from unsold bread. Changing a number in the model shows what would happen, without any risk to the real bakery.

Management science works best on problems that have clear numbers: how much stock to hold, how to schedule work, how to plan deliveries, how to choose between projects. Computers made the calculations quick, and the approach spread. We use some of its tools in Chapter VIII, where we choose between alternatives using probability.

Its weakness is the other side of its strength. Many managerial problems cannot be reduced to numbers. How should a manager handle a worker who feels unfairly treated? What mathematics decides that? Management science is a strong tool, but it is not a whole theory of management.

TWO DIFFERENT KINDS OF SCIENCE

Beginners often confuse the two terms. Scientific management is Taylor's method of studying work. Management science is the later use of mathematical models. Both claim to use science, and the names are close, so we put them side by side in a table. The

The researchers used a small test room to watch a group of workers closely.

In a second study, the workers' own group set a limit on output.

The approach corrects Taylor's view that pay is the main reason people work.

The approach has limits: later critics doubted the studies and the idea that contented workers are always productive.

Management science. Using mathematical models and calculation to help managers solve problems.

Management science uses mathematics and computers to solve problems that can be measured.

Model. A simplified picture of a real problem, made with numbers and rules.

A model is a simplified picture of a problem that uses numbers.

Management science suits problems with clear numbers, such as stock, scheduling and transport.

Management science helps with measurable problems but says little about people.

Scientific management and management science have similar names but different meanings.

first column names the feature, and the next two columns describe each approach.

THE TWO APPROACHES THAT USE THE WORD SCIENCE

FEATURE	SCIENTIFIC MANAGEMENT	MANAGEMENT SCIENCE
When it appeared	Early twentieth century	Middle twentieth century
Main figure or source	Frederick Taylor	Wartime scientific teams and later researchers
What it studies	How a worker does a task	How a manager solves a problem
Main tool	Stopwatch and movement study	Mathematical model and computer study
Typical question	What is the best way to pack a loaf?	How many loaves should we bake tomorrow?
Main weakness	Treats workers as machines	Hard to use where numbers are missing

The table shows that one studies tasks and the other studies decisions.

Read the table across a row to compare the two on one feature. The pattern is plain. Scientific management studies the task at the level of the worker. Management science studies the problem at the level of the manager. Both rely on measurement, and that is why both carry the word science.

THE SYSTEMS APPROACH

Systems approach. Seeing an organisation as a set of linked parts that work together.
The systems approach sees an organisation as a set of linked parts working toward a goal.

The third later approach asks us to look at the firm as a whole. A **system** is a set of parts that are linked and that work together toward a purpose. A bicycle is a system. The wheels, chain, pedals and frame each do a different task, and the bicycle moves only when all of them work together. The **systems approach** treats an organisation in the same way: as a set of linked parts, such as departments, that cannot be understood one by one.⁸

Feedback. Information about results that returns to the firm to guide later work.
A system takes inputs, changes them, and delivers outputs.

The figure below shows how a business works as a system. First, the firm takes in inputs, the resources we met in Chapter I: people, money, materials, equipment and information. Next, the firm transforms them by baking, selling, or some other work. Finally, it delivers outputs, such as bread and profit. The results return to the firm as information, and this return is called **feedback**. The firm uses feedback to correct its next round of work.

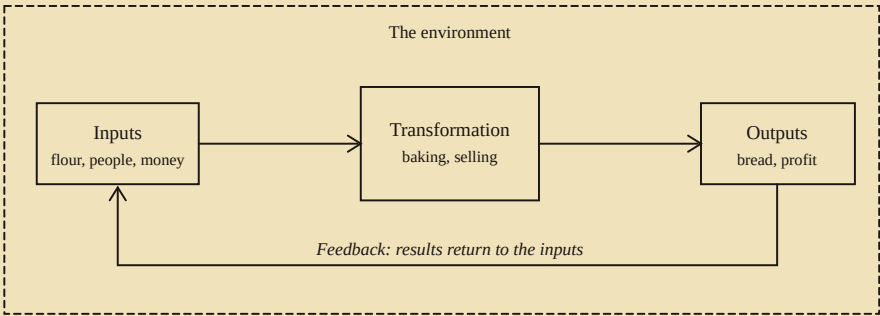


FIG. 7. A firm seen as a system, with its environment around it

The lesson of the systems approach is that a change in one part affects the others.

The lesson of this approach is that the parts depend on each other. If the baking team makes more bread, the packing bench must pack more. If the packing bench is slow, bread waits and goes stale. A manager who improves one part and

⁸ A Austrian-born biologist, Ludwig von Bertalanffy, developed general systems theory in the middle of the twentieth century. Management writers later applied it to organisations.

ignores the others may make the whole system worse. The approach also reminds us that the firm is surrounded by an environment, which we study in Chapter IV.

THE CONTINGENCY APPROACH

The last approach answers a question that all the earlier ones left open. Is there one best way to manage? The **contingency approach** says no. It holds that the best method depends on the situation. The word contingent means depending on something else.

The idea grew from research in the middle of the twentieth century. Researchers studied firms in different industries and found that successful firms were not organised in the same way. A firm in a fast-changing market did better with flexible roles and quick talk between people. A firm in a steady market did better with fixed roles and clear rules.⁹ Neither form was best everywhere. Each fitted its situation.

Contingency thinking turns every managerial question into a pair of questions. What is the situation? Which method fits it? Size, technology, the environment and the people involved all change the answer. A rule that works in a large factory may fail in a small bakery. The approach is more demanding than the earlier ones, because it offers no single rule. It offers a way of thinking.

The next table gives three situations and the method that would probably fit each. It is an illustration of the way of thinking, not a rule. Read each row as one pair: the situation first, and then the method that fits it.

SITUATION AND FITTING METHOD

SITUATION	A METHOD THAT PROBABLY FITS	WHY
A bakery makes the same bread every day, in a steady market	Fixed roles and written routines	The work repeats, so routines save effort
A bakery must create new products quickly for a festival	Flexible roles and frequent talk	The work changes, so rigid roles slow it
A bakery of four people compared with one of forty	Direct talk with the owner, then more levels and rules	A larger group needs more structure to keep order

Notice that the same bakery appears in all three rows, and the fitting method changes. The manager's task is to read the situation first. A manager who applies one favourite method to every case will sometimes be right and sometimes be wrong, and will not know why.

Do not suppose that the contingency approach has defeated all the others. It has joined them. A manager facing a slow packing bench uses Taylor's study. A manager facing two bosses uses Fayol's rule. A manager facing unhappy staff uses the human relations idea. A manager facing a stock problem uses management science. A manager considering all of these together uses the systems approach, and then the contingency approach decides which to prefer. The ideas in this chapter that are old are not dead. They form a toolbox.

Contingency approach. The view that the best way to manage depends on the situation.

The contingency approach says that the best method depends on the situation.

Researchers found that firms in changing markets and in steady markets need different structures.

Contingency thinking asks a manager to match the method to the situation.

A table shows how different situations call for different methods.

The table shows that the right method changes when the situation changes.

Each approach is useful, and a good manager draws on all of them.

⁹ Studies by Tom Burns and G. M. Stalker, by Joan Woodward, and by Paul Lawrence and Jay Lorsch are often named here. Check each source before citing it.

One more idea belongs in this history, though it is not one of the four approaches.

A table helps a student keep the schools in order.

The table shows a pattern: each school corrected a weakness in an earlier one.

We now ask what kind of knowledge all these ideas amount to.

People disagree about what management is, so we need tests.

We need three tests, one for each word.

A subject is a science when it has organised knowledge that has been tested.

A NOTE ON BUREAUCRACY

At about the same time as Taylor and Fayol, a German scholar named Max Weber (1864–1920) studied how large organisations, such as governments and armies, were run. He described a form of organisation built on written rules, fixed ranks and written records, in which people are chosen for their skills and not for their family or friends. Weber saw that such a form could be fair and steady. Later writers pointed out that it could also become slow and rigid. We study this form in Chapter IX, where we look at the ways in which organisations are designed.

THE SCHOOLS SIDE BY SIDE

Five schools have now been described, and they are easy to confuse. The next table sets them in order and gives for each the main belief, a strength and a weakness. Read it across, one row at a time. Use it as a map when you revise the chapter.

THE MAIN SCHOOLS OF THOUGHT COMPARED

SCHOOL	MAIN BELIEF	A STRENGTH	A WEAKNESS
Scientific management	Study work to find the best method	Measurement and method	Treats workers as machines
Fayol's principles	A firm needs general rules	A checklist for any firm	Rules can conflict
Human relations	Feelings and groups matter	Attention to people	Doubts about the studies
Management science	Models can solve problems	Precision with numbers	Weak where numbers are missing
Systems and contingency	Parts are linked and the situation decides	Sees the whole and the setting	Offers no single rule

Look at the last column and the first. Each weakness was the opening for a later school. Treating workers as machines opened the way for human relations. A lack of attention to numbers opened the way for management science. A search for one best rule opened the way for contingency thinking. This is the pattern that the timeline at the start of the chapter promised.

We have now told the story. One question remains. When we gather these ideas together, what kind of subject is management? We take the question next.



IS MANAGEMENT A SCIENCE, AN ART OR A PROFESSION?

THREE WORDS, THREE TESTS

People call management by three names. Some say it is a science, some an art, and some a profession. A dispute of this kind is settled by tests, not by opinion. We shall state one test for each word, and then apply it.

The first word is **science**. A subject is a science when its knowledge is organised, built on facts that have been tested, and able to explain cause and effect. It should also give results that others can repeat. Physics is a

science. When you heat water in the same conditions, you get the same result each time.

The second word is **art**. A subject is an art when success depends on personal skill, practice and judgement, and not only on knowledge. A good cook and a good musician use rules, but their skill comes from practice and from a feeling for the situation. Two cooks with the same recipe may produce different food.

The third word is **profession**. A profession is an occupation that rests on a body of knowledge learned through long training, a code of conduct that members accept, and a duty to serve others. Doctors and lawyers are professions. Their members must pass tests, and a licensing body may remove those who break the code.

A subject is an art when success depends on personal skill and judgement.

Profession. An occupation with trained skill, a code of conduct and set standards.

A subject is a profession when it has trained members, a code and a standard.

TESTING MANAGEMENT AGAINST EACH WORD

First, is management a science? In part, it is. Managers use organised knowledge, such as the principles and models in this chapter, and many ideas have been tested by research. But management deals with people, who differ and change. The same method may succeed in one firm and fail in another. Management therefore does not give the repeatable results of physics.

Management meets the test of science in part.

Second, is management an art? Here the answer is stronger. Two managers with the same knowledge may do very different work. One knows when to give an order, and when to wait. One notices an unhappy worker before the worker complains. This sense of the right moment comes from practice. The contingency approach says as much: the manager must judge the situation.

Management meets the test of art fully, because success depends on judgement.

Third, is management a profession? In part again. Management has a growing body of knowledge, and universities teach it. Many managers belong to associations, and some have codes of conduct. But in most countries a person may become a manager with no licence and no set course of study. A manager who behaves badly is not removed by a professional body. Management is moving toward a profession but has not reached it.

Management meets the test of profession only in part.

MANAGEMENT MEASURED AGAINST THE THREE TESTS

TEST	WHAT IT ASKS	HOW FAR MANAGEMENT MEETS IT
Science	Is knowledge organised, tested and repeatable?	In part: knowledge is organised, but people vary
Art	Does success depend on personal skill and judgement?	Strongly
Profession	Is there trained entry, a code and a standard?	In part: growing, but no licence in most places

The table gives a fair verdict. Management uses science, because it draws on tested knowledge. It is practised as an art, because skill and judgement decide the result. It is moving toward a profession, because training and codes are growing. A person who asks which one word is right has asked the wrong question. The better question is how much of each is present.

The table summarises the answer: management uses science, practises as art and moves toward a profession.

IN SHORT

Management is not only a science, only an art, or only a profession. It applies organised knowledge, with personal judgement, in an occupation that is moving toward professional standards.

Why the answer matters: it tells a student what to learn.

The answer also tells the student what to learn. Knowledge of the principles is necessary, because it is the science. Practice and reflection are necessary, because they build the art. A sense of duty to staff, customers and society is necessary, because it belongs to a profession. Chapter V takes up that duty.



A WORKED CASE: TWO TROUBLES AT THE BAKERY

The case applies two ideas from this chapter to the bakery's third month.

This case applies the chapter to one situation. It has five parts: the situation, the question, the reasoning, the conclusion and a check. It is not a model answer to copy. It shows one way of thinking that you can use on other cases.

The case states the facts of month 3 before any reasoning begins.

Case file, illustrative. Month 3. The Morning Loaf Bakery has four staff: two bakers, one counter assistant and one rider. Output is 250 loaves a day, produced in 32 staff hours. Packing a loaf at the counter bench takes 40 seconds. The owner moves the bags, labels and string close to the packer, and packing then takes 30 seconds. Until now, two people have given orders to the counter assistant. The owner decides that from now on only one will.

The first step is to name each trouble and match it to an idea.

Question. What do the ideas of Taylor and Fayol say about each trouble? And how much does the change at the bench gain?

The packing trouble belongs to scientific management.

Reasoning, first trouble. The packing bench is a task, and Taylor studied tasks. The owner timed the work, found movements that were not needed, and moved the supplies. That is a time-and-motion study in small form. Now we measure the gain. Before the change, one loaf took 40 seconds, so one person could pack $3,600 \div 40 = 90$ loaves in an hour. After the change, one loaf takes 30 seconds, so the same person can pack $3,600 \div 30 = 120$ loaves in an hour. Output per hour has risen by 30 loaves, which is a rise of one third.

At 250 loaves a day, the time saved at the bench is about 42 minutes.

We can also count the time saved in a day. At 250 loaves a day, packing took $250 \times 40 = 10,000$ seconds before the change. It takes $250 \times 30 = 7,500$ seconds after the change. The saving is 2,500 seconds, which is about 42 minutes. The counter assistant can use that time to serve customers.

The second trouble belongs to Fayol's unity of command.

Reasoning, second trouble. The counter assistant received orders from two people. This breaks unity of command, which says that each person should have one superior. The cost is confusion. When one person says to pack first and another says to serve first, the assistant must choose, and whichever choice the assistant makes will displease someone. By naming one person to give orders, the owner restores the principle.

The conclusion is that two different ideas solved two different troubles.

Conclusion. The bench trouble was a trouble of method, and the cure came from studying the task. The command trouble was a trouble of authority, and the cure came from a clear line of command. The two troubles needed two different ideas, which shows why a manager needs more than one approach.

The check confirms that the arithmetic and the logic agree.

Check. The time per loaf fell from 40 to 30 seconds, a cut of $10 \div 40 =$ one quarter. If the time falls by one quarter, the output per hour rises by one third, because $1 \div 0.75 = 1.33$. This agrees with the rise from 90 to 120 loaves, which is also one third. The figures are consistent. In words, the check on the second trouble is that each person now has exactly one boss.



SUMMARY

A school of thought is a group of ideas that share one main belief. New schools appear because earlier ones left problems unsolved. Scientific management, from Frederick Taylor, studies work by measurement to find the best method, and its main tool is the time-and-motion study. Fayol's principles are fourteen general rules for running a firm, and they fall into groups about work and order, authority and command, and people and spirit. Unity of command means one boss for each person.

The human relations approach added feelings and groups. Management science added mathematical models. The systems approach added the view of the firm as linked parts with feedback. The contingency approach added the idea that the best method depends on the situation. Management is partly a science, strongly an art, and moves toward a profession.

The summary restates the main ideas of the chapter in a few lines.

The later approaches added people, models, wholes and situations.

THINK IT THROUGH

Consider a small clinic with three staff. A clinic assistant takes orders about patient files from two doctors. The files are kept in boxes with no labels, and no one knows where a file is. Name two of Fayol's principles that the clinic breaks. For each, say what trouble it causes and what the clinic could do. Then ask which approach from this chapter would help most with the files.

Now consider a firm of four people. Of Fayol's fourteen principles, which matters least in so small a firm? You could argue for scalar chain, because the owner speaks to everyone directly. You could argue for division of work, because four people may share tasks. Choose one, give your reasons, and then consider what would change if the firm grew to forty people.

The first problem asks you to apply two principles to a new case.

The second problem asks you to decide which principle matters least in a small firm.

WHERE PEOPLE GO WRONG

Error: The two terms mean the same thing. *Correction:* Scientific management is Taylor's study of how workers perform tasks. Management science is the later use of mathematical models to solve managerial problems. Their names are alike, but their subjects and tools differ.

Error: Fayol's principles are laws that every firm must obey. *Correction:* Fayol said that the principles are flexible and must be weighed with judgement. A hospital ward cannot apply unity of command in the way a small shop does.

Error: The newest approach is the best, and the old ones are out of date. *Correction:* Each approach solves a different kind of problem. Managers still use time study, still follow many of Fayol's rules, still pay attention to feelings, still build models, and still ask what fits the situation.

Error: Unity of command and unity of direction are two names for one rule. *Correction:* Unity of command concerns one person and one boss. Unity of direction concerns one group and one plan.

A common error is to think that scientific management and management science are the same.

A second error is to treat Fayol's principles as laws that never bend.

A third error is to believe that a later approach cancels the earlier ones.

A fourth error is to mix up unity of command and unity of direction.

IN HISTORY

In 1888 Henri Fayol became managing director of a French mining and iron company that was in serious financial difficulty. He later wrote that the firm recovered, and that the cause was not luck but method: a clear plan, a careful choice of managers, and sound rules of

Fayol's firm recovered under his direction, and he gave the credit to management.

administration.¹⁰ Whatever the exact figures, the story shows why Fayol's work mattered. He tested his ideas in a real firm, in a real crisis, and then he wrote them down so that others could use them.

SOURCES

Taylor's book of 1911, *The Principles of Scientific Management*, is the landmark source for scientific management. It matters because it set out the idea that work can be studied and improved by measurement. Fayol's book, *Administration industrielle et générale* (1916), is the landmark source for the principles of management. It matters because it treated management as a body of teachable rules.

REVIEW QUESTIONS

Taylor and Fayol are the two landmark sources of this chapter.

The review questions test recall and understanding of the chapter.

Recall the idea of a school.

Recall Taylor's method.

Recall three terms with examples.

Recall the principles by group.

Separate two similar principles.

Explain one principle in depth.

Compare two approaches in a table.

Recall three later approaches.

Judge the three names for management.

1. What is a school of thought, and why does a new school usually appear?

2. Explain scientific management in your own words. What was Taylor's main tool?

3. Define soldiering, rule of thumb and piece rate, and give a fresh example of each.

4. List Fayol's fourteen principles in their three groups.

5. Distinguish unity of command from unity of direction, with an example of each.

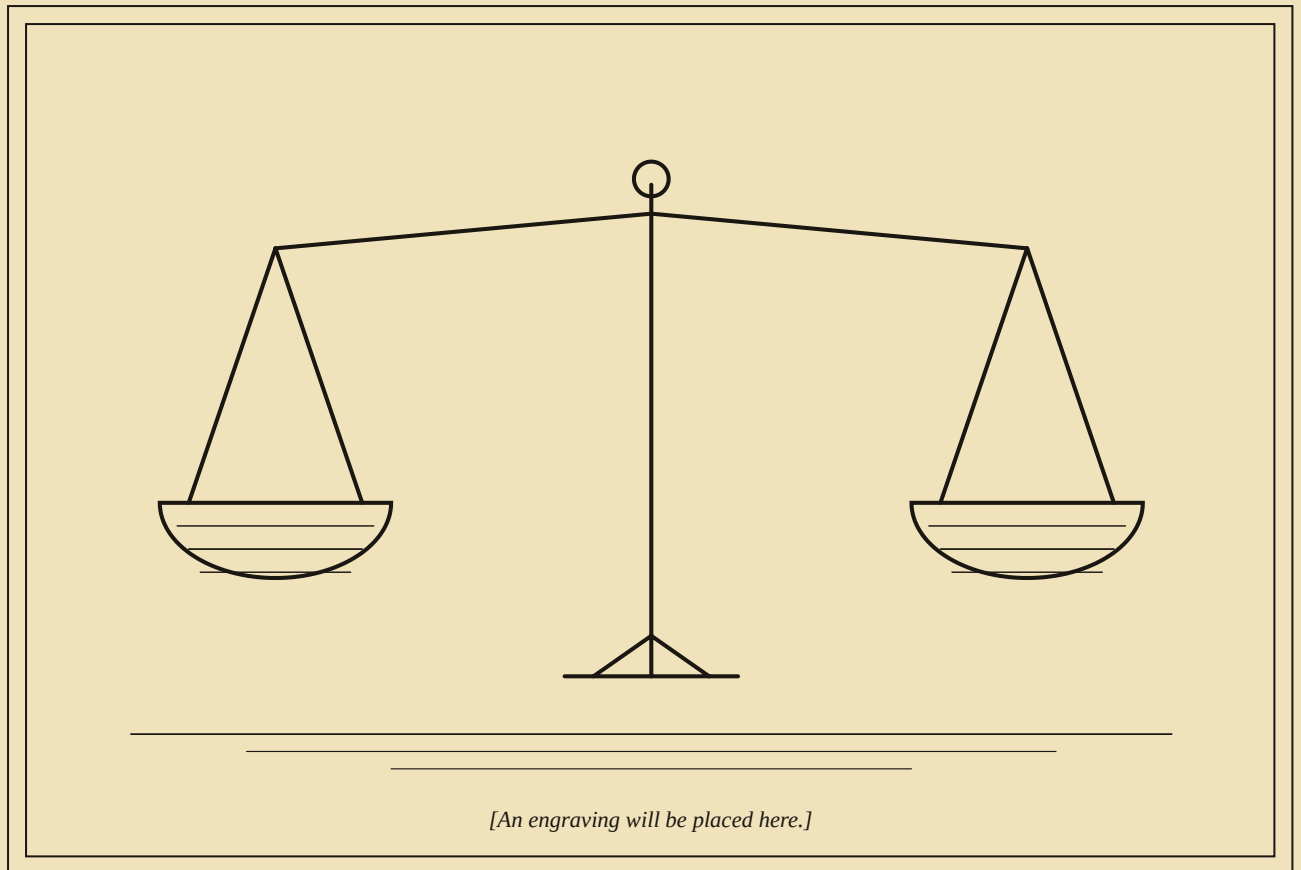
6. Explain the scalar chain, and say why Fayol allowed a gang plank.

7. Compare scientific management and management science in a short table.

8. Describe the human relations approach, the systems approach and the contingency approach, each in two sentences.

9. Explain why management can be called a science, an art and a profession, and say which description fits least well.

¹⁰ This account rests mainly on Fayol's own writing. Historians have not always been able to check the details independently.



CHAPTER IV

The Outside World

Where every organisation meets forces it did not choose



An organisation does not work alone. Customers, prices, laws, new machines and rival firms act on it every day, whether the manager watches them or not. This chapter names these outside forces, shows how they reach the work inside, and describes the two ways a manager can answer them: to adapt, or to influence.

A stall owner loses half her customers in one day and cannot find any mistake of her own.

She checks her stall, finds nothing wrong, and then looks outside it.

Every organisation has such a Monday, because results depend on what happens outside as well as inside.

The chapter asks three questions: what is outside, how does it reach us, and what can a manager do?

An organisation has a boundary: managers direct what lies inside it and cannot direct what lies outside it.

Boundary. The line between what managers can direct and what they cannot.

The word environment has a wider meaning here than in daily speech.

GO SLOWLY HERE.

Environment ■. Everything, outside and inside the organisation, that affects its work and results. In daily speech: nature.

A second picture from the market shows how a seller reads the surroundings every day.

The environment has two parts: the outside part, which the manager cannot control, and the inside part, which the manager can.

External environment. The forces outside the organisation's control.

Hasina sells tea and biscuits from a small stall beside a bus stop. On Sunday she sells about 120 cups. On Monday she sells about 60. Her tea is the same as before. Her price is the same. Her helper arrives on time, and the kettle works well. Nothing in her stall has changed, yet half of her customers have gone.

She checks her cups, her stock and her cash box. All are in order. Then she looks up the road. The road authority has moved the bus stop to the next street, 200 metres away. The people who waited beside her stall every morning now wait somewhere else. Her work was good. The cause of her loss lay outside the stall.

Every organisation, small or large, meets a day like this one. A manager who watches only the work inside the organisation will be surprised again and again. Good tea does not protect a stall from a moved bus stop. Good bread does not protect a bakery from a rise in the price of flour. Results come from two sources: the decisions made inside, and the forces that act from outside.

This chapter asks three questions. What lies outside an organisation, and how can we sort it? How does a change outside reach the work inside? And what can a manager do about it? The answers prepare the ground for planning in Chapter 6, because a plan is only as good as the manager's view of the future outside.

4.1 INSIDE AND OUTSIDE

In Chapter 1 we defined an organisation as a group of people working together for shared goals. Every organisation has a **boundary**, which is the line between what its managers can direct and what they cannot. Inside the line are the people, money, materials and equipment the manager controls. Outside the line is everything else. For Hasina, the kettle and the helper are inside. The bus stop is outside.

Managers use the word **environment** in a special way. In daily speech, the word often means nature: air, trees and rivers. In this book it has a wider meaning. The environment of an organisation is everything, outside it and inside it, that affects its work and its results.¹ The moved bus stop belongs to Hasina's environment. So does the skill of her helper. Both affect how many cups she sells.

Think of a vegetable seller in a street market. Each morning she looks at the sky, the number of people in the street and the prices that other sellers write on their boards. She cannot change the rain. She can bring a cover for her goods, which is a change inside her stall. She can also stand near the busiest corner, or agree with her neighbours to sell at the same price. In this way she already reads the outside, tests what lies in her power, and chooses an answer. A manager does the same work with more care and with more at stake.

We divide the environment into two parts. The **external environment** is the set of forces outside the organisation's control. These are the forces that begin beyond the boundary: customers, prices in the market, laws, rival firms and new technology. The manager did not choose them and cannot direct them. For a bakery, the price of flour in the market belongs here.

¹ The two meanings are related. In both, the environment is what surrounds a living thing or a business and shapes how it fares. A manager uses the word for the whole surrounding set of forces, not only for nature.

The **internal environment** is the set of owners, staff and culture inside the organisation. The owners decide how much money to risk and what to aim for. The staff bring skill and effort. The culture is the set of habits and shared beliefs, such as the habit of starting work at five in the morning. A manager can change these parts, though not always quickly. A habit that has lasted ten years does not change in a week.

The internal environment holds the parts the manager can direct: owners, staff and culture.

Internal environment. The owners, the staff and the culture inside the organisation.

The figure shows the boundary, the internal environment and the outside forces that cross the line.

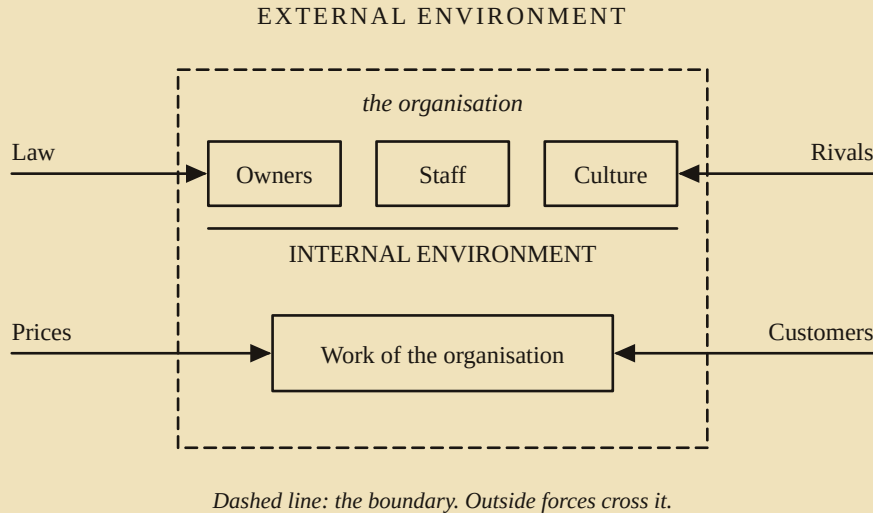


FIG. 8. The organisation, its boundary, and the forces that cross it

Of the three inside parts, culture is the hardest to see. It is the set of habits and beliefs that the staff share, such as whether they report a mistake at once or hide it. Culture decides how the staff respond when an outside force arrives. In one bakery, a worker who sees that the flour is of poor quality tells the owner immediately. In another, the worker says nothing. The same outside force, a poor delivery, then produces two different results. The internal environment therefore shapes the effect of the external one.

Culture is the least visible inside part, yet it shapes how the staff meet every outside force.

The table below compares the two parts. Read it across each row. The first row says where each part lies. The second says what the manager can do about it. The third gives an example from the bakery that we shall follow through the book.

The two environments differ in where they lie, in who controls them, and in how a manager can respond.

THE TWO ENVIRONMENTS COMPARED

QUESTION	INTERNAL ENVIRONMENT	EXTERNAL ENVIRONMENT
Where does it lie?	Inside the boundary	Outside the boundary
Can the manager direct it?	Yes, though change may be slow	No; the manager can only try to influence it
Parts	Owners, staff, culture	Economic, social, political-legal, technological, competitive and global forces
Bakery example	Four staff who start at five in the morning	The price of flour in the market

The boundary is not a wall: forces cross it all the time, and some inside parts are hard to change.

A stakeholder is any person or group affected by the organisation, and some stakeholders also affect it.

Stakeholder. Anyone who is affected by the organisation, and may also affect it.

A stakeholder is a party, while a force is an influence; each stakeholder carries one or more forces.

The line matters because it shows where effort can repair a problem and where only a response is possible.

The bakery of the case file already has both environments in its first month.

The boundary is not a wall. Forces cross it in both directions, as the arrows in the figure show. A flour mill raises its price, and the change enters the bakery. The bakery sells a loaf, and a customer carries an opinion away. Notice also that the dividing line is one of control, not of place. The bakery's rented shop is a place outside the owner's mind, yet the owner controls its use. A customer may stand inside the shop and still belong to the external environment, because the owner cannot direct the customer's choice.

Managers also need a word for the people and groups who stand around the organisation. A **stakeholder** is anyone who is affected by what the organisation does, and who may also affect it in return. The bakery has many stakeholders: its customers, its staff, its owner, its flour supplier, its neighbours and the tax office. Some stakeholders lie inside the boundary, such as staff and owners. Others lie outside it, such as customers and suppliers.

Do not confuse a stakeholder with a force. A stakeholder is a person or a group. A force is a pressure that acts on the organisation. The two are linked. The flour supplier is a stakeholder, and the supplier carries a force: the price of flour. The customers are stakeholders, and they carry another force: their choice to buy or not to buy. In Chapter 5 we shall ask what an organisation owes to its stakeholders. Here we only use them to find the forces.

Why does the line matter? It tells a manager where effort can change a result. If the cause of a problem lies inside, the manager can repair it: train the staff, mend the oven, correct the schedule. If the cause lies outside, no repair is possible, and the manager must choose a response. A manager who mistakes one for the other wastes effort. Hasina could have blamed her helper for the fall in sales. She might have replaced the helper, and nothing would have improved, because the cause was the bus stop.

The bakery of our case file shows both environments from its first month. Its owner opened it as a sole proprietor, with four staff: two bakers, one counter assistant and one rider. The owner's savings of Tk 600,000, the staff and the habit of early baking are inside. The price of flour, the customers who pay Tk 70 for a loaf, the owner of the building who receives Tk 40,000 in rent each month, and the rules for food shops are outside. Notice that rent is a cost inside the books but a force outside the boundary: the owner cannot direct how the rent is set.

IN SHORT

The environment is everything that affects an organisation. The part outside its control is the external environment. The part inside it is the internal environment. Stakeholders are the people and groups who carry the forces.

Test your grasp: sort five items into inside and outside.

You should now be able to say: the environment has two parts; the manager can direct the internal part but only influence the external part; and a stakeholder is a party that carries a force. As a test, sort five items into the two parts: the skill of the baker, a new tax on packaged food, the habit of cleaning the oven each night, the price of sugar, and the owner's savings. The skill, the habit and the savings are internal. The tax and the price of sugar are external.



4.2 THE EXTERNAL ENVIRONMENT

Outside forces are many and mixed. A manager who tries to watch them all at once will see none of them well. So managers sort them into kinds. This book uses six: economic, social, political-legal, technological, competitive and global. The sorting is a tool. It does not describe six separate boxes in the real world. Real forces overlap, and one event can belong to two kinds. The figure below shows the six kinds around one organisation.

Outside forces are many, so managers sort them into six kinds to see them more easily.

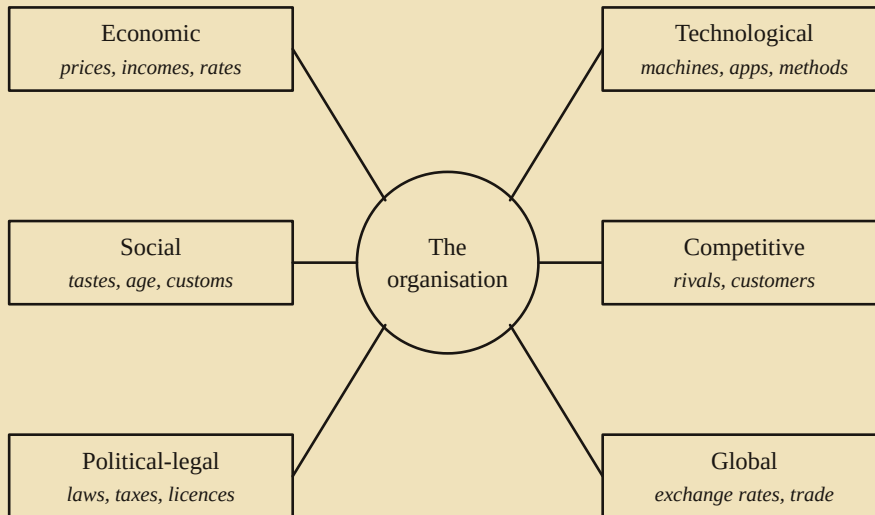


FIG. 9. The six kinds of outside force acting on one organisation

How does a manager sort a new force? Three questions are enough. What has changed? Who or what carries the change? And to which of the six kinds does it belong? Apply them to Hasina's Monday. What changed? The place of the bus stop. Who carries it? The road authority, which is a public body that makes decisions for everyone. Which kind is it? A political-legal force, because it is a decision of government. The force then reached the stall through its customers, as the next section explains.

A force can be sorted by asking three short questions: what changed, who carries it, and which kind it is.

ECONOMIC FORCES

Economic forces are the outside conditions of money and trade. They include the general level of prices, the incomes of households, the cost of borrowing money, and the number of people who have work. They affect what the organisation pays and what its customers can pay. Think of a family whose income falls. It buys fewer treats and chooses cheaper food. A shop in that street feels the change within a week.

Economic forces are the conditions of money and trade that change what things cost and what people can buy.

Economic forces. Outside conditions of money and trade, such as prices, incomes and interest rates.

Economic forces reach a firm through its costs and through its customers' spending.

The same forces act in two directions. They act on cost: when the price of flour rises, the bakery pays more for each loaf. They act on demand: when household incomes rise, more families can afford fresh bread, and fewer bake at home. A general rise in prices is called inflation. When inflation is high, the bakery's costs rise, and its customers also feel poorer. The two effects can arrive together.

The cost of borrowing is also an economic force. When banks charge more interest on loans, a firm that wishes to buy an oven on credit must pay more for the same oven. Some firms then delay their purchases. When banks charge less, more firms buy. The bakery of our case file used its owner's savings to start, so it

The cost of borrowing is another economic force, and it reaches firms that need money for machines.

An economic force is neither good nor bad in itself; the same change can help one firm and harm another.

Social forces are the changing numbers, tastes, habits and values of the people around the organisation.

Social forces. Outside changes in people's numbers, ages, tastes, habits and values.

Tastes and customs change what people buy, and they also change who is available to work.

Social forces usually change slowly, so a manager who watches them early has time to respond.

Political-legal forces are the laws, rules, taxes and policies of governments that apply to the organisation.

Political-legal forces. Outside laws, rules, taxes and government policies that apply to the organisation.

A new law reaches an organisation as a new duty, a new cost, or a new limit.

Political-legal forces are not only limits; governments also support firms, and stable rules help planning.

feels this force less than a firm that borrowed. A manager must remember that the same force can be large for one firm and small for another.

An economic force is not good or bad in itself. A rise in the price of flour harms the bakery. The same rise helps the farmer who grows the wheat and the mill that holds a large stock. When managers classify a force, they ask a second question: for whom is this a threat, and for whom is it an opportunity? For many forces, the answer is different for different firms.

SOCIAL FORCES

Social forces are the outside changes in how people live and what they value. They include the number of people in an area, their ages, their education, their tastes and their customs. A bakery near a new housing estate gains customers because the number of people nearby has grown. A bakery in a district where young people leave for work in other cities may lose them.

Customs change demand in regular ways. Festival seasons, such as Eid, change what families buy and when. Tastes change more slowly. If more families begin to prefer brown bread because they think it is healthier, a bakery that bakes only white bread loses part of its market. Social forces also reach the labour market. The education and the skills of the young people in a town decide who can be hired as a baker or a counter assistant.

Social forces usually move slowly. A change in age or taste takes years. This is useful to the manager, because early observation leaves time to respond. There are exceptions. A fashion or a rumour can spread in days. A report that a certain food is unsafe can empty a shop in a week, even if the report is false. Managers therefore watch both the slow movements and the sudden ones.

POLITICAL-LEGAL FORCES

Political-legal forces are the laws, rules, taxes and government policies that apply to the organisation. They also include the stability of the government itself. A law may forbid an action, such as selling unsafe food. It may require an action, such as keeping written records. It may charge the organisation money, as a tax does. Laws differ from one country to another. This book takes Bangladesh as its default and says so when another system differs.

Consider a clinic, a new setting for the idea. Suppose a new law requires every clinic to keep a written record of each patient visit. The law is a political-legal force. The clinic did not choose it and cannot ignore it. Each nurse must now spend minutes on every visit writing the record. The clinic may need a clerk. Its costs rise, and the time for care falls. In the bakery, the corresponding force is a labelling rule for packaged goods, which the case file records in Month 4.

Political-legal forces are not only limits. A government may support a type of business with training schemes or lower charges, and clear and stable rules help a manager to plan. A government that changes its rules often makes planning hard, even when each rule is fair. A manager therefore watches two things: what the rules are, and how often they change.

TECHNOLOGICAL FORCES

Technological forces are the new machines, methods and knowledge that exist outside the organisation. The organisation can use them, and its rivals can use them too. Technology reaches a firm through purchase. The bakery's new mixer in Month 2 is an example. The mixer was designed and built elsewhere. The bakery bought it, and its output rose from 200 to 250 loaves a day with the same 32 staff hours.

The same technology can be an opportunity or a threat. A bakery that adopts a mixer gains speed. A bakery that does not adopt it may fall behind its rivals. Technology also changes how customers behave. Mobile payment applications let a customer pay without cash. Ordering by telephone or by message lets a customer buy without visiting the shop. Each change creates a possible gain for the firm that responds early.

At its strongest, new technology can make an old product unnecessary. A firm that makes a product by an old method may find that customers no longer want the product at all. The history at the end of this chapter tells of one famous firm that faced such a change. For now, remember that technology is a force that managers must watch even when their own product seems safe.

COMPETITIVE FORCES

Competitive forces are the outside pressures from the firms and groups that share the market with the organisation. The most plain pressure comes from rivals, who sell the same product to the same customers. A second pressure comes from customers, who can compare prices and move to another seller. A third comes from suppliers, who may raise their prices if the buyer has no other source. A fourth comes from new firms that enter the market, and a fifth from substitutes.²

The bakery in the case file meets such a force in Month 4: a competitor opens 300 metres away. The competitor does not enter the bakery. Its effect arrives through the customers, who now have a second choice at a short walk. If the competitor sells a similar loaf at a lower price, the bakery may lose customers. If the competitor sells a different product, the effect may be small. The manager must find out which case applies.

Competition has exceptions. Rivals sometimes gain from one another. Several bakeries in one street may attract more buyers than a single bakery would, because customers know where to look for bread. Rivals may also buy flour together to obtain a lower price. A manager who sees every rival only as an enemy may miss these gains.

GLOBAL FORCES

Global forces are the outside influences that come from other countries. They include the prices of goods traded across borders, the exchange rate between currencies, the rules of trade, and events in other lands. A firm need not sell abroad to feel them. Suppose the flour used by the bakery is made from wheat bought from other countries. Then a change in the world price of wheat will reach the bakery's flour price after some time.

The exchange rate is the price of one currency in another. If the taka falls in value against the dollar, then an item priced in dollars costs more taka. Imported machines, spare parts and ingredients all become dearer. A factory that sells

Technological forces are the new machines, methods and knowledge available outside the organisation.

Technological forces. Outside advances in machines, methods and knowledge that an organisation can use or must face.

The same new technology can be an opportunity for the firm that adopts it and a threat for the firm that does not.

New technology can destroy an old business, as the history at the end of this chapter shows.

Competitive forces are the pressures from rivals, and also from customers, suppliers and others who shape the market.

Competitive forces. Outside pressures from rivals, customers, suppliers and others who shape the market.

A rival opening nearby, such as the competitor 300 metres from the bakery, makes it easier for customers to leave.

Competition is not always a fight: firms may also share costs or buy together, and a rival may increase the number of customers in the area.

Global forces are the influences that come from other countries and from trade between countries.

Global forces. Outside influences from other countries and from trade between countries.

A change in the exchange rate changes the taka price of every imported item.

² A substitute is a different product that meets the same need, such as rice or biscuits for bread. The idea that these five pressures together shape an industry comes from Michael Porter (1979).

Global forces overlap with the other five, so one event often belongs to two kinds.

The table gives one example of each force for the bakery and for a clinic.

Forces often act together and affect one another, so managers must watch the whole set.

A manager gives each force a second label: opportunity, threat, or both.

Opportunity. An outside change that could help the organisation if it acts.

Threat. An outside change that could harm the organisation if it does not act.

The same force can be an opportunity for one firm and a threat for another, as three examples show.

clothes to buyers in other countries depends on global forces more directly still, since its customers live abroad. Even a one-shop bakery, however, is not free from them.

Global forces overlap with the others. A foreign firm that opens in Dhaka is a global force and also a competitive one. A foreign government that changes a tax is a political-legal force that acts as a global one. The sorting into six kinds is a help to thought, not a rule of nature. When an event fits two kinds, the manager writes both names and moves on to the more useful question: what will it do to our work?

The table below shows the six kinds side by side. Read each row from left to right. The first column names the force. The second states what it covers. The third and fourth give an example for the bakery and for a clinic. The clinic column shows that the same kinds of force act on very different organisations.

SIX KINDS OF OUTSIDE FORCE, WITH EXAMPLES

FORCE	WHAT IT COVERS	BAKERY EXAMPLE	CLINIC EXAMPLE
Economic	Prices, incomes, interest rates	Flour price rises	Patients' incomes fall
Social	Numbers, ages, tastes, customs	Families prefer brown bread	The town's population grows older
Political-legal	Laws, taxes, policies	Labelling rule for packaged goods	Written record required for each visit
Technological	Machines, methods, knowledge	New mixer	New test equipment
Competitive	Rivals, customers, suppliers	Competitor opens 300 m away	A second clinic opens nearby
Global	Other countries and trade	Wheat price abroad	Imported medicine costs more

Forces seldom act alone. A drought in another country (global) raises the price of wheat (economic). A government may respond with a new tax or a limit on exports (political-legal). Families may then buy less bread (social). The manager sees only the final change in the flour price, but the cause was a chain of forces. For this reason, a manager who studies one kind of force and ignores the others will often misread the situation.

OPPORTUNITY OR THREAT?

After sorting a force into its kind, the manager gives it a second label. An **opportunity** here means an outside change that could help the organisation, if it acts. A **threat** means an outside change that could harm it, if it does not act. Many forces are both. The label depends on the firm, not on the force alone.

Three examples show this. A rise in the price of flour is a threat to a bakery and an opportunity to the farmers who grow wheat. A new mixer on the market is an opportunity to the bakery that buys it and a threat to a bakery that stays with hand kneading. A new bus stop near a shop is an opportunity for the shop that gains customers and a threat for the shop that loses them. When managers speak of forces, they should therefore always name the firm they have in mind.

An opportunity is real only if the organisation can use it. A bakery that hears of a large festival order has an opportunity only if it has the ovens, the staff and the flour to fill the order. Otherwise the order is a risk: it may be taken, failed and lost. The manager therefore compares each force with the organisation's strengths and weaknesses inside. This comparison joins the two environments in one picture, which is why managers study both in the same chapter.

A clinic gives a new setting for the method. Suppose three things happen in one year. A new law requires written records of each visit. A larger hospital opens a branch two streets away. And the town's population grows older. For each change we ask what changed, who carries it, and which kind it is. The law is political-legal, and it carries a threat to the clinic's time and cost. The branch is competitive, and it carries a threat to the clinic's patients. The older population is social. It carries both a threat, because waiting times may grow, and an opportunity, because more people need care. A manager who writes these three lines on a page has begun to understand the clinic's year.

Environments also differ in speed. A tea stall in a quiet village meets few changes in a year. A shop that sells telephones in a city meets new models, new rivals and new prices every month. The first manager can scan seldom. The second must scan often. We cannot say that one environment is better than the other. We can say that the speed of change should decide how often the manager looks outside and how much stock, money and time the organisation keeps ready for surprises.

You should now be able to say: the external environment contains six kinds of force; each kind reaches the organisation in a different way; and the same change may be a threat to one firm and an opportunity to another. Try this on a new case. A private school learns that a new school has opened nearby, that a new rule limits school fees, and that parents now prefer lessons that include computer skills. Name the kind of force in each part. The answers are competitive, political-legal and social.



4.3 HOW OUTSIDE FORCES REACH INSIDE

An outside force does no harm to the work of an organisation until it reaches something inside. The moved bus stop did not touch Hasina's kettle. It reached her through her customers. This is the idea of this section: every outside force travels along a route, arrives at an inside part, and changes a result. We shall follow the idea in four steps, with small numbers first.

An organisation has four common routes through its boundary. The first route is its **inputs**, the resources it buys or receives from outside, such as materials, money, people and information. The second is its **outputs**, the goods or services it sends out to customers. The third is the **rules** that the law and the market place on how the work is done. The fourth is the **people and ideas** that cross the boundary every day, such as staff who bring new habits from home and information that brings news of new methods.

Take the first route. The bakery buys flour, sugar, fuel and labour. When any of these changes in price or supply, the change enters through the purchase. The mill raises the price of flour, and the bakery's cost of each loaf rises on the day it buys. Inputs carry the economic forces and many of the global ones.

A force is also an opportunity only if the organisation can use it, so capacity matters.

A short study case applies the three sorting questions and the two labels to a clinic.

Environments differ in how fast they change, and a manager scans more often where change is fast.

You should now be able to name the six forces and give one example of each.

A force outside the boundary is harmless to the work until it reaches an inside part.

GO SLOWLY HERE.

There are four common routes: inputs, outputs, rules, and people with ideas.

Inputs carry changes in price and supply: a rise in the price of flour enters the bakery through its purchases.

Outputs carry changes in demand: a customer's choice to leave enters the organisation through its sales.

Rules carry the law into the daily work; people and ideas carry tastes and technology inside.

Inside the organisation, different parts feel different forces, so the same route may reach several parts.

A force follows a chain: outside force, route, inside part, result, and then the manager's choice.

The second route is the output. The bakery sells loaves, and the customers' choices come back to it as sales. A rival opens, and the customers change their choice. The bakery sees the effect as fewer sales at the counter. Outputs carry the social forces and the competitive forces, because both act through what customers decide.

The third route is the rules. A new law does not need a customer or a supplier to carry it. It applies to the work directly, at the moment it begins to hold. The fourth route is people and ideas. A new baker arrives and knows a faster method. A helper reads a message about a new product and proposes it. Technological and social forces often arrive in this way.

Inside the organisation the force meets different parts. In a bakery we may name four: the buying of materials, the baking, the selling and the keeping of accounts. A rise in the price of flour first reaches the buying and the accounts. A rival's new shop reaches the selling. A new labelling rule reaches the baking and the packing. A manager who knows which part a force meets can send the right person to deal with it, and can warn that person early.

We can now state the idea as a chain. An outside force acts on a route. The route carries it to an inside part. The inside part changes its result. The manager, seeing the result, must choose a response. The figure below shows the chain for the bakery's flour. Notice that the boundary lies between the first box and the second. Only the first box is outside.

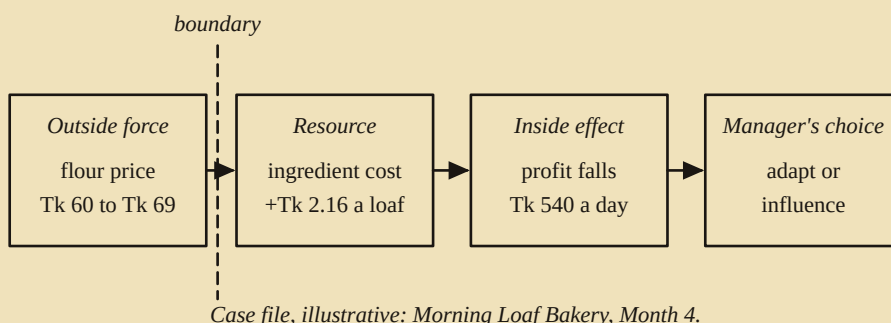


FIG. 10. One outside change followed through the organisation

The first example uses small numbers: a stall's flour price rises by Tk 10 a kilogram.

The first worked example sets out the data, the method, the arithmetic and the check.

Three common slips: the wrong base for a percentage, mixed periods, and missing units.

Now we follow the chain with numbers. We begin with small, clean figures so that the method is the only new thing.

Given. A bread stall uses 0.2 kg of flour for each loaf and sells 100 loaves a day. The price of flour rises from Tk 60 to Tk 70 per kg. **Find.** The extra cost of flour for each loaf, and for each day. **Formula.** Extra cost per loaf = flour used per loaf $\times$ rise in price per kg. Extra cost per day = extra cost per loaf $\times$ loaves per day. **Steps.** The rise in price is Tk 70 – Tk 60 = Tk 10 per kg. Extra cost per loaf = 0.2 kg $\times$ Tk 10 per kg = Tk 2. Extra cost per day = Tk 2 $\times$ 100 loaves = Tk 200. **Answer.** The rise adds Tk 2 to the cost of each loaf and Tk 200 to the cost of each day. **Check.** The stall uses 0.2 kg $\times$ 100 = 20 kg of flour each day. At Tk 10 more for each kilogram, 20 $\times$ Tk 10 = Tk 200. The two methods agree.

Three slips are common. First, a percentage rise is measured on the old price. Here the rise is Tk 10 on Tk 60, which is $10 \div 60 = 0.167$, or about 16.7 per cent. Dividing by the new price gives 14.3 per cent, which is wrong. Second, a cost per loaf and a cost per day must not be mixed. Third, always write the unit, because Tk 2 and Tk 200 mean very different things.

Our second example comes from a clinic, a new setting for the same idea. The route here is the rules, and the result involves time rather than money. We use the productivity ratio from Chapter 2: output divided by input.

Given. A nurse works 8 hours, or 480 minutes, a day. Each patient visit takes 20 minutes. A new law adds 5 minutes of writing to each visit. *Find.* The change in patients seen each day, and in productivity. *Formula.* Patients seen = minutes available ÷ minutes per visit. Productivity = patients seen ÷ hours worked. *Steps.* Before the law: $480 \div 20 = 24$ patients, so productivity is $24 \div 8 = 3.0$ patients per hour. After the law each visit takes $20 + 5 = 25$ minutes, so $480 \div 25 = 19.2$ patients. Productivity is $19.2 \div 8 = 2.4$ patients per hour. *Answer.* The nurse sees about 19 patients instead of 24. Productivity falls from 3.0 to 2.4 patients per hour, a fall of $0.6 \div 3.0 = 20$ per cent. *Check.* Five extra minutes on a 20-minute visit is one quarter more time. One quarter more time gives one fifth fewer patients: $24 \times 0.8 = 19.2$. The answers agree.

Two short traces show that the method is general. For the bakery, a competitor opens 300 metres away. The force is competitive, and the route is outputs. The inside part is the selling. The effect is fewer sales at the counter, and perhaps lower prices later. For the clinic, the town's population grows older. The force is social, and the route is outputs, since the patients are the clinic's customers. The inside part is the clinic's rooms and staff. The effect is more visits and longer waiting, unless the clinic adds staff.

Two firms facing the same force may feel it very differently. The strength of the effect depends on how much the firm depends on the thing that changed. A bakery that spends a large part of its costs on flour feels a flour price rise sharply. A firm that uses little flour barely notices it. Speed matters too. A change in the price of flour arrives within days. A change in the age of the town's people may take ten years. Managers therefore ask of each force: how soon will it arrive, and how much do we depend on it?

The chain in the figure is short on purpose. Real chains are longer. A change in the price of flour may first reduce the owner's profit, then delay a plan to buy a new oven, then slow the rise of output, and at last change the staff's chance of a bonus. The effects may also arrive late. A manager who traces only the first step sees only a small part of the result. For this reason it is useful to ask a second question after each step: and what does that change next?

You should now be able to say: an outside force reaches the organisation through inputs, outputs, rules, or people and ideas; the force changes an inside part and then a result; and the strength of the effect depends on dependence and on speed. Try it on a new case. A bank raises the interest rate on loans. Which route does the change use to reach a bakery that has borrowed money for an oven? It uses the inputs route, because borrowed money is an input.



4.4 MANAGING THE ENVIRONMENT

A manager who sees an outside force has two kinds of answer. The manager can change something inside the organisation so that the force does less harm or brings more gain. Or the manager can try to change the force itself. Both answers have names, and both are needed. Before we describe them, we must ask how the manager sees the force at all.

The second example is a new setting: a clinic, where a law reaches the work through the rules.

The second worked example applies the same method to time and to productivity.

Two more traces show that the method works for competitors and for social change.

How strongly a force strikes depends on how much the firm depends on the thing that changed.

Real chains are often longer and slower than the simple chain, and the effects may be indirect.

You should now be able to follow any outside change along its route to a result.

A manager who sees an outside force can answer it in two ways: change the inside, or try to change the outside.

Environmental scanning is the habit of watching outside forces for early signs of change.

Environmental scanning.

Watching outside forces for early signs of change.

Scanning uses many cheap sources, and the manager chooses a few to watch every week.

Scanning feeds the planning premises of Chapter 6: assumptions about the future that a plan rests on.

To adapt means to change the inside of the organisation so that it fits the outside force.

Adapt. To change the organisation so that it fits an outside force.

Adaptation can change four things: the price, the product, the process and the people.

Waiting and watching is also a choice, and it is right when a force is small or uncertain.

Adapting early costs less than adapting late, because early action leaves time to choose.

Adapting is fast and within the manager's power, but it leaves the cause in place and may not be enough.

To influence means to try to change the outside force itself, by actions that stay within the law.

Influence. To try to change an outside force.

The first task is to see. **Environmental scanning** is the habit of watching the outside forces for early signs of change. A baker who scans talks to customers at the counter, asks the flour supplier about coming prices, reads the news about wheat and tax, and notes when a new shop opens. Each small signal is useful. Together the signals warn the manager early. A warning that arrives late leaves no choice. A warning that arrives early leaves many.

The sources of scanning are many and cheap. A manager may listen to customers and staff, who see changes first. The supplier knows about coming prices. Newspapers report new laws and new rivals. Trade groups circulate information among their members. A manager cannot watch everything, so it is wise to choose a few sources and to review them every week. A short written list is better than a memory, because a list shows which forces have gone unwatched.

Scanning has a further use. In Chapter 6 we shall meet the planning premise, which is an assumption about the future on which a plan rests. A plan that assumes flour will cost Tk 60 for a year is a poor plan if flour is already rising. The premises come from the environment, so good scanning supports good planning. A manager who scans seldom writes premises that are far from the facts.

The first answer is to **adapt**, which means to change the organisation so that it fits the outside force. The force stays as it is. The organisation moves. When flour becomes dearer, the bakery may raise its price, change its recipe, find a cheaper supplier or reduce waste. When a new law requires records, the clinic may hire a clerk. All of these changes are inside the boundary, so they lie within the manager's power.

An organisation can adapt in four main ways. It can change its price, as the bakery may do when flour rises. It can change its product, for example by offering a smaller loaf at the old price. It can change its process, by using less flour with a better recipe or by wasting less. And it can change its people, by training the staff for a new rule or by hiring a clerk. Each of the four lies inside the boundary. A manager who lists all four before choosing is less likely to choose only the first one that comes to mind.

There is a fifth possibility that deserves its place: to wait and watch. If a force is small, or if its path is uncertain, an early answer may cost more than the force itself. A manager who waits should still scan, and should set a point, such as a price of flour or a loss of sales, at which waiting will end. Waiting with such a point is a decision. Waiting without one is only delay.

Timing changes the cost of adapting. A bakery that learns in advance that flour will rise can buy extra stock, negotiate with its supplier or change the recipe in an orderly way. A bakery that learns on the day of the rise must pay at once. The same answer, applied early, may cost less and cause fewer errors. This is the practical value of scanning: it moves the manager's answer from the day of the shock to the days before it.

Adapting has strengths. It is within the manager's power, and it can begin at once. It also has limits. The cause remains, so the pressure may return. An adaptation may cost money, as a new oven does. A rival may adapt faster. And some adaptations are not possible: a bakery cannot lower its price to match a rival if its costs are higher.

The second answer is to **influence**, which means to try to change the outside force. Here the organisation acts across the boundary. It cannot control the force, so it can only try to change it. A bakery may advertise, so that more customers prefer its bread. It may sign a long contract with a mill, to fix the price of flour

for a year. Bakeries may form a trade group and speak together to officials about a tax. A clinic may explain to the authority how a rule affects patients.

Influence takes several forms. Advertising tries to change the tastes of customers. A long contract tries to fix a price or a supply. Joint action, through a trade group, tries to change a rule or a price by speaking with many voices. Good relations with officials, suppliers and neighbours also count, because a firm that is trusted is more likely to be heard. Each form has a cost in time or money, and none can promise the result.

Influence takes several forms, such as advertising, agreements and joint action with other firms.

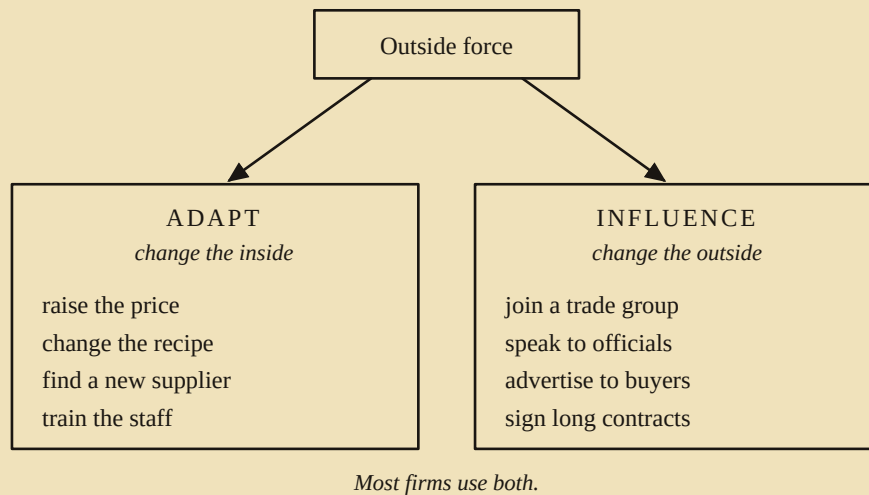


FIG. 11. *The two answers to an outside force*

Influence also has strengths and limits. It can remove the cause instead of only the effect. Its limits are real. A single small firm has little power. Results arrive slowly, and they are uncertain. Honesty is a firm rule: to influence an outside force by lies or by gifts to officials is wrong and may be against the law. Chapter 5 discusses such limits under the name of ethics.

Influence can remove the cause, but it needs power or numbers, it is slow, and it must stay honest and lawful.

The table below compares the two answers. Read down each column. Notice that the two columns describe different places of action. Adapting acts inside the boundary. Influence acts across it.

The table compares the two answers in four ways.

ADAPTING AND INFLUENCING COMPARED

QUESTION	ADAPT	INFLUENCE
What changes?	The inside of the organisation	The outside force
Bakery example	Raise the price; change the recipe	Join a bakers' group to speak with flour mills
Strength	Within the manager's power; can be quick	May remove the cause itself
Limit	The cause stays; may cost money	Slow, uncertain, needs power; must stay lawful

Most organisations use both answers. A manager chooses the mix by asking four questions. Can the force be changed at all? A law may be changed by many voices, but the weather cannot. Does the firm have power alone, or can it join others? A single stall cannot move a bus stop, but fifty stalls may be heard. How soon is the answer needed? Influence is slow, so an urgent problem needs adaptation first. And what does each answer cost, in money and in right conduct? The cheapest answer is not always acceptable.

Most firms use both answers, and four questions help a manager to choose the mix.

Timing and cooperation often decide which answer is better, and both answers can be used together.

The word control must be used with care: the manager can influence the outside, not control it.

You should now be able to name the two answers and to say when each is suitable.

The case file for Month 4 records three outside changes that reached the bakery together.

Three separate changes arrive in one month, and the owner must understand each before choosing.

The question asks which forces act, how they arrive, and how large the cost of the flour change is.

The first step sorts the three changes into kinds and routes.

The second step measures the flour change, one line at a time.

The bakery's flour change is worked in the same order as the earlier examples.

Adaptation and influence do not exclude one another. A bakery may adapt this month by raising its price a little, and may also join a group of bakeries to seek a better price from the mills next year. The first answer protects the profit today. The second tries to reduce the problem for the future. A manager who uses only one of the answers, always, is likely to be surprised sooner or later.

Here we meet an exception to the earlier definition. We said that the external environment lies outside the organisation's control. Influence does not contradict this. To influence is to try, and to try is not to control. A large firm may succeed often, and a small firm rarely. Even a large firm cannot control the weather, the exchange rate or the tastes of a whole country. The accurate statement is that the manager controls the inside, influences a small part of the outside, and must adapt to the rest.

You should now be able to say: scanning lets the manager see forces early; to adapt is to change the inside, and to influence is to try to change the outside; and most firms use both, choosing by power, speed and cost. Try this on a new case. Rain in the week before a festival reduces the number of people in the market. The firm cannot change the rain. Which answer remains? Only adapting, for example by selling through telephone orders for that week.



WORKED CASE: THREE FORCES AT ONCE

Case file, illustrative. Morning Loaf Bakery, Month 4. Flour rises from Tk 60 to Tk 69 per kg, a rise of 15%. Output is 250 loaves a day at 0.24 kg of flour per loaf, so the bakery uses 60 kg of flour a day. It trades 26 days a month. A competitor opens 300 metres away. A new labelling rule applies to packaged goods. The loaf price is Tk 70 and the ingredient cost is about Tk 24 per loaf. Rent is Tk 40,000 a month.

SITUATION

Three outside changes arrive in the same month. The price of flour has risen. A competitor has opened close by. A rule has changed the way packaged bread must be labelled. We shall use the chapter's tools in order: sort the forces, trace each to the inside, measure what can be measured, and list the answers.

QUESTION

Which kind of force is each change? By which route does each reach the bakery? What does the flour change cost in a day and a month? Which answers are open to the owner?

REASONING

First, sort. The rise in the price of flour is an economic force, and it arrives through the inputs route. The competitor is a competitive force, and it arrives through the outputs route, because the effect shows in the choices of customers. The labelling rule is a political-legal force, and it arrives through the rules route, because it changes how packaged bread is prepared.

Second, measure the flour change.

Given. Flour rises from Tk 60 to Tk 69 per kg. The bakery uses 0.24 kg per loaf, makes 250 loaves a day and trades 26 days a month. *Find.* The extra cost per loaf, per day and per month, and the percentage rise. *Formula.* Extra cost per

loaf = flour per loaf $\times$ rise in price per kg. Extra cost per day = extra cost per loaf $\times$ loaves a day. Extra cost per month = extra cost per day $\times$ trading days. *Steps.* The rise is Tk 69 – Tk 60 = Tk 9 per kg, which is $9 \div 60 = 0.15$, or 15 per cent. Extra cost per loaf = $0.24 \text{ kg} \times \text{Tk } 9 = \text{Tk } 2.16$. The ingredient cost of a loaf moves from about Tk 24 to about Tk 26.16. Extra cost per day = $\text{Tk } 2.16 \times 250 = \text{Tk } 540$. Extra cost per month = $\text{Tk } 540 \times 26 = \text{Tk } 14,040$. *Answer.* The flour rise adds Tk 2.16 to each loaf, Tk 540 to each day and Tk 14,040 to each month. *Check.* The bakery uses 60 kg a day. At Tk 9 more per kilogram, $60 \times \text{Tk } 9 = \text{Tk } 540$. The two methods agree. The monthly cost is also about 35 per cent of the monthly rent ($14,040 \div 40,000 = 0.351$), which shows that the change is large for a shop of this size.

Fourth, test the price answer. The next figures are suppositions made for study. They are not facts of the case file. Before the flour rise, the money left from each loaf after ingredients was Tk 70 – Tk 24 = Tk 46, and the bakery made 250 loaves, so the daily total was $250 \times \text{Tk } 46 = \text{Tk } 11,500$. After the rise it is $250 \times \text{Tk } 43.84 = \text{Tk } 10,960$, a fall of Tk 540, as before. If the price rises to Tk 72.16, each loaf again leaves $\text{Tk } 72.16 - \text{Tk } 26.16 = \text{Tk } 46$. The price answer pays only if enough customers stay. The break-even number of loaves is $\text{Tk } 10,960 \div \text{Tk } 46 = 238.3$. So the bakery can lose up to $250 - 238.3 = 11.7$ loaves a day, or about 4.7 per cent of its sales, before the price rise makes things worse than doing nothing. With a rival 300 metres away, the owner must judge whether a loss of that size is likely.

Third, list the answers. The owner may adapt. A price rise that recovers the whole extra cost would be Tk 2.16 on each loaf, which is $2.16 \div 70 = 0.031$, or about 3.1 per cent of the price. Other answers are a change of recipe, a search for a cheaper source of flour, and less waste. But the competitor 300 metres away limits the first of these. If the rival keeps its price, a higher price may send customers across the road. The owner may also try to influence. A single bakery has little power over a flour mill, but several bakeries buying together may obtain a better price. For the labelling rule, adaptation is the only answer, because the owner must follow the law. These are possible answers only. The case file records no decision for this month.

CONCLUSION

The three forces arrive by three different routes. Each allows a different answer. They also act together: the flour rise cuts the profit of every loaf, and the competitor makes it harder to raise the price. The bakery must therefore read the three forces as a set. This habit, to see forces together, prepares the planning of Chapter 6.

SUMMARY

The environment is everything, outside and inside, that affects an organisation. The external environment lies outside the manager's control and has six kinds of force: economic, social, political-legal, technological, competitive and global. The internal environment consists of owners, staff and culture. Stakeholders are the people and groups who carry the forces. A force reaches the work inside by four routes: inputs, outputs, rules, and people and ideas. A manager sees forces early by scanning. The manager then adapts, by changing the inside, or tries to influence, by changing the outside. Most firms do both. The manager should also remember three further points. The same force may be an opportunity for one firm and a threat for another. Forces seldom act alone, and real chains of

A fourth step tests the price answer against the competitor, using stated suppositions.

The third step lists the answers open to the owner, with the limits that the competitor creates.

The three forces differ in route and in the answers they allow, and they act together on one set of results.

The chapter's main ideas are listed in order.

effect are longer than the simple chain. And the choice between adapting and influencing depends on power, speed, cost and right conduct.

THINK IT THROUGH

The problem asks when influence is better than adaptation.

A second problem repeats the bakery calculation with a smaller rise in the price of flour.

Forty small clinics in one district each face a new law that requires costly equipment. A clinic that adapts alone must buy the equipment or close. Answer these questions. Which kind of force is the law? By which route does it reach each clinic? Which of the four questions in section 4.4 favour trying to influence the law, and which favour adapting? When should a firm try to change its environment instead of adapting to it?

Now try a second problem on the bakery. Suppose flour rises by Tk 5 per kg instead of Tk 9. Use the figures of the case file: 0.24 kg of flour in each loaf, 250 loaves a day and 26 trading days. Find the extra cost for each loaf, each day and each month, and check your answer by a second method. Then decide which answers of section 4.4 would still be worth considering at this smaller size, and which would no longer be worth their cost.

WHERE PEOPLE GO WRONG

Five common errors are stated and corrected.

A force can be an opportunity for one firm and a threat for another.

A force acts whether or not the manager watches it.

Customers are only one kind of stakeholder.

Influence uses honest and lawful means.

Adapting does not mean surrender, since it is often the quickest way to protect the goal.

The environment means nature. In this book it means everything, outside and inside, that affects the organisation. Nature is only one small part.

Outside forces are always threats. A force is a threat to one firm and an opportunity to another. The manager asks for whom it is which.

A firm can ignore what it cannot control. A force acts whether the manager watches it or not. What the manager cannot control, the manager can still see and answer.

A stakeholder is the same as a customer. Customers are one kind of stakeholder. Staff, owners, suppliers, neighbours and officials are others.

To influence the environment means to pay for favours. Influence uses honest and lawful means such as advertising, contracts and speaking together. Bribes and lies are wrong and may be against the law.

To adapt means to give in. Adapting is a choice made to protect the goal. A bakery that changes its recipe to meet a flour price has not given in. It has kept the goal and changed the method.

IN HISTORY

Kodak shows how a technological force can reach a firm that has itself created the technology.

The firm later sold digital cameras, yet people's habits and the arrival of phone cameras reduced the value of its older business.

The case also shows that a firm may hold the knowledge of a change long before it holds the will to answer it.

In 1975 an engineer at Eastman Kodak, Steven Sasson, built a camera that recorded pictures as digital data. The prototype was about the size of a toaster, and it stored black-and-white pictures on a cassette tape. Sasson later recalled that managers doubted that people would wish to view photographs on a television screen. Printed photographs were selling well, and the firm had a large film business to protect.

Over the following decades, the habits of people changed. Digital pictures spread, and cameras appeared inside telephones. Kodak made digital cameras itself, but its older business of film and prints declined. In January 2012 Kodak filed for Chapter 11 protection, a legal process in the United States that lets a firm reorganise its debts. It left the process in 2013 as a smaller company.

The case also shows a difference between knowing and answering. A firm may hold the knowledge of a change long before it decides to respond. In Kodak's story the knowledge existed in 1975. The difficulty lay in the response,

which would have changed a large and profitable business. Managers who read the environment well must therefore ask two questions: what is changing, and are we willing to change in answer?

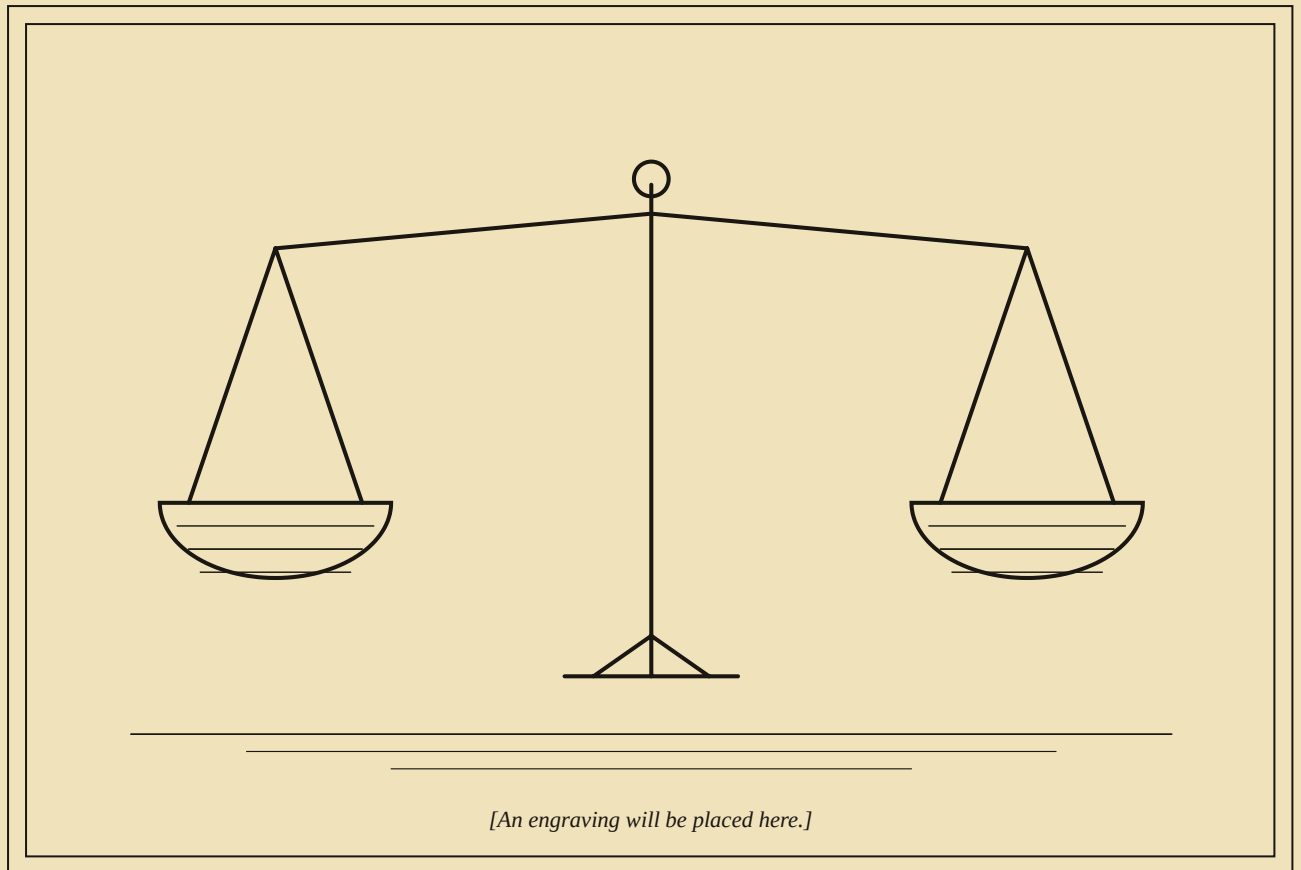
The case does not give one simple cause, and observers still disagree about the reasons. It does show three of the ideas in this chapter. A technological force can reach even the firm that created the technology. Social change, in how people took and kept pictures, travelled with it. And an early warning is useful only if the organisation answers it. Scanning shows the change, and adapting or influencing is the answer that follows.

The lesson is about seeing and answering outside change early; the causes of the firm's decline are still debated.

REVIEW QUESTIONS

1. Define the environment of an organisation, and say how it differs from the everyday meaning of the word. 2. State the difference between the external and the internal environment, with one example of each. 3. Who is a stakeholder? Name four stakeholders of a bakery, and say which lie inside the boundary. 4. Name the six kinds of external force and give one example of each for a school. 5. A new bank rule raises the interest rate on loans. Name the kind of force and the route by which it reaches a firm that has borrowed. 6. A stall uses 0.5 kg of flour for each loaf and sells 80 loaves a day. Flour rises by Tk 8 per kg. Find the extra cost per loaf and per day. Show the check. 7. Compare adapting and influencing in a table with at least three rows. 8. A single small firm faces a tax change it dislikes. State two reasons why it may be wise to adapt and one reason why it may try to influence. 9. Name the four common routes by which an outside force reaches the work inside an organisation, and give one example of each for a clinic. 10. A clinic sees 30 patients in 600 minutes. A new rule adds 4 minutes to each visit, so each visit now takes 24 minutes. Find the new number of patients and the change in productivity, and show a check.

Ten questions test the chapter.



CHAPTER V

Responsibility, Sustainability, Ethics

What an organisation owes to others, beyond its profit



A firm may obey every law and still harm the people near it. This chapter asks what a firm owes to society, how far its duty reaches, and how a manager can tell right from wrong when the law is silent. It presents the case for and against social responsibility, a model by Keith Davis, the idea of sustainability, a plain test for ethical choices, and four ways in which a firm may answer pressure from society.

A workshop owner holds a valid licence, yet his neighbours cannot sleep.

The owner has broken no law, but the neighbours are harmed.

The puzzle asks whether a lawful act can still be wrong, and what a manager owes to people outside the firm.

The chapter follows from Chapter 4: forces and stakeholders around the firm now ask what the firm should give back.

A firm needs society, and society needs the firm, so each affects the other.

Social responsibility is the duty to act for the good of society as well as for profit.

Social responsibility. The duty of an organisation to act for the good of society as well as for its own profit.

Responsibility has layers: the firm must first earn, then obey the law, then act rightly, then give freely.

Mr Karim owns a small metal workshop in a residential lane. He has a licence from the city authority, and he pays his taxes on time. His workers beat sheets of metal until eleven o'clock at night, because customers want the orders early. A family lives next door. Their eldest daughter studies for an important examination, and the noise comes through the wall until midnight.

The family asks Mr Karim to stop at nine o'clock. He looks at his licence. It says nothing about the hour. No official has complained, and no rule has been broken. If he stops at nine, he will lose about one fifth of his orders. His work is lawful, and his workers need their wages. Yet the family suffers, and the suffering is a result of his own decisions.

Here is the puzzle of this chapter. The law does not forbid what Mr Karim does. Is it enough that an act is lawful? What does an organisation owe to the people near it, and to people not yet born? And how can a manager decide, when no rule gives the answer? These questions lead to three ideas: social responsibility, sustainability and ethics. They are related, but they are not the same, and a manager who mixes them will reason badly about all three.

In Chapter 4 we studied the forces around an organisation and the stakeholders who carry them. That chapter asked how the firm should answer these forces for its own sake. This chapter asks a different question: what does the firm owe to its stakeholders, and to society, whether or not they can force it to pay? The answer will shape the plans of Chapter 6, because a plan that ignores what is owed to others is a plan that may later fail.

5.1 BUSINESS AND SOCIETY

An organisation does not stand apart from society. It uses roads, public order, trained workers and the trust of customers, all of which society provides. In return it supplies goods, jobs, taxes and income to the families who work for it. The bakery of our case file gives bread to its customers, wages to its staff and rent to its landlord. It also uses the lane in front of its door, makes smoke and noise, and leaves waste. Each firm therefore gives and takes. The question of this section is how far its duty extends beyond what it gives in the normal course of business.

The name for this duty is **social responsibility**. It means the duty of an organisation to act for the good of society, as well as for its own profit. The words "as well as" matter. Social responsibility does not ask a firm to give up profit. It asks the firm to consider the effects of its actions on others, and to accept some cost when the effects are harmful. A small example is a shop owner who sweeps the street in front of the shop. No law requires it. The owner does it because a clean street is good for everyone.¹

How wide is this duty? One useful answer divides it into four layers, which the figure below shows.² The first layer is economic: the firm must make goods or services that people want, and it must earn enough to survive. The second is legal: the firm must obey the laws of the country. The third is ethical: the firm must do what is right, even where no law demands it. The fourth is discretionary, which means a matter of free choice: the firm gives time or money to a good purpose because it wishes to, such as a bakery that gives bread to a school.

¹ Some writers use the longer name corporate social responsibility, often shortened to CSR. In this book we say social responsibility for any organisation, whether or not it is a company.

² This four-part division follows Archie Carroll (1979); he later drew it as a pyramid. The wording here is our own.

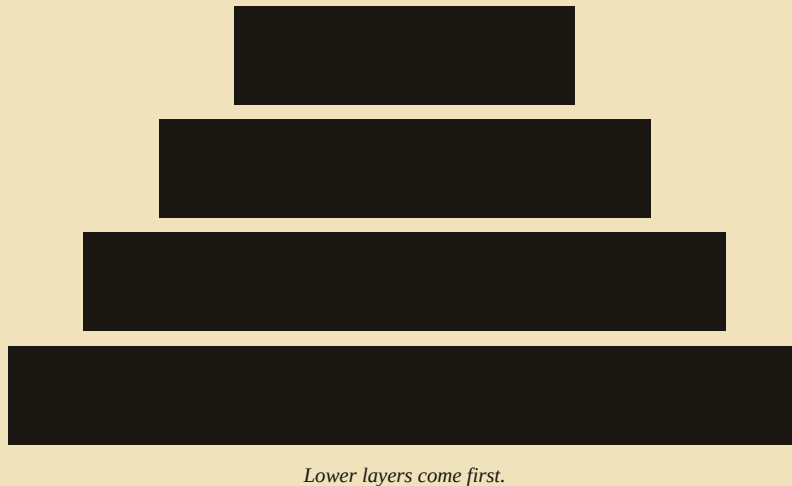


FIG. 12. *The four layers of a firm's responsibility*

The order of the layers is important. A firm that sells unsafe bread cannot repair the fault by giving money to a charity. The lower layers come first. A firm must be economically sound, or it will not survive to help anyone. It must obey the law, or it injures others and loses its right to trade. Only then can the ethical and discretionary layers be judged. The two upper layers are the field in which Mr Karim's choice lies. His workshop is economic and legal. The question left open is whether it is ethical.

Not everyone accepts that a firm owes anything beyond profit and law. The debate is old, and each side has serious arguments. A manager should know both. We shall state the case for social responsibility first, then the case against, in the form that each side's defenders would present it. At the end we shall say how most managers settle the matter in practice.

THE CASE FOR

The first argument for is self-interest over a long period. A firm that harms its neighbours, deceives its customers or pollutes its district loses goodwill, which means the good opinion of the people around it. Customers leave, workers do not stay, and officials look at the firm more closely. A firm that treats people well gains trust, and trust is a form of capital that does not appear in the accounts. The gain may take years to arrive, but it is real.

The second argument is that a firm holds power, and power brings duty. A large employer decides the fortunes of a whole town. A factory decides whether the river stays clean. Those who hold such power affect many people who had no voice in the decision. The third argument is practical. If firms act responsibly of their own will, governments need not pass strict laws to force them. The fourth is that firms have resources and skills that society lacks, and the fifth is that a healthy society is a better place in which to do business. A district with clean water, good schools and public order is a better market than one without them.

THE CASE AGAINST

The case against begins with a question of duty. Managers act for the owners of the firm. The owners have put their money at risk, and they expect managers to use it to earn a return. Milton Friedman (1970) argued that a manager who spends the firm's money on social aims spends the owners' money without their

The layers are ordered, and a firm that fails a lower layer cannot make up for it by giving at a higher one.

The idea of social responsibility has two opposed views, and a manager should know the best argument of each.

The first argument for is that long-term profit depends on the goodwill of society.

The second argument is that firms hold power, and power brings duties; the third is that firms can prevent stricter laws.

The first argument against is that managers are agents of the owners, and owners hire them to produce profit.

Further arguments against concern cost, skill and accountability.

The table sets the two cases side by side so that they can be compared point by point.

consent.³ In his view the responsibility of business is to use its resources to increase profit, within the law and without deceit. Other duties belong to individuals and to governments.

There are three further arguments. The first is cost. Money spent on social aims raises the firm's costs, and costs reach customers as higher prices or fall on workers as lower wages. The second is skill. A bakery knows how to bake, and it may not know how to run a school or to clean a river. The third is accountability, which means being answerable for results. A government is elected and can be removed. A manager who decides social goals for a district has not been elected and cannot easily be called to account.

The table below compares the two cases. Read each row across. The first column names the point at issue. The second gives the argument of those who favour social responsibility, and the third gives the reply of those who oppose it.

THE CASE FOR AND THE CASE AGAINST SOCIAL RESPONSIBILITY

POINT AT ISSUE	THE CASE FOR	THE CASE AGAINST
Profit	Goodwill brings profit over a long period	Managers are paid to earn profit for the owners
Power	Those with power over many lives owe a duty	Duties of this kind belong to governments
Law	Responsible firms make strict laws less likely	Laws, not managers, should set the limits
Cost	Harm to society becomes a cost to all, including the firm	Social aims raise prices and reduce wages
Skill	Firms hold money, staff and knowledge	Firms are trained for business, not for social work
Answerability	A firm that is open about its acts can be held to account	A manager who sets social goals has not been chosen to do so

In practice most managers accept a middle position: do the core work well, obey the law, avoid harm, and give where the firm can.

The position is not a rule of nature; exceptions exist, and the best argument depends on the size and power of the firm.

How do managers settle the debate? Most accept a middle position. They agree with the opponents that a firm must first earn its living and obey the law. They agree with the supporters that a firm must avoid harming others and should deal honestly with all who depend on it. The disagreement that remains is about the upper layer, the free giving of money and effort. There, reasonable people differ, and each firm decides for itself.

This middle position has exceptions. A very small firm has little power and little money to give, and its chief duty is to avoid harm and deal honestly. A very large firm has great power, and its duty to use the power with care is greater. A firm in a poor district may find that the needs of society are large and the money available is small. In such a case honesty and avoidance of harm are the main duties, and gifts are small. We shall return to power and duty in the next section.

IN SHORT

Social responsibility is the duty to act for society's good as well as for profit. It has four layers: economic, legal, ethical and discretionary. Those in favour point to goodwill, power and the cost of harm. Those against point to the duty to owners, the cost, and the question of who has chosen the manager to set social aims.

³ Friedman wrote this in an essay in *The New York Times Magazine*, 13 September 1970.

You should now be able to say: social responsibility means a duty to society beyond profit; it has four layers in a fixed order; and the case for and the case against each rest on strong arguments. As a test, place each of the following in a layer: paying the tax that the law requires, giving bread to a school, selling a good loaf at a fair price, and refusing to sell a product that is known to be unsafe, even though no law forbids it. The answers are legal, discretionary, economic and ethical.



5.2 DAVIS'S MODEL OF SOCIAL RESPONSIBILITY

The debate in 5.1 asks whether a firm should be responsible. Keith Davis, a professor of management in the United States, asked a different question: how much responsibility goes with how much power? His answer is the **Davis model**, which is a set of propositions linking a firm's social power to its duties.⁴ In this chapter we use his model in its common form: one central law and five propositions.

The central idea is simple. Social power means the ability of an organisation to affect the lives of other people and the state of the places where they live. A firm that employs ten workers has little social power. A firm that employs ten thousand has a great deal. Davis said that responsibility should grow with power. The more a firm can affect society, the more it owes to society. We shall call this the power-responsibility link.

The model also carries a warning, which Davis named the iron law of responsibility. It says that over a long period, those who do not use their social power in a way society sees as responsible will tend to lose that power.⁵ The loss may come through new laws, through lost customers or through public anger. The word "iron" means that the effect cannot be avoided in the long run. Consider a river that a factory pollutes year after year. Sooner or later, neighbours complain, officials act, and the factory may be forced to close or to pay heavily.

We now state the five propositions. Each is followed by an example from the bakery or from a new setting, so that the meaning is clear.

One: social responsibility arises from social power. This is the central link. A firm owes more because it can do more. A bakery with four staff and a firm with four thousand staff do not owe the same. A small bakery should not harm its customers or its neighbours. A large textile factory that is the main employer of a district must also think about the safety of its buildings, the quality of the water and the health of its workers, because it can change the lives of all of them.

Two: the firm should work as an open system. In Chapter 3 we met the idea that an organisation is made of linked parts. An open system is one that takes in information from outside and gives information back. For Davis, a firm should listen to the opinions of the public and should tell the public truthfully what it does. The bakery listens when a customer complains about a label. It tells the truth when the label states the ingredients. A firm that hides its acts, or that does not listen, cannot know what society expects of it.

A short test shows whether the reader can tell the layers apart.

Keith Davis offered a model that links responsibility to power, and it explains why responsibility grows with the size of a firm.

Davis model. A set of propositions linking the social power of a firm to its social responsibility.

The central idea is that responsibility rises and falls with social power.

GO SLOWLY HERE.

The iron law says that a firm that fails to use its power responsibly will, over time, lose it.

The five propositions describe what a firm should do if it accepts that its power brings duty.

Proposition one: responsibility arises from social power.

Proposition two: the firm should be open to society, listening and telling.

⁴ Davis set out five propositions in an article in *Business Horizons* (1975), and developed the ideas with Robert Blomstrom in their book *Business and Society*. The wording of the propositions below is our own paraphrase.

⁵ Davis put this idea forward in the 1960s.

Proposition three: the firm should count the social costs and benefits of each act before it begins.

Proposition four: social costs should be priced into the product, so that the buyer pays for the harm the product causes.

Proposition five: a firm, as a citizen, should help with some social problems outside its usual work.

The table sets out the five propositions with one example each from a bakery and from a textile factory.

Three: social costs and benefits should be counted before the firm acts. A social cost is a harm that falls on people other than the firm and its customers: noise, smoke, waste, risk. A social benefit is a gain that falls on them: jobs, training, cleaner air. Davis said that managers should estimate both before deciding whether to go ahead. Mr Karim's workshop would count the lost sleep of the family and its effect on their studies, along with his own gain from late orders. Such counting is not easy, because many social costs cannot be measured in money. But even a list of costs, without numbers, improves the decision.

Four: social costs should be built into the price. If a product causes harm to others as it is made or used, the cost of limiting that harm should be added to the price, so that the buyer pays it. This is sometimes called making the user pay. If the factory does not do this, the cost does not disappear. It falls on others: on the people who drink the river water, or on the government that must clean it. We shall work through a numerical case in 5.3.

Five: firms should take part in some social problems outside their normal work. Davis argued that a firm is a member of society, as a person is. A firm that has the skill, the money or the staff to help with a serious problem near it, and that is better placed to help than others, should do so. A bakery near a school can supply bread for a midday meal. A construction firm can lend engineers after a flood. Notice the limits that Davis set: the firm should have the means, and the problem should be one that the firm is able to help to solve.

The table below sets out the five propositions. Read each row from left to right. The first column gives the proposition in a few words. The second and third give a bakery example and a factory example, so that you can see the same idea in two different settings.

THE FIVE PROPOSITIONS, WITH EXAMPLES

PROPOSITION	BAKERY EXAMPLE	TEXTILE FACTORY EXAMPLE
1. Responsibility rises with power	A small shop avoids harm; a chain of shops owes more	The main employer of a district owes more than a small workshop
2. Be open: listen and tell	Truthful labels; a note for complaints	Publish water and safety reports; meet local people
3. Count social costs and benefits first	List the effect of a new oven's smoke on neighbours	List the effect of a new dye plant on the river
4. Build social costs into the price	Pay for bags that will rot in the soil and add the cost to the price	Pay for water treatment and add the cost to the price of cloth
5. Help with some problems outside the firm's work	Supply bread to a school meal	Lend a building for local training

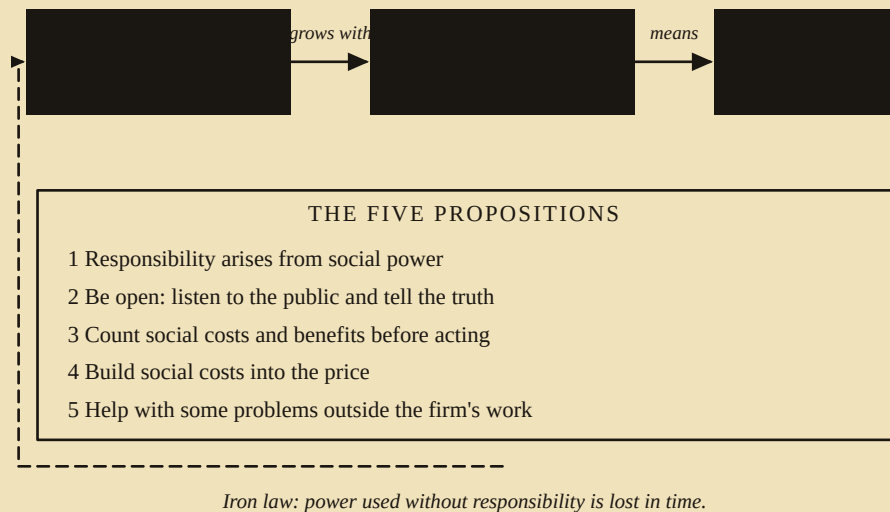


FIG. 13. *Davis's model: power, responsibility and the five propositions*

The figure gathers the model in one picture. Social power stands on the left. The power-responsibility link leads to social responsibility, and the five propositions show what responsibility means in practice. The arrow that runs below warns of the loss of power when responsibility is neglected. Read the figure from left to right, then read the warning.

The model has limits, and a careful reader should know them. First, "social power" is hard to measure. A firm may have great power in one town and little in another. Second, the propositions state what firms should do, but they do not say how much. Proposition five, for instance, gives no amount. Third, those who oppose social responsibility, as in 5.1, reject the model's basis, because they deny that power brings duty beyond the law. Fourth, in some cases the costs in proposition three cannot be known before the act. The model is therefore a guide to reasoning and not a formula that gives a single answer.

Let us apply the model to Mr Karim. Proposition one says that his responsibility is limited because his power is small. Yet his power over the family next door is large, since he decides how well they can sleep. Proposition two says that he should listen. The family has asked him to stop at nine, and he has heard the request. Proposition three says that he should count the cost: one fifth of his orders on one side, the rest and the studies of a young person on the other. Proposition four suggests that he might build the cost of quieter work into his prices. We shall return to this in the closing pack.

You should now be able to say: responsibility rises with social power; a firm that fails to use power responsibly may in time lose it; and the five propositions describe how a responsible firm acts. Try this on a new case. A large clinic in a small town is the only provider of care for forty kilometres. Which proposition most clearly applies to its power? The answer is the first, because its power over the town's health is great, and its responsibility should be great also.

The figure shows the model as one chain: power, then responsibility, then the five acts, with the iron law as a warning below.

Several exceptions and criticisms limit the model.

Davis's model fits our case: Mr Karim's workshop, though small, still holds some power over the people next door.

You should now be able to state the central link, the iron law and the five propositions.



5.3 SUSTAINABILITY IN ORGANISATIONS

A village that catches every fish this year will have none next year.

Sustainability means meeting today's needs without harming the ability of later generations to meet theirs.

Sustainability. Meeting the needs of today without harming the ability of future people to meet their own needs.

Sustainability looks at time, and social responsibility looks at duty, so the two ideas are related but different.

Sustainability has three parts: economic, social and environmental, and the three must be considered together.

Think of a village beside a large pond. Every family catches fish. In a good year, each family catches more than before, and they sell the surplus. If they catch all the fish, including the young ones, the pond is empty by the next season. Fishing was a good way to live, but it could not continue. The families gained today and lost tomorrow. The same lesson applies to a bakery that uses up its profit, a factory that uses up a river, or a firm that uses up its workers' health.

The word for this idea is **sustainability**. It means meeting the needs of today without harming the ability of future people to meet their own needs.⁶ The word comes from "to sustain", which means to keep going. A practice is sustainable if it can continue for a long time without destroying what it depends on. The village fishing was not sustainable. A rule that each family may take only half the fish each year would be.

How does sustainability differ from social responsibility? Social responsibility asks what the firm owes to others. Sustainability asks whether the firm's way of working can continue without harm to the future. A firm can be socially responsible this year and still use methods that cannot last. A firm can also have methods that last and still treat its neighbours badly. The two ideas overlap, and a good manager considers both. We shall keep them apart in our definitions.

Many writers divide sustainability into three parts, as the figure shows.⁷ The **economic** part asks whether the firm can earn enough to continue. A firm that loses money for years will close, however good its intentions. The **social** part asks whether the firm treats its people and the community fairly, so that workers are healthy and trained and neighbours are not harmed. The **environmental** part asks whether the firm uses land, water, energy and materials in a way that does not leave them damaged or exhausted.

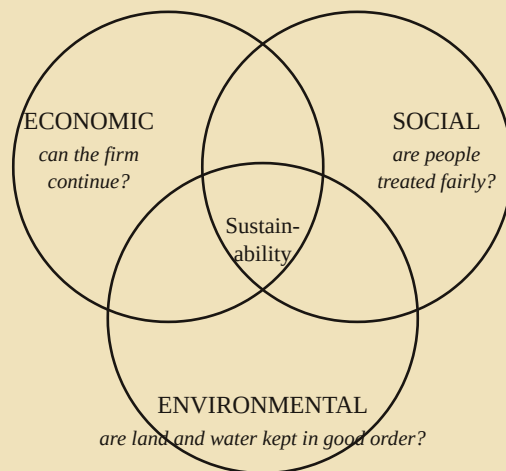


FIG. 14. *The three parts of sustainability*

The three parts overlap, and the middle is where all three are served.

The three circles in the figure overlap, and the middle is the place where all three parts are served. A firm that sits in the middle earns a profit, treats its people well and keeps its surroundings in good order. Many firms are not there.

⁶ This definition follows the report *Our Common Future* (1987), written by a commission of the United Nations led by Gro Harlem Brundtland. We state it in our own words.

⁷ The idea of three parts is often called the triple bottom line, a phrase from John Elkington (1994). The three-part form is common in teaching books. Our wording is our own.

A firm may gain in two parts and lose in the third. A factory may earn well and treat its workers well, yet pour untreated water into a river. A manager should ask of each choice: which parts does it help, and which does it harm?

The table below gives one practice and one measure for each part. Read across each row. The measures are chosen because a manager can count them. What cannot be counted is hard to improve, and measures turn a general wish into a task for the week.

The table gives examples of each part for a bakery and for a textile factory.

THREE PARTS OF SUSTAINABILITY, WITH PRACTICES AND MEASURES

PART	BAKERY PRACTICE	BAKERY MEASURE	TEXTILE FACTORY PRACTICE	FACTORY MEASURE
Economic	Keep waste low and prices fair	Waste as a share of loaves baked	Plan orders to keep machines busy	Profit per month
Social	Safe ovens; fair hours; honest labels	Injuries; staff who leave	Safe buildings; fair wages	Injuries; hours worked
Environmental	Compostable bags; less waste	Bags used; bread thrown away	Treat used water before release	Litres treated per metre of cloth

Why would a firm act on sustainability, apart from duty? There are four practical reasons. First, saving: less waste means lower cost. A bakery that bakes only what it can sell saves flour. Second, supply: a flour mill needs good harvests, and a bakery needs flour, so the health of farmland is part of the bakery's future. Third, customers: many buyers now prefer firms that look after people and land. Fourth, law: rules on waste, water and safety grow stricter over time, and a firm that is ready early meets them at lower cost.

Firms have practical reasons to act on sustainability, apart from their duty.

Sustainable acts often have a cost. We shall measure two of them with the six-step layout of earlier chapters. The first uses small numbers and the second uses numbers a little less tidy. Both link to Davis's fourth proposition, that social costs should be priced into the product.

Sustainable acts have a cost, so we measure it with the same care that we use for any other cost.

Given. The bakery uses 250 bags a day, one for each loaf. A compostable bag costs Tk 0.80 more than the plastic bag it replaces. The bakery trades 26 days a month. The loaf price is Tk 70. *Find.* The extra cost per day and per month, and the extra cost as a share of the price of a loaf. *Formula.* Extra cost per day = bags per day $\times$ extra cost per bag. Extra cost per month = extra cost per day $\times$ trading days. Share of price = extra cost per loaf $\div$ price of a loaf. *Steps.* Extra cost per day = $250 \times \text{Tk } 0.80 = \text{Tk } 200$. Extra cost per month = $\text{Tk } 200 \times 26 = \text{Tk } 5,200$. Share of price = $\text{Tk } 0.80 \div \text{Tk } 70 = 0.0114$, which is 1.14 per cent. *Answer.* The bags add Tk 200 to each day and Tk 5,200 to each month. This is about 1.1 per cent of the price of a loaf. *Check.* The bakery sells $250 \times 26 = 6,500$ loaves a month. $\text{Tk } 5,200 \div 6,500 = \text{Tk } 0.80$ for each loaf. The answers agree.

The first example measures the cost of compostable bags in the bakery of the case file.

Given. A textile factory makes 120,000 metres of cloth a month. It runs a plant that cleans its used water before release. The plant costs Tk 24,000 a month to run. Cloth sells for Tk 100 a metre. These figures are made up for study. *Find.* The cost of cleaning per metre of cloth, and its share of the price. *Formula.* Cost per metre = monthly cost $\div$ metres made. Share of price = cost per metre $\div$ price per metre. *Steps.* Cost per metre = $\text{Tk } 24,000 \div 120,000 = \text{Tk } 0.20$. Share of price = $\text{Tk } 0.20 \div \text{Tk } 100 = 0.002$, which is 0.2 per cent. *Answer.*

The second example is a textile factory, in a new setting with the same method.

Common slips: using the wrong base for a share, mixing days with months, and thinking that an unpaid cost has vanished.

A manager can start with a short list of measures and improve it each year.

Sustainability has limits and exceptions: trade-offs between the three parts, and claims that are not true.

You should now be able to define sustainability and give its three parts.

A baker finds a damp sack of flour, and the law does not say whether to use it.

Ethics means standards of right and wrong, and business ethics applies them to the work of firms.

Business ethics. The standards of right and wrong that guide the conduct of people in business.

Ethics, law and social responsibility are three different things, and a table can keep them apart.

Cleaning the water adds Tk 0.20 to a metre of cloth, which is 0.2 per cent of its price. *Check.* Tk 0.20 $\times$ 120,000 metres = Tk 24,000. The answers agree.

Three slips are common. First, a share must use the right base: here the price of a loaf, not the cost of the bag. Second, a cost per day and a cost per month must not be mixed. Third, and most important, a cost that the firm does not pay has not vanished. If the factory does not clean its water, the Tk 24,000 is not saved. It becomes a cost for the people downstream, who must find clean water elsewhere or become ill. This is exactly what Davis's fourth proposition means.

A firm need not measure everything at once. A small firm can begin with three or four figures that it can count without cost: the bread thrown away each week, the bags used each day, the injuries in a month, and the staff who leave in a year. A short list that is kept up is worth more than a long list that is abandoned. After a year, the manager compares the figures with those of the year before, decides which one matters most, and sets a target for it. Chapter 7 will show how to set such a target in a form that can be checked.

Sustainability has limits. The three parts sometimes pull against one another. A cleaner method may cost more and put a small firm out of business, which harms its workers. A cheaper method may harm a river and the future of the village. The manager cannot avoid the choice, but can make it openly, with the costs listed. A second danger is a false claim. A firm may say that its product is kind to the land when it has not checked the claim. Such a claim deceives customers and harms honest firms. Honest claims must be checked, as the bakery's label rule in 5.4 requires.

You should now be able to say: sustainability is meeting today's needs without harming the future; it has an economic, a social and an environmental part; and a cost that a firm does not pay does not vanish, for it falls on others. Try this on a new case. A tailoring shop throws away fabric scraps, and a neighbour uses them to make quilts. Which part of sustainability does the shop's choice serve? The answer is the environmental part, since waste is reduced, and the social part, since a neighbour gains work.



5.4 BUSINESS ETHICS

A baker opens a sack of flour on a wet morning and finds that one corner is damp. The flour does not smell bad. The owner is away, and the day's orders are large. The law forbids selling food that is unfit to eat, but it does not say whether this particular flour is fit. Throwing away the sack costs Tk 1,500. Using it carries a small chance that a customer will be harmed. The baker must decide alone, with no rule to read. How should the baker think?

The subject of this section is **business ethics**, which means the standards of right and wrong that guide the conduct of people in business. Ethics asks what we ought to do, and not only what we must do or what we can do. An ethical choice is one that people with good judgement would call right, after thinking about the effects on all those affected.⁸

Ethics is often mixed with law and with social responsibility. The three are related but different. The law is what the government requires, and the courts enforce it. Social responsibility is what an organisation owes to society, as 5.1

⁸ The word ethics has two uses. It names the study of right and wrong, and it names the standards that a person or group follows. This book uses the second meaning.

explained. Ethics is the standard of right and wrong that guides a person's choices. The three may agree. They may also disagree, which is why we study each. The table below gives the difference.

LAW, SOCIAL RESPONSIBILITY AND ETHICS COMPARED

QUESTION	LAW	SOCIAL RESPONSIBILITY	ETHICS
What is it?	Rules made by government	Duty of the organisation to society	Standards of right and wrong
Who enforces it?	Courts and officials	Public opinion, stakeholders, the firm's own aims	Conscience, habit, the example of others
Who holds it?	Everyone in the country	The organisation as a whole	Each person, and also the organisation
Bakery example	Selling food that is fit to eat	Giving unsold bread to those who need it	Not claiming "sugar-free" on a label unless it is checked

A legal act can be unethical. Mr Karim's late work is legal, and it may be unethical. A firm may sell a product that is legal but known to harm the buyer if used as sold. The reverse also holds. An act may be right and yet not required by law, such as telling a customer about a fault that the customer has not noticed. There are also rare cases in which a law itself is unjust. Most managers meet these cases seldom. The usual lesson is that the law gives the minimum, and ethics asks for more.

Where do ethical standards come from? They come from the family, from religion, from culture, from schools, from the example of colleagues and from the law. Because the sources differ, standards differ among persons and among places. In one place it may be usual to give a gift to an official who helps a firm. In another, the same gift is a bribe. Even so, some standards are shared widely: be honest, treat people fairly, do not harm those who trust you, and keep promises. These four are the base on which a manager can build.

Managers often face an **ethical dilemma**, which is a choice in which each option has something right about it and something wrong or costly. The baker's choice is one. To throw away the flour protects the customers and costs Tk 1,500. To use it saves the money and risks harm. Most dilemmas are not as plain. Honesty may seem to conflict with loyalty to a colleague, and fairness to one group may seem to conflict with fairness to another. The test that follows is meant for these cases.

There are three common ways of judging a choice.⁹ The first looks at **results**: which option does the most good and the least harm to all who are affected? The second looks at **duties and rights**: which option keeps our promises and respects the rights of others, such as the right to truthful information? The third looks at **fairness**: which option treats people equally, or gives to each what is due? The three often agree. When they do not, the manager must think more carefully, and should say which approach decided the choice.

For daily use the three ways can be turned into four questions.¹⁰ The first is: is it legal? A choice that fails this question is rejected at once. The second is: who is affected, and how? Here the manager lists everyone affected, including

An act can be legal and unethical, or ethical and not required by law, so the two must be asked separately.

Ethical standards come from many places, and they differ among cultures, though some are shared widely.

An ethical dilemma is a choice in which each option has a right side and a cost.

Ethical dilemma. A choice in which each option has something right about it and something wrong or costly.

Three ways of judging a choice are by its results, by duties and rights, and by fairness.

The ethical test turns the three ways of judging into four short questions that a manager can ask in a few minutes.

⁹ These three correspond to the main schools of moral thought: those that judge by results (Bentham, Mill), by duty (Kant) and by justice. We give them in plain words.

¹⁰ The four-question form is our own summary of common teaching practice; it is a guide to judgement and not a formula.

people outside the firm, and notes the help or the harm to each. The third is: is it fair? The fourth is: could I tell those affected, openly, what I decided and why? A choice that one would be ashamed to explain is a warning. The figure shows the four questions in order.

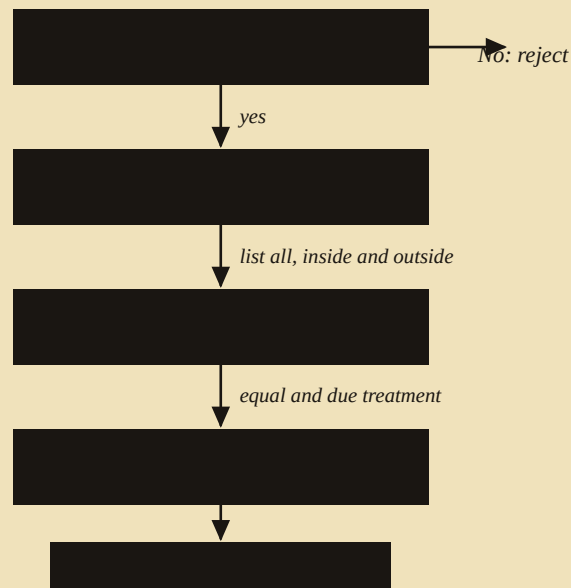


FIG. 15. The four-question ethical test

The four questions apply to the baker's damp flour.

People act unethically for common reasons, and managers who know them can reduce them.

Managers can support ethical conduct with a written code, with example, with training and with a safe way to report.

Code of ethics. A written statement of the standards of conduct that an organisation expects of its people.

The standards in a code can be tested in the same way: by asking the four questions.

Apply the test to the damp flour. Is it legal? Selling unfit food is not, and it is not certain that the flour is fit, so the answer is doubtful. Who is affected? The customers may be harmed, the business may lose its name, and the owner loses Tk 1,500 if the flour is thrown away. Is it fair? Customers cannot judge the flour, so they depend on the baker. Could the baker tell the customers? Few would say, "We used flour that we feared was damp." The test points toward throwing the flour away, or at least asking the owner and the supplier. An answer reached in this way can be explained to others.

If most people know the difference between right and wrong, why do some act unethically? There are common reasons. One is pressure to meet a target. When a manager is rewarded only for a number, some people will reach the number by any means, as Chapter 7 will show. A second is an order from a superior. A third is habit: everyone in the group acts in the same way, so no one asks whether it is right. A fourth is the belief that no one is harmed. A fifth is fear: a person who sees wrong-doing may fear the loss of a post for reporting it.

What can a manager do? A **code of ethics** is a written statement of the standards of conduct that an organisation expects of its people. A code gives people something to point to when they are pressed to do wrong. The bakery of our case file has a short code of one rule: no claim on a label that is not checked. A code alone is not enough. People copy what their leaders do, so a manager's own example matters more than any paper. Training helps people to see dilemmas early. A safe way to report a concern, so that a person can speak without fear, lets problems come to light while they are small. Chapter 13 will show how control can check that the code is kept.

Do not make a code too long. A long code is not read. A code of three to five plain rules, each of which can be applied to a real case, is better than a hundred pages. Each rule should answer a question that the staff really meet: what do we do when a label is wrong, when a customer offers a gift, when a supplier offers a favour.

The

bakery's single rule answers the first of these. It can grow as new questions appear.

You should now be able to say: ethics is the standard of right and wrong; it differs from law and from social responsibility; and a manager can test a choice by asking four questions. Try this on a new case. A shop owner knows that a rival's product is safer than his own, but his sales staff are told to say that his is the best. Which of the four questions does the instruction fail most plainly? The answer is the fourth: could the staff tell the customers openly what they know?



5.5 RESPONDING TO SOCIAL, POLITICAL AND LEGAL PRESSURE

Firms do not choose all the demands made on them. Neighbours complain, newspapers report, officials inspect, and new laws arrive. In Chapter 4 we named these forces and the stakeholders who carry them. This section asks how a firm should answer when the pressure is about its duty to others. We call the three kinds **social pressure**, which comes from public opinion, groups and the press; **political pressure**, which comes from government policy and officials; and **legal pressure**, which comes from laws and courts.

A firm can answer the same pressure in four ways. We name them obstructive, defensive, accommodative and proactive, and place them on a scale from the least active to the most active.¹¹ The figure shows the scale. Notice that the four names describe a manner of response, not a person's character. The same firm may be obstructive on one matter and proactive on another.

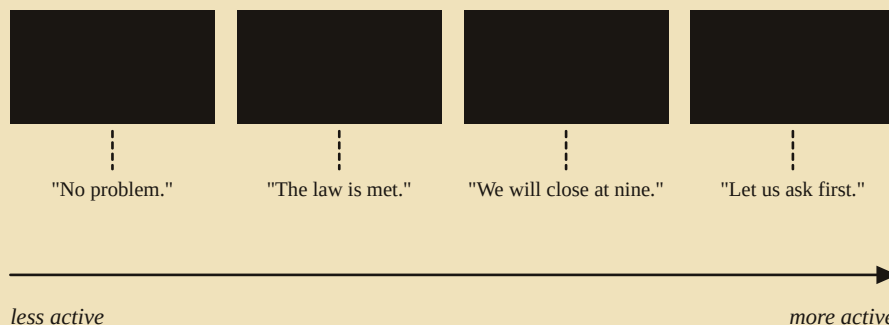


FIG. 16. Four answers to pressure, from least to most active

An **obstructive** firm denies that a problem exists, hides the facts and resists those who press it. If neighbours complain of smoke, an obstructive firm says there is no smoke, or closes its gate to visitors. This stance may save money for a time. Its danger is the iron law of 5.2: those who resist society's wishes may lose their power. The obstructive firm also breaks the second and fourth questions of the ethical test.

You should now be able to define business ethics, distinguish it from law, and apply the test.

A firm meets pressure from society, from government and from the law, and it must choose how to answer.

The same pressure can be answered in four ways, which we place on a scale from least to most active.

An obstructive firm denies the problem and resists the pressure.

Obstructive. A stance in which the firm denies a problem and resists those who press it to act.

¹¹ Several writers describe a scale of this kind, including Archie Carroll (1979), with slightly different names. The four names here are those used in many teaching books on management.

A defensive firm does the legal minimum and argues that it need do no more.

Defensive. A stance in which the firm does only what the law requires and argues that more is not its duty.

An accommodative firm does what the law asks and also what society reasonably requests, when asked.

Accommodative. A stance in which the firm meets legal duties and also meets reasonable requests from society when it is asked.

A proactive firm looks ahead, acts before it is asked, and tries to lead others.

Proactive. A stance in which the firm looks ahead, acts before it is pressed, and tries to lead others.

The table shows how the four stances answer the same event: a complaint about the water released by a textile factory.

A **defensive** firm admits the problem but does only what the law requires, and argues that nothing more is its duty. If a rule says that smoke must be below a stated level, the defensive firm brings its smoke exactly to that level and no lower. It acts within the law, so it is not a wrong-doer. It earns little goodwill, and when the law grows stricter it must change quickly and at greater cost.

An **accommodative** firm meets its legal duties and also meets reasonable requests from society when it is asked. Neighbours ask the firm to close its doors at nine, and the firm agrees. It does not wait to be forced, yet it does not look for problems. Most responsible small firms stand here for most matters, and the stance is adequate in most cases.

A **proactive** firm looks ahead, finds the problems that its work may cause, and acts before it is pressed. It may meet people who live nearby before a new oven is built and ask what worries them. It may publish its own reports. It may help to make a standard that other firms in its trade then follow. The proactive firm accepts a cost early and hopes for trust and fewer shocks later.

The table below shows the four stances on one event. A textile factory is told by neighbours that the water in the stream beside it has changed colour. Read each row from left to right. The first column names the stance, the second states what the firm does, and the third gives the likely result.

FOUR STANCES APPLIED TO ONE COMPLAINT

STANCE	WHAT THE FACTORY DOES	LIKELY RESULT
Obstructive	Denies the change and refuses entry to visitors	Anger grows; officials may act; trade may be lost
Defensive	Tests the water; cleans only to the legal level; argues that nothing more is due	Peace for a time; new rules may force costly change later
Accommodative	Cleans to the legal level and agrees to neighbours' reasonable requests	Good relations; limited extra cost
Proactive	Cleans beyond the legal level; shares the test results; helps other factories do the same	High early cost; trust and a stronger name

No stance is always best, because cost, size and the matter at hand differ.

Is the proactive stance always best? No. It is the costliest. A small firm cannot be proactive on every matter, and it need not be. The sensible course is to choose the matters where the firm's power or its harm is greatest, and to be proactive there. On other matters, an accommodative stance is enough. The obstructive stance is rarely wise, for it risks all that the other stances protect. An exception is a demand that is unlawful or plainly unfair, which a firm may rightly resist by lawful means.

Political pressure can be answered by influence, as Chapter 4 described, but the means must stay open and honest.

Political pressure needs a word of its own. A government may ask firms to follow a policy, such as training young workers, before it makes the policy into law. A firm may answer by adapting, or it may try to influence the policy, as Chapter 4 explained. Influence is acceptable when it is open and honest: a letter, a meeting with officials, a statement through a trade group. It is not acceptable when it uses gifts or threats. The ethical test of 5.4 applies here with full force, and the fourth question is the best guide: could the firm describe its approach to the officials in public?

Pressure does not always come from outside. Staff may ask for safer machines, and owners may ask the firm to give to a cause. These requests arise from the internal environment of Chapter 4, and they deserve the same careful answer. A firm that listens to its own people early often avoids a larger conflict later, since staff who see a wrong are the first to see it.

We can now state a procedure for answering pressure, in six steps. A procedure is a fixed order of steps that a manager follows each time.

Step 1: Find who presses and what they ask. Name the stakeholder and state the request in one sentence. Do not guess the request; ask.

Step 2: Check the law. Find which rule applies, and whether the firm now meets it. *Step 3: Apply the ethical test.* Ask the four questions of 5.4. The law sets the minimum, and the test shows what lies above it.

Step 4: Count the costs. For each stance, list the cost now and the risk later. *Step 5: Choose a stance, act, and tell the stakeholders openly.* *Step 6: Review.* After some weeks, ask whether the pressure has eased and whether the cost was as expected. Chapter 13 will show how such a review fits into control.

Pressure from society and government has changed whole industries. In April 2013 the Rana Plaza building in Savar, near Dhaka, collapsed, and more than 1,100 people died, most of them garment workers. In the weeks that followed, many foreign buyers and trade unions agreed to an accord on fire and building safety in the country's garment factories. The accord is a case of pressure from society leading to new rules that firms had to follow. Factories that had treated safety as someone else's problem now had to inspect their buildings and repair faults.

We cannot know what would have happened if factories had acted earlier. But we can see the pattern that the iron law describes. Where firms did not give safety the attention that their power over workers demanded, the pressure that arrived later was far stronger than any earlier request would have been. The cost of a proactive stance is real, but it is small beside the cost of a collapse.

You should now be able to say: pressure on a firm may be social, political or legal; a firm may answer in four stances, from obstructive to proactive; and a manager chooses by the six-step procedure and not by habit. Try this on a new case. Neighbours ask a bakery to stop deliveries before six in the morning. The owner agrees at once, and no law requires it. Which stance is this? The answer is accommodative, since the firm meets a reasonable request when asked.



WORKED CASE: MONTH 5 AT THE BAKERY

Case file, illustrative. Morning Loaf Bakery, Month 5. Unsold bread, about 10 loaves a day, is donated at closing time. Compostable bags cost Tk 0.80 more each, and the bakery uses 250 a day, a cost of Tk 200 a day. Labels that did not match the product were corrected. The owner sets a written ethics rule: no claim on a label that is not checked. Loaf price Tk 70; ingredient cost about Tk 24 before the flour rise of Month 4 and about Tk 26.16 after it; 26 trading days a month; output 250 loaves a day.

Pressure also comes from inside the firm, because staff and owners hold views about what is right.

A short procedure helps a manager to choose the stance in a reasoned way.

Step one is to find who is pressing and what they ask.

Step two is to check the law, and step three is to apply the ethical test.

Steps four to six are to count the costs, to choose and tell, and to review the result.

Pressure from society can cause large changes, as when buyers and unions responded to a building collapse near Dhaka in 2013.

The case shows that the cost of being obstructive can be far greater than the cost of acting early.

You should now be able to name the four stances and place a response on the scale.

The case file for Month 5 records four actions by the owner and a cost figure for the bags.

Four acts are recorded, and the case file gives no reasons for them, so we show how the chapter's tools describe them.

The questions ask which layer, which Davis proposition, and which stance each act shows, and what the acts cost.

The first step sorts each act by layer, by proposition and by stance.

The second step measures the donation and the bags with the standard layout.

The third step puts the total beside the flour rise of Month 4, using the figure carried forward from Chapter 4.

The four acts show different layers, propositions and stances, and they carry a cost that the owner has accepted.

SITUATION

The case file records four acts. The file does not record why the owner chose them. We therefore use the tools of this chapter to describe each act, and to measure what can be measured. We make no claim about the owner's motives.

QUESTION

Which layer of responsibility and which of Davis's propositions does each act show? Which stance does each show? What do the bags and the donated bread cost in a month? How does the cost compare with the flour rise of Month 4?

REASONING

First, sort the acts. The donation of unsold bread is a matter of free choice and falls in the discretionary layer. It shows proposition five, help with a social problem outside the firm's usual work. The compostable bags are a sustainability act in the environmental part, and they show proposition four, since the cost is built into the price of production. The correction of the labels is first a matter of law, because labelling rules came in Month 4, and then of ethics, since the labels did not match the product. The written rule shows proposition two, that the firm should be open, and it is a code of ethics. As for stance, the donation and the bags go beyond the law, so they are accommodative or proactive. The label correction meets a legal duty and, with the rule, goes beyond it.

Second, measure the costs. *Given.* About 10 loaves a day are donated. After the flour rise of Month 4, the ingredient cost of a loaf is about Tk 26.16. The bags cost Tk 200 a day. The bakery trades 26 days a month. *Find.* The monthly ingredient value of the donated bread, the monthly cost of the bags, and the monthly total. *Formula.* Value of donated bread per month = loaves per day $\times$ ingredient cost per loaf $\times$ trading days. Bag cost per month = bag cost per day $\times$ trading days. *Steps.* Donated bread: $10 \times \text{Tk } 26.16 \times 26 = \text{Tk } 6,801.60$, or about Tk 6,802. Bags: $\text{Tk } 200 \times 26 = \text{Tk } 5,200$. Total: $\text{Tk } 6,802 + \text{Tk } 5,200 = \text{Tk } 12,002$. *Answer.* The donation represents about Tk 6,802 of ingredients a month, and the bags cost Tk 5,200 a month. The two together come to about Tk 12,002 a month. *Check.* $\text{Tk } 12,002 \div 26$ trading days is about Tk 462 a day. The parts are Tk 261.60 and Tk 200, which sum to Tk 461.60. The answers agree.

Third, compare. In Chapter 4 the flour rise cost the bakery Tk 14,040 a month. The bags alone cost Tk 5,200, which is $5,200 \div 14,040 = 0.370$, or about 37 per cent of that amount. The bags and the donated bread together are Tk 12,002, which is $12,002 \div 14,040 = 0.855$, or about 85 per cent. These figures show that the acts of Month 5 are not small beside the pressures of Month 4. The donated bread is a cost only in the sense that it is bread that could not be sold, so the sum is an upper bound on what is given up, not an amount paid.

CONCLUSION

The four acts of Month 5 touch four different parts of the model of this chapter. They go beyond the law in three cases. They carry a monthly cost that is a large share of the flour rise in the same period. The owner has accepted the cost, so far as the case file shows. The chapter offers a way to describe such choices and to examine them, whether or not the owner used it.

CHECK

Count the acts: donation, bags, label correction and written rule. All four are sorted. The figures used are those of the case file: Tk 0.80 a bag, 250 bags, Tk 200 a day and about 10 loaves a day. The ingredient cost of Tk 26.16 is derived from Chapter 4, as the case note states.

A check confirms that no act has been left unsorted and that the figures match the case file.

SUMMARY

Social responsibility is the duty to act for society's good as well as for profit. It has four layers: economic, legal, ethical and discretionary. Arguments for it rest on goodwill, power and the cost of harm. Arguments against rest on duty to owners, cost and answerability. Davis's model links responsibility to social power, adds the iron law, and gives five propositions. Sustainability means meeting today's needs without harming the future, and has an economic, a social and an environmental part. Business ethics is the standard of right and wrong in business; it differs from law and from social responsibility, and a four-question test helps a manager to apply it. A firm answers pressure in four stances, obstructive, defensive, accommodative and proactive, and a six-step procedure helps it to choose.

The chapter's ideas, in order.

THINK IT THROUGH

A textile factory in a district earns a good profit. Its treatment plant is old and works at only half its power. The law sets a limit for the colour of released water, and the factory meets the limit on the days when officials test it. On other days the water is darker. Cleaning the water on every day would cost Tk 20,000 a month, which is 0.4 per cent of the factory's monthly sales. Downstream families wash and bathe in the stream. Answer these questions. Which layers of responsibility does the choice touch? Which of Davis's propositions apply? How does the choice fare on the four-question test? Which stance does the factory now show, and which would you advise? And can an action be legal and still unethical? Give a reason for your answer.

The problem sets profit, law and ethics against one another in a textile factory.

WHERE PEOPLE GO WRONG

Social responsibility means giving money to charity. Giving is only the top layer. The first duties are to do the core work well, to obey the law and to avoid harm.

Six common errors are stated and corrected.

Giving is only the top layer of responsibility.

If it is legal, it is ethical. The law sets the minimum. Mr Karim's late work is legal and may still be wrong.

The law sets the minimum, and ethics asks for more.

Sustainability means only protecting nature. It has three parts, and the economic and social parts matter as much as the environmental one.

Sustainability has three parts, not one.

A cost that a firm does not pay does not exist. It falls on others, and in time it may return to the firm.

An unpaid cost falls on other people.

Ethics is a matter of opinion, so no test is possible. People differ on some questions, but they agree on many, and a four-question test brings the reasoning into the open.

A four-question test brings ethical reasoning into the open.

The proactive stance is always best. It is the most costly. A firm should be proactive where its power and its harm are greatest and accommodative elsewhere.

The best stance depends on power, harm and cost.

The Tylenol recall of 1982 is often cited as an example of a firm acting for the public before it was forced to.

The firm told the public, took the loss, and later new rules for safe packaging followed.

The main sources of the chapter's defining ideas are listed with the reason each matters.

Davis's propositions have their source in two works.

Friedman's essay states the case against.

Carroll's article is the source of the layers and the scale.

The 1987 report is the source of the definition.

Ten questions test the chapter. Defining social responsibility. Arguments for and against.

Davis's central idea. The five propositions.

Defining sustainability.

A bag calculation.

Ethics, law and responsibility.

IN HISTORY

In late September 1982, seven people in the Chicago area died after taking capsules of Tylenol, a painkiller made by a subsidiary of Johnson & Johnson. Investigators found that someone had put cyanide in capsules after they left the factory. The company recalled about 31 million bottles across the United States, and later sold the product in new sealed packages. The recall was reported to have cost about 100 million United States dollars.

Commentators have often praised the firm for putting public safety ahead of short-term profit, for speaking openly to the public and for working with officials. New federal rules for tamper-resistant packaging followed. The case shows several ideas of this chapter: a firm acting before the law required it, and openness of the kind that Davis's second proposition describes. The case also has a limit: the poisoner was never found, and the firm was not at fault for the crime. Its conduct is judged on how it answered.

SOURCES

The sources are listed below, one to a paragraph, with a note on why each matters.

Keith Davis (1975), "Five propositions for social responsibility", *Business Horizons*. The source of the five propositions. Davis and Robert Blomstrom, *Business and Society* (several editions), develops the power-responsibility link and the iron law.

Milton Friedman (1970), "The Social Responsibility of Business Is to Increase Its Profits", *The New York Times Magazine*. The best-known statement of the case against social responsibility.

Archie Carroll (1979), "A three-dimensional conceptual model of corporate performance", *Academy of Management Review*. The source of the four layers and the scale of responses.

World Commission on Environment and Development (1987), *Our Common Future*. The source of the usual definition of sustainability.

REVIEW QUESTIONS

1. Define social responsibility, and name its four layers with one example of each for a bakery.

2. State two arguments for and two arguments against social responsibility. Which do you find stronger, and why?

3. State the central idea of Davis's model, and say what the iron law warns.

4. List Davis's five propositions in your own words, and give one example of the fourth.

5. Define sustainability and name its three parts. Give a measure for each part in a school.

6. A bakery uses 300 bags a day. Each compostable bag costs Tk 0.60 more than a plastic one, and the bakery trades 26 days a month. Find the extra cost per day and per month, and the extra cost as a share of a Tk 70 loaf if there is one bag for each loaf. Show a check.

7. Distinguish business ethics from law and from social responsibility, using a table.

8. State the four questions of the ethical test and apply them to a shop that wishes to sell goods past their expiry date at a lower price.

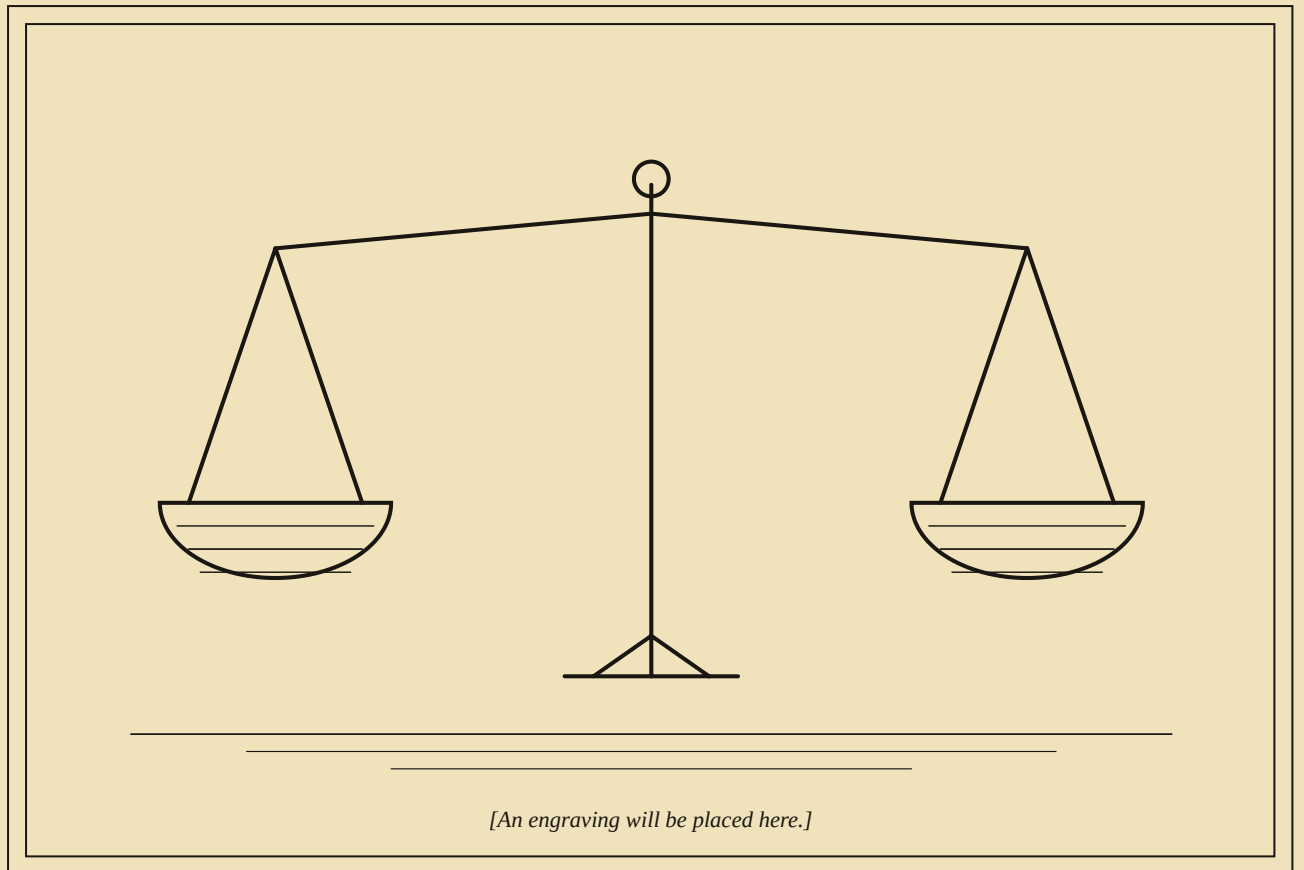
The ethical test.

9. Define obstructive, defensive, accommodative and proactive responses. Give one example of each for a clinic that is told its waiting room is unsafe.

The four stances.

10. A factory cleans its water at a cost of Tk 18,000 a month and makes 90,000 metres of cloth. Find the cost per metre, and say whether the cost should be added to the price, using Davis's fourth proposition.

A cost per metre.



CHAPTER VI

Planning

Before a journey, a sensible traveller asks where to go and how to get there. A manager asks the same two questions.



This chapter explains how managers decide in advance what to do.

It names the kinds of plans, from long-range strategies to daily rules, and it walks through the steps of making a plan, including the assumptions on which every plan rests. It ends by asking why plans fail and what makes planning succeed.

A JOURNEY WITHOUT A MAP

A bakery owner has six months to grow the shop, and no plan yet.

The owner can wait for events, or prepare for them.

This chapter studies the second choice, which is called planning.

You need three ideas from earlier chapters.

Return to the Morning Loaf Bakery at the start of its sixth month. Sales are steady and the staff work well together. Yet the owner feels uneasy. Customers keep asking for more bread. Flour has become dearer. A festival will soon bring many buyers to the street. If the owner does nothing special, each of these events will arrive as a surprise.

The owner has two choices. One choice is to wait and then react to each event as it comes. The other choice is to look ahead, to decide what the shop should become, and to prepare for what is likely to happen. The second choice takes time now. It saves time and money later.

Chapter II named planning as the first of the four functions. We now study it closely. We ask what a plan is, what kinds of plan exist, how a plan is made, why plans fail, and how a manager can make planning work.

You will need three earlier ideas. From Chapter II, the four functions and the meaning of a decision. From Chapter IV, the idea of the environment, because a plan must be made with the outside world in view. From Chapter I, the idea of resources, because a plan decides how resources will be used.



WHY PLAN: MEANING, NATURE AND PURPOSES

PLANNING IN EVERYDAY LIFE

Planning is common in daily life, though people seldom use the word.

A journey shows the main parts of planning.

The student's planning has three parts: a place to reach, a way to get there, and thought about the road.

Think of a student who must travel to a university in another city for an examination. The student does not leave the house without a thought. First the student decides to arrive the evening before. Then the student chooses the bus, buys a ticket, and packs a bag. The student also thinks about what might go wrong, such as a late bus, and leaves early to allow for it. Every one of these acts is planning, although none is called by that name.

Look closely at what the student did. There was a place to reach, which was the city. There was a way to get there, which was the bus. And there was thought about the road, which was the allowance for delay. These three parts appear in every plan, from a student's journey to a firm's strategy for ten years.

WHAT A PLAN IS

Plan. A chosen route from where we are to where we want to be.

A plan is a chosen route to a goal.

Planning. Deciding in advance what to do, how, when and by whom.

Planning is the act of making plans.

Planning looks to the future, but it uses information from the past and the present.

The word **plan** has a clear meaning in management. A plan is a chosen route from where we are to a result that we want to reach. The result is called a **goal**, and we met that word in Chapter I. In the journey, the goal is to reach the city and the plan is the bus and the early start.

Planning is the act of deciding in advance what to do, how to do it, when to do it, and who will do it. A plan is the result. Planning is the activity that produces it. Keep the two words apart: a manager does planning, and a firm holds a plan.

Planning looks forward. Yet it does not guess in the dark. The manager studies what happened last month, what is happening today, and what is likely to happen next. The bakery owner, for example, looks at last year's sales, this week's orders and the date of the festival. Out of all this comes a picture of the months ahead.

THE NATURE OF PLANNING

Five features describe the nature of planning. We take them one by one.

First, planning comes before the other functions. A manager cannot arrange people (organising), guide them (leading) or check results (controlling) until it is known what they are supposed to do. Plans give the other three functions something to work on. This is why planning is called the first function, although in practice all four overlap.

Second, planning happens at every level. The owner of a large firm plans for years ahead. A first-line supervisor plans the next shift. The scale differs, but the activity is the same. A bakery with only one manager still needs plans, and the owner is the planner.

Third, planning never ends. Conditions change, and a plan made in January may be out of date by April. The manager therefore returns to the plan, tests it, and revises it. A good plan is not a document in a drawer. It is a living guide.

Fourth, planning aims at goals. A plan with no goal is like a bus with no destination. Chapter VII studies goals and objectives in detail. For now, remember that every plan asks first: what are we trying to reach?

Fifth, planning means choosing among alternatives. There are almost always several routes to a goal. The student may go by bus, by train or by a friend's car. Planning asks the manager to weigh the choices and to pick one. A problem with only one possible answer needs no planning.

THE PURPOSES OF PLANNING

Why should a manager spend time on plans? Five purposes explain it.

The first purpose is direction. When everyone knows the goal and the route, people pull in the same direction. In the bakery, bakers, counter staff and the rider all know that the shop will bake more bread by a certain date. Each can then decide what to do without asking the owner every hour.

The second purpose is to prepare for change. Nobody knows the future, but managers can think about what is likely. A plan that says flour may become dearer prepares the shop to buy flour early or to change its recipes. Planning does not remove uncertainty. It reduces the damage that surprise can do.

The third purpose is to avoid waste. When the work is planned, resources are used where they are needed. A baker does not wait for flour that was never ordered. A rider does not make two trips where one would do. Planned work wastes less time, money and material.

The fourth purpose is to make control possible. Chapter XIII shows that managers check results against a standard. A plan supplies the standard. Without a plan, a manager can see what happened but cannot say whether it was good or bad.

The fifth purpose is to make managers think ahead. A manager who must write a plan has to look at the future, and that habit is useful in itself. Even when a plan changes, the thinking behind it leaves the manager better prepared.

Planning has five features that distinguish it from other work.

First, planning comes first among the functions.

Second, planning is done at every level.

Third, planning is a continuing process.

Fourth, planning is aimed at goals.

Fifth, planning involves choice.

Planning serves five purposes in a firm.

The first purpose is to give direction.

The second purpose is to reduce the effect of uncertainty.

The third purpose is to cut waste and overlap.

The fourth purpose is to set up control.

The fifth purpose is to encourage forward thinking.

IN SHORT

A plan is a chosen route to a goal. Planning is the activity of deciding in advance what to do, how, when and by whom. It gives direction, prepares for change, cuts waste, makes control possible, and trains managers to think ahead.

Planning joins each of the other three functions.

A manager spends more time planning at higher levels.

The time covered by a plan is called its horizon.

The manager should look as far ahead as the commitments that are being made.

You should now be able to say what planning is and why it matters.

PLANNING AND THE OTHER FUNCTIONS

Planning touches every other function, and it helps to see how. When a manager organises, the structure is built to carry out a plan. A plan to bake 300 loaves needs more bakers and a clearer division of tasks, and organising supplies them. When a manager leads, the plan tells the manager what to ask of people and why. When a manager controls, the plan supplies the standard of comparison. The four functions are not four separate boxes. They form one cycle, and planning is the place where the cycle starts again each time.

The amount of planning in a manager's day depends on the level. A top manager spends much time on long-range plans and little on daily tasks. A first-line supervisor does the reverse. Yet every manager plans, and every manager must plan with the plans of the level above in view. A supervisor who ignores the firm's strategy may plan a perfect shift that serves no useful purpose.

HOW FAR AHEAD SHOULD A MANAGER LOOK?

Every plan covers a stretch of time. We call this stretch the **planning horizon**. A short horizon covers days. A long horizon covers years. The right horizon depends on how quickly the firm's world changes and on how long its commitments last. A firm that buys an oven expected to last fifteen years must think many years ahead. A firm that sells fresh bread each day must plan hours ahead.

A useful rule is to look ahead as far as the present decision will have effects. If the bakery signs a five-year lease on a new shop, it is making a five-year commitment, and it should test the plan against five years of possible change. If the bakery buys flour for tomorrow, the horizon is one day. A manager who chooses too short a horizon is surprised by the consequences. A manager who chooses too long a horizon wastes effort on a future that cannot be known.

You should now be able to say this: a plan is a chosen route to a goal, planning is the continuing work of choosing it, and planning gives direction, prepares for change, cuts waste, makes control possible and encourages thought about the future.



KINDS OF PLANS

Plans differ by how far ahead they look and by how often they are used.

Not all plans are alike. A plan to open a second shop is very different from a plan for cleaning the ovens each night. To keep the many kinds in order, we sort plans in two ways. First we sort them by level, which means by how far ahead they look and how much of the firm they cover. Then we sort them by use, which means whether the plan serves again and again or only once.

PLANS BY LEVEL

Strategic plan. A long-range plan for the whole firm, setting its main direction.

A strategic plan sets the main direction of the whole firm for a long period.

The first level is the **strategic plan**. A strategic plan is a long-range plan for the whole firm. It answers the largest questions. What business are we in? Where do we wish to be in several years? Which markets will we serve? Top managers make strategic plans, because only they see the whole firm. A bakery that decides to become a chain of five shops within ten years has made a strategic plan.

The second level is the **tactical plan**. A tactical plan is a middle-range plan that shows how one department will carry out its part of the strategy. If the strategy is to open more shops, a tactical plan for the baking department might say how many bakers to train and which new ovens to buy in the next year. Middle managers usually make these plans.

The third level is the **operational plan**. An operational plan is a short-range plan for the daily or weekly work of one unit. It says who bakes what, at which hour, with which ingredients. First-line managers make these plans, because they are closest to the work.

The three levels fit together. The strategic plan sets the direction. Each tactical plan carries part of it forward. Each operational plan turns a tactical plan into daily tasks. The diagram below shows the shape. At the top there are few plans, and each is broad. At the bottom there are many plans, and each is narrow.

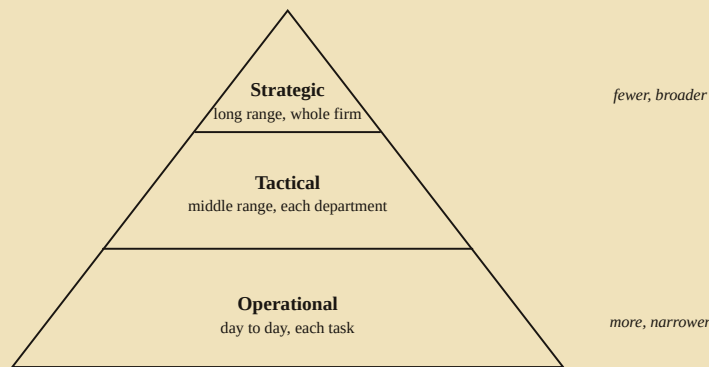


FIG. 17. *The three levels of plans*

We can compare the three levels in a table. Each row names one feature, and each column describes one level.

THREE LEVELS OF PLANS

FEATURE	STRATEGIC	TACTICAL	OPERATIONAL
Time covered	Several years	One year or less	Days or weeks
Part of the firm covered	The whole firm	One department	One unit or task
Usually made by	Top managers	Middle managers	First-line managers
Typical question	Where should the firm be going?	How will this department help?	Who does what today?
Bakery example	Open a second shop	Train bakers for the new shop	Set tomorrow's baking list

The time spans in the table are guides. A very fast-moving firm may call one year long-range. A slow-moving firm may call five years middle-range. What matters is the order: strategic first, tactical next, operational last, each serving the one above.

You should now be able to say this: a strategic plan is long-range and covers the whole firm, a tactical plan is mid-range and covers a department, and an operational plan is short-range and covers daily work.

Tactical plan. A middle-range plan that turns part of a strategic plan into work for one department.

A tactical plan turns part of the strategy into work for one department.

Operational plan. A short-range plan for daily or weekly work in a single unit.

An operational plan covers daily or weekly work.

The three levels fit together like a ladder.

The levels differ in time, in width and in who makes them.

The time spans are guides and not fixed limits.

You should now be able to name the three levels of plan.

One example can show how a strategic plan becomes daily work.

The tactical plans in turn become operational plans.

A mismatch between levels is a common weakness.

GO SLOWLY HERE. THIS SORTING IS A NEW IDEA.

Some plans are used again and again, and others are made for one occasion only.

Standing plan. A plan made once and used again and again for situations that repeat.

A standing plan is used again and again.

Single-use plan. A plan made for one occasion or one project, and not used again in the same form.

A single-use plan serves one occasion.

A table helps to keep the two apart.

HOW THE LEVELS FIT TOGETHER: A BAKERY CHAIN OF PLANS

An example will show how the three levels join. Suppose the owner of the bakery decides on a strategy: within five years the firm will serve three districts of the city. That is the strategic plan. The owner then asks each part of the firm to prepare a tactical plan. The baking department decides to train two new bakers each year and to buy a second oven in the second year. The sales department decides to open a counter in a second district in the third year. Each of these is a tactical plan.

The head baker now makes an operational plan. It says that this week, the new trainee will work beside an experienced baker on the morning shift, and that on Thursday the oven will be cleaned. These small plans look unimportant by themselves. Yet each is a small step toward the strategy. If the operational plans did not serve the tactical plans, and the tactical plans did not serve the strategy, the firm would be busy without arriving anywhere.

A common weakness is a mismatch between levels. A strategy may promise growth, while the daily plans cut every cost that growth would need. The manager who finds such a mismatch should not blame the people at the lower level. The fault lies in the plans, and the cure is to fit them together.

PLANS BY USE: STANDING AND SINGLE-USE

Now we sort plans in a second way. Think of a school. The school has a rule that every student must wear a uniform. This rule applies every day, to every student, year after year. The school also has a plan for its annual sports day. That plan is made once for one event and put away. The two plans serve different purposes, and management gives them different names.

A **standing plan** is a plan made once and used again and again for situations that repeat. It saves the manager from deciding the same matter each time. The school's uniform rule is a standing plan.

A **single-use plan** is a plan made for one occasion or one project. When the occasion ends, the plan ends. The school's plan for sports day is a single-use plan.

We put the two kinds side by side in a table. Read it across to compare them.

STANDING PLANS AND SINGLE-USE PLANS

FEATURE	STANDING PLAN	SINGLE-USE PLAN
Purpose	Handle matters that repeat	Handle one occasion or project
Life	Long, until changed	Ends with the occasion
Effort to make	Made once, used many times	Made fresh each time
Examples	Policy, procedure, rule	Programme, project, budget
Bakery example	The daily hygiene procedure	The festival-week order schedule

KINDS OF STANDING PLAN

Policy. A general guide to thinking and decision, which leaves room for judgement.

A policy is a general guide that leaves room for judgement.

Three standing plans need careful definition, because beginners confuse them. The first is the **policy**. A policy is a general guide to thinking and decision. It tells people the limits within which they may decide, and it leaves room for judgement. A bakery policy might say: we use only fresh ingredients and we

treat customers fairly. The policy does not say what to do in each case. It guides the decision of the person who must act.

The second is the **procedure**. A procedure is a fixed series of steps for a task that repeats. It tells people what to do and in what order. The bakery's hygiene procedure may say: wash hands, put on a clean apron, clean the table, and only then begin. Where a policy leaves room for judgement, a procedure leaves little.

The third is the **rule**. A rule is a fixed instruction that says that something must or must not be done, with no choice. A rule might say: no one enters the baking room without a head covering. A rule has no steps and no order. It is a single instruction. Note the difference. A procedure is a list of steps, and a rule is one instruction.

The next table shows the three side by side, with one bakery example for each.

POLICY, PROCEDURE AND RULE

PLAN	WHAT IT GIVES	ROOM FOR JUDGEMENT	BAKERY EXAMPLE
Policy	A general guide	Much	Use only fresh ingredients
Procedure	Steps in order	Little	Hygiene steps before baking
Rule	One fixed instruction	None	Head covering in the baking room

Read the last column from top to bottom. The same shop uses all three, and each plays a different part. The policy points the way. The procedure shows the steps. The rule draws a firm line.

KINDS OF SINGLE-USE PLAN

The first single-use plan is the **programme**. A programme is a large single-use plan that groups the goals, steps, resources and timing for a whole set of tasks. A bakery's programme to open a second shop would include finding a site, buying ovens, hiring staff and advertising.

The second is the **project**. A project is a smaller single-use plan, often one part of a programme, with its own tasks and deadline. Buying and installing the new ovens would be a project within the programme.

The third is the **budget**. A budget is a plan written in numbers. It states how much money, material or time will be used or earned over a period. A budget may support any plan. For a manager, it is the plan's common language, because numbers can be added, compared and checked.

A word of care is needed. In some books the budget is listed as a single-use plan, because it is made for one period and then replaced. In others it is listed separately. We list it with the single-use plans, since each budget covers one period only, and a new one is made for the next.

You should now be able to say this: policy, procedure and rule are standing plans, and programme, project and budget are single-use plans, and each of the six can be explained with its own example.

Procedure. A fixed series of steps for doing a repeated task.

A procedure is a fixed series of steps in order.

Rule. A fixed instruction that allows no choice.

A rule is a fixed instruction with no room for choice.

We can sum up the three in a table.

The three differ in how much choice they leave to the person.

Programme. A single-use plan that groups the goals, steps, resources and timing for a large set of tasks.

A programme is a large single-use plan made of many tasks.

Project. A smaller single-use plan, often one part of a programme, with its own tasks and deadline.

A project is a smaller plan, often one part of a programme.

Budget. A plan expressed in numbers: money, materials or time to be used over a period.

A budget is a plan in numbers.

We take care to say that budgets may be standing or single-use depending on how they are used.

You should now be able to tell the six plans apart.



THE PLANNING PROCESS

Making a plan follows an order of steps, and each step depends on the one before.

A good plan is not written in one burst of thought. It is built step by step, and each step gives the next one something to work on. We shall describe six steps. They are shown in the figure below, followed by the act of carrying the plan out and checking it.

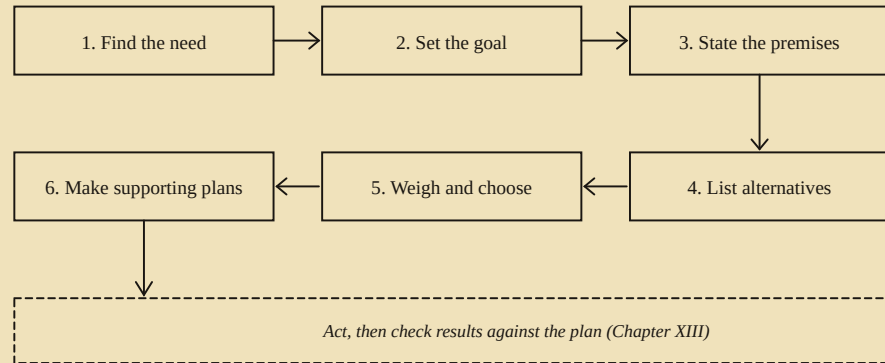


FIG. 18. *The six steps of planning*

The steps are a guide, and real planners often move back and forth between them.

Real planning is not always so tidy. A manager may find at step five that the goal of step two was too bold, and go back to change it. The six steps show the logic of planning. They do not forbid a return to an earlier step.

STEP ONE: FIND THE NEED OR THE OPPORTUNITY

Planning begins when a manager notices that something needs to be done.

Planning begins with awareness. A manager sees a problem, or an opportunity, before it is fully clear. A fall in sales is a problem. A new housing estate near the shop is an opportunity. The manager who notices early has more time to plan. At the bakery, the owner sees three things: rising demand, a rising price of flour, and the coming festival.

To judge the need, the manager must know the present position. Where does the firm stand today? What is it good at, and where is it weak? Without an honest answer, the plan may aim at something the firm cannot reach.

STEP TWO: SET THE GOAL

The goal says what the plan is trying to reach, and by when.

The second step is to set the goal. A goal states what the plan is trying to reach, and the time by which it should be reached. A weak goal says: we should bake more bread. A better goal says: we shall bake 300 loaves a day by the end of the twelfth month. The second can be checked, and the first cannot. We study goals more closely in Chapter VII.

Everything that follows depends on this step. If the goal is vague, the manager cannot tell which alternative is better. If the goal is wrong, the best plan in the world will carry the firm to the wrong place.

STEP THREE: STATE THE PREMISES

Planning premise. An assumption about the future conditions under which a plan will be carried out.

The third step is one that beginners often skip. The manager states the **planning premises**. A planning premise is an assumption about the future conditions under which the plan will be carried out. It answers the question: what do we expect the world to look like while the plan runs?

A planning premise is an assumption about the future that a plan rests on.

Think of a family that plans a picnic. The plan rests on a belief that the weather will be dry. That belief is a premise. If the day is wet, the plan fails, and the family fails with it. The family did not know the weather. It assumed. Every plan has premises of this kind, and the careful planner writes them down.

Premises come from two sources. Some come from the outside environment, which Chapter IV describes: prices, laws, competitors, the economy, the habits of customers. Some come from inside the firm: how many staff there will be, how much money is available, how well the equipment will work. The bakery owner writes two premises: flour will rise in price by about 5 per cent, and demand for bread will rise by about 8 per cent.

Premises should be stated openly. If the head baker plans on the belief that flour will cost the same, and the owner plans on a rise, the two plans will not fit. Written premises keep everyone planning on one picture of the future. They also let the manager find the cause of a failure, since a plan fails when a premise proves false.

The list of possible premises is endless, so the manager must choose. A good test is to ask: if this assumption were false, would the plan fail? If the answer is yes, the assumption is a key premise and must be written down. If the answer is no, it can be left out.

STEP FOUR: LIST THE ALTERNATIVES

The fourth step is to list the alternatives. An **alternative** is one possible way to reach the goal. The student's journey had three: bus, train, or a friend's car. The bakery has several ways to reach 300 loaves a day: buy a second oven, extend the opening hours, add a second shift, or buy dough from another supplier.

A manager should look widely. The first idea is not always the best, and a plan chosen from one alternative is not really a choice. It is better to list many and then to reduce the list.

STEP FIVE: WEIGH THE ALTERNATIVES AND CHOOSE

The fifth step is to weigh the alternatives and choose one. The manager measures each alternative against two standards: how well it will reach the goal, and how well it will work under the premises. Cost, risk, time and the skills of the staff all enter the comparison.

Some factors can be counted. The cost of a second oven is a number. Others must be judged. How will bakers feel about a second shift? How will customers respond to a new bread? A good decision uses both kinds of information. Chapter VIII studies this step, the making of a decision, in detail.

STEP SIX: MAKE SUPPORTING PLANS

The last step of planning is to turn the chosen plan into supporting plans. A plan to produce 300 loaves a day needs a hiring plan, a purchasing plan, a training plan and a schedule. It also needs a budget, which puts the whole plan into numbers. The budget shows whether the firm has the money, and it provides the figures against which results will later be compared.

Without supporting plans, a plan stays an idea. With them, each person knows the part to play.

AFTER THE PLAN: ACT, THEN CHECK

Planning does not end when the plan is written. The manager organises the work, leads the people and then checks results against the plan. This is the

GO SLOWLY HERE. A PREMISE IS NOT A FACT. IT IS A BELIEF ABOUT THE FUTURE.

The word premise needs everyday words to make it clear.

Premises come from the environment inside and outside the firm.

A premise should be stated openly so that everyone plans on the same assumptions.

A manager should choose the premises that matter most.

There are nearly always several ways to reach a goal.

The manager should search widely before choosing.

Each alternative is measured against the goal and the premises.

Some factors can be counted, and others must be judged.

The chosen plan needs smaller plans and a budget to make it work.

A plan is not finished until supporting plans and a budget exist.

The plan is carried out and then checked against its results.

Planning and controlling depend on each other.

The steps can be practised in four settings, from familiar to complex.

First setting: a journey.

Second setting: a week at the bakery.

Third setting: a school fair.

Fourth setting: premises from the outside world.

You should now be able to say the six steps in order.

function of controlling, studied in Chapter XIII. If the results differ from the plan, the manager corrects the work or changes the plan. The check completes the loop, and its findings feed the next round of planning.

Planning and controlling are therefore like two sides of one coin. Planning sets the standard. Controlling compares the results with the standard. A manager who plans without controlling does not learn whether the plan worked. A manager who controls without a plan has nothing to compare against.

THE LADDER: ONE PROCESS, FOUR SETTINGS

To make the process familiar, we apply it in four settings. We begin with an everyday case and end with one that joins ideas from earlier chapters.

A journey. A student must reach an examination in another city. The need is to arrive rested. The goal is to arrive by six in the evening before the examination. The premises are that buses run on time and the road stays open. The alternatives are bus, train and a friend's car. The student weighs cost and risk, chooses the train, and then makes supporting plans: buy the ticket, pack the bag, set an alarm.

A week at the bakery. The need is a larger order from a school on Thursday. The goal is to deliver 120 extra rolls on Thursday morning. The premises are that the ovens work all week and that flour arrives on Tuesday. The alternatives are to bake on Wednesday night or on Thursday before dawn. The owner chooses Wednesday night, since it leaves time to fix a problem. The supporting plans are a staff roster and a flour order.

A school fair. A school wishes to raise money at its yearly fair. The need is a new library cupboard. The goal is to raise a set sum by the day of the fair. The premises are good weather and a certain number of visitors. The alternatives are a food stall, a games stall or a book sale. The committee weighs cost and profit, chooses the food stall, and makes supporting plans: a list of helpers, a shopping list and a budget.

Premises from the environment. A clinic expects a new law to raise the cost of its medicines. This is an outside force of the kind we studied in Chapter IV, and it becomes a premise: medicine costs will rise by a certain share. The clinic's plan to keep services free for poor patients must now be tested against that premise. If the rise is larger than expected, the plan must change. This case shows how the outside world feeds the inside of a plan through its premises.

You should now be able to say this: planning has six steps, which are to find the need, set the goal, state the premises, list the alternatives, weigh and choose, and make supporting plans, and after them the plan is carried out and checked.



LIMITS TO PLANNING AND WHY PLANS FAIL

Planning is valuable, but it has limits, and many plans fail.

The first limit is that the future cannot be known.

Planning is not magic. A manager can plan with care and still be disappointed. To use planning well, we must know where it stops being useful, and the common reasons why plans fail. We take the limits first, and then the failures.

THE LIMITS OF PLANNING

The first limit is the unknown future. Every plan rests on premises, and premises are beliefs about a future that nobody can see. A flood, a sudden law or a rival's new shop may destroy a premise overnight. The further ahead a plan looks, the greater this risk. A plan for tomorrow is far safer than a plan for ten years.

The second limit is cost. Gathering information, holding meetings and writing plans use time and money. A small decision does not deserve a long plan. A manager who spends a week planning a task that takes an hour has wasted effort. The value of planning must be larger than its cost.

The second limit is that planning takes time and money.

The third limit is rigidity. A written plan can become a habit that people follow even when the world changes. A firm that clings to its plan may miss a chance that nobody expected. Plans are guides, and they should bend when the facts bend.¹

The third limit is that a plan can make a firm rigid.

The fourth limit is delay. A manager may keep studying and never decide, hoping for more certainty. By the time the plan is ready, the chance may be gone. There is a point beyond which more study adds little and waiting costs much.

The fourth limit is that planning can delay action.

The fifth limit is incompleteness. No plan covers every event. Staff will meet cases that the plan never imagined. The firm therefore needs people who can use judgement when the plan is silent.

The fifth limit is that plans cannot cover everything.

WHY PLANS FAIL

The limits above are built into planning. Plans also fail for reasons that lie within the manager's power. We name six of them.

Plans fail for reasons that a careful manager can often prevent.

The first reason is unclear goals. If the people who must carry out the plan cannot say what the goal is, they cannot work toward it. The plan then wanders. A manager should be able to state each goal in one plain sentence.

Reason one: the goals are unclear.

The second reason is false premises. A plan is only as sound as the beliefs under it. If the owner assumed a steady price of flour and the price rose sharply, the budget fails. The cure is to state premises openly, to check them from time to time, and to keep watching the outside world.

Reason two: the premises are wrong or never examined.

The third reason is lack of involvement. A plan made in the office and handed down to the workers is often misunderstood or resisted. The people closest to the work know things that planners may not know. When they help to make the plan, they understand it and they feel it is their own.

Reason three: the people who carry out the plan did not help to make it.

The fourth reason is missing resources. A plan to bake 300 loaves needs ovens, flour, staff and money. If any one is missing, the plan stops. Managers must check, before they commit, that the necessary resources can be found.

Reason four: the plan lacks resources or support.

The fifth reason is the lack of follow-up. A plan that nobody checks drifts. Chapter XIII shows that controlling is the other half of planning, and a plan without control is a wish.

Reason five: the plan is not followed up.

The sixth reason is the wrong balance. A plan that is too rigid cannot adapt. A plan that is too loose gives no direction. The art of the manager lies in a plan that is firm about the goal and flexible about the route.

Reason six: the plan is too rigid or too weak.

The next table lists the common causes of failure, with a short cure for each. Read each row as a problem and its remedy.

We can collect the limits and failures in a table.

¹ Some writers describe this danger as people becoming loyal to the plan and forgetting the goal that the plan was made to serve.

COMMON CAUSES OF FAILURE AND THEIR REMEDIES

CAUSE OF FAILURE	WHAT GOES WRONG	A REMEDY
Unclear goals	People do not know where to go	State each goal in one plain sentence
False premises	The plan rests on a wrong belief	Write premises down and check them
No involvement	People resist or misunderstand	Ask those who do the work to help plan
Missing resources	The plan stops half way	Check resources before committing
No follow-up	The plan drifts	Compare results with the plan regularly
Wrong balance	Too rigid or too loose	Fix the goal, keep the route flexible

You should now be able to give limits and causes of failure.

You should now be able to say this: planning is limited by an unknown future, by cost, by rigidity, by delay and by incompleteness, and plans commonly fail through unclear goals, false premises, lack of involvement, missing resources, no follow-up and the wrong balance between firmness and flexibility.

MAKING PLANNING EFFECTIVE

Effective planning follows a few simple conditions.

If we turn the causes of failure upside down, we find the conditions for success. They are few and they are practical.

First, make sure that planning has the support of top managers.

First, planning needs the support of top managers. When the owner or the senior team takes planning seriously, others do the same. When top managers ignore their own plans, nobody else will respect them.

Second, state goals and premises in plain words and share them.

Second, goals and premises should be stated in plain words and shared with all who need them. A plan that stays in one head cannot guide ten people.

Third, involve the people who will carry out the plan.

Third, involve the people who will do the work. Consult them while the plan is being made. They will improve it, and they will support it.

Fourth, keep plans simple and linked.

Fourth, keep plans simple. A plan that nobody can understand will not be used. The plans of different levels and departments should also fit together, so that the operational plans serve the tactical plans and the tactical plans serve the strategy.

Fifth, build in flexibility.

Fifth, allow for change. Good planners ask what they will do if a key premise proves false. A simple second plan, ready in advance, can save a firm that would otherwise be caught unprepared.

Sixth, check results and revise the plan.

Sixth, check results and revise. Planning is never finished. The manager compares what happens with what was planned, learns from the gap, and plans again with better knowledge.

PLANNING IN SMALL AND LARGE FIRMS

Small firms plan too, though often in the owner's head.

Beginners sometimes think that planning belongs only to large firms with planning departments. This is not so. A small firm plans, but the planning is often informal. The owner keeps the plan in mind and talks it over with a few people. This works while the firm is small and the owner knows every detail. As the firm grows, the plan must be written, because one head can no longer hold it, and because new staff cannot read the owner's thoughts.

A written plan has other uses for a small firm. A bank that is asked for a loan will wish to see one. A new manager will use it to understand the firm's direction. And the owner, in writing it, will find weak places that were hidden when the plan existed only in thought. For these reasons, even the smallest firm gains from putting its main plans on paper.

In a large firm, planning is shared among many people and often has its own staff, who collect information and help the departments to plan. The main danger there is distance: the planners may not know the work, and the workers may not know the plans. The conditions we listed apply with even greater force. Goals must be shared, and the people who do the work must be heard.

A last question remains: when is a plan not worth making? Where the matter is small, where the cost of planning exceeds the possible gain, or where a quick action is needed and a delay would cause harm, the manager may reasonably act at once. But where the matter is large, repeated or costly if wrong, planning pays for itself many times.



A WORKED CASE: A SIX-MONTH PLAN AT THE BAKERY

This case applies the six steps to one situation. It has five parts: the situation, the question, the reasoning, the conclusion and a check.

Case file, illustrative. Month 6. The Morning Loaf Bakery sells a loaf at Tk 70. Ingredients cost about Tk 24 a loaf when flour is Tk 60 a kilogram, and each loaf uses 0.24 kilograms of flour. Flour now costs Tk 69 a kilogram. The shop trades 26 days a month and pays Tk 40,000 monthly rent. The owner expects flour to rise by about 5 per cent and demand to rise by about 8 per cent over months 7 to 12. A festival will bring a week of heavy orders. The owner wishes to plan for 300 loaves a day by month 12.

Question. How should the owner plan months 7 to 12? What premises, what standing plan, what single-use plan and what budget does the plan need?

Reasoning. Step one: the owner has noticed three things, which are rising demand, dearer flour and a festival. Step two: the goal is 300 loaves a day by the end of month 12. Step three: the premises are that flour will rise by about 5 per cent, so that it costs about Tk 72.45 a kilogram, and that demand will rise by about 8 per cent. Step four: the alternatives are a second oven, longer hours or buying dough elsewhere. Step five: the owner weighs the three, a task that Chapter VIII treats in detail, and carries the choice forward. Step six: the owner makes supporting plans, and we describe three of them.

A standing plan is needed for hygiene. As output rises, more people handle more bread, and mistakes become more likely. The owner writes a hygiene procedure of four steps: wash hands, put on a clean apron, clean the table, and begin work. A rule is added: nobody enters the baking room without a head covering. Both will be used every day.

A single-use plan is needed for the festival week. The owner writes a festival-week order schedule that says how much extra dough to prepare on each day, when extra flour must arrive, and who works late. It will be used once and then put away.

A budget is needed in numbers. At 300 loaves a day for 26 days, the shop will bake 7,800 loaves a month. At Tk 70 a loaf, sales will be $7,800 \times 70 = \text{Tk}$

A written plan helps a small firm to borrow, to hire and to grow.

The plans of a large firm are more formal and more numerous.

A final test helps a manager to judge whether to plan at all.

The case applies the whole planning process to the bakery's sixth month.

The case states the facts first.

The question asks for the steps of the plan.

The reasoning takes the steps in order.

The first supporting plan is a standing plan for hygiene.

The second supporting plan is a single-use plan for the festival.

The third supporting plan is a budget.

546,000. Flour at the new price costs $0.24 \times 72.45 =$ about Tk 17.39 a loaf. The other ingredients cost $24 - 14.4 =$ Tk 9.60. Ingredients are therefore about Tk 27 a loaf, which gives $7,800 \times 27 =$ about Tk 210,600 a month. Rent stays at Tk 40,000.

The conclusion states the plan in a few words.

Conclusion. The plan has a goal of 300 loaves a day by month 12, two stated premises, one standing plan with a procedure and a rule, one single-use plan for the festival, and a budget that shows sales of about Tk 546,000, ingredients of about Tk 210,600 and rent of Tk 40,000 a month at full output.

The check tests the premise against the budget.

Check. We test the premise. If flour were to rise by 10 per cent instead of 5, it would cost Tk 75.90 a kilogram and Tk 18.22 a loaf. Ingredients would then be $18.22 + 9.60 =$ about Tk 27.8 a loaf, or about Tk 217,000 a month. The difference from the plan is about Tk 6,500 a month, which the budget can bear. The plan survives a wrong premise of this size, and the owner now knows how far the premise may be wrong before the plan must change.



SUMMARY

The summary restates the main ideas in a few lines.

A plan is a chosen route to a goal, and planning is the continuing work of choosing it. Planning gives direction, prepares for change, cuts waste, makes control possible and encourages forward thinking. Plans are sorted by level into strategic, tactical and operational, and by use into standing plans (policy, procedure, rule) and single-use plans (programme, project, budget).

The process has six steps and rests on premises.

The planning process has six steps: find the need, set the goal, state the premises, list the alternatives, weigh and choose, and make supporting plans. After the plan comes action, and then a check of results. A planning premise is an assumption about the future that the plan rests on. Plans fail through unclear goals, false premises, lack of involvement, missing resources, no follow-up and the wrong balance of firmness and flexibility. They succeed when these causes are removed.

THINK IT THROUGH

The first problem asks when a plan is not worth making.

A street seller opens a stall each morning and sells tea. A friend advises the seller to write a full plan, with goals, premises and a budget, for each day. Is this advice sound? Consider the cost and the possible gain of planning for such a small task. Then say what kind of planning, if any, would still help the seller.

The second problem asks you to find a premise.

A school plans an outdoor sports day for next month. List three premises on which this plan rests. For each, say what would happen to the plan if the premise proved false, and what second plan could be kept ready.

WHERE PEOPLE GO WRONG

A first error is to confuse a plan with a goal.

Error: A goal and a plan are the same thing. **Correction:** The goal is the result to reach. The plan is the chosen route to it.

A second error is to treat a premise as a fact.

Error: A premise is something we know to be true. **Correction:** A premise is an assumption about the future. It may prove wrong, and the plan should say what will be done if it does.

A third error is to mix up a policy, a procedure and a rule.

Error: A policy, a procedure and a rule are three names for one thing. **Correction:** A policy guides judgement, a procedure gives steps in order, and a rule is a single fixed instruction.

Error: A good plan is never changed. *Correction:* A good plan is firm about the goal and flexible about the route. When premises change, the plan should change.

A fourth error is to think that a plan, once made, must not change.

IN HISTORY

In May 1961 the American president, John F. Kennedy, told his country's lawmakers that the nation should commit itself, before the decade was out, to landing a man on the Moon and returning him safely to Earth. The goal was clear and carried a date. The space agency then built the plan in stages. It flew crews in orbit, practised meeting and joining spacecraft in space, and tested each part before the next. In July 1969 the Apollo 11 mission landed two astronauts on the Moon, within the time set.² The case shows a firm goal, a plan made of many linked smaller plans, and the careful testing of each step.

A national goal set in 1961 shows planning at a very large scale.

SOURCES

Fayol, whose work Chapter III described, placed planning first among the elements of management and stressed the value of a written plan of action in his book of 1916. The ideas of strategic, tactical and operational plans, and of standing and single-use plans, were developed by later management writers in the middle of the twentieth century. A reader who wishes to learn more should consult a standard management textbook and verify the exact origin of each term.

The idea that managers plan first comes from Fayol's work.

REVIEW QUESTIONS

1. Define plan and planning, and explain how they differ.
2. State and explain the purposes of planning.
3. Distinguish strategic, tactical and operational plans, with an example of each.
4. Distinguish standing plans from single-use plans, with a table.
5. Define policy, procedure and rule, and give a fresh example of each.
6. Describe the six steps of the planning process.
7. What is a planning premise? Why should premises be written down?
8. Explain the limits of planning and the reasons why plans fail.
9. State the conditions that make planning effective.

The review questions test recall and use of the chapter.

Recall the meaning of a plan.

Recall the purposes of planning.

Sort plans by level.

Sort plans by use.

Separate three standing plans.

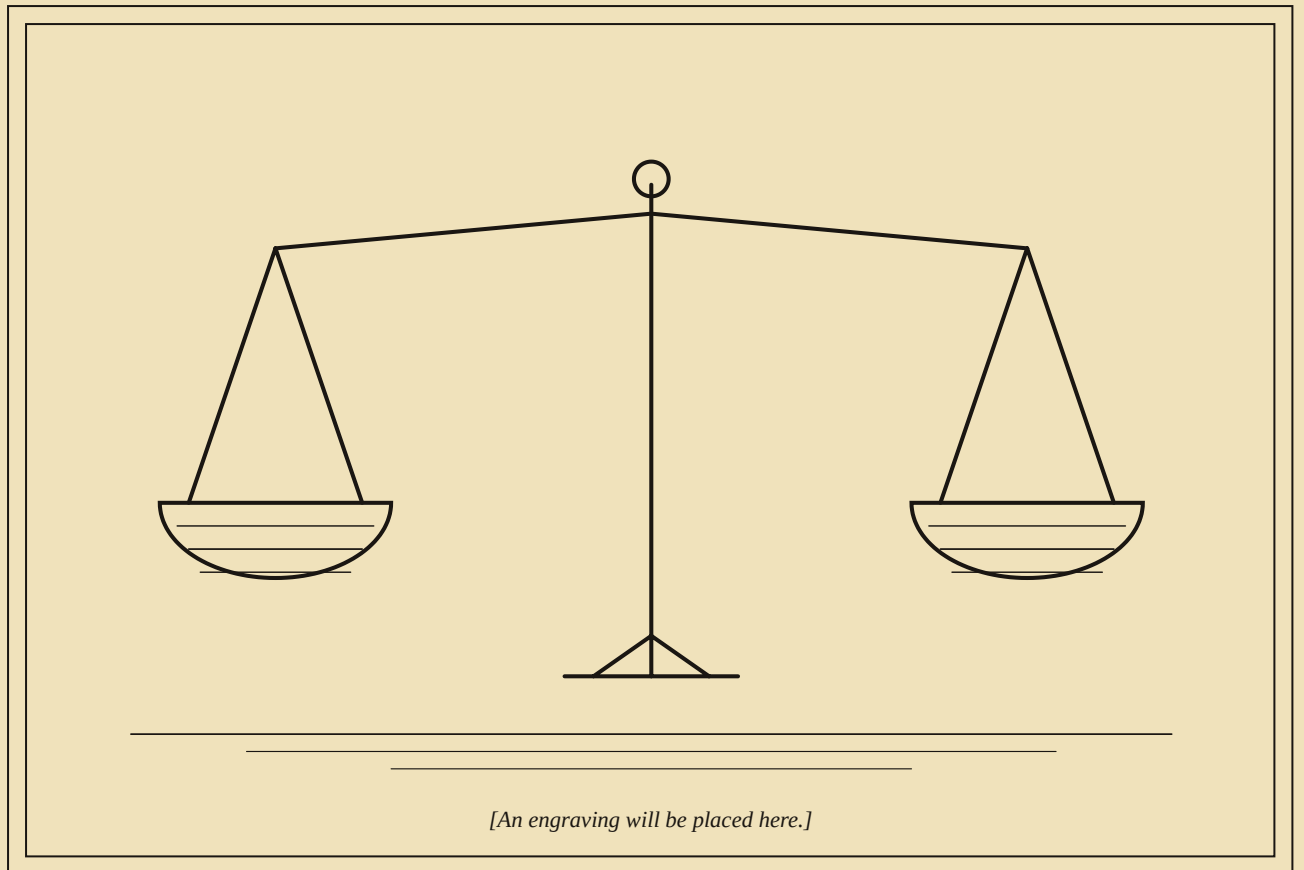
Recall the steps.

Explain premises.

Explain failure.

Explain effectiveness.

² Check all dates, the programme stages and the president's words against reliable sources before this passage is printed.



CHAPTER VII

Objectives and Management by Objectives

A traveller who cannot name the destination cannot say whether the journey went well.



This chapter explains what an objective is and why every plan needs one. It shows how objectives are arranged from the top of a firm to each worker, how they are set so that they can be checked, and how management by objectives joins the goals of managers and staff in a repeating cycle. It ends by weighing the gains and the dangers of the method.

BUSY, BUT GOING WHERE?

A bakery owner works hard all day and still cannot say whether the month went well.

Effort is not the same as progress.

The missing piece is a clear and shared statement of what must be reached.

You need three ideas from earlier chapters.

Return to the Morning Loaf Bakery in its seventh month. The plan of Chapter VI is made. Yet at the end of the first week the owner meets a puzzle. The staff have worked from dawn until night. The ovens have never been cold. Was it a good week?

The owner cannot answer, because the plan names a direction but does not say how far the shop should have come by Friday. The bakers feel they worked well. The rider feels tired. The counter assistant feels that customers were rude. Each person has a different idea of success, and no one is wrong, because no one has been told what success means.

The missing piece is the objective: a clear statement of the result to be reached, and the date by which it should be reached. Chapter VI gave the first step of planning as finding a need and the second step as setting the goal. This chapter studies the second step in depth, and then shows how a whole firm can set objectives together.

You will need three earlier ideas. From Chapter VI, the plan and the six steps of planning. From Chapter II, the four functions. And from Chapter VI again, the planning premise, because objectives are set with the future in mind.



WHAT OBJECTIVES ARE

GOAL AND OBJECTIVE

Everyday speech uses goal and objective as if they meant the same.

Objective. A specific result to be reached by a set time, stated so that it can be checked.

An objective is a specific result with a deadline.

A school exam shows the difference between a wish and an objective.

In daily speech, people use the words goal and objective for nearly the same thing. In management we keep a small but useful difference. A **goal** is a general result that someone wants to reach: to grow, to serve customers well, to be known for good bread. An objective is a more exact version of a goal. It says how much, and by when.

An **objective** is a specific result to be reached by a set time, stated so that it can be checked. The goal says: we want to bake more bread. The objective says: we shall bake 300 loaves a day by the end of the twelfth month. The goal gives a direction. The objective gives a place and a date.

Think of a student. A student who says, "I want to do well in the examination," has a goal. A student who says, "I shall score at least 70 per cent in the economics paper in June," has an objective. The second statement can be checked on the day, and it shapes what the student does each week.

THE NATURE OF OBJECTIVES

Objectives have four main features.

First, objectives are specific.

Four features describe good objectives. We take them in turn.

First, an objective is specific. It names exactly what will be reached. "Better service" is a goal. "Ninety-five orders in every hundred delivered on the day promised" is specific.

Measurable. Able to be counted or checked by a number or a plain test.

Second, objectives are measurable.

Second, an objective is **measurable**. Measurable means that it can be counted or checked by a number or a plain test. If nobody can say how much has been reached, nobody can say whether the objective has been met.

Third, objectives have a time limit.

Third, an objective has a time limit. "300 loaves a day" has no end point.

"300	loaves	a	day	by
------	--------	---	-----	----

the end of month 12" has one. The date tells everyone when the check will be made.

Fourth, an objective is realistic and agreed. An objective that nobody believes can be reached will be ignored. An objective forced on a person without discussion will be resented. The best objectives stretch people a little, and are accepted by those who must reach them.

WHY OBJECTIVES MATTER

Objectives serve four uses. The first is direction: they tell each person where to go. The second is coordination: when people share objectives, their work fits together. The third is motivation: people work harder toward a result that they understand and have helped to choose. The fourth is control: objectives give the standard against which results are measured, a link that Chapter XIII follows up.

Notice how objectives and plans fit. An objective says where to go. The plan says how to get there. A plan without an objective has no destination, and an objective without a plan has no road.

THE HIERARCHY OF OBJECTIVES

A large firm has many objectives. To keep them in order, managers arrange them in a **hierarchy** of objectives. A hierarchy is an arrangement in levels, each below the one above. At the top are the objectives of the whole firm. Below them are the objectives of each department. At the bottom are the objectives of each person.

Each level serves the level above. The firm's objective is divided into parts, and each department takes a part. Each department's objective is divided again, and each person takes a part. If every person reaches the personal objective, the department reaches its objective. If every department reaches its own, the firm reaches its own. The figure below shows the shape.

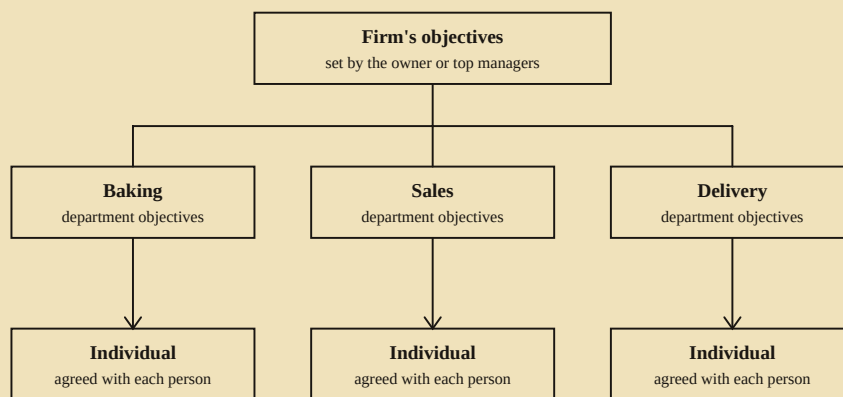


FIG. 19. The hierarchy of objectives, from the firm to the individual

The hierarchy works only when the levels fit. A department that sets its own objective without looking at the firm's may pull the firm in another direction. A good hierarchy is built both from the top down, which gives direction, and from the bottom up, which gives knowledge of what is possible.

Fourth, objectives are realistic and agreed.

Objectives serve four uses in a firm.

Objectives and plans are linked, because a plan is the route to an objective.

Objectives are arranged in levels, from the whole firm to each person.

Each level serves the one above it.

A hierarchy works only if each level is agreed with the level above.

Key result area. A part of the firm's work where results matter so much that objectives must be set there.

A firm cannot set objectives for everything, so it chooses the key areas.

Drucker proposed eight areas that fit most firms.

WHERE OBJECTIVES ARE NEEDED: KEY RESULT AREAS

A firm cannot set objectives for every action. It chooses the parts of its work where results matter most. We call these the **key result areas**. A key result area is a part of the firm's work where results matter so much to its success that objectives must be set there.

Peter Drucker, whose work we meet later in this chapter, suggested eight key areas for a business.¹ We give them below in plain words, with a bakery example for each.

EIGHT KEY RESULT AREAS, WITH A BAKERY EXAMPLE

KEY AREA	WHAT IT CONCERNS	BAKERY EXAMPLE
Market standing	The firm's place among competitors and customers	Share of bread sold in the district
Innovation	New products and better methods	Number of new breads launched
Productivity	Output compared with input	Loaves baked per staff hour
Resources	Money, equipment, materials	Ovens in working order
Profitability	Profit compared with money invested	Profit each month
Manager performance and development	How well managers do and grow	Training days completed
Worker performance and attitude	How well staff work and feel	Staff who stay in the job
Public responsibility	The firm's duty to society	Unsold bread given away

The table helps a manager decide where objectives are needed.

Read the table down the first and third columns. A bakery would not need an objective for every row at once. A manager looks at the table, decides which areas matter most this year, and sets objectives there. A firm that neglects one area, such as staff attitude, may meet its goals for sales and then lose its best workers.

OBJECTIVES BEYOND PROFIT

Many people think that a firm has only one objective, which is profit.

Many beginners think that a firm exists for one purpose only, which is to earn profit. Profit matters a great deal, because a firm that loses money cannot survive. Yet a firm that sets only a profit objective may lose its way. A bakery that cuts the size of its loaves to raise profit may find that customers go elsewhere. Profit is better seen as the reward for doing the other things well.

A firm needs several objectives that support one another.

For this reason, a firm sets objectives in several key areas at once. Some concern customers, some concern products, some concern staff and some concern the community. These objectives support one another over time. Pleased customers return, and returning customers bring profit. Skilled and settled staff bake better bread. The eight key areas of the table above are a reminder that a sound firm looks in many directions.

¹ Check the exact list and wording in Drucker's *The Practice of Management* (1954) before this chapter is printed.

OBJECTIVES OVER DIFFERENT PERIODS

Objectives, like plans, cover different lengths of time. Chapter VI separated strategic, tactical and operational plans, and objectives follow the same pattern. A long-range objective states what the firm wishes to be in several years. A middle-range objective states what a department must reach in a year. A short-range objective states what a person must reach this month or this week. The long-range objective gives meaning to the short ones, and the short ones are the steps by which the long one is reached.

A long objective that is not broken into stages is hard to manage. If the bakery wishes to reach 300 loaves a day in six months, the owner can add a milestone at the end of each month, such as 10 more loaves a day. A milestone is a mark on the way. It shows whether the work is on schedule, and it allows a correction while there is still time.

WHO SETS THE OBJECTIVES?

There are three ways to set objectives. In the first, top managers set them and pass them down. This gives direction but may ignore what is possible. In the second, each unit sets its own and sends them up. This uses local knowledge but may lose sight of the whole. In the third, both happen together: the top proposes, the units respond, and the two sides agree. Management by objectives uses the third way. It takes more time than the first two, but it combines direction with knowledge, and it gives each person a part in the decision.

You should now be able to say this: an objective is a specific, measurable result with a time limit, objectives are arranged in a hierarchy from the firm to each person, and a firm sets them in its key result areas.



SETTING OBJECTIVES

Writing an objective is a skill. A careless objective is vague, and a vague objective does no work. In this section we give a short method and four tests, and then we practise on examples.

A METHOD IN FOUR QUESTIONS

A manager can turn a wish into an objective by asking four questions in order. We state them in the form of a method.

The first question is: what exactly do we want? The manager names the result in plain words, as a thing that could be seen. "Happy customers" is hard to see. "Fewer complaints" is easier. "No more than two complaints in a week" can be seen exactly.

The second question is: how will we measure it? The manager chooses a number or a plain test. If no sensible number exists, the manager asks what would be true, and could be seen by anyone, if the result were reached.

The third question is: by when? The manager fixes a date, and if the objective is large, the dates of steps along the way.

The fourth question has two parts. Is the result reachable with the people and resources we have? And who will own it? An objective owned by no one is reached by no one. Each objective should have one person who answers for it.

Objectives look different over short, middle and long periods.

A long objective should be broken into stages.

Objectives may be set from the top, from the bottom, or by both.

You should now be able to say what an objective is.

Objectives must be made with care, and a few simple tests help.

Four questions turn a wish into an objective.

The first question is: what exactly do we want?

The second question is: how will we measure it?

The third question is: by when?

The fourth question is: is it reachable, and who will own it?

Five mistakes are common among beginners.

Mistake one: describing an activity and not a result.

Mistake two: stating no time limit.

Mistake three: setting too many objectives.

Mistake four: setting an objective that the person cannot influence.

Mistake five: setting an objective that is too easy or too hard.

A good objective passes four tests.

Test one: could a stranger check it?

Test two: does it serve the level above?

Test three: is it within the person's control?

Test four: does it fit with the other objectives?

Practice on vague goals shows how the four questions work.

COMMON MISTAKES IN WRITING OBJECTIVES

Beginners tend to make the same few mistakes when they write objectives. It helps to name them so that you can avoid them.

The first mistake is to describe an activity and not a result. "Hold weekly meetings" is an activity. "Reduce late orders to five in every hundred" is a result. An activity may be done well and still produce nothing. A result tells you whether the work succeeded.

The second mistake is to leave out the date. Without a date, an objective can be delayed for ever. The date also lets people plan their effort.

The third mistake is to set too many. When a person has twenty objectives, all seem equally important, and none gets full attention. A person usually does well with a few: perhaps three to six. Fewer, better chosen objectives work better than many.

The fourth mistake is to ask for a result that the person cannot influence. A baker cannot be held to the price of flour. A baker can be held to the care of the dough.

The fifth mistake is to aim too low or too high. An objective that is too easy is reached without effort, and teaches nothing. An objective that is too hard is abandoned early. A good objective is a stretch: hard enough to need effort, but possible for a capable person.

FOUR TESTS FOR A GOOD OBJECTIVE

When the objective is written, the manager applies four tests. Each test is a question that can be answered yes or no.

Test one: could a stranger check whether it was reached? If two people would disagree about whether the objective was met, the objective is too vague. Return and make it more exact.

Test two: does it serve the objective at the level above? An objective that helps nobody higher up wastes effort.

Test three: can the person who owns it influence the result? To ask a rider to raise sales is unfair, since the rider does not sell. To ask the rider to deliver on time is fair.

Test four: does it clash with other objectives? A firm that asks for the highest output and the lowest waste at once should ask whether these pull in opposite directions. We return to this point later in the chapter.

PRACTICE: MAKING VAGUE GOALS MEASURABLE

The best way to learn is to practise. The table gives four vague goals, as a manager might first state them, and a better version of each.

FROM VAGUE GOAL TO OBJECTIVE

VAGUE GOAL	WHAT IS WRONG	BETTER OBJECTIVE
Improve our service	No measure, no date	At least 95 orders in every 100 delivered on the day promised, by the end of month 12
Waste less bread	No amount, no date	Unsold and spoiled bread no more than 5 loaves in every 100 baked, by the end of month 9
Be a good place to work	Cannot be checked	Lose no more than one of eight staff in the next twelve months
Sell more	No amount, no date	Raise sales from Tk 400,000 to Tk 450,000 a month by the end of month 10

Look at the right-hand column. Each objective names a result, an amount and a date. A stranger could check each one. Notice also that each objective could be reached in more than one way, so the objective leaves room for the people to choose how.

Each better objective names a measure, an amount and a date.

You should now be able to say this: to set an objective, ask what exactly we want, how we will measure it, by when, and who will own it, then test it for checking, fit, control and clashes.

You should now be able to turn a vague goal into an objective.

WHEN OBJECTIVES CLASH

Objectives do not always agree. A bakery may wish to raise output and also to cut waste. When the ovens work faster, more loaves are burnt or undercooked. When the baker takes more care, output slows. The two objectives are each sensible, and together they pull against each other.

Sometimes two good objectives pull in opposite directions.

The manager should find such clashes before the work begins. There are three ways to deal with them. The first is to rank the objectives, and say which matters more if both cannot be reached. The second is to set the objectives together, so that the people who must reach them can say what is possible. The third is to set one objective that joins both, such as the number of good loaves per hour. A joint measure removes the clash.

A manager should find the clash early and decide what matters more.

The next table sets the three ways in order, with the bakery example.

A table shows the three ways in use.

THREE WAYS TO SETTLE A CLASH BETWEEN OBJECTIVES

WAY	WHAT THE MANAGER DOES	BAKERY EXAMPLE
Rank them	Decides which matters more	Waste is no more than 5 per cent, even if output is slightly lower
Set them together	Asks the people involved what both allow	The head baker says the ovens can reach 300 with 5 per cent waste
Join them	Writes one measure that includes both	Good loaves per staff hour

MANAGEMENT BY OBJECTIVES

Management by objectives. A method in which managers and staff agree objectives together and review results at fixed times (MBO).

Management by objectives is a way of running a firm by agreed objectives at every level.

Drucker gave the method its name in 1954.

MBO changes the usual picture of how objectives are set.

MBO has four features that distinguish it from other methods.

First, objectives are set together.

Second, objectives cascade through the firm.

Third, each person is judged on results.

Fourth, there are regular reviews.

One conversation shows how MBO differs from giving orders.

THE IDEA AND ITS ORIGIN

Setting objectives for a firm is one thing. Running a firm through objectives is another. The second idea is called **management by objectives**, and it is usually shortened to **MBO**. Management by objectives is a method in which managers and staff agree on objectives together, work toward them, and review the results at fixed times.

The idea is linked to the writer Peter Drucker (1909–2005). In his book *The Practice of Management*, published in 1954, he argued that a firm should be managed by agreed objectives in every key area, and that each manager should know the objectives of the whole and the objective of his or her own part.² Other writers, and some firms, helped to develop the method. We treat Drucker's book as the landmark source.

In the older picture, a senior manager decides what each person must do and tells them. In MBO, the manager and the person talk, and agree on the objective together. The manager brings the needs of the firm. The person brings knowledge of what is possible. Out of the talk comes an objective that both accept.

FOUR FEATURES OF MBO

Four features mark management by objectives.

First, objectives are set together. The manager and the person agree. This agreement is more than a courtesy. A person who helped to set an objective understands it and feels responsible for it.

Second, objectives cascade. The word cascade means to flow down in steps, like water over a series of rocks. The firm's objectives flow down to the departments, and the departments' objectives flow down to each person, as the figure of the hierarchy showed.

Third, people are judged on results against the agreed objectives. The manager asks: did you reach what we agreed? The manager does not ask only how many hours were worked. This puts attention on what the work achieves.

Fourth, reviews are held at fixed times. At each review the manager and the person look at the results together, discuss the reasons for any gap, and agree on what to do next.

AN MBO CONVERSATION

A short example shows how MBO sounds. The owner meets the person who will become head baker. In the older method, the owner might say: "You will bake 300 loaves a day." In MBO, the owner says: "The shop plans to reach 300 loaves a day by month 12, and to keep waste low. What can the baking team reach, and what will you need?" The candidate replies that 300 loaves are possible if a second oven is bought and a helper is hired, and that waste can fall to 5 per cent. The two agree on the objective, and on the help that will be given. The objective is now a shared promise and not an order.

² Check the date, the title and the argument against the book itself before printing.

Notice that the owner promised something: an oven and a helper. In MBO the manager also has duties. A manager who sets an objective and gives no support has used the language of MBO without its spirit.

GO SLOWLY HERE. IN MBO THE MANAGER ALSO PROMISES SOMETHING: HELP AND RESOURCES.

In MBO, the manager has duties too.

THE MBO CYCLE

MBO is not done once. It runs in a repeating round that we call the **MBO cycle**. The MBO cycle is the round of agreeing objectives, planning the action, acting, reviewing progress, and then appraising and resetting. The figure below shows the five stages.

MBO cycle. The repeating round of agreeing objectives, planning, acting, reviewing and resetting.

MBO is not a single event but a repeating cycle of five stages.

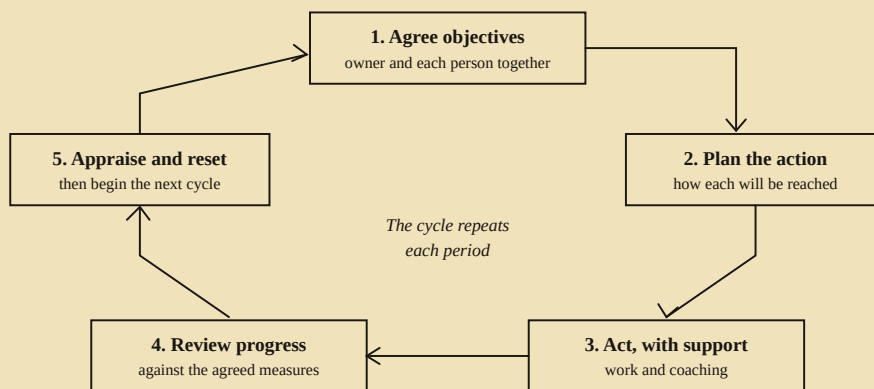


FIG. 20. *The MBO cycle*

STAGE ONE: AGREE OBJECTIVES

The owner or top manager first sets the firm's objectives, using the key result areas. Then each manager meets each person to agree the person's objectives. The objectives pass the four tests of the last section.

The first stage is to agree objectives from the top down and from the bottom up.

STAGE TWO: PLAN THE ACTION

Each person then writes a short plan: the steps, the resources needed and the dates. This stage links MBO to the planning of Chapter VI. The manager reads the plan, discusses it, and agrees on any help that will be given.

The second stage is to plan how each objective will be reached.

STAGE THREE: ACT, WITH SUPPORT

The people now do the work. The manager's role is to support: to remove obstacles, to supply what was promised, to coach, and to leave the person room to decide how to proceed. A manager who stands over every task defeats the purpose of MBO.

The third stage is to do the work, with help from the manager.

STAGE FOUR: REVIEW PROGRESS

At agreed intervals, the manager and the person look at the results. They compare actual results with the objective. If the results fall short, they ask why. The cause may be in the person, the plan, the resources, or the objective itself. Chapter XIII treats this comparison as the heart of controlling.

The fourth stage is the review of progress at fixed times.

STAGE FIVE: APPRAISE AND RESET

At the end of the period, the manager and the person appraise the whole. Appraise means to judge the worth of something. They ask what was reached,

The fifth stage appraises the period and sets new objectives.

what was learned and what should change. New objectives are then agreed, and the cycle begins again. Each round starts with better knowledge than the last.

HOLDING A GOOD REVIEW MEETING

The review is where MBO succeeds or fails. A poor review feels like a trial, and people hide problems. A good review feels like a shared study of how the work is going. A few habits help.

First, prepare. The manager reads the figures before the meeting, and asks the person to do the same. Second, begin with what went well. People who hear nothing but faults stop speaking. Third, listen. The person closest to the work often knows the true cause of a gap. Fourth, keep to facts and avoid blame. The question is not "who failed?" but "what stood in the way?" Fifth, end with agreement on what will happen next, and write it down in a few lines.

A last habit is often forgotten. The manager and the person should ask whether the objective itself is still right. If flour has doubled in price, or an oven has failed, an objective set last month may no longer be fair. A good review is willing to change the objective, and does not insist on a number that has lost its meaning.

MEASURING AIMS THAT ARE HARD TO COUNT

Some aims, such as friendliness, skill or trust, are hard to count. Yet a manager should not drop them. There are three ways to measure them. One is to use a sign that can be counted: the number of customers who return, or the number of compliments received, can stand in for good service. A second is to use a plain test: a customer who orders by telephone is greeted within three rings. A third is to ask the people concerned, through a short question put to customers or staff each month.

Remember that a sign is not the thing itself. A count of compliments may rise because staff ask customers for praise, and not because the service improved. The manager should look at the sign with some doubt, and check it against what is seen and heard on the shop floor.

MINI-CASE: A CLINIC SETS OBJECTIVES

Consider a small clinic with a doctor, two nurses and a receptionist. The clinic's purpose is to treat patients well, and its income only has to cover its costs. The doctor and staff meet to set objectives for the next six months.

Three objectives are agreed. First, no patient with an appointment should wait more than 20 minutes in at least 90 visits in every 100. The receptionist and nurses share this objective. Second, every patient who needs a follow-up visit should be given a date before leaving, in all cases. The receptionist owns this one. Third, the records of every visit should be completed on the same day. Each nurse owns this one.

We apply the four tests. A stranger can check each objective, since waiting times and records can be counted. Each serves the clinic's aim of good care. Each owner can influence the result. The first objective, however, shows a danger. A nurse who wishes to meet the 20-minute limit might hurry a patient. The doctor therefore adds a fourth objective for quality, by asking patients each month whether they felt rushed. The clinic keeps both numbers in view. This shows how a manager can guard against the distortion that a single measure may cause.

You should now be able to say this: management by objectives is a method in which managers and staff agree objectives together, and

The review meeting is the heart of the cycle, and it needs care.

The manager should prepare, listen and keep to the facts.

The meeting should also ask if the objective is still right.

Some important aims resist counting, but a plain test can still be found.

A sign is not the aim itself, and care is needed.

A clinic shows MBO in a service where profit is not the aim.

The staff agree three objectives that can be checked.

The objectives pass the tests, and show a danger.

You should now be able to describe the MBO cycle.

it runs as a cycle of agreeing objectives, planning, acting, reviewing and resetting.



BENEFITS AND WEAKNESSES OF MBO

No method is perfect. We list the gains first, and then the dangers, and then we show how a careful manager can reduce the dangers.

MBO has real gains and real dangers, and a manager should know both.

THE BENEFITS

The first benefit is clear direction. Each person knows what is expected, and can see how it connects to the aims of the firm. People waste less time on work that serves no purpose.

The first benefit is clearer direction.

The second benefit is commitment. People who help to set their objectives feel that the objectives are their own. They are likely to try harder than people who obey orders.

The second benefit is stronger commitment.

The third benefit is better control. Because objectives are measurable, the manager can see where results fall short, and can act early. Judgement of people is also fairer, since it rests on agreed results and not on the manager's mood.

The third benefit is better control and fairer judgement.

The fourth benefit is better communication. The regular talks give people a chance to say what they need and to raise problems. The manager learns what is happening at the working level.

The fourth benefit is better talk between manager and staff.

The fifth benefit is development. People who set objectives and review them learn how to plan, to measure and to judge their own work. Many learn to take on larger tasks.

The fifth benefit is better development of people.

THE WEAKNESSES

The first weakness is that objectives may distort behaviour. People tend to do what is measured. If an objective counts loaves baked, people may bake carelessly to raise the count. If it counts speed of delivery, the rider may take dangerous risks. The measure then replaces the purpose that it was meant to serve.

The first weakness is that objectives can take over from the real purpose.

The second weakness is that easy things get attention. Some important aims, such as friendliness or creativity, are hard to measure. Under MBO they may be neglected, because they do not appear in the objectives.

The second weakness is that what is easy to measure gets attention.

The third weakness is the burden of time. Setting objectives with each person, writing plans and holding reviews take many hours. In a large firm, the paper may grow until it hides the work.

The third weakness is that it takes time and paper.

The fourth weakness is misuse. A manager may impose the objectives and call it MBO. Then the method has the name but not the agreement, and people feel pressed and not helped.

The fourth weakness is that it can be used as a way to push people harder.

The fifth weakness is rigidity. If the world changes during the year, the objectives set at the start may no longer make sense. A firm that holds to them can drift away from what it now needs.

The fifth weakness is that a changing world can make objectives out of date.

REDUCING THE DANGERS

A careful manager can lower these dangers. Set a small number of objectives, so that no one is overloaded. Choose measures that reflect the true purpose. Include some objectives for quality, not only for amount. Review objectives during the year and change them when the facts change. And above all, keep the agreement real: if the person has no voice, the method does not work.

Most of the dangers can be reduced by care.

A table sets benefits and weaknesses side by side.

The table below puts the gains and the dangers side by side, with the remedy.

BENEFITS, WEAKNESSES AND REMEDIES IN MBO

BENEFIT	MATCHING WEAKNESS	A REMEDY
Clear direction	Measures replace the purpose	Choose measures that reflect the true aim
Commitment	Objectives imposed and called MBO	Keep the agreement real
Better control	Easy things get the attention	Include quality, not only amounts
Better talk	Heavy time and paper	Few objectives, short forms
Development	Objectives go out of date	Review during the year and revise

You should now be able to weigh MBO.

You should now be able to say this: MBO gives direction, commitment, control, communication and development, but it can distort behaviour, favour what is easy to measure, use much time, be misused, and become out of date, and these dangers can be reduced by care.

MBO AND THE OTHER PARTS OF MANAGEMENT

MBO connects to the other functions and to later chapters.

Management by objectives is not a separate world. It connects to many of the ideas that appear in other chapters. We name three links here, and later chapters develop them.

The first link is with planning.

The first link is with planning, which we studied in Chapter VI. An objective is the second step of the planning process, and MBO is a way of carrying that step through the whole firm. Objectives come first, and plans follow. If the objectives change, the plans must change as well.

The second link is with motivation.

The second link is with motivation. People work harder when they take part in choosing the result and see how it serves a purpose they respect. Chapter XI studies what drives people to work, and it will show that this effect has a long record in research.

The third link is with controlling.

The third link is with controlling. An agreed objective is a standard, and the monthly review compares results with the standard. This is the process of Chapter XIII. When the comparison shows a gap, the manager acts to close it. MBO may thus be seen as planning, motivating and controlling joined in one repeating method.

THE COST OF MEASURING

Measuring takes effort, and the manager should weigh the cost.

Every measure has a cost. Someone must collect the figure, write it down and read it. A measure that costs more than it is worth should be dropped. At the bakery, counting loaves at the end of the day takes a minute and is worth it. Timing every customer at the counter might take hours and tell the owner little. A good manager measures the few things that matter most, in the simplest way that gives a reliable answer.

A measure should be reliable and easy to understand.

A good measure has two further qualities. It is reliable, which means that two people counting the same thing would reach the same answer. And it is easy to understand, which means that the people being measured know what is counted and how. Nobody can work toward a number that they do not understand.

THE LADDER: ONE IDEA, FOUR SETTINGS

Four settings show the ideas of the chapter at work.

We now apply the chapter in four settings, from the simplest to a case that joins several ideas.

One objective, one measure. A baker wishes to waste less dough. The goal is vague. The baker states the objective: no more than 5 loaves in every 100 spoiled, by the end of month 9. The measure is the count of spoiled loaves each day. A stranger can check it. This is the clean case: one result, one number, one date.

Clashing objectives. The owner also asks the baker to reach 300 loaves a day. The baker points out that fast baking spoils loaves. The two objectives clash. The owner and the baker agree to a joint measure: good loaves per staff hour. The joint measure lets the baker work at a pace that gives both output and care.

A charity. A charity feeds poor families in a town. It cannot count profit, so its measures differ. The director agrees with the volunteers: serve 400 meals a week by the end of month 6, and keep complaints at no more than one in every hundred meals. The measures are not money, yet they can be checked. This setting shows that MBO is not limited to firms that sell for profit.

A cascade. The owner of a bakery sets a firm-level objective for month 12. The head baker agrees the part for the baking team. The head baker then agrees a part with each baker, who decides how to reach it. The three levels fit like the figure of the hierarchy. If each person reaches the personal objective, the firm reaches its own. The next section shows this cascade in a worked case.



A WORKED CASE: AGREEING OBJECTIVES AT THE BAKERY

This case has five parts: the situation, the question, the reasoning, the conclusion and a check.

Case file, illustrative. Month 7. The Morning Loaf Bakery has a plan to reach 300 loaves a day by month 12, described in Chapter VI. The owner decides to turn the plan into objectives and to agree them one by one with the head-baker candidate, the counter assistant and the rider. Three firm objectives are set: 300 loaves a day by month 12, waste of no more than 5 per cent of loaves baked, and at least 95 per cent of orders delivered on time. The owner will review progress with each person every month.

Question. How should the owner agree these objectives with each person, and do they pass the tests?

Reasoning. Stage one: agree. The owner meets the head-baker candidate first, and proposes the output and waste objectives. The candidate agrees to both on condition that a second oven is bought, which is a matter for the decision of Chapter VIII. The owner then meets the rider and agrees the objective for on-time delivery. The owner meets the counter assistant and agrees that orders will be packed and labelled by an agreed hour each day, since late packing causes late delivery. The last objective is the counter assistant's own part of the on-time objective.

Stage two: each person writes a short plan. The head baker's plan names the oven, the trainee helper and the dates. Stage three: the work begins, and the owner supports it by buying the oven and not by checking every tray. Stage four: at the end of each month, the owner and each person compare results with the objective. Stage five: at the end of the six months, the owner and staff appraise the period and set new objectives.

Now we apply the four tests. Could a stranger check each objective? Yes: loaves can be counted, waste can be counted, and deliveries on time can be

First setting: one objective with one measure.

Second setting: clashing objectives.

Third setting: a charity.

Fourth setting: a cascade from owner to staff.

The case applies the chapter to the bakery's seventh month.

The case states the facts first.

The question asks how the objectives are agreed and tested.

The reasoning follows the MBO cycle.

The reasoning continues through the other stages.

We test the objectives against the four tests.

counted. Does each serve the level above? Yes, each supports the firm's plan. Can each owner influence the result? Yes: the baker controls baking and waste, the rider controls delivery, the counter assistant controls packing. Do any clash? Possibly: higher output may raise waste. The owner therefore asks the baker to watch both numbers together.

The conclusion states the result.

Conclusion. The bakery now has three firm objectives, each owned by a named person, each measurable, each dated, and each agreed in conversation. The objectives are linked to the plan, and a monthly review is fixed.

The check measures what the waste objective is worth.

Check. We test what the waste objective is worth. At 300 loaves a day, 5 per cent waste means 15 loaves, since $300 \times 0.05 = 15$. If waste were 8 per cent, it would be 24 loaves, since $300 \times 0.08 = 24$. The difference is 9 loaves a day, or $9 \times 26 = 234$ loaves in a month of 26 trading days. At about Tk 27 of ingredients a loaf, the saving is about $234 \times 27 = \text{Tk } 6,318$ a month. The objective is worth the effort of measuring it.



SUMMARY

The summary restates the main ideas in a few lines.

An objective is a specific, measurable result with a time limit, set so that it can be checked. Objectives are arranged in a hierarchy from the whole firm to each person, and firms set them in key result areas such as market standing, productivity and profitability. A good objective passes four tests: a stranger could check it, it serves the level above, the owner can influence it, and it does not clash with others.

MBO and its cycle are the central ideas.

Management by objectives is a method in which managers and staff agree objectives together, work toward them, and review the results at fixed times. It runs as a cycle of five stages: agree objectives, plan the action, act with support, review progress, and appraise and reset. MBO gives direction, commitment, control, communication and development. It can also distort behaviour, favour what is easy to measure, use much time, be misused and become out of date. Care in use reduces these dangers.

THINK IT THROUGH

The problem asks why too many targets can distort behaviour.

A delivery firm gives each rider ten objectives: speed, fuel use, number of parcels, number of complaints, cleanliness of the van, politeness, early arrival, no accidents, punctual return and low costs. After a few months, the firm finds that complaints are rising and two vans have been damaged. Explain why too many targets might distort behaviour. Then choose the three objectives that you would keep, and say why.

A second problem asks you to rewrite two vague goals.

Rewrite these two goals as objectives that pass the four tests: "Make the shop cleaner" and "Keep our regular customers happy". For each, name the measure, the date and the person who would own it.

WHERE PEOPLE GO WRONG

A first error is to treat a goal and an objective as the same.

Error: Any wish is an objective. **Correction:** A goal gives a direction. An objective says how much and by when, and can be checked.

A second error is to think MBO means the manager sets targets and the staff obey.

Error: MBO means the manager sets the targets and the staff reach them. **Correction:** In MBO the objectives are agreed, and the manager owes help and resources as well.

Error: If a result is hard to measure, it does not belong in the objectives.
Correction: Hard-to-measure aims such as quality or friendliness may need a plain test, and a firm that ignores them can win its numbers and lose its purpose.

Error: Once objectives are agreed, they stay fixed for the year. *Correction:* Objectives are reviewed at fixed times and revised when the facts change.

A third error is to measure only what is easy.

A fourth error is to set objectives once and forget them.

IN HISTORY

In 1954 Peter Drucker published *The Practice of Management*. In it he argued that a manager should not be judged by effort or by long hours, but by results measured against objectives that the manager had helped to set. He also argued that every business needs objectives in several key areas, and not in profit alone.³ The book was widely read, and the method of managing by objectives spread to many firms and public bodies in the following decades. Later writers added criticism, and firms changed the method, but the central idea that people should work toward agreed, measurable results remains in wide use.

Drucker's 1954 book made objectives a central idea of management.

SOURCES

The landmark source for this chapter is Peter Drucker's *The Practice of Management* (1954). It matters because it set out management by objectives as a general method and named the key areas in which a business needs objectives. A reader who wishes to learn more may consult a standard management textbook, and should verify the origin of each term.

Drucker's book of 1954 is the landmark source.

REVIEW QUESTIONS

1. Define objective, and explain how it differs from a goal.
2. State and explain the main features of a good objective.
3. What is a hierarchy of objectives? Draw and explain it for a small firm.
4. Name five key result areas and give one example of each.
5. Describe the four questions and four tests used to set an objective.
6. Define management by objectives and describe its main features.
7. Draw the MBO cycle for a new case and explain each stage.
8. State the benefits and the weaknesses of MBO, and say how the weaknesses may be reduced.

The review questions test recall and use of the chapter.

Recall the meaning of an objective.

Recall the features of objectives.

Recall the hierarchy.

Recall the key areas.

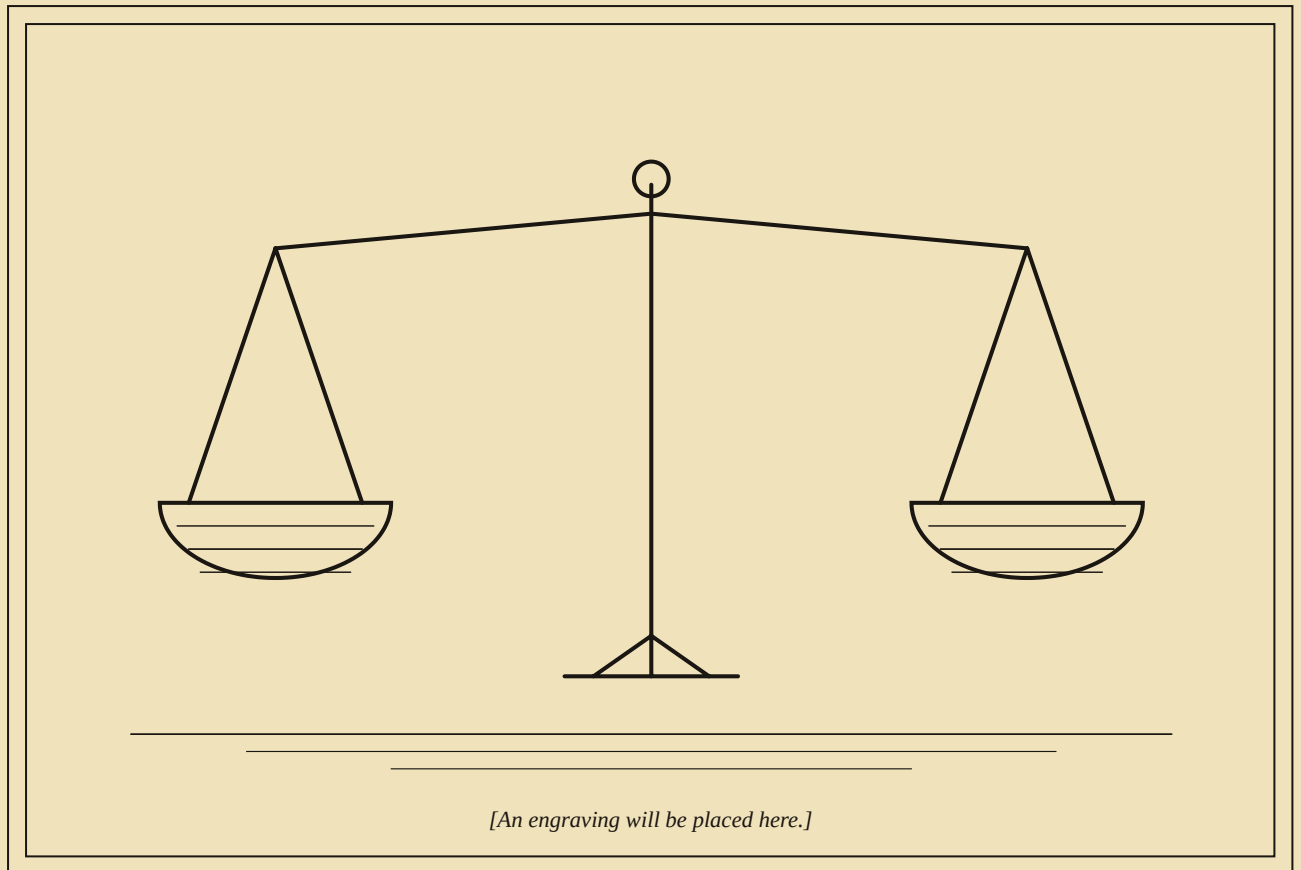
Recall the method.

Define MBO.

Draw the cycle.

Weigh the method.

³ Verify the publication details, the key areas and the exact claims against Drucker's own text before this passage is printed.



CHAPTER VIII

Decision Making

*Every choice closes some doors; a sound method shows us
which doors we are closing.*



This chapter explains how managers choose between alternatives.

*It describes the parts of a decision, the three conditions under
which decisions are made, and the steps of a rational process. It
then teaches probability, expected value and the decision tree, and
ends with the limits that shape real decisions, made alone and in
groups.*

TWO STALLS, TWO RESULTS

A puzzle: the careful trader loses and the careless trader gains.

Karim gathers facts and chooses umbrellas; Salma checks nothing and chooses sun hats.

The month is dry, so Karim loses and Salma gains.

The question: who decided better, the one with the better method or the one with the better result?

The chapter builds the needed language in five sections and then returns to the traders.

You will need four earlier ideas, each from an earlier chapter.

Imagine two traders, Karim and Salma, who sell goods from stalls in the same market. Each has Tk 10,000 to spend on goods for the coming month. Each must choose between buying umbrellas and buying sun hats. The monsoon is near, and rain is likely, but no one can promise it.

Karim studies the last three rainy seasons. He asks two suppliers for prices and estimates how many umbrellas he can sell in a wet month. He compares this plan with the sun hats and decides on umbrellas, for good reasons. Salma checks no figures. She sees a pile of bright sun hats at a low price and buys them.

The month is dry. Rain falls on only two days. Karim sells few umbrellas and ends the month with a loss. Salma sells nearly all her sun hats and earns a profit. Karim made a careful choice and received a poor result. Salma made a careless choice and received a good result.

Now ask a plain question: which trader decided better? Some readers answer "Salma", because her result was better. Others answer "Karim", because his method was better. Both answers seem reasonable, and the difference between them is the subject of this chapter. To settle the matter we must separate three things: the choice, the method of choosing, and the result. We also need words to describe what Karim knew when he chose. He knew a great deal, but he did not know the weather.

The chapter builds this language step by step. Section 8.1 names the parts of every decision. Section 8.2 shows that decisions are made under three conditions, according to what the chooser knows. Section 8.3 states the seven steps of a careful, logical method. Section 8.4 teaches three tools for comparing chancy alternatives: probability, expected value and the decision tree. Section 8.5 asks how managers really decide, alone and in groups. At the end, we return to Karim and Salma.

You will need. This chapter uses four earlier ideas: the resources a manager works with (Chapter I), the four functions of management (Chapter II), planning premises (Chapter VI), and objectives (Chapter VII). Each is recalled briefly where it is used.



8.1 THE DECISION SITUATION

Everyone decides; managers decide for many people at once.

A decision is a choice from two or more possible actions.

Decision. A choice made from two or more possible actions.

Each possible action is an alternative, and doing nothing counts as one.

Alternative. One of the possible actions from which a choice is made.

Every day you make choices. This morning you chose how to travel to your classes: by bus, by rickshaw, or on foot. You compared cost, time and comfort, perhaps without noticing, and then you chose. In this chapter we study the same activity as managers perform it. A manager spends the money, time and effort of an organisation, and so the choice carries more weight.

We begin with exact words. A **decision** means a choice made from two or more possible actions. Your choice of bus, rickshaw or walking is a decision. If only one action were possible, there would be no choice to make, and so no decision.

Each possible action is an **alternative**. An alternative means one of the possible actions from which a choice is made. Bus, rickshaw and walking are three alternatives. Doing nothing is also an alternative, and managers often forget this. A shop owner who delays a purchase has chosen to delay, and that choice has results.

Chapter I defined management as getting things done through people and resources. Every function in Chapter II contains decisions. Planning asks which goal to pursue and which route to take. Organising asks who will perform which task. Leading asks how to guide a group. Controlling asks what to correct. Some writers therefore call decision making the central part of a manager's work. We need not accept that claim fully to agree that no manager works for a day without choosing.

A manager's decision differs from your own in three ways. It commits resources that belong to the organisation, not to the manager. It affects other people: staff, customers, suppliers and owners. And it must often be explained afterwards to those who are affected. These features make the quality of the method important, and not only the quality of the result. The opening puzzle pointed to this idea, and we shall return to it.

Every decision, large or small, contains four elements. We name them one at a time and use Karim's choice as the example.

The first element is the **decision maker**. A decision maker means the person or group that chooses and is accountable for the choice. Chapter I defined accountability as being answerable for results. In the opening puzzle, the decision maker is Karim. In a large firm, it may be a committee.

The second element is the goal. Chapter I defined a goal as a result someone wants to reach. The goal gives the decision maker a standard for judging. Karim wants the highest profit from Tk 10,000. Without a goal, no alternative can be called better than another. Chapter VII showed how a vague goal becomes a measurable objective, and a measurable objective is the best standard of all.

The third element is the set of alternatives, which we have defined. Karim's alternatives are umbrellas, sun hats, or buying nothing. A decision can be only as good as the best alternative on the list, so the list needs care. Section 8.3 returns to this point.

The fourth element is the outcome. An **outcome** means what happens after an alternative is chosen. The same alternative can lead to different outcomes, because events outside the decision maker's control also act. If Karim buys umbrellas, the outcome in a wet month is good and the outcome in a dry month is poor. Karim controls his choice. He does not control the weather. A **payoff** means the value of an outcome to the decision maker, usually measured in taka. If the umbrellas bring a loss of Tk 3,000, the payoff of that outcome is -Tk 3,000.

The table below summarises the four elements. Read each row from left to right: it names the element, states its meaning, and shows the element in the traders' case.

Every management function contains decisions, so no manager works a day without choosing.

A manager's decision uses the organisation's resources, affects others, and must be explained.

Every decision has four elements, which we now name one by one.

The first element is the person or group who chooses and is accountable.

Decision maker. The person or group that chooses and is accountable for the choice.

The second element is the goal, which gives a standard for judging alternatives.

The third element is the set of alternatives: umbrellas, sun hats, or neither.

The fourth element is the outcome, and each outcome has a payoff.

Outcome. What happens after an alternative is chosen.

Payoff. The value of an outcome to the decision maker, usually measured in taka.

The table below summarises the four elements; read each row from left to right.

THE FOUR ELEMENTS OF A DECISION

ELEMENT	MEANING	IN THE TRADERS' CASE
Decision maker	The person or group that chooses and is accountable	Karim, or Salma
Goal	The result wanted; the standard for judging	The highest profit from Tk 10,000
Alternatives	The possible actions	Buy umbrellas; buy sun hats; buy nothing
Outcomes and payoffs	What happens, and its value	Profit or loss at the month's end, which depends on the rain

Not every decision needs the same effort: routine choices can be settled in advance.

The same decision looks different from the owner's, the employee's and the customer's seat.

Top, middle and first-line managers decide matters of different size.

Recap: a decision is a choice, and it has four elements.

Not every decision needs the same effort. Some decisions repeat often, such as ordering the usual flour each week. A manager can settle these in advance with a policy or a procedure, as Chapter VI described, so that no new thought is needed each time. Other decisions are new, large, or hard to reverse, such as buying an oven. These deserve the full effort that this chapter describes. Two questions help the manager to judge: how costly would a mistake be, and can the choice be undone? The higher the cost and the harder the reversal, the more care the decision deserves.

The same decision looks different from different seats. From the owner's seat, it is a use of savings. From the employee's seat, it is a change that may affect work and pay. From the customer's seat, it is a price or a service that suits or does not suit. A good decision maker pictures all three seats before choosing.

Decisions also differ by level. Top managers decide matters that shape the whole organisation for years, such as entering a new market. Middle managers decide how to divide a budget among departments. First-line managers decide daily matters, such as who covers for an absent worker. Chapter I described these three levels. The method in this chapter serves all three, but the effort grows with the size of the matter.

Before moving on. A decision is a choice among alternatives. Every decision has a decision maker, a goal, alternatives, and outcomes with payoffs. The decision maker controls the choice but not the outside events that shape the outcome.



8.2 CONDITIONS: CERTAINTY, RISK AND UNCERTAINTY

What a decision maker knows about outcomes changes how the choice is made.

Begin with a journey: the fare is known, the delay has a known chance, the new road is unknown.

GO SLOWLY HERE. THREE NEW TERMS FOLLOW, ONE IDEA AT A TIME.

Return to Karim's umbrellas. Karim knew the price from his supplier. He knew roughly how often it rained in past years. He did not know whether the coming month would be wet. He knew each of these three facts to a different degree, and the degree decides which method of choosing is suitable. This section gives a name to each degree.

Begin with a journey. Suppose you must travel to a friend's house. If the bus fare is fixed at Tk 20, you know the cost before you leave. If one road is blocked on about one day in five, you know the chance of delay on that road. If you have never travelled on a road, and no one can describe it, you know almost nothing about the delay. These are three different positions, and managers meet all three.

The first position is certainty. **Certainty** means that the decision maker knows exactly what outcome each alternative will bring. Suppose a baker may buy flour from supplier A at Tk 60 per kilogram or from supplier B at Tk 63, with both prices fixed in writing for this week. The cost of each alternative is known before the choice. The decision is easy: if quality is equal, choose the lower cost.

The second position is risk. In daily life, risk means danger. In decision making, **risk** means that the decision maker knows the possible outcomes and can state the chance of each. Imagine a shop owner with ten years of records. The first week of June was wet in seven of those ten years. The owner cannot know whether next June will be wet, but the owner knows the chance. We measure such a chance with a number called a probability, which section 8.4 teaches.

The third position is uncertainty. **Uncertainty** means that the decision maker cannot state the chances of the outcomes, and may not know all the possible outcomes. A firm that launches a new kind of product in a city where it has never sold has no records to guide it. Customers may welcome the product, ignore it, or react in a way that nobody foresaw.

One question separates the three conditions: how much does the decision maker know about the outcomes? Under certainty, the decision maker knows everything about them. Under risk, the decision maker knows the outcomes and their chances. Under uncertainty, the decision maker knows little or nothing about the chances.

Notice that the condition belongs to the decision maker's knowledge, and not to the world. The same decision may be a risk for an experienced trader, who has records, and an uncertainty for a newcomer, who has none. Gathering information can therefore move a decision from uncertainty toward risk. This is one reason why Step 3 of the process in section 8.3 matters so much.

The figure below shows the three conditions. In each picture the square is the choice. Each line leads to a possible outcome, and the label beside the outcome says what is known about its chance.

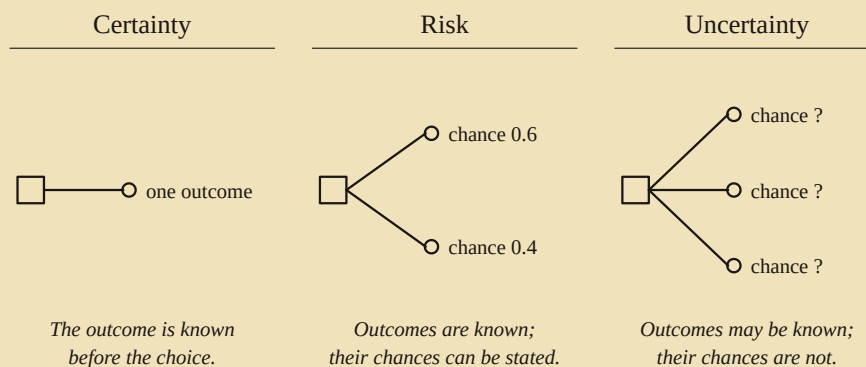


FIG. 21. The three conditions: under certainty the outcome is known; under risk its chance is known; under uncertainty its chance is unknown.

The table below compares the three conditions. Read across each row: it states what the decision maker knows, gives one example, and names the usual method of choosing.

Certainty means that the outcome of every alternative is known in advance.

Certainty. The condition in which the outcome of each alternative is known in advance.

Risk means that outcomes are known and their chances can be stated.

Risk. In daily life, danger. In decisions, a condition in which the outcomes are known and their chances can be stated.

Uncertainty means that the chances, and perhaps the outcomes, are not known.

Uncertainty. The condition in which the chances of outcomes cannot be stated, and some outcomes may be unknown.

One question separates the three conditions: how much does the decision maker know?

The condition belongs to the decision maker's knowledge, not to the world.

The figure below shows the three conditions as a choice leading to outcomes.

The table below compares the conditions by knowledge, example and usual method.

THREE CONDITIONS COMPARED

CONDITION	WHAT IS KNOWN	EXAMPLE	USUAL METHOD
Certainty	The outcome of each alternative	Two suppliers with written prices	Compare the outcomes and choose the best
Risk	The outcomes and the chance of each	A wet or dry month, with a chance from records	Expected value and decision tree (section 8.4)
Uncertainty	Little or nothing about the chances	A new product in an unfamiliar city	Gather information, test on a small scale, then use judgement

Under certainty the task is simple comparison, but true certainty is rarer than it looks.

Under risk, each payoff is weighted by its chance, as section 8.4 shows.

Under uncertainty, the first task is to reduce the uncertainty.

When uncertainty remains, managers use judgement, often by a hopeful or a cautious view.

Neither view is correct; a third approach keeps the choice flexible.

Real decisions lie along a scale between certainty and uncertainty, and delay is itself a choice.

Pause and check: name the condition in each of three cases.

Recap: the three conditions differ by what is known about outcomes and their chances.

Under certainty, the task is to compare known outcomes against the goal. Even so, true certainty is rarer than it looks. A written price can change, a supplier can fail to deliver, and a customer can cancel. Most business decisions contain at least some risk, and a careful manager asks what could change.

Under risk, the decision maker weighs each possible payoff by its chance. Section 8.4 shows exactly how. The key idea is that a large payoff with a small chance may matter less than a modest payoff with a large chance.

Under uncertainty, the first task is to reduce the uncertainty. The manager can collect information, ask people with experience, and run a small trial before a large commitment. The manager can also state assumptions openly. A planning premise, which Chapter VI described, is such an assumption.¹ By agreeing a premise, planners turn an unknown future into a working basis, while accepting that the premise may prove wrong.

Some uncertainty always remains, and then judgement is needed. Two simple views are common. Suppose alternative P may bring +Tk 20,000 at best and -Tk 5,000 at worst. Suppose alternative Q may bring +Tk 8,000 at best and +Tk 2,000 at worst. A hopeful decision maker compares the best outcomes, 20,000 against 8,000, and chooses P. A cautious decision maker compares the worst outcomes, -5,000 against +2,000, and chooses Q. The data are the same. The choices differ because the attitudes differ.

Neither view is correct. Each describes an attitude, and the right attitude depends on what the organisation can afford to lose. A third approach is to prefer alternatives that keep the choice flexible. For example, a firm may rent a shop for one year before it buys a building. If the trial goes badly, the firm has lost little and can change direction.

In practice the three conditions are not sharp boundaries. Real decisions lie along a scale from certainty to uncertainty. One warning applies to every point on the scale: uncertainty does not excuse a manager from deciding. Waiting for more information is itself an alternative, and it has costs.

Pause and check. Name the condition in each case. (a) A student knows that the fee for a course is Tk 5,000. (b) A farmer knows from long records that floods damaged the crop in two years of every ten. (c) A company enters a country whose laws and customers it has never studied. The answers are at the back of the book.

Before moving on. Certainty, risk and uncertainty differ by what the decision maker knows about outcomes and their chances. The condition decides the method. You should now be able to say: "Under certainty I know the outcomes, under risk I know their chances, and under uncertainty I do not."

¹ Chapter VI, section 6.3, explains planning premises.

8.3 THE RATIONAL DECISION PROCESS

Suppose the debating society of a college must choose a place for its annual dinner. Forty members will attend, and the committee has Tk 12,000. The committee could argue for an hour and accept the first idea. It could also follow a method. A method gives the committee a record of its reasons, and a record is valuable when members disagree afterwards.

Management writers describe such a method as a **rational decision process**. A rational decision process means a series of logical steps that leads from a problem to a chosen action. In daily life, rational means sensible. Here it has a narrower meaning: each step follows from the one before, and the final choice can be justified by the goal. The process has seven steps. We follow the society's committee through all of them.

Step 1: identify the problem or opportunity. A decision begins when someone sees a gap between what is and what should be. The gap may be a problem, such as falling sales, or an opportunity, such as a new group of customers. State it exactly, in one sentence, and separate the symptom from the cause. Falling sales are a symptom. The cause may be a higher price, poorer quality, or a new competitor. A decision that treats only the symptom leaves the cause untouched. The society states its problem in this way: choose a venue for forty members, within Tk 12,000, on a date already fixed.

Step 2: set the criteria. A **criterion** means a standard used to judge an alternative; the plural is criteria. Criteria come from the goal and, inside an organisation, from its objectives, as Chapter VII explained. A good criterion is measurable. The word "cheap" is vague, but the phrase "costs no more than Tk 12,000" can be checked.² Criteria are of two kinds: those that every alternative must meet, and those that make one alternative better than another. The society sets three must-have criteria: total cost not above Tk 12,000, seats for at least forty, and a distance of not more than 5 km from the college. Among venues that meet all three, the lower cost is better.

Step 3: gather information and develop alternatives. Information costs time and money. A manager gathers enough to judge, and stops when more would cost more than it helps. The manager also builds a list of several alternatives and does not accept the first idea. The list should include doing nothing, because it shows what the other alternatives must beat. Chapter XI, section 11.6, treats the making of new ideas. The society finds three venues. The college hall costs Tk 4,000 to hire, and catering costs Tk 6,500; it seats 60 and stands on the campus. A restaurant charges Tk 11,800 in total, seats 45, and lies 3 km away. A member's rooftop needs only catering, at Tk 7,000; it seats 30 and lies 1 km away.

Step 4: evaluate the alternatives. The manager now compares each alternative with the criteria. The table below sets each venue against the three must-have criteria. Read each row across; the last column states the result.

A method gives a group a record of its reasons.

The rational decision process is a series of logical steps from problem to chosen action.

Rational decision process. A series of logical steps that leads from a problem to a chosen action.

GO SLOWLY HERE. SEVEN STEPS FOLLOW; WE USE ONE EXAMPLE THROUGHOUT.

Step 1 states the gap between what is and what should be, and separates symptom from cause.

Step 2 sets the criteria, the standards that every alternative will be judged by.

Criterion. A standard used to judge an alternative; the plural is criteria.

Step 3 gathers information and builds a list of several alternatives, including doing nothing.

Step 4 compares each alternative with the criteria, and first removes any that fail a must-have.

² Legal and ethical standards are also criteria. Chapter V explains how managers judge right from wrong.

THE THREE VENUES AGAINST THE CRITERIA

VENUE	TOTAL COST (TK)	SEATS	DISTANCE	MEETS ALL THREE?
College hall	10,500	60	On campus	Yes
Restaurant	11,800	45	3 km	Yes
Rooftop	7,000	30	1 km	No: too few seats

The rooftop is the cheapest, but it fails a must-have criterion and leaves the list.

Step 5 chooses the alternative that best serves the goal; a choice must be made.

Step 6 puts the choice into action: who does what, by when, and who is told.

Step 7 compares results with the goal and feeds what is learned back to Step 1.

The figure below shows the seven steps as a chain with a return arrow.

The hall's total cost is Tk 4,000 + Tk 6,500 = Tk 10,500. Two venues pass. The rooftop is the cheapest, but it fails on seats and so leaves the list. This shows a first rule of evaluation: remove any alternative that fails a must-have criterion, and only then compare the rest.

Step 5: choose. The manager chooses the alternative that best serves the goal. The hall and the restaurant both pass. The hall costs Tk 11,800 – Tk 10,500 = Tk 1,300 less, and it has more seats. The society therefore chooses the hall. Sometimes no alternative is perfect. The manager then chooses the best of the imperfect ones, or improves one by combining features of two. A choice must be made, because delay is itself an alternative, with its own costs.

Step 6: implement the choice. A choice changes nothing until someone acts on it. The manager decides who will do what, by what date, with which resources. The manager also tells the people who are affected, and explains the reasons. People support a choice that they understand, and they resist one that arrives without explanation; Chapter XI studies this further. The society's treasurer books the hall, and its secretary informs the members.

Step 7: follow up. After the dinner, the committee compares the results with its goal. Did the cost stay within Tk 12,000? Did the members attend? This comparison is the same logic that Chapter XIII calls control: measure the result, compare it with the standard, and correct. If the result is poor, the committee asks why. The information may have been wrong, the criteria may have been weak, or the action may have been poor. Follow-up also turns experience into learning, and the learning improves the next decision.

The figure below shows the seven steps as a chain. The arrow on the right returns from the last step to the first. It shows that the results of one decision become information for the next.

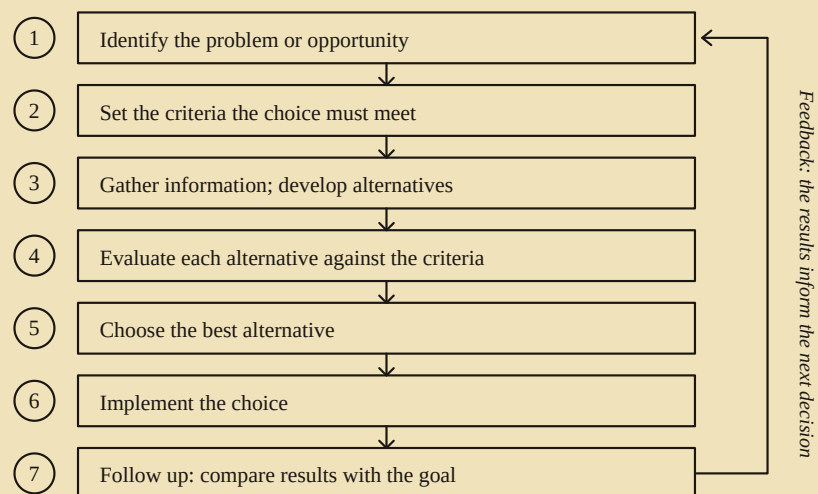


FIG. 22. The rational decision process: seven steps, with feedback from the results to the next decision.

The seven steps are a guide, and they do not bind. They often overlap and loop. New information in Step 3 may send the committee back to change a criterion in Step 2. Not every decision needs all seven steps in full. A routine decision may need only a glance at Step 4, while an expensive and permanent decision deserves every step. The effort should match the cost of a mistake, as section 8.1 explained.

The process works under all three conditions of section 8.2, but Step 4 changes. Under certainty, the manager compares known outcomes. Under risk, the manager uses the tools of section 8.4. Under uncertainty, the manager relies on judgement after reducing the uncertainty as far as possible.

One more point returns us to the opening puzzle. A rational process improves the chance of a good result. It does not guarantee one. Karim followed the steps well and still received a poor result, because the weather was beyond his control. The process also assumes that the decision maker has time, information and a clear goal. Section 8.5 asks what happens when these are missing.

Before moving on. The rational decision process has seven steps: identify the problem, set the criteria, gather information and develop alternatives, evaluate, choose, implement, and follow up. You should now be able to say: "Each step builds on the one before, and the last step feeds the first."



8.4 TOOLS FOR CHOOSING UNDER RISK

Step 4 of the process, evaluation, is easy when outcomes are certain: compare the numbers. It is harder under risk, because each alternative has several possible outcomes, each with its own chance. Three tools help. Probability measures chance. Expected value combines chances and payoffs into one number. The decision tree shows the whole situation as a picture. A fourth tool, the decision support system, applies the first three by computer to large problems. We take them in this order, with small numbers first.

PROBABILITY: MEASURING CHANCE

The mathematics of chance began with a question about a game. In 1654, two French scholars, Blaise Pascal and Pierre de Fermat, exchanged letters about how to divide the money fairly when a game of chance stops early.³ Their reasoning grew into the theory of probability, which managers now use to weigh uncertain outcomes.

Begin with a coin. Toss a fair coin. It can land heads or tails, and neither is favoured. We say that each face has a chance of one in two. **Probability** means a number from 0 to 1 that measures how likely an outcome is. The probability that a fair coin lands heads is 0.5. The same chance can be written as 50 per cent, because $0.5 \times 100 = 50$.

Three rules govern probabilities. First, a probability is never below 0 and never above 1. Second, a probability of 0 means the outcome cannot happen, and a probability of 1 means it is certain. Third, the probabilities of all the possible outcomes of one event total exactly 1. For the coin, $0.5 + 0.5 = 1$. If the chance of rain is 0.7, the chance of no rain is $1 - 0.7 = 0.3$.

³ Historians usually treat this exchange of letters as the start of probability theory. Ian Hacking's book, listed under Sources, tells the story.

The steps overlap and loop; they guide the manager and do not bind.

The process works under all three conditions; only the evaluation step changes.

A good process raises the chance of a good result but does not guarantee it.

Recap: the process has seven steps, from the problem to the follow-up.

Step 4 is easy under certainty, but chancy outcomes need tools.

The mathematics of chance began with a question about a game.

Toss a fair coin: each face has a chance of one in two, written 0.5.

Probability. A number from 0 to 1 that measures how likely an outcome is.

GO SLOWLY HERE. PROBABILITY IS NEW; READ EACH EXAMPLE TWICE.

Three rules: probabilities lie between 0 and 1, 0 means impossible, and all outcomes total 1.

The figure below shows the scale from 0 to 1.

The figure below shows the scale. Read it from left to right: the further right a point lies, the more likely the outcome.

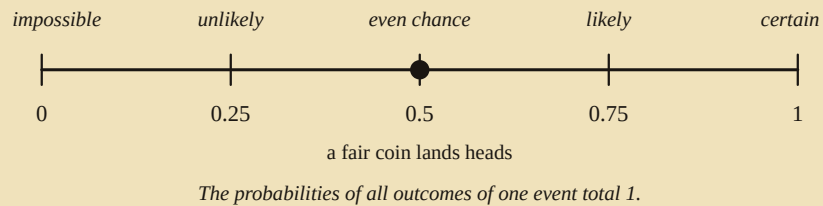


FIG. 23. The probability scale: 0 means impossible, 1 means certain, and 0.5 means an even chance.

Probabilities come from records, or from careful judgement when no records exist.

Worked example 1: given, find and formula for a stall owner's sell-out chance.

Worked example 1: steps, answer and check.

Common slips: writing 6 for 0.6, totals that miss 1, and too few records.

Recap: a probability is a number from 0 to 1, and all outcomes of one event total 1.

A coin game poses the puzzle: should you play once, or many times?

GO SLOWLY HERE. THIS IDEA IS NEW AND ABSTRACT.

Expected value is the average payoff if a decision were repeated many times.

Expected value. In daily life, expected means thought likely. In decisions, an average payoff found by weighting each payoff by its probability.

Managers obtain probabilities in two ways. When records exist, the manager counts how often the outcome occurred and divides by the number of times it could have occurred. When no records exist, as under uncertainty, an experienced person makes a best estimate. The estimate is imperfect, but it is better than a hidden guess, because it can be recorded, discussed and corrected.

We now work an example, in the standard order. *Given.* A stall owner counted her market days. Over 40 days, she sold out of her goods on 10 days. *Find.* The probability that she sells out on a day, and the probability that she does not. *Formula.* In words: probability equals the number of times the outcome occurred, divided by the number of times it could have occurred. In symbols: $P = n \div N$, where n is the number of times it occurred and N is the number of times it could have occurred.

Steps. The probability of selling out is $P = 10 \div 40 = 0.25$. She did not sell out on $40 - 10 = 30$ days, so the probability of not selling out is $30 \div 40 = 0.75$. *Answer.* The chance of selling out is 0.25, or 25 per cent, and the chance of not selling out is 0.75. *Check.* The two probabilities must total 1, and $0.25 + 0.75 = 1$. Both lie between 0 and 1.

Three slips are common. Students write 6 when they mean 0.6, and so break the first rule. They give probabilities that do not total 1, which shows an error or a missing outcome. And they trust too few records: three sell-outs in four days gives 0.75, but four days is too small a sample to rely on.

Before moving on. You should now be able to say: "A probability is a number from 0 to 1, and the probabilities of all the outcomes of one event total 1."

EXPECTED VALUE: ONE NUMBER FOR A CHANCY CHOICE

A friend offers you a coin game. If the coin lands heads, you win Tk 100. If it lands tails, you lose Tk 40. Should you play? A single game is a gamble. But suppose you could play 100 times. About half the games, 50, would give +Tk 100, and about half would give -Tk 40. You would win about $50 \times 100 = \text{Tk } 5,000$ and lose about $50 \times 40 = \text{Tk } 2,000$. The net gain is Tk 3,000, or Tk 30 per game on average. This average is the idea we need.

The average has a name. **Expected value** means the average payoff of a decision if it could be repeated many times: each payoff is multiplied by its probability, and the results are added together. In daily life, "expected" means what you think will happen. Here it means a weighted average, and it may never occur in a single trial.

The formula is as follows. In words: expected value equals the sum of each payoff multiplied by its probability. In symbols: $EV = (p_1 \times v_1) + (p_2 \times v_2) + \dots$, where p_1 is the probability of the first outcome, v_1 is its payoff, and the dots show that the sum continues for every outcome. For the coin game, $EV = (0.5 \times 100) + (0.5 \times (-40)) = 50 - 20 = \text{Tk } 30$.

In one game you will never receive Tk 30. You win 100 or you lose 40. The expected value describes the average of many games. Managers face many decisions over time, so choosing the highest expected value each time tends to give the best total result. For one very large decision that cannot be repeated, expected value is a guide and not a promise.

Given. A student club may run one stall at a school fair. Alternative A is cold drinks. Alternative B is snacks. The weather is sunny with probability 0.5 and rainy with probability 0.5. Cold drinks earn Tk 1,000 on a sunny day and Tk 200 on a rainy day. Snacks earn Tk 700 on a sunny day and Tk 400 on a rainy day. *Find.* The expected value of each alternative, and the better choice. *Formula.* $EV = (p_1 \times v_1) + (p_2 \times v_2)$.

Steps. For A: $0.5 \times 1,000 = 500$, and $0.5 \times 200 = 100$, so $EV(A) = 500 + 100 = \text{Tk } 600$. For B: $0.5 \times 700 = 350$, and $0.5 \times 400 = 200$, so $EV(B) = 350 + 200 = \text{Tk } 550$. *Answer.* Cold drinks have the higher expected value, Tk 600 against Tk 550, so the club chooses A. *Check.* Each expected value must lie between its worst and best payoffs: 600 lies between 200 and 1,000, and 550 lies between 400 and 700. The probabilities total $0.5 + 0.5 = 1$.

Notice what the expected value does not show. Cold drinks may earn as little as Tk 200 or as much as Tk 1,000, while snacks stay between Tk 400 and Tk 700. A club that cannot bear a poor day might prefer snacks, although their expected value is lower. Expected value ignores the spread of the payoffs and the attitude of the decision maker toward risk. It is a strong tool, and it is not the only consideration.

Real data are messier. *Given.* A food-cart owner must choose one site for next month. Each site has a fee, and demand may be high, medium or low, with different chances at each site. Site X, at a university gate, has a fee of Tk 4,000, chances of 0.5, 0.3 and 0.2, and profits before the fee of Tk 30,000, Tk 20,000 and Tk 6,000. Site Y, at a bus terminal, has a fee of Tk 9,000, chances of 0.4, 0.4 and 0.2, and profits before the fee of Tk 36,000, Tk 22,000 and Tk 4,000. Site Z, at a weekly market, has a fee of Tk 2,000, chances of 0.3, 0.5 and 0.2, and profits before the fee of Tk 28,000, Tk 18,000 and Tk 10,000. *Find.* The expected value of each site.

Steps, part one. The payoff must be the net amount, so we subtract each fee first. For X: $30,000 - 4,000 = 26,000$; $20,000 - 4,000 = 16,000$; $6,000 - 4,000 = 2,000$. For Y: $36,000 - 9,000 = 27,000$; $22,000 - 9,000 = 13,000$; $4,000 - 9,000 = -5,000$. For Z: $28,000 - 2,000 = 26,000$; $18,000 - 2,000 = 16,000$; $10,000 - 2,000 = 8,000$.

Steps, part two. For X: $(0.5 \times 26,000) + (0.3 \times 16,000) + (0.2 \times 2,000) = 13,000 + 4,800 + 400 = \text{Tk } 18,200$. For Y: $(0.4 \times 27,000) + (0.4 \times 13,000) + (0.2 \times (-5,000)) = 10,800 + 5,200 - 1,000 = \text{Tk } 15,000$. For Z: $(0.3 \times 26,000) + (0.5 \times 16,000) + (0.2 \times 8,000) = 7,800 + 8,000 + 1,600 = \text{Tk } 17,400$.

The table below gathers the results. Read each row across: it shows the best and worst payoffs at the site, and the expected value.

The formula: multiply each payoff by its probability and add the results.

In one game you never receive the expected value; it describes the average of many games.

Worked example 2: given, find and formula for two stalls at a school fair.

Worked example 2: steps, answer and check.

Expected value ignores the spread of payoffs and the decision maker's attitude to risk.

Worked example 3: a messier case with three sites, three levels of demand and a fee for each site.

Worked example 3: first subtract each fee, because payoffs must be net.

Worked example 3: multiply each net payoff by its chance and total.

The table below gathers the best, worst and expected values.

EXPECTED VALUE OF EACH SITE (TK PER MONTH)

SITE	BEST OUTCOME	WORST OUTCOME	EXPECTED VALUE
X (university gate)	26,000	2,000	18,200
Y (bus terminal)	27,000	-5,000	15,000
Z (weekly market)	26,000	8,000	17,400

Answer and check: site X has the highest value, and the chances sum to 1 at every site.

Testing the estimates: a small change in one chance can reverse the ranking of X and Z.

Common slips: forgetting costs, mixed periods, and treating the expected value as a result.

Recap: expected value is each payoff multiplied by its probability, then totalled.

When choices have many branches, a picture is easier to follow than a list.

Decision tree. A branching diagram that shows choices, chance events, outcomes and payoffs.

GO SLOWLY HERE. WE DRAW THE SCHOOL-FAIR CHOICE AS A TREE.

Answer. Site X has the highest expected value, Tk 18,200. *Check.* The chances at each site total 1: $0.5 + 0.3 + 0.2 = 1$; $0.4 + 0.4 + 0.2 = 1$; $0.3 + 0.5 + 0.2 = 1$. Each expected value lies between its worst and best payoffs. Y is the only site with a possible loss, and it has the lowest expected value, so rejecting it is easy.

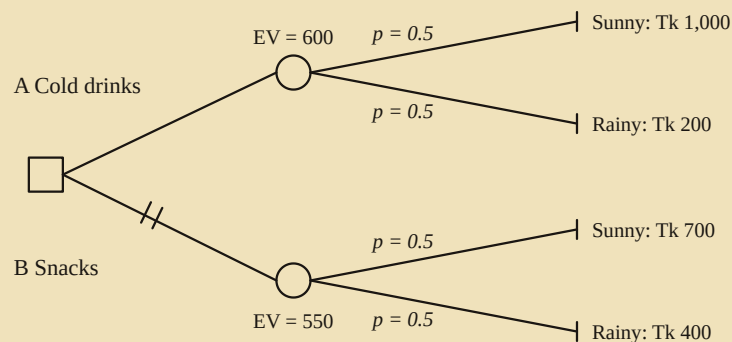
The choice between X and Z deserves a second look. They differ by only Tk 800. Suppose the owner doubts the estimate for X, and changes the chances to 0.4, 0.4 and 0.2. Then $EV(X) = (0.4 \times 26,000) + (0.4 \times 16,000) + (0.2 \times 2,000) = 10,400 + 6,400 + 400 = \text{Tk } 17,200$. Now Z, at Tk 17,400, is higher. A change of 0.1 in one chance reversed the ranking. When two alternatives lie this close, the estimates matter more than the arithmetic, and the safer worst case of site Z may decide the choice.

Several slips are common here. Students forget to subtract costs before multiplying. They mix periods, such as a monthly payoff beside a yearly one. They compare payoffs measured in different units. And they treat the expected value as the result that will happen, when it is only the average of many trials.

Before moving on. You should now be able to say: "Expected value is each payoff multiplied by its probability and then added together; it is an average of many trials, and not a promise for one."

THE DECISION TREE: SEEING THE WHOLE CHOICE

As choices grow more complex, numbers in a list become hard to follow. A picture helps. A **decision tree** means a branching diagram that shows choices, chance events, outcomes and payoffs. The figure below draws the school-fair choice of worked example 2 as a tree. We read it from left to right, in time order: first the choice, then the chance event, then the payoff.



Square: a choice. Circle: a chance event. Figures: payoffs in taka.

FIG. 24. A decision tree for the school-fair stall. The square is a choice, the circles are chance events, and the figures at the right are payoffs in taka.

A tree has two kinds of point. A **decision node** is a point, drawn as a square, where the decision maker chooses between branches. A **chance node** is a point, drawn as a circle, where an outside event decides which branch follows. Each branch that leaves a chance node carries a probability, and the probabilities leaving one chance node total 1. At the end of each branch we write the payoff.

We solve the tree by moving backwards, from the payoffs toward the choice. First, we write the probabilities on the branches and the payoffs at the ends. Second, we calculate the expected value at each chance node and write it above the node. Third, at the decision node we compare these values and choose the largest, or the smallest if the payoffs are costs. Fourth, we mark each rejected branch with two short strokes. In the figure, $EV(A) = (0.5 \times 1,000) + (0.5 \times 200) = 600$, and $EV(B) = (0.5 \times 700) + (0.5 \times 400) = 550$. We choose A, as before, because the tree is the same calculation drawn as a picture.

Advanced note, which a beginner may defer. A tree may contain a second decision node after a chance event. For example, an owner may first decide whether to pay for a survey, and then, after seeing the result, decide whether to buy. The method is the same: we move backwards from the payoffs, and calculate each node in turn.

Pause and check. Draw a decision tree for this choice, and find the better alternative. Alternative 1: open a stall at a fair. If the crowd is large, with probability 0.4, the profit is Tk 9,000. If the crowd is small, with probability 0.6, the profit is Tk 2,000. Alternative 2: work as a paid helper for a certain Tk 4,000. The answer is at the back of the book.

Before moving on. You should now be able to say: "A decision tree shows choices as squares and chance events as circles, and I solve it by moving backwards from the payoffs."

DECISION SUPPORT SYSTEMS

Real decisions often involve many alternatives, many outcomes and large amounts of data. Calculating by hand then becomes slow. A **decision support system** means a computer tool that helps a manager study alternatives by using data and models. A simple example is a spreadsheet in which a bakery owner types flour prices and expected demand, and the sheet calculates the expected profit of each plan.

The main strength of such a system is the "what if" question. What if flour costs 10 per cent more? What if the chance of strong demand is 0.5 and not 0.6? Each time, the system recalculates in seconds. The test that we performed by hand on site X, earlier in this section, becomes quick and easy. A manager can then see which estimates matter most, and can spend effort on improving those.

The system has limits. It supports a decision, and it does not make one. Its answers are only as good as the data and assumptions put into it. A mistaken probability, entered by a person and handled by a machine, gives a confident but wrong answer. The user must therefore understand the model well enough to question it. Such systems also cost money, for equipment, software and training, and a small firm should compare that cost with the benefit. The information that a system stores and processes is one of the resources of Chapter I.

Before moving on. Probability measures chance from 0 to 1. Expected value turns chances and payoffs into one number. A decision tree draws the same

Squares show where the decision maker chooses, and circles show where chance decides.

Decision node. A point on a tree, drawn as a square, where the decision maker chooses a branch.

Chance node. A point on a tree, drawn as a circle, where an outside event decides which branch follows.

Solving the tree: calculate each chance node, compare at the square, and mark the rejected branch.

Advanced note: a tree may hold a second choice after a chance event; defer it if needed.

Pause and check: draw and solve a tree for a stall and a certain wage.

Recap: a tree shows choices as squares and chance as circles, and we solve it backwards.

Computers apply the same tools to large problems quickly.

Decision support system. A computer tool that helps a manager study alternatives by using data and models.

The main strength is the "what if" question, answered in seconds.

A system supports the decision but does not make it, and its answers depend on its inputs.

Recap: the four tools of this section answer different needs.

calculation as a picture. A decision support system performs it quickly on a computer.



8.5 HOW MANAGERS REALLY DECIDE

The rational process assumes a perfect decision maker, and no one is perfect.

Bounded rationality is the set of limits that stops anyone from weighing every alternative.

Bounded rationality. The limits of time, information and mental capacity that stop a person from weighing every alternative.

Four limits bound a decision: information, time, the mind and the future.

Satisficing means choosing the first alternative that is good enough.

Satisficing. Choosing the first alternative that meets all the criteria, instead of searching for the best.

Satisficing is often sensible because search costs time, but low criteria give poor choices.

Experience gives managers a sense of judgement that helps when time is short but can mislead.

Three habits of thought distort decisions: anchoring, sunk costs and overconfidence.

Section 8.3 described a decision maker who knows the goal, finds every alternative, gathers all the information, and weighs everything perfectly. Does any manager decide in this way? Almost none does. This section asks why, and what managers do instead. It also asks whether one person or a group should decide.

Herbert Simon, an American scholar who studied how organisations decide, argued from 1947 onward that human beings cannot decide in the perfect way.⁴ He called the limit **bounded rationality**. Bounded rationality means the limits of time, information and mental capacity that stop a person from weighing every alternative. The word "bounded" means limited, as a field is limited by a fence. The decision maker still tries to be rational, but works inside the fence.

Four limits form the fence. First, information is often incomplete, late or costly. Second, time is short, because decisions have deadlines. Third, the mind is limited: a person can hold only a few factors in attention at once, and memory fails. Fourth, the future is unknown, and the goals of different people may conflict. Because of these limits, no manager can list every alternative or foresee every outcome.

Because of these limits, managers often do not search for the best alternative. They search until they find one that is good enough. Simon named this habit **satisficing**, a word that joins "satisfy" and "suffice". Satisficing means choosing the first alternative that meets all the criteria, instead of searching for the best. In section 8.3, two venues passed the criteria. A committee short of time might have booked whichever venue it examined first.

Satisficing is not laziness. Search has a cost, and after a point the cost of searching exceeds the gain from a slightly better alternative. A manager who satisfices wisely has judged that point. The danger lies elsewhere: if the criteria are set too low, the first alternative that passes may be a poor one. The criteria of Step 2 therefore deserve the most care.

Experienced managers also use judgement. After many years, a manager may sense that a plan is unsound before being able to say why. This sense comes from many past cases stored in memory. It is valuable when time is short. It can mislead when the new situation differs from the old ones. The best managers combine analysis with judgement, and use each to check the other.

Some habits of thought distort decisions, and managers should know them. The first is *anchoring*, which means relying too heavily on the first number heard. An early price quotation then sets a standard that it does not deserve. The second is the sunk cost error. A *sunk cost* is money already spent that cannot be recovered. A manager who continues a failing project because so much has been spent lets the past control the future. The third is *overconfidence*, which means feeling surer than the evidence allows. A written record of the reasons for a choice helps to correct all three.

⁴ Simon received the Nobel Memorial Prize in Economic Sciences in 1978, partly for his study of decision making in organisations.

Decisions are also shaped by the organisation and its surroundings. Rules, culture and politics may rule out an alternative that the manager prefers. Law may forbid it, as Chapter IV explained when it described the external environment. A budget may not allow it. The decision maker therefore chooses within limits set by others, and a skilled manager learns these limits early.

Many decisions are made by one manager, and many by groups, such as committees, teams and boards. Which is better? The answer depends on the situation. The table below lists the strengths and the limits of group decisions. The two columns are separate lists: the rows are not pairs, and each row stands alone.

GROUP DECISIONS: STRENGTHS AND LIMITS

STRENGTHS	LIMITS
More information and more skills are present	Discussion takes more time and costs more
More alternatives are suggested	Pressure to agree can silence doubts
Members understand and support the choice	One strong member may dominate
Errors are more likely to be noticed	Compromise may give a weak choice
Responsibility is shared	No one may feel accountable

Read the strengths first. A group holds more knowledge than one person, and so it can suggest more alternatives and notice more errors. People also support a choice that they helped to make, and so implementation in Step 6 is easier. Now read the limits. A meeting of eight people for two hours uses $8 \times 2 = 16$ hours of staff time, and that time has a cost. Members may stay silent to avoid conflict, or one forceful member may control the discussion. A compromise that pleases everyone may be weaker than the best choice. And when everyone shares responsibility, no one may feel accountable.

These strengths and limits suggest a guide. A group suits a complex problem, a choice that needs the support of many people, and a case in which time allows discussion. One manager suits a routine matter, an emergency, and a case in which one person holds all the needed knowledge. A manager may also consult: ask the group for views, then decide alone and remain accountable. How far a leader should involve others is a question of leadership style, which Chapter XII studies.⁵

The most serious limit deserves a name. Irving Janis, an American psychologist, studied failed group decisions and described the problem in 1972. He called it **groupthink**. Groupthink means the habit of a group of seeking agreement so strongly that its members stop testing ideas. Members who have doubts keep silent, and silence is taken for agreement. The group then feels more confident than the evidence allows. The section on history, at the end of this chapter, describes a well-known case.

A group can guard against groupthink by building disagreement into its method. The leader can speak last, so that members are not influenced by the leader's view. Each member can write a view before the discussion begins. One member can be given the task of questioning the plan. People from outside the group can be invited to comment. The group can divide into two subgroups, work separately, and then compare. Chapter XI, section 11.6, describes methods for producing ideas in groups.

Rules, culture, law and competitors also limit which alternatives a manager can choose.

Many decisions are made by groups, which raises the question of when a group is better.

A group gains information and acceptance but loses time and may suffer from pressure to agree.

One person suits routine matters, emergencies and cases where one person holds the knowledge.

Groupthink is the group's habit of seeking agreement so strongly that ideas are not tested.

Groupthink. The habit of a group of seeking agreement so strongly that its members stop testing ideas.

Groups can guard against groupthink by protecting disagreement in the way they work.

⁵ Chapter XII, section 12.3 onward, discusses how far a leader should involve others in choices.

Recap: real decisions are bounded, managers satisfice, and groups bring strengths and limits.

Before moving on. Real decision makers face bounded rationality, and so they often satisfice. Groups add information and acceptance but cost time and risk groupthink. You should now be able to say: "A rational process is an ideal, and real managers work inside limits."



A WORKED CASE: CHOOSING THE SECOND OVEN

The bakery faces a choice under risk and settles it with a tree.

Case file, illustrative. Month 8. Demand for Morning Loaf Bakery has grown, and the single oven limits output. The owner has two alternatives. Alternative A is to buy a second oven for Tk 180,000. Alternative B is to extend the opening hours. The case file records a probability of 0.6 that demand is strong and 0.4 that demand is weak. Under A, the monthly gain is Tk 30,000 if demand is strong and Tk 8,000 if it is weak. Under B, the monthly gain is Tk 15,000 if demand is strong and Tk 9,000 if it is weak.

The question: which alternative should the owner choose, and how soon does the oven repay its cost?

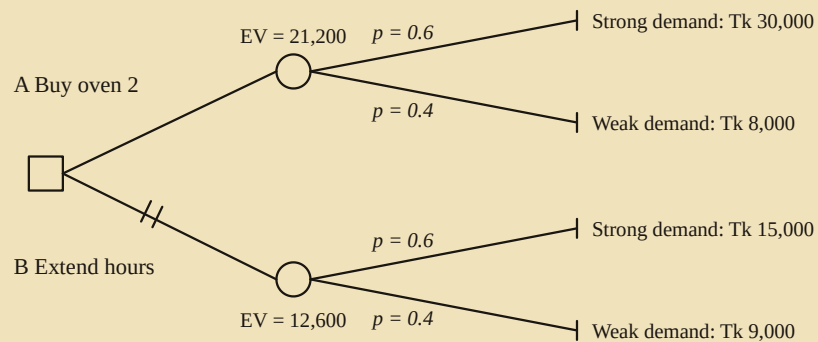
Question. Which alternative should the owner choose, and how long would the oven take to repay its cost? This worked case applies the idea of the chapter to a situation. It is not a model answer to any examination, and it shows one reasoning that you can adapt.

Reasoning begins with the condition: risk, because the chances are recorded.

Reasoning: the condition. The case file records the chances of strong and weak demand, and the possible gains are known. The owner knows the outcomes and their chances, so the condition is risk and not uncertainty. The tools of section 8.4 therefore apply. The goal is the highest expected monthly gain, and the criteria are the expected value and the time needed to repay the cost.

The figure below shows the owner's choice as a tree.

The figure below draws the choice as a tree. The square is the owner's choice, the circles are the two chance events, and the figures at the right are the monthly gains.



Square: a choice. Circle: a chance event. Figures: gain per month, in taka.

FIG. 25. The owner's choice at Morning Loaf Bakery. Payoffs are monthly gains in taka.

The expected values are Tk 21,200 for A and Tk 12,600 for B.

Reasoning: the expected values. For A: $(0.6 \times 30,000) + (0.4 \times 8,000) = 18,000 + 3,200 = \text{Tk } 21,200$. For B: $(0.6 \times 15,000) + (0.4 \times 9,000) = 9,000 + 3,600 = \text{Tk } 12,600$. Alternative A is higher by $\text{Tk } 21,200 - \text{Tk } 12,600 = \text{Tk } 8,600$ per month. The expected value compares monthly gains only, and the oven also costs Tk 180,000 at the start. The next step takes that cost into account.

Reasoning: the payback period. The **payback period** means the time that a purchase takes to earn back its cost. In words: payback equals the cost divided by the monthly gain. For alternative A, the payback is $180,000 \div 21,200 = 8.49$, or about 8.5 months.⁶ The result depends on demand. If demand is strong, the payback is $180,000 \div 30,000 = 6$ months. If demand is weak, it is $180,000 \div 8,000 = 22.5$ months. The owner can therefore see both the likely benefit and the cost of being wrong.

Reasoning: the people. The case file records that the owner spoke with the head baker and the counter lead before choosing. These two staff members know the kitchen and the counter better than anyone else, and so they add information that section 8.5 values in a group. The choice still belonged to the owner, who stays accountable for the result.

Conclusion. The owner chooses alternative A, the second oven. It has the higher expected gain, Tk 21,200 against Tk 12,600, and an expected payback of about 8.5 months. In the strong case, A beats B by Tk 15,000 a month. In the weak case, A falls below B by only Tk 1,000 a month. Step 7, the follow-up, belongs to the control of results, which Chapter XIII describes.

Check. The chances total $0.6 + 0.4 = 1$ on every branch. Each expected value lies between its worst and best gains: 21,200 lies between 8,000 and 30,000, and 12,600 lies between 9,000 and 15,000. The payback is consistent: $21,200 \times 8.5 = 180,200$, which is close to the cost of 180,000.



SUMMARY

A decision is a choice among alternatives. Every decision has a decision maker, a goal, alternatives, and outcomes with payoffs. The decision maker controls the choice but not the outside events that shape the outcome.

Decisions are made under three conditions. Under certainty, the outcomes are known. Under risk, the outcomes and their chances are known. Under uncertainty, the chances are not known. The condition belongs to the decision maker's knowledge, and the best first response to uncertainty is to gather information.

The rational decision process has seven steps: identify the problem, set the criteria, gather information and develop alternatives, evaluate, choose, implement, and follow up. The process raises the chance of a good result. It does not guarantee one.

Probability measures chance from 0 to 1. Expected value multiplies each payoff by its probability and adds the results. A decision tree draws the same calculation, and we solve it by moving backwards from the payoffs. A decision support system performs such work by computer, but it supports the decision and does not make it.

Real decision makers face bounded rationality and often satisfice. Groups add information and acceptance, but they cost time and risk groupthink. A group can guard against groupthink by protecting disagreement in its method.

The payback period is the cost divided by the expected monthly gain.

Payback period. The time that a purchase takes to earn back its cost.

The owner consulted two staff members before choosing, and remained accountable.

Conclusion: alternative A is chosen.

Check: the chances total 1, each value lies in its range, and the payback is consistent.

A decision is a choice among alternatives, with a decision maker, a goal, and outcomes.

The condition of the decision depends on what is known: certainty, risk or uncertainty.

The rational process has seven steps, and it raises the chance of a good result without guaranteeing one.

Probability, expected value and the tree help to choose under risk; computers apply them quickly.

Real decisions are bounded, and groups bring both strengths and risks.

⁶ The payback period ignores gains after the cost is repaid, and it ignores the change in the value of money over time. Later courses in finance add tools for both points.

Return to the two traders and judge the decisions with the ideas of this chapter.

Mistake 1: judging a decision only by its result.

Mistake 2: believing that the expected value is the amount you will receive.

Mistake 3: believing that certainty, risk and uncertainty belong to the world.

Mistake 4: believing that a rational process needs all possible information.

Mistake 5: believing that more people always give a better decision.

Mistake 6: arithmetic slips, such as costs not subtracted or chances that do not total 1.

A famous failure of a group decision, studied to learn how groups can fail.

Janis argued that the advisers failed to test the plan, and drew the lesson of groupthink.

THINK IT THROUGH

Return to Karim and Salma. Karim chose umbrellas and lost money in a dry month. Salma chose sun hats and gained. Using the ideas of this chapter, decide which trader decided better, and give your reasons. Consider these questions. Which condition did each trader face, and what could Karim not know? Would your judgement change if the same two traders chose again in each of ten months? What would you wish to learn from Karim's loss? Does a rational process guarantee a good outcome? Write your own answer before you discuss it with others.

WHERE PEOPLE GO WRONG

Mistaken idea. "A decision is good if it produces a good result." *Correction.* A result depends on the decision and on events outside the decision maker's control. A fair judgement looks at the information available and the method used, as well as the result.

Mistaken idea. "The expected value is the amount I will receive." *Correction.* The expected value is an average over many repeated trials. In a single trial, you receive one of the possible payoffs, and it may be far from the average.

Mistaken idea. "Each decision is certain, risky or uncertain by its nature." *Correction.* The condition depends on how much the decision maker knows. Gathering information can move a decision from uncertainty toward risk.

Mistaken idea. "A rational process means collecting every fact before choosing." *Correction.* Information has a cost, and time is limited. A rational manager gathers enough information to judge, and stops when more would cost more than it helps.

Mistaken idea. "A group always decides better than one person." *Correction.* A group adds information and acceptance, but it also costs time and risks groupthink. The situation decides which is better.

Mistaken idea. "The arithmetic is not important if the method is right." *Correction.* A forgotten cost or a set of chances that does not total 1 can reverse the ranking of alternatives. Always check that the chances total 1 and that costs are subtracted.

IN HISTORY

In April 1961, a few months after John F. Kennedy became President of the United States, a force of about 1,400 Cuban exiles landed at the Bay of Pigs, on the southern coast of Cuba. The American Central Intelligence Agency had trained and equipped them. The plan aimed to start a revolt against the Cuban government. No revolt followed, and Cuban forces defeated the landing within about three days.

Irving Janis studied the meetings in which the plan was approved. He argued, in a book of 1972, that the group failed to test the plan. Members who had doubts kept them private, and the group took silence for agreement. Janis used this case to describe groupthink. He contrasted it with the meetings of October 1962, during the Cuban missile crisis, when the President changed the method. In some meetings the President did not attend, so that members could speak freely. The advisers were also divided into subgroups, and outside views were invited.

The lesson for managers is practical. The advisers of 1962 were largely the same people as those of 1961. The quality of the decision improved when the method protected disagreement. The remedy lies in the process of the group, as section 8.5 described, and not only in the character of its members.

The lesson: a better method, not better people, improved the later decisions.

SOURCES

Herbert A. Simon, *Administrative Behavior* (1947), is the origin of the idea that decision makers are bounded in their rationality. Simon's article "Rational Choice and the Structure of the Environment" (1956) develops the idea of satisficing. These works matter because they replaced the picture of a perfect decision maker with a picture of a limited one.

Simon, 1947 and 1956: the origin of bounded rationality and of satisficing.

Irving L. Janis, *Victims of Groupthink* (1972), studies how groups of capable people made poor decisions. It matters because it showed that the method of a group, and not only its members, shapes its decisions.

Janis, 1972: the study of groupthink in failed decisions.

Howard Raiffa, *Decision Analysis* (1968), introduced many managers and students to decision trees. Ian Hacking, *The Emergence of Probability* (1975), tells the history of the mathematics of chance, including the work of Pascal and Fermat.

Raiffa, 1968, and Hacking, 1975: decision analysis and the history of probability.

REVIEW QUESTIONS

1. Define a decision, and name the four elements that every decision contains. Give an example from your own household, and identify each element in it.

Question 1: define a decision and name its four elements. Hint: section 8.1.

2. Distinguish certainty, risk and uncertainty. For each, give a business example that is not used in this chapter.

Question 2: compare the three conditions with new examples. Hint: section 8.2.

3. Name the condition in each case, and give a reason. (a) A school buys paper at a price fixed by contract for the whole year. (b) A shopkeeper has records showing that power cuts stopped work on 6 days of every 30. (c) A firm plans to sell a new kind of medicine device in a market that no one has studied.

Question 3: classify cases by condition. Hint: section 8.2.

4. Explain why the same decision may be a risk for one manager and an uncertainty for another.

Question 4: explain why the condition belongs to knowledge. Hint: section 8.2.

5. List and describe the seven steps of the rational decision process. Use a decision of your own choice, such as choosing a course or a house, to show each step.

Question 5: describe the seven steps of the process on a decision of your own. Hint: section 8.3.

6. Explain why the criteria are set before the alternatives are evaluated. What is the difference between a must-have criterion and a preferred criterion?

Question 6: explain why criteria come before alternatives are evaluated. Hint: section 8.3.

7. Define probability and state its three rules. A shopkeeper recorded power cuts on 6 days out of 30. Find the probability of a power cut on a day, and the probability of no power cut. Check your answers.

Question 7: define probability and compute from records. Hint: section 8.4.

8. A tailor may launch one of two products. Product 1 is a school-uniform line: a strong order has probability 0.3 and gives +Tk 50,000; a normal order has probability 0.5 and gives +Tk 20,000; a weak order has probability 0.2 and gives -Tk 10,000. Product 2 is ready-made shirts: the same three chances give +Tk 30,000, +Tk 22,000 and +Tk 10,000. Find the expected value of each product, and choose. Then comment on the spread of the payoffs.

Question 8: compute and compare expected values with a spread. Hint: section 8.4.

9. A dairy manager must choose between hiring a delivery van and using a courier. If the van is hired, a busy season (probability 0.7) gives +Tk 12,000 and a quiet season (probability 0.3) gives -Tk 3,000. If the courier is used, a busy

Question 9: draw and solve a tree. Hint: section 8.4.

Question 10: *payback period.*
Hint: *section 8.5 case.*

Question 11: *explain expected value to a friend. Hint: section 8.4.*

Question 12: *bounded rationality and satisficing. Hint: section 8.5.*

Question 13: *groups, strengths, limits and groupthink. Hint: section 8.5.*

Question 14: *decision support systems. Hint: section 8.4.*

season gives +Tk 7,000 and a quiet season gives +Tk 3,000. Draw the decision tree, find the expected value at each chance node, and state the choice.

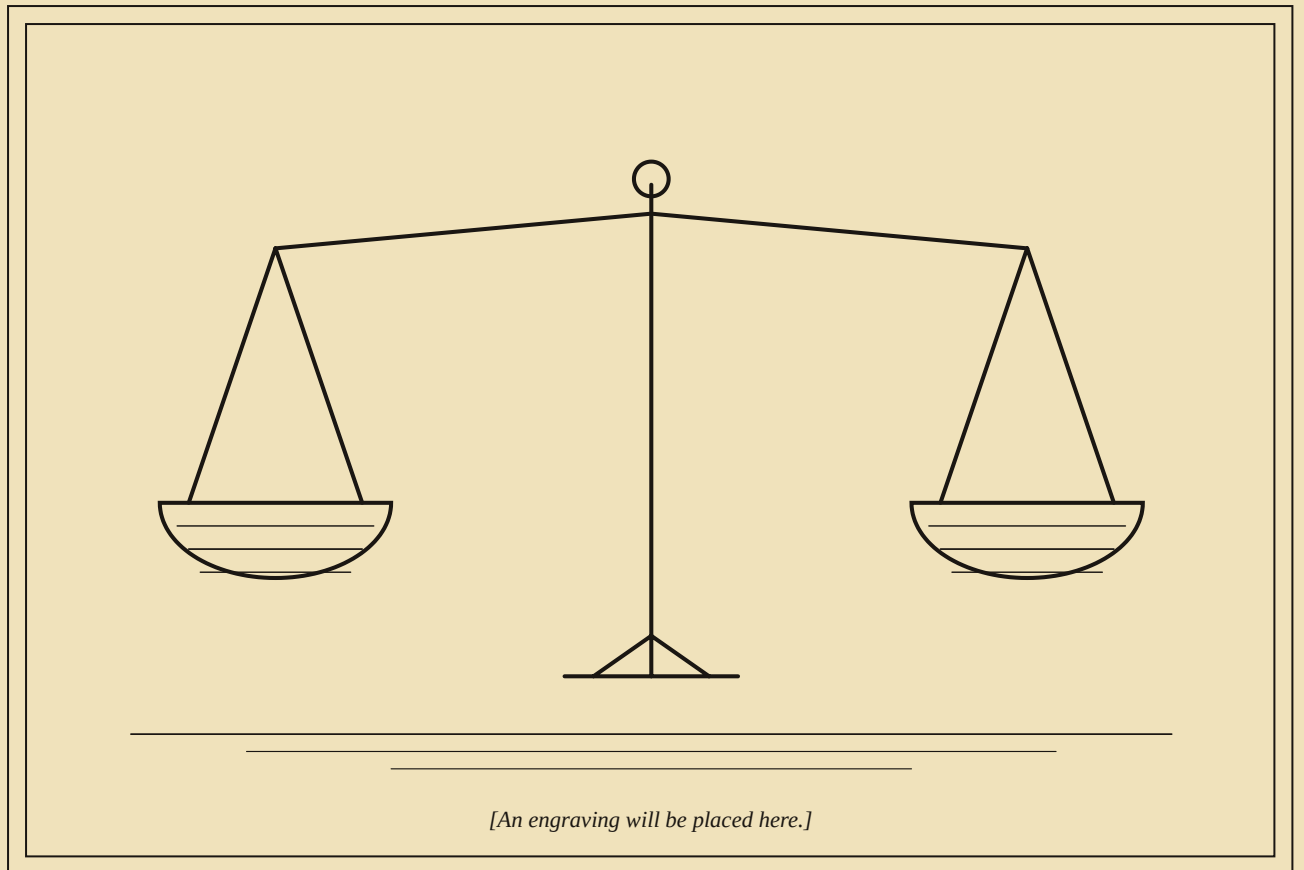
10. A clinic may buy a machine that costs Tk 240,000 and is expected to add Tk 20,000 to its profit each month. Calculate the payback period. State one thing that the payback period ignores.

11. A friend says: "An expected value of Tk 600 means I will receive Tk 600." Explain why this is mistaken, in words that your friend can follow.

12. Explain bounded rationality and satisficing. State one situation in which satisficing is a sensible choice and one in which it is dangerous.

13. State three strengths and three limits of group decisions. Define groupthink, and describe two ways in which a group can guard against it.

14. Describe what a decision support system does, and state two of its limits.



CHAPTER IX

Structure and Design

Who does what, and who answers to whom?



Work does not arrange itself. This chapter shows how an organisation divides its work, groups its people, takes a shape, and keeps its parts moving together, and why a shape that suits one firm can fail another.

Three cousins run a stall, and every task is shared without effort.

Growth brings five new workers, and the order disappears.

The work is the same; only the number of people has changed, and that raises four questions.

The answers to the four questions form the structure of the organisation.

Structure. The way an organisation divides its work among people and links those people together.

Design. Choosing a structure on purpose, so that it fits the organisation's needs.

The chapter answers the four questions in order.

A family preparing a wedding feast must divide the work before anyone cooks.

Three cousins open a tea-and-snack stall beside a bus stop in Dhaka. On the first day, each cousin does everything in turn. One buys the tea leaves, then boils the water, then serves the customers, then washes the cups. The cousins talk all day, so each one knows what the others have done. Nobody needs a rule, because the whole business can be seen from the place where any one of them stands.

After a year the stall is busy, and the cousins hire five more people. Within a week the trouble begins. Two workers buy sugar on the same morning, and nobody buys tea leaves. One cousin tells a new worker to serve, and another cousin tells her to fry samosas; she does neither well. Two people argue over whose turn it is to wash the cups while customers wait. The stall sells the same tea as before, to the same people, at the same price.

Nothing about the tea has changed. What has changed is the number of people, and with it the distance between one person and the whole business. With three people, each could see everything. With eight, nobody can. Four questions now need an answer. What work must be done? Who does which part? Who gives orders to whom? How do the parts fit together?

The answers to these four questions together give the stall its **structure**. Structure means the way an organisation divides its work among people and links those people to one another. The stall's first structure was three cousins sharing everything. Its second structure must be chosen with care, and choosing a structure on purpose is called **design**. A good design lets eight people work as smoothly as three. A poor design makes eight people work worse than three.

This chapter answers the four questions in turn. It begins with what organising is and the steps it follows. It then shows how work is divided and grouped, how formal and informal arrangements differ, and which forms an organisation can take. Next it explains how to choose a design that fits the situation. It ends with the means of keeping the parts working together.

YOU WILL NEED

Chapter 2: the four functions of management, and productivity as output divided by input. Chapter 3: unity of command, which means one boss for each person; that chapter also pointed ahead to bureaucracy, which this chapter teaches in full. Chapter 4: the environment around an organisation. Chapter 6: plans. Chapter 7: objectives, and why they must be measurable.

WHAT ORGANISING IS

Begin with something familiar. A family is preparing a wedding feast for two hundred guests. The plan is clear: the food must be ready by seven in the evening. A plan alone does not cook the rice. Someone must decide who buys the food, who cooks it, who prepares the hall, who welcomes the guests, and who answers if the fish runs short. Until those choices are made, willing people will do some tasks twice and leave other tasks undone.

That arranging is what management calls **organising**. Organising means arranging people and work so that plans become action. The family's plan said what must happen; the organising said who would do each part. Chapter 2 named organising as the second of the four functions, after planning. Here we study it fully, one idea at a time.

Four features describe the nature of organising. First, it follows planning: without a plan, nothing tells us what to arrange. Second, it is continuing: when plans, people or conditions change, the arrangement must change too. Third, it deals with people as well as tasks, so it must take account of their skills, habits and feelings. Fourth, it is a means and not an end: nobody organises for its own sake, but only to reach the goals in the plan.

Organising serves five purposes. It makes plans become daily work. It gives every task an owner, so that no task is left undone and none is done twice. It allows resources to be used well, because people and equipment are placed where they help most; this raises productivity, the measure taught in Chapter 2. It makes each person's duties known, so that anyone can say who is answerable for a result. And it makes growth possible, because a business that can add people without confusion can add output without confusion.

The work of organising follows six steps, in this order. Step one is to study the plan and the objectives, because the work to be done comes from them. Step two is to list all the work, in small parts: buying, mixing, baking, selling, delivering, and so on. Step three is to divide that work into jobs that one person can do. Step four is to group the jobs under managers; the next section calls these groups departments.

Step five is to give each person and each manager the authority and the resources needed to do the job. Authority means the right to give orders and use resources; Chapter 10 teaches it in full. Step six is to link the parts, so that they work together as one. The figure below shows the six steps in order. Steps two to four are the subject of the next section, and step six is the subject of the last section of this chapter.

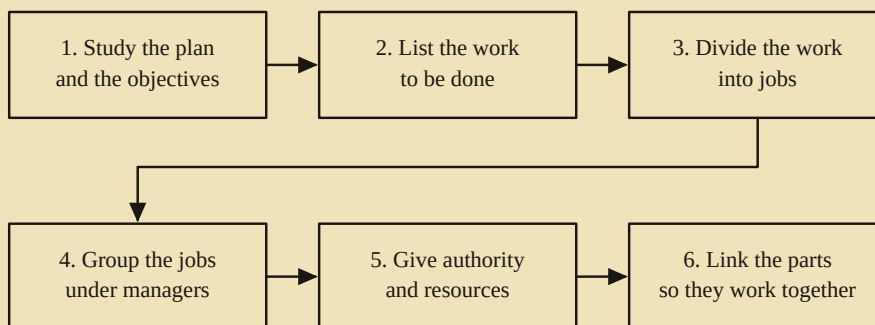


FIG. 26. The six steps of organising, from the plan to the working arrangement.

The steps match the four questions from the stall. Steps two and three answer *what work must be done* and *who does which part*. Steps four and five answer *who gives orders to whom*. Step six answers *how the parts fit together*. When a business is in trouble, it helps to ask which of the four questions has been left unanswered, because the missing answer usually points to the step that was skipped.

Organising looks different from each seat. The owner sees a way to reach the plan with fewer mistakes. The employee sees an answer to two daily questions:

Arranging people and work so that plans become action is called organising.

GO SLOWLY HERE.

Organising. Arranging people and work so that plans become action.

Organising follows planning, continues over time, concerns people, and serves goals.

Organising has five purposes, and each removes one kind of waste.

Six steps lead from a plan to a working arrangement.

The last two steps give people the means to act and join the parts.

The steps match the four questions raised by the stall.

Owner, employee and customer each see organising from a different seat.

No arrangement is final, and a very small firm needs little of it.

A short recap fixes the idea before moving on.

The reader tests the idea before going on.

In 1776 Adam Smith described ten workers who made far more pins together than apart.

A short calculation shows how large the gain was.

Working alone, one person could make perhaps one pin a day, not thousands.

The idea behind the pin workshop is called division of labour.

Division of labour. Splitting a job into parts so that each person does one part.

what exactly is my job today, and whom do I ask when I am unsure? The customer notices organising only when it fails: a long wait, a wrong order, or two people giving different answers. A good arrangement is mostly invisible, which is why its value is easy to forget.

There are limits. A one-person shop needs almost no organising, because one mind holds the whole plan. Arrangements also grow old: a structure that suits ten people may block fifty. And organising can be overdone. Too many jobs, too many levels and too many rules can slow the very work they were meant to help. The right amount depends on the situation, and the section on designing a structure returns to this point.

Before moving on. Organising means arranging people and work so that plans become action. It follows planning, continues as conditions change, concerns people, and serves goals. Its six steps run from studying the plan to linking the parts. *You should now be able to say:* "Organising is arranging people and work so that plans become action, and it follows six steps."

Pause and check. (a) Which step of organising comes directly before giving authority and resources? (b) Why does a one-person shop need almost no organising? The answers are at the back of the book.

DIVIDING THE WORK AND GROUPING THE JOBS

In 1776 a Scottish thinker named Adam Smith (1723 to 1790) published a famous book on the wealth of nations. Early in it he describes a small pin workshop.¹ Making a pin, he says, takes about eighteen separate operations. One worker draws out the wire, another straightens it, a third cuts it, a fourth points one end, and a fifth grinds the other end to receive the head. Making the head takes several more operations, and so do whitening the pins and putting them in paper. In the workshop he describes, ten workers shared the operations among them.

Smith reports that the ten workers made more than 48,000 pins in a day. We can find the share of each worker with the measure from Chapter 2. *Given:* total output 48,000 pins a day; 10 workers. *Find:* pins per worker per day. *Formula:* output per worker equals total output divided by the number of workers. *Steps:* $48,000 \div 10 = 4,800$ pins. *Answer:* each worker's share was 4,800 pins a day. *Check:* $4,800 \times 10 = 48,000$, which matches the total.

Smith then asks what would happen if each worker made pins alone, doing every operation. He answers that one such worker could hardly make twenty pins in a day, and perhaps not even one. Take the generous figure of twenty. The shared work gave 4,800 pins per worker, and $4,800 \div 20 = 240$. So the divided work produced at least 240 times as much per worker. Check: $20 \times 240 = 4,800$. The same ten people, the same wire, and a far larger output: the gain came from the way the work was arranged.

The idea behind Smith's workshop has a name: **division of labour**. Division of labour means splitting a job into parts, so that each person does one part. Each worker then becomes a specialist, a person who does one narrow kind of work and does it well. The cousins' stall had no division of labour in its first year, because each cousin did every task. After hiring, the stall must decide how to divide its work.

¹ Smith wrote that he had seen a small workshop of this kind with ten workers. His figures are his own account and not a measured study.

Two small terms keep the idea exact. A **task** means one piece of work, such as drawing out wire or serving a customer. A **job** means the set of tasks that one person is asked to do. In Smith's workshop, drawing out wire was a task, and the worker who did it all day held a job. Dividing labour changes the jobs: it cuts big jobs into smaller ones.

Smith gave three reasons for the gain. First, a worker who repeats one task becomes quick and skilled at it. Second, no time is lost in changing from one task to another, such as changing tools or moving to another place. Third, a worker who studies one task often finds a better tool or method, and some tasks can in time be given to machines. The third reason links to Chapter 3, where we saw that studying work can find the best method.

Case file, illustrative. Month 1. Morning Loaf Bakery opens with four employees: two bakers, one counter employee and one rider. Output is 200 loaves a day from 32 working hours a day, which is 6.25 loaves per hour. Labour is already divided into three kinds of job: making bread, selling bread and delivering bread. The owner is the only manager. No employee is asked to bake in the morning and ride in the afternoon.

Division also has costs. Work cut into very small parts can bore the person who repeats one part all day, and boredom can lower quality and raise absence. A worker may learn only one narrow skill, and so find it hard to cover for a colleague who is ill. Workers who see only their small part may not care about the finished product. Finally, every division creates a need to join the parts together again. When ten workers make pins in separate steps, someone must make sure that the straightened wire reaches the cutter. This last cost is called coordination, and the final section of the chapter teaches it.

Managers have found ways to reduce the costs of narrow jobs without giving up the gains of skill. In job rotation, employees move between jobs at fixed times, so that each learns more than one task and none repeats one task all day. In job enlargement, a job is made wider by adding tasks of a similar kind. Both ideas keep most of the benefit of division and lower the boredom. Chapter 11 returns to them when it studies what drives people to work well. The point to hold here is that division of labour is a matter of degree, and the right degree can be changed.

Division of labour has natural limits. Smith himself saw one: it cannot go further than the size of the market allows. A workshop that sells only two thousand pins a week cannot employ ten specialists, because most of them would have nothing to do. A second limit lies in the costs named above. At some point another division costs more in waiting, boredom and coordination than it gains in skill. The rule is to divide the work while the gain from skill is larger than the cost of joining the parts, and to stop there.

Before moving on. Division of labour means splitting a job into parts so that each person does one part. It raises output through skill, saved time and better methods. It costs boredom, narrow skills and a need for coordination, and it is limited by the market and by those costs.

A task is one piece of work; a job is the set of tasks one person holds.

Task. One piece of work, such as cutting wire or serving a customer.

Job. The set of tasks that one person is asked to do.

Smith gave three reasons why dividing the work raises output.

At the bakery, even four people divide their labour into three kinds of job.

Division of labour has costs as well as gains.

Job rotation and wider jobs reduce the boredom of narrow work.

Division of labour has two natural limits: the market and the cost of joining the parts.

A short recap before the next idea.

Once many jobs exist, they must be gathered under managers.

Department. A group of jobs placed under one manager.

Departmentalisation. Grouping jobs into departments.

The first choice is what the jobs in each group should have in common.

Function. In Chapter 2, one of the four management activities. Here, one specialist area of work, such as selling.

GROUPING JOBS INTO DEPARTMENTS

Dividing the work creates many jobs. One manager cannot direct fifty jobs one by one, so the jobs are gathered into groups. A **department** is a group of jobs placed under one manager. **Departmentalisation** means grouping jobs into departments. The choice is important, because a department decides who works beside whom, who shares a manager, and which skills sit together.

Four bases for grouping are common. Grouping by **function** puts together jobs that need the same kind of skill, such as production, selling or finance. Grouping by product puts together all the jobs needed to make and sell one product. Grouping by territory puts together all the jobs in one place. Grouping by customer puts together all the jobs that serve one kind of buyer. The table below compares the four. Read each row from left to right: the basis, what it gathers, when it works well, and where it is weak.

FOUR COMMON WAYS TO GROUP JOBS

BASIS	GATHERS TOGETHER	WORKS WELL WHEN	WEAKNESS
Function	jobs needing the same skill	the firm has one main product	departments may put their own goals before the whole firm's goals
Product	all jobs for one product	the firm sells several different products	the same skills are repeated in each group, which costs more
Territory	all jobs in one place	customers are spread over a wide area	it is harder to keep one standard everywhere
Customer	all jobs for one kind of buyer	buyers differ greatly in their needs	groups overlap in skill, and costs rise

Function is the usual first choice for a firm with one main product.

Grouping by function is the usual first choice for a firm that makes one main product. Skills sit together, so people learn from one another and equipment is shared. Its weakness is that each department may care more for its own work than for the whole. A clinic with twelve employees could group by function: a medical department, a pharmacy, and a front desk. A company with clinics in four cities might group by territory instead, so that each city has its own manager who knows the local patients.

Factories also group by process or by time.

Two further bases appear in factories. Grouping by process puts together all jobs that use the same machine or method, such as painting. Grouping by time creates shifts, such as a day shift and a night shift, when the work must continue around the clock.

Large firms use more than one basis, at different levels.

Large firms often use more than one basis. A firm may group by product at the top, giving each product its own division, and then group by function inside each division. The choice is not made once for all time. When a firm adds a new product, enters a new city, or doubles in size, the best basis may change.

Grouping by function helps specialists but can hide the customer from view.

Each basis has a hidden risk, and the risk of function is worth a closer look. When all the bakers sit in one department and all the sellers in another, each group learns its own craft well. Yet the customer who wants fresh bread at eight o'clock deals with both groups, and neither group sees the whole request. If the bread is late, production blames the counter for a wrong order, and the counter blames production for slow baking. The remedy is not a different chart alone. It is

link between the departments, which is the subject of the last section of this chapter.

Before moving on. A department is a group of jobs under one manager. Departmentalisation groups jobs by function, product, territory, customer, process or time. Function is the usual first choice for a small firm with one main product. *You should now be able to say:* "Division of labour splits the work into jobs; departmentalisation gathers the jobs into departments."

Pause and check. (a) A shirt maker sells to schools, to hospitals and to shops, and each buyer needs a different product and service. Which basis of grouping might suit, and why? (b) Name one cost of dividing work into very small tasks. The answers are at the back of the book.

A short recap of grouping.

The reader tests both ideas.

FORMAL AND INFORMAL ORGANISATION

Most organisations draw their structure as a diagram of boxes and lines, called an **organisation chart**. Each box is a job, and each line joins a manager to the people who report to that manager. The path of orders from the top box down to the bottom box is the **chain of command**. A new nurse in a ward looks at the chart and learns that she reports to the ward sister. In her first days she does exactly that.

A chart shows who should report to whom, but not how people really work together.

Organisation chart. A diagram of the jobs in an organisation and the reporting lines between them.

Chain of command. The path along which orders pass from the top of an organisation to the bottom.

The structure that the chart shows, together with the written rules and job descriptions that go with it, is the **formal organisation**. Formal organisation means the structure that is officially described, with its jobs, departments and reporting lines. It is planned in advance. It is written down. It belongs to the jobs, not to the people who hold them: when one nurse leaves, the box remains and another nurse fills it.

The chart, the written rules and the job descriptions together form the formal organisation.

Formal organisation. The structure that is officially described, with its jobs, departments and reporting lines.

But people are not boxes. In her first week, the same nurse learns that the quickest way to find a missing medicine trolley is to ask an older nurse who is not her manager. She learns who eats lunch with whom and which doctor likes to be asked early. These ties make up the **informal organisation**. Informal organisation means the friendships, habits and unwritten rules among people that the chart does not show. No manager ordered them, and no document records them.

People form ties that no chart shows.

Informal organisation. The friendships, habits and unwritten rules among people, outside the chart.

The table below sets the two side by side. Read each row as one question and its two answers.

The table sets the two kinds of organisation side by side.

FORMAL AND INFORMAL ORGANISATION COMPARED

QUESTION	FORMAL ORGANISATION	INFORMAL ORGANISATION
How does it begin?	It is planned by managers	It grows among people
What is it built on?	Jobs and positions	Persons and friendships
Is it written down?	Yes, in charts and rules	No
How fast does it change?	Slowly, by decision	Quickly, with the people

Informal organisation has good and bad effects. It can speed work, because people share knowledge across departments without waiting for permission. It can give people a sense of belonging, which helps them stay. It can also spread rumours, protect poor work from criticism, or unite people against a change. Whether the effect is good or bad depends on whether the informal groups share the goals of the organisation.

Informal ties can speed up work or spread resistance.

A ward with a friendly informal network shows both the benefit and the danger.

A manager should know the informal organisation, use it, and not try to remove it.

Four checks help a manager read the informal organisation.

A short recap of the section.

Six forms show the main ways to arrange jobs and authority.

The simplest form is a single chain of command from top to bottom.

Line. In daily life, a row of people or things. In an organisation, the positions that do its core work, joined in one chain of command.

Advisers who support the line, but do not command it, are added as a firm grows.

Staff. In daily life, all the employees of a firm. In organisation design, advisers who support the line with special knowledge but do not direct it.

Line-and-staff organisation gains expertise but can cause friction.

Consider a hospital ward again. The ward sister has told her nurses to record each medicine in a new written book. An older nurse, whom the others trust, says the book wastes time. Within a week most nurses have stopped using it. The formal organisation gave the order, and the informal organisation decided whether it was followed. Had the sister first asked the trusted nurse to try the book and report its faults, the same group might have carried the change instead of blocking it.

Managers cannot abolish informal organisation, because it comes from people's need for company. A sensible manager learns who the trusted people are, listens to what they say, and explains changes before announcing them. Managers who ignore the informal organisation are often surprised when a plan that looks perfect on the chart fails in daily practice. Chapter 11 returns to informal groups and explains why they form.

A manager can learn a great deal by asking four plain questions. Who do people ask when they are unsure? Who eats and rests together? Whose opinion changes the opinions of others? Which unwritten habits help the work, and which hinder it? The answers do not appear on any chart, but they explain why some orders are carried out quickly and others are quietly ignored.

Before moving on. The formal organisation is the structure shown on the chart and in the rules. The informal organisation is the web of friendships and habits outside it. Both exist in every organisation, and a manager works with both.

FORMS OF ORGANISATION

Structures can be arranged in several recognisable forms. Six are described here: line, line-and-staff, functional, committee, project and matrix. Few real organisations use one form alone. Most mix several, as a house may be built with brick, wood and glass. Learning the pure forms helps us to see what each part of a mixed structure is for.

LINE ORGANISATION

The simplest form uses only **line** positions. Line positions are the jobs that do the organisation's core work directly, joined in one chain of command. In a bakery, the line jobs are baking, selling and delivering. In a line organisation every person has one manager, and orders pass straight down the chain. This gives clear duties, fast decisions and unity of command, the idea from Chapter 3. Its weakness is that each manager must know everything about the work below. As the firm grows, the managers become overloaded, and no specialist is there to help them. Line organisation therefore suits small firms.

LINE-AND-STAFF ORGANISATION

As a firm grows, line managers need expert help with matters such as law, accounts or hiring. The firm adds **staff** positions. Staff means advisers who support the line with special knowledge but do not direct line work. Take care with the word. In daily speech, staff means all the employees of a firm. In this chapter it means advisers only, and we use the word employees when we mean everyone. A legal adviser who tells a manager whether a contract is safe holds a staff position. The adviser gives advice; the manager decides.

A **line-and-staff organisation** keeps the chain of command and adds advisers beside it. Its strength is expertise without overloading the line managers. Its weaknesses are cost and friction. Line managers may ignore advice they did not ask for, and advisers may feel that their good ideas are wasted. Chapter 10

explains the kinds of authority that line and staff positions hold. The figure below shows the line form and the line-and-staff form side by side.

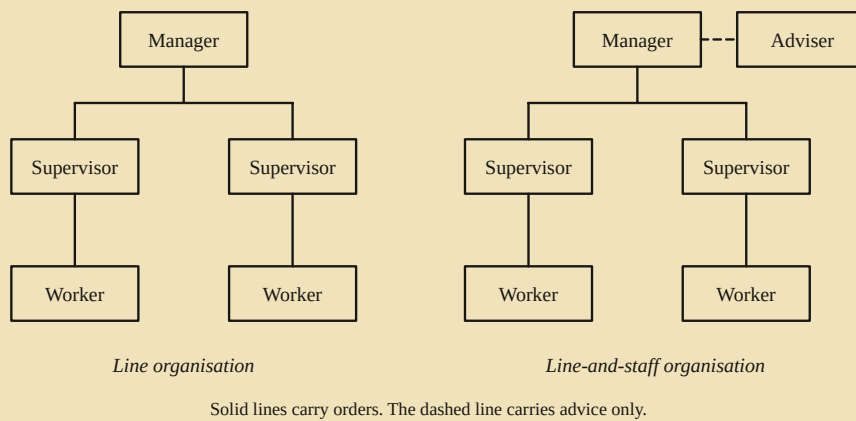


FIG. 27. A line organisation (left) and a line-and-staff organisation (right).

FUNCTIONAL ORGANISATION

Frederick W. Taylor, whose scientific management we met in Chapter 3, proposed a different arrangement for the workshop (Taylor, 1903). In a **functional organisation**, each worker takes orders from several specialist bosses, and each boss is an expert in one part of the work. One boss might direct the speed of work, another the quality checks, and another the repair of machines.² The strength is expert guidance at every point. The weakness is serious: the worker has more than one boss, so unity of command is broken, orders may conflict, and nobody can say who is answerable for the result. Few firms use the pure form today. A mild version remains where specialists have the right to instruct others on their own subject, and Chapter 10 returns to it.

Taylor proposed that a worker take orders from several specialist foremen.

COMMITTEE

A **committee** is a group of people formally chosen to study a matter or to decide it together. Some committees are standing, meaning they continue; others are temporary, meaning they end when the task ends. Committees bring together different knowledge and different views, and they share responsibility for a decision. Their weaknesses are slowness, compromise and cost, since many people spend time on one matter. They can also leave nobody answerable. Chapter 8 listed the strengths and weaknesses of group decisions, and the same points apply here. A committee is best used for matters that need many views, and worst used for matters that need speed.

A committee gives a task to a group and not to one person.

PROJECT ORGANISATION

A **project organisation** is a temporary team formed to complete one task that has a fixed end, such as opening a new shop. The team is led by a project manager, and its members come from the departments. When the task is finished, the team ends and the members return to their departments. Its strengths are focus and speed on the one task. Its weaknesses are disruption to the departments that lend their people, and uncertainty for members who do not know where they will go next. Keep this form apart from the matrix below: the project team is temporary, while a matrix is permanent.

A project team brings people from several departments for one task, then ends.

² The word *functional* has two meanings in this chapter. Here it names a form of organisation with several specialist bosses. Earlier it described grouping jobs by function, which means by specialist area.

A student in a school play has a class teacher and a play director at once.

GO SLOWLY HERE.

The matrix crosses a line of skill with a line of tasks.

Matrix. Two lines of command crossed, so that one person reports to two managers.

MATRIX ORGANISATION

Begin with something familiar. In a school, a student acts in the school play. During lessons the student answers to the class teacher. During rehearsals the same student answers to the play director. Both give orders, and both are right to do so, each in their own area.

A **matrix** is the organisation form that does this on purpose. A matrix means two lines of command crossed, so that one person reports to two managers. One line follows the specialist area, such as production, sales or finance. The other follows the task, such as one project or one product. The figure below shows the idea as a grid. Each employee sits where a column meets a row: the head of the column controls the skill, and the manager of the row controls the task.

	Head of Production	Head of Sales	Head of Finance
Manager of Project A	employee	employee	employee
Manager of Project B	employee	employee	employee

Each employee answers to the head of the column and to the manager of the row (heavy box: one example).

FIG. 28. A matrix organisation: each employee answers to a head of function and to a project manager.

Matrix organisation shares scarce experts but breaks unity of command.

A recap of the matrix before the comparison of forms.

The table compares the six forms in one place.

The matrix has real strengths. Scarce experts can be shared among many projects instead of being locked into one. The firm can respond quickly when a new project begins. Technical standards and project needs both have a voice. It also has real weaknesses. Recall unity of command from Chapter 3, which gives each person one boss. The matrix breaks that rule, so an employee may receive two conflicting orders on the same morning. Decisions can be slow, power struggles are common, and the structure works only where people communicate openly. It suits firms with many projects and scarce experts, such as building and engineering firms. It rarely suits a small firm with one product.

Before moving on. A matrix gives each person two bosses, one for the skill and one for the task. It shares experts and speeds new projects, but it breaks unity of command and needs open communication. *You should now be able to say:* "A matrix crosses a line of skill with a line of tasks, so one person has two managers."

CHOOSING AMONG THE FORMS

The table below compares the six forms. It does not name a winner, because none exists. Read each row for the main idea, one strength and one weakness.

SIX FORMS OF ORGANISATION COMPARED

FORM	MAIN IDEA	STRENGTH	WEAKNESS
Line	one chain of command	simple, fast	managers overloaded
Line-and-staff	line plus advisers	expert help	friction, cost
Functional	several specialist bosses	expert direction	breaks unity of command
Committee	a task given to a group	many views	slow, unclear answerability
Project	temporary team for one task	focus on the task	disrupts departments
Matrix	two bosses, skill and task	shares experts	conflicting orders

Each form also looks different from each seat. The owner of a small firm sees a line organisation as control and speed. An employee in a matrix sees it as interesting work and also as two sets of demands on one day. A customer of a firm that relies on committees sees it as delay, because every decision waits for a meeting. The same structure can therefore be a strength to one person and a burden to another, and a wise manager asks all three how the design feels in practice.

No form is best in every case. A bakery of four employees may be a line organisation. When it grows, it may add a part-time adviser and become line-and-staff. It may form a temporary project team to open a second shop, and a committee to review hygiene. The skill lies in matching each form to the part of the work it suits, and in knowing the weakness that comes with each choice.

The same form looks different from the owner's, the employee's and the customer's seat.

No form is best; real firms mix them to suit different parts of the work.

DESIGNING A STRUCTURE

Consider two organisations. A tax office handles thousands of similar forms. Mistakes are costly, and every citizen must be treated by the same rules. A small web design studio makes a different website each month, and its work changes every week. Give the tax office the studio's loose arrangement, and each clerk will handle the same form in a different way. Give the studio the tax office's rules, and every new idea will wait for three signatures. The two structures differ because the two situations differ. Design means choosing the structure that fits.

A tax office and a design studio need very different structures, and neither is wrong.

THE BUREAUCRATIC MODEL

Max Weber (1864 to 1920) was a German thinker who studied how large organisations, such as state offices and armies, did their work.³ He described a form that he called bureaucracy. In everyday speech the word now suggests delay, but Weber used it for a particular design, and we use it in his sense.

Max Weber described a design built on rules, ranks and written records.

The **bureaucratic model** is a design built on written rules, fixed ranks and records. Weber named its main features. Duties are fixed, through a clear division of labour. Positions are arranged in ranks, and each rank is controlled by the one above. Written rules say how tasks are done, and written records keep the results. Everyone is treated by the same rules, whoever they are. And people are chosen and promoted for their skill, not for their family or friends.

The bureaucratic model rests on written rules, fixed ranks and records.

Bureaucratic model. A design built on written rules, fixed ranks and records.

³ Weber's work on bureaucracy was published in German after his death, in 1922; English readers know it through a translation published in 1947.

The bureaucratic model gives fairness and accuracy, but it responds slowly to change.

The behavioural model builds the structure around the needs and abilities of people.

Behavioural model. A design that stresses people, teamwork and flexibility more than rules and ranks.

The behavioural model gives speed and creativity, but weaker order and fairness.

The table compares the two models on features, strengths and weaknesses.

The two models are ends of a scale, not two separate boxes.

The model has strong points. It is fair, since the same rule applies to all. It is predictable, since people know what to expect. It is accurate, because records and procedures reduce errors. It can run very large organisations. Its weaknesses appear when conditions change. Rules written for yesterday may not fit today. Employees may follow a rule even when it plainly harms the customer. Decisions climb through many ranks and arrive late. People who are not asked for ideas stop offering them. The model suits stable conditions, repeated work and large size.

THE BEHAVIOURAL MODEL

The **behavioural model** is a design that stresses people, teamwork and flexibility more than rules and ranks. It draws on the human relations approach from Chapter 3, which stressed people's feelings and groups. In this design, jobs are broad and can change. Work is done in teams. There are few written rules. Decisions are taken by the people who have the knowledge, whatever their rank. Talk moves freely across levels, and influence comes from expertise as well as from position.⁴

This model also has strong points. It responds quickly to change, it encourages new ideas, and it often raises commitment, since people feel trusted. Its weaknesses are the reverse of the first model's strengths. Duties can be unclear, and conflicts can grow without clear rules to settle them. Routine decisions may take longer because more people join the talk. The design is hard to manage when the organisation is large, and it depends on skilled and willing people. It suits changing conditions, creative work and expert employees.

COMPARING THE TWO MODELS

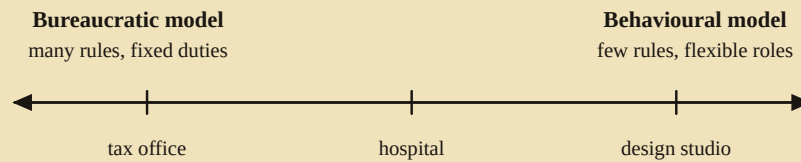
The table below compares the two models. Each row names a feature, and the two columns show how each model treats it.

THE BUREAUCRATIC AND THE BEHAVIOURAL MODEL COMPARED

FEATURE	BUREAUCRATIC MODEL	BEHAVIOURAL MODEL
Rules	many, written	few, flexible
Jobs	narrow and fixed	broad and changing
Decisions made by	managers at higher ranks	people with the knowledge
Main strength	fairness, accuracy, order	speed, ideas, commitment
Main weakness	slow and rigid	unclear duties, weaker order
Suits	stable, repeated work	changing, creative work

The two models are the ends of a scale, not two separate boxes. Most organisations stand somewhere between them, and many use one model in one part and the other model elsewhere. A hospital may apply strict rules to medicine records and give flexible teamwork to patient care. The figure below shows the scale.

⁴ Some writers call the bureaucratic model *mechanistic* and the behavioural model *organic* (Burns and Stalker, 1961). This book keeps to the first two names.



Illustrative positions only. Most organisations mix both models.

FIG. 29. The two design models as the ends of a scale; positions are illustrative.

SITUATIONAL FACTORS

If no model is best, what decides the choice? Chapter 3 introduced the contingency approach, which says that the best action depends on the situation. Applied to structure, it names factors that push a design one way or the other. Five factors are taught here: size, technology, environment, strategy and people. We take them one at a time.

Size. A firm of four can work by talk alone, because everyone sees everything. A firm of four hundred cannot. As size grows, the firm needs more departments, more levels of managers, and more written rules, because no one can speak with everyone. Growth therefore pushes a design toward the bureaucratic end of the scale, and a firm that grows must review its structure.

Technology. Technology here means the methods and equipment used to do the work. Routine work, such as making thousands of identical items, fits fixed jobs and clear rules. One-of-a-kind work, such as designing a special machine, needs flexible teams. Joan Woodward studied manufacturing firms in England in the 1950s and found that structures differed with the production method (Woodward, 1965).

Environment. Chapter 4 explained the forces outside an organisation. When they change slowly, rules written today will still serve tomorrow. When they change fast, through new technology, new competitors or new laws, rules age quickly and people must adapt on their own. Tom Burns and G. M. Stalker studied firms in Britain in the 1950s and found that firms in stable conditions did well with rule-bound structures, while firms in fast-changing conditions did better with flexible ones (Burns and Stalker, 1961).

Strategy. Strategy means the firm's chosen way to reach its goals, the long-range kind of plan from Chapter 6. Alfred Chandler studied large American firms and found that when a firm changed its strategy, for example by adding many products, its old structure began to fail, and a new one followed (Chandler, 1962). His conclusion is often stated in four words: structure follows strategy.

People. The skills and attitudes of employees also matter. Skilled professionals who value freedom usually work well under the behavioural model. New employees who need clear instruction may do better with defined duties and rules. A structure that ignores the people who must work in it will be resisted, however logical it looks on the chart.

APPLYING THE FACTORS

Take a new setting. A clinic has twelve employees and is told by a new law to keep written records of every patient. How should its structure respond? We apply the five factors in turn. **Size:** twelve is small, so little formality is needed in general. **Technology:** patient care differs from one patient to the next, but billing is routine. **Environment:** the new law adds a fixed outside demand for

No single design suits every firm, so the design must fit the situation.

GO SLOWLY HERE.

Size changes how many rules and levels a firm needs.

Technology means the way the work is done, from routine to one-of-a-kind.

A stable environment fits rules; a changing one needs flexibility.

Structure should follow strategy, the firm's chosen way to reach its goals.

People matter, because skilled and willing employees need less supervision.

A small clinic shows the five factors working together.

The same factors also tell a manager when to change the design.

The factors point to a mixed design.

records. *Strategy*: the clinic aims at good care, which depends on the doctors' judgement. *People*: doctors and nurses are skilled and value freedom to decide.

The factors also tell a manager when to change a design. Suppose the clinic doubles to twenty-four employees. Size has changed, so more people need clear reporting lines, and a single manager can no longer hear every request. Suppose instead that a new technology lets patients book by telephone application. Technology has changed, so the front desk must learn new routines and the rules for records must change. A manager who reviews the five factors once a year will notice such changes before they cause trouble, and will adjust the structure in small steps and not in a crisis.

The factors point in different directions, and the answer is a mixed design. For records and billing, the clinic should adopt bureaucratic features: written rules, forms and a person answerable for accuracy. For patient care, it should keep behavioural features: teamwork and decisions by those who know the patient. Mixed designs are common because most organisations contain both routine and changing work.

IN SHORT

A good design divides the work into jobs, groups the jobs under managers, links the groups, and fits the firm's size, technology, environment, strategy and people. When one of these changes, the design should be reviewed.

A short recap of the section.

Before moving on. The bureaucratic model is built on written rules, fixed ranks and records. The behavioural model stresses people and flexibility. Five factors, namely size, technology, environment, strategy and people, decide which mix fits. *You should now be able to say*: "The best structure depends on the situation, so a manager studies five factors before choosing."

COORDINATION: FITTING THE PARTS TOGETHER

Division creates parts, and the parts must then be fitted together.

Coordination. Fitting the parts of an organisation together so that they work toward one goal.

The need for coordination grows faster than the number of people.

Return to the stall. The cousins have divided the work: one makes tea, one fries snacks, one washes cups. If the cups are not ready when the tea is ready, the tea grows cold and customers wait. Each worker has done the assigned task, and the stall still fails. The missing element is **coordination**. Coordination means fitting the parts of an organisation together so that they work toward one goal. Division makes the parts, and coordination makes them one.

Coordination becomes harder as the organisation grows, and the growth is faster than it first appears. Any two people may need to speak to each other, so the number of possible pairs matters. *Given*: the stall has 3 people, then 8. *Find*: the number of possible pairs. *Formula*: pairs equal $\text{people} \times (\text{people} - 1) \div 2$. *Steps*: for 3 people, $3 \times 2 \div 2 = 3$ pairs; for 8 people, $8 \times 7 \div 2 = 28$ pairs. *Answer*: the stall has 3 possible pairs at first and 28 after hiring. *Check*: for 3 people called A, B and C, the pairs are AB, AC and BC, which is 3. More than doubling the people (3 to 8) multiplied the possible links by nearly ten.

Two other conditions raise the need for coordination. The first is dependence between the parts. Dependence can be of three kinds. In sequence, the output of one part becomes the input of the next, as dough becomes bread and bread reaches the counter. In sharing, the parts work separately but together make the whole, as two shops of one firm add their sales. In two-way dependence, each part keeps needing something from the other, as a kitchen and a counter keep exchanging changes to orders (Thompson, 1967). The more the dependence, the

Two other conditions raise the need: dependence between parts, and uncertainty.

more the coordination. The second condition is uncertainty. When plans change often, fixed rules fail, and people must talk.

Managers have six main means of coordination. *The chain of command*: a manager above resolves disputes between two parts; Chapter 10 teaches it. *Plans, schedules and rules*: these fix in advance who does what and when, as Chapter 6 showed. *Shared objectives*: when every part aims at one measurable target, as in Chapter 7, the parts adjust themselves to it. *Direct contact*: meetings, briefings and conversation let people settle matters as they arise. *Linking roles and teams*: a person or team whose task is to connect departments. *Shared records and information systems*: a written order sheet, or a computer system, that all can read.

Return to the wedding feast from the start of the chapter. The family can coordinate in two ways. It can hold a short meeting at noon, where each helper reports what is ready and what is late; that is direct contact. It can also keep a written list on the kitchen wall, where each task is ticked when finished; that is a shared record. The meeting settles surprises, and the list prevents forgotten tasks. Most organisations, like the family, use more than one means at once, because each covers a weakness in the other.

No means is free. Rules and plans cost little to use but suit stable, repeated work. Direct contact suits changing work but takes time, and a firm that holds too many meetings may coordinate its parts and exhaust its people. A manager chooses by asking how much dependence and how much uncertainty there is, and then selects the cheapest means that works. Coordination also differs from control. Coordination fits the parts together as work proceeds, while control, taught in Chapter 13, checks results against standards afterward.

Many firms count the saving from dividing work and forget the price of joining it again. A firm that splits one job into four should ask what it will spend on meetings, forms and managers to fit the four parts together. If the answer is larger than the gain in skill, the division has gone too far. This is why the question at the end of the chapter asks whether more division always needs more coordination. The answer lies in the balance between the two, and a good manager keeps both in view.

Before moving on. Coordination means fitting the parts of an organisation together so that they work toward one goal. The need grows with the number of people, the dependence between the parts, and uncertainty. Six means serve it: chain of command, plans and rules, shared objectives, direct contact, linking roles, and shared records.



WORKED CASE: THE BAKERY AT MONTH NINE

Case file, illustrative. Month 9. *Situation.* Morning Loaf Bakery has grown from four employees to eight: production has four, sales has three, and delivery has one. The plan for months 7 to 12 is 300 loaves a day, with objectives agreed in Month 7. Buying and money stay with the owner for now. Until now the owner told each employee what to do each day. With eight employees, that has become too much.

Question. How should the work be divided and grouped, and how should the parts be fitted together?

Coordination uses six means, from the chain of command to shared records.

The wedding feast shows how a briefing and a written list work together.

Each means suits some conditions and costs something.

Coordination is the main cost of a design, and it is easy to underpay.

A short recap of the section.

The case applies the whole chapter to one growing business.

The question asks for a division, a grouping and a link between the parts.

The reasoning starts with the work and the plan.

Reasoning. First, we list the work from the plan: mixing and baking, selling at the counter, delivering, and buying and money. Second, we divide the work into jobs. Baker, helper, counter employee and rider are jobs, and the buying and money tasks stay with the owner. Third, we choose a basis for grouping. The bakery has one main product, one shop and one kind of customer, so grouping by function suits it. Fourth, we form three departments: production, sales and delivery. Production is headed by the head baker and sales by the counter lead. The rider reports directly to the owner.

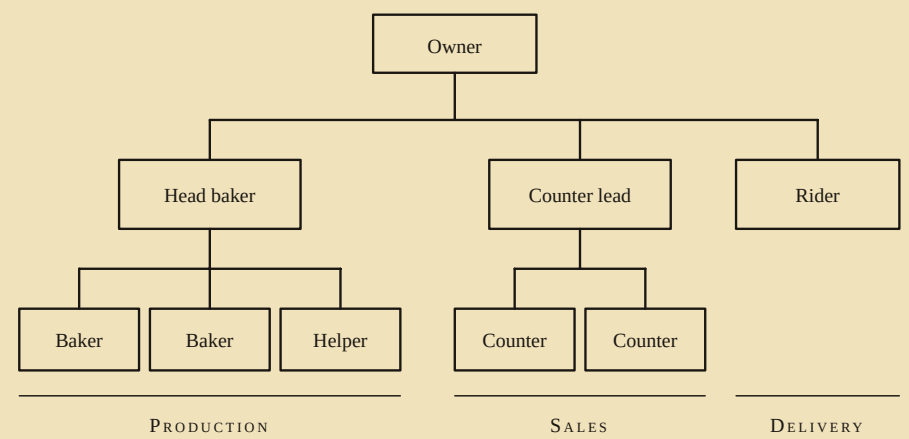


FIG. 30. *Morning Loaf Bakery, Month 9: three departments grouped by function. Case file, illustrative.*

The parts are linked by a daily briefing and an order sheet.

Reasoning, continued. Fifth, we ask how the parts fit together. Production must know today's orders, sales must know when the bread is ready, and delivery must know which orders to carry. These are dependencies of the sequential and two-way kinds. The bakery coordinates them with two means: a briefing at 7 am each day, which is direct contact, and an order sheet, which is a shared record. The owner also notices an informal organisation: a lunch group has formed among four of the eight employees. It appears on no chart. The owner decides to listen to it, since it may carry news early, and not to forbid it.

The conclusion states the design in one sentence.

Conclusion. The bakery is grouped by function into three departments under the owner, coordinated by a daily briefing and an order sheet, and it has an informal lunch group beside the formal structure.

The check counts people and tests the design against the rules of the chapter.

Check. Count the employees: production 4 plus sales 3 plus delivery 1 equals 8, which matches. Each employee has one manager: the head baker directs three, the counter lead directs two, and the owner directs the head baker, the counter lead and the rider. Every task on the work list has an owner. Chapter 10 returns to these counts. Since the plan rises to 300 loaves a day, the owner should review the design when the bakery next grows.



SUMMARY

A summary of the whole chapter.

Organising means arranging people and work so that plans become action, in six steps. Division of labour splits the work into jobs, which raises output but needs coordination. Departmentalisation gathers jobs into departments by function, product, territory, customer, process or time. Every organisation has a formal structure on its chart and an informal organisation among its people. The main forms are line, line-and-staff, functional, committee, project and matrix. The

bureaucratic model suits stable, routine work; the behavioural model suits changing, creative work; and size, technology, environment, strategy and people decide the mix. Coordination fits the parts together by the chain of command, plans, shared objectives, direct contact, linking roles and shared records.

THINK IT THROUGH

A tailor working alone makes three shirts a day. A workshop of six tailors, each doing one step, makes twenty-four shirts a day. Find the output per worker in the workshop, and compare it with the lone tailor. Then notice that the cutter finishes early on some days and the stitchers wait for one another on others. Does more division of labour always need more coordination? What new work appears when the workshop grows to twelve? Think about the number of possible pairs, the kinds of dependence, and the means of coordination, and form your own answer.

A tailoring workshop tests whether more division always needs more coordination.

WHERE PEOPLE GO WRONG

"A good design lasts for ever." A design is a decision made for a situation, and situations change. Size, technology, environment, strategy and people all move over time, so the design must be reviewed from time to time.

A last mistake: a design chosen once stays right for ever.

"The chart shows the organisation." The chart shows only the formal organisation. Friendships, habits and unwritten rules also shape how work is done, and a plan that ignores them may fail.

Mistake one: the chart is the whole organisation.

"More departments mean better order." Each department adds a manager and a boundary that the work must cross. Departments should be added when the gain from specialist skill exceeds the cost of coordination.

Mistake two: more departments always mean better order.

"Staff means all employees." In daily speech it does. In organisation design it means advisers who support the line but do not command it. The same word has two meanings, so check which one is in use.

Mistake three: staff means employees.

"Informal groups are a problem." They are a fact. They can spread rumours, and they can carry knowledge and loyalty. A manager studies and uses them and does not try to remove them.

Mistake four: informal organisation is a problem to be removed.

"There is one best structure." The tax office and the design studio show that the best design depends on the situation. A structure that was right at ten employees may be wrong at a hundred.

Mistake five: one design is best for all firms.

"Coordination and control are the same." Coordination fits parts together while the work goes on. Control checks results against standards afterward and corrects them.

Mistake six: coordination and control are the same.

IN HISTORY: ORDER AT GENERAL MOTORS

General Motors was founded in 1908 by William Durant, who built it by buying many car companies. By 1920 the company was in crisis. Its units were poorly joined. Durant resigned as president in November 1920.

General Motors was rescued by a better structure and not by a better car.

Alfred P. Sloan (1875 to 1966), an executive of the company, had written a plan for a new structure. He proposed that each car division should run as a separate business under its own general manager, grouped by product. A central office would set broad policy, gather figures, and judge each division by the return it earned. Sloan became president in 1923. Committees, such as a technical committee formed in that year, helped the divisions to share knowledge without losing their independence.

Sloan proposed independent divisions under a central office.

The structure became a model for large firms.

The design worked. It gave divisions the freedom to respond to their own markets, and it gave the centre the information to judge them. The historian Alfred Chandler later described it as a pattern copied by many large firms. Sloan's own account appeared in *My Years with General Motors* (1963). The case shows this chapter's main lessons in one story: group by product when products multiply, coordinate by information and committees, and review the structure when the firm grows.

SOURCES

Smith, 1776: the origin of the idea of division of labour.

Smith, Adam (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*. Book I, chapter 1 gives the pin workshop, and chapter 3 gives the limit set by the size of the market. It matters because it first explained why divided work raises output.

Weber: the origin of the bureaucratic model.

Weber, Max. *Economy and Society*, published in German after his death in 1920, with an English translation in 1947. It gave the first careful account of bureaucracy as a form of organisation.

Studies that tie structure to the situation.

Woodward, Joan (1965), *Industrial Organization: Theory and Practice*; Burns, Tom and Stalker, G. M. (1961), *The Management of Innovation*; Chandler, Alfred D. (1962), *Strategy and Structure*. Together they showed that technology, environment and strategy shape the best structure, and they support the contingency approach.

Studies of dependence and of the General Motors design.

Thompson, James D. (1967), *Organizations in Action*, for the kinds of dependence between parts; Taylor, Frederick W. (1903), *Shop Management*, for functional foremanship; Sloan, Alfred P. (1963), *My Years with General Motors*, for the divisional design.

REVIEW QUESTIONS

Start from the plan and list the steps in order.

1. Define organising in your own words, and list its six steps in the order they are taken.

Think of one gain, one cost and the limit.

2. A canteen sells tea, snacks and lunch plates, and one worker now does everything. The owner will hire five more people. Explain division of labour, and name two gains and two costs it would bring to the canteen.

Compare strengths and weaknesses, not only names.

3. A company has pharmacies in three cities, and each city has different customers. Compare grouping its jobs by function with grouping them by territory, giving one strength and one weakness for each.

Give a different example for each kind.

4. Distinguish the formal organisation from the informal organisation, with an example of each from a school or a hospital.

Remember what staff means in organisation design.

5. A clinic has these twelve employees: a manager, two doctors, three nurses, a pharmacist, two receptionists, a laboratory technician, a cleaner and a part-time accountant. Draw an organisation chart. State which positions you treat as line and which as staff, and why.

Match the situation to the form.

6. Name the form that best fits each case, and give one reason for each: a firm opening one new shop in six months; a firm that needs the opinions of five departments on a new policy; a firm whose experts must serve three projects at once.

Use a table with strengths and weaknesses.

7. Compare the bureaucratic model and the behavioural model in a table. Include one strength and one weakness of each, and name one organisation that suits each.

Name the five factors and apply each one.

8. A firm of thirty employees sells from one shop and now plans to take orders online from the whole

country. Name the five situational factors and say how each might change its structure.

9. A tea stall grows from 3 people to 12. Find the number of possible pairs of people in each case, and explain why the need for coordination rises faster than the number of people.

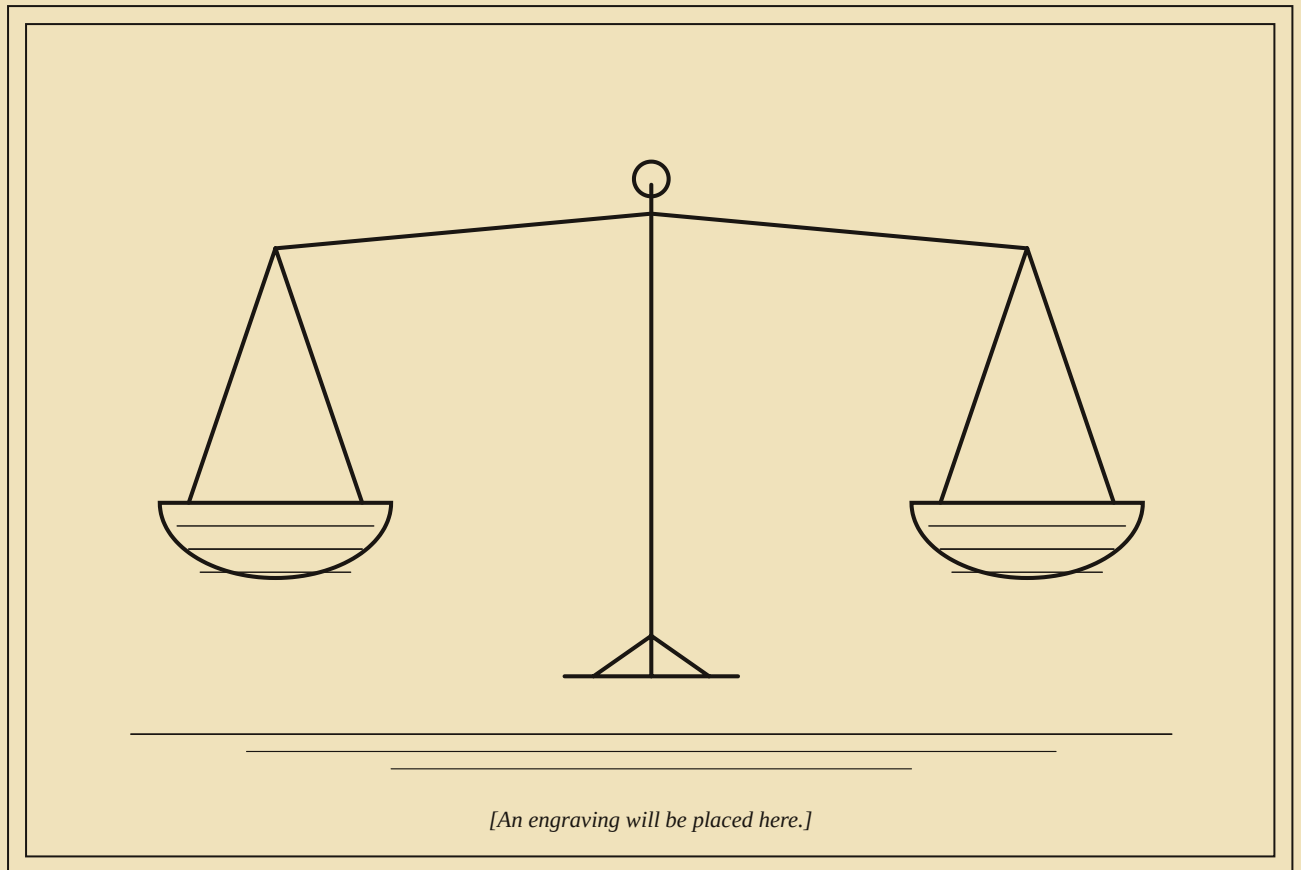
Use the pairs formula from this chapter.

10. Name four means of coordination. Which would you choose for a workplace where plans change every day, and why?

Think of the cheapest means that works.

11. Explain what a matrix organisation is, and give one benefit and one problem that it brings.

Link the idea to unity of command.



CHAPTER X

Authority, Delegation and Span

How much may one person decide, and for how many?



Structure gives people places; authority gives them power to act in those places. This chapter explains authority and responsibility, the three kinds of authority, the art of delegating, the number of people one manager can direct, and the choice between deciding at the top and deciding far below.

A mother gives her son a task but not the means to do it.

The mother then makes the opposite mistake and checks every step.

Both failures come from one problem: matching the task to the power to do it.

The chapter answers these questions with five ideas.

A mother is expecting guests for dinner. She tells her sixteen-year-old son to cook the meal. He agrees, and then finds that the store cupboard is locked and the housekeeping money is in her purse. He cannot buy rice. He cannot even start. The task was given, but the means were kept back.

The next week she tries again. This time she gives him the key and the money. But she stands in the kitchen and corrects every cut of the knife, every pinch of salt. After twenty minutes he leaves, saying that she may cook it herself. Now the means were given, but the freedom to use them was taken away.

Both failures have one cause. The mother did not match the task she handed over to the power she handed over with it. In a family, the cost is a spoiled evening. In a business, the same mistake costs sales, wasted time and employees who stop caring. Three questions follow. How much power should go with each job? How many people can one manager direct? And which decisions should be taken at the top, and which far below?

This chapter answers the three questions in turn. It begins with authority and its partners, responsibility and accountability. It then separates three kinds of authority, teaches the steps and obstacles of delegation, and shows how to judge the number of people one manager can direct. It ends with the choice between centralising and decentralising decisions.

YOU WILL NEED

Chapter 9: structure, the chain of command, the forms of organisation, and coordination. Chapter 1: accountability, which means being answerable for results. Chapter 3: unity of command, which means one boss for each person. Chapter 8: the idea of a decision.

AUTHORITY AND RESPONSIBILITY

A plan, a job and a chart do not by themselves let anyone act.

GO SLOWLY HERE.

Authority is the right to give orders and use resources.

Authority. The right to give orders and to use resources, given by one's position.

Subordinate. A person who reports to a manager.

Responsibility is the duty to do the assigned task.

Responsibility. The duty to do the task that has been assigned.

Accountability means answering for results; it stays with the person who delegates.

Chapter 9 gave every job a place on the chart. A place on a chart does not let the holder buy, hire, refuse or approve. Something more is needed, and that something is the subject of this section. Three ideas work together here, so we take them one at a time and then join them.

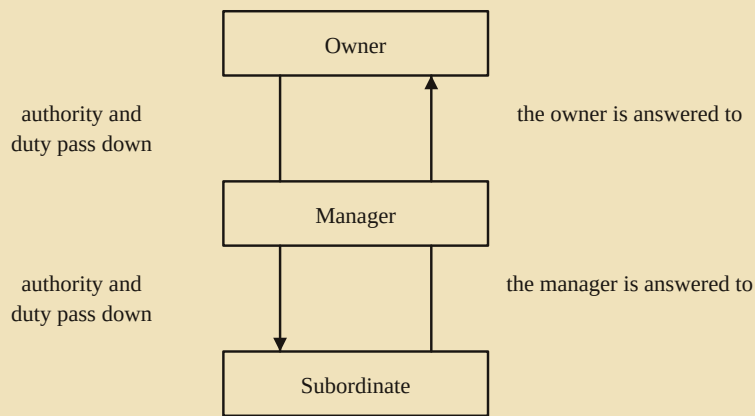
The first idea is **authority**. Authority means the right to give orders and to use resources, given by one's position. A shop manager may order stock and tell the counter employees which tasks to do. The right belongs to the position, not to the person: when a new manager takes the post, the right passes to the new person. A person who reports to a manager is a **subordinate**, and the manager's authority is the right to direct the subordinate's work.

The second idea is **responsibility**. Responsibility means the duty to do the task that has been assigned.¹ When the head baker is told to see that the bread is ready by six o'clock, she holds a responsibility. It is an obligation toward the person who gave the task. It is also an obligation that cannot be forgotten, since the task still waits if she does nothing.

The third idea was met in Chapter 1: accountability, being answerable for results. It differs from responsibility. Responsibility looks forward, to the task to be done. Accountability looks back, to the result that was reached, and it asks who must explain it. The head baker is responsible for the bread; the owner remains accountable to customers and lenders for all the bread the

¹ In everyday speech, *responsible* can also mean sensible or trustworthy. In this chapter it means only having a duty to perform an assigned task.

bakery sells. The figure below shows how the three pass along the chain of command.



Accountability for the result stays with the one who delegates.

FIG. 31. How authority and duty pass down the chain of command, while answerability passes up.

The three ideas must fit together. The rule is that authority should match responsibility. The mother's first failure gave responsibility without authority; the son could not do what he was told to do. Her second failure gave authority and then took it back. When responsibility is larger than authority, the holder is blamed for things beyond control, and morale falls. When authority is larger than responsibility, the holder may act carelessly, because no duty follows the act. Good organising keeps the two in step.

Where does a manager's authority come from? There are four common sources. The first is ownership: the owner of a business may direct it, and may pass part of that right to others. The second is law, which gives certain officers, such as a company's directors, rights and duties that no one may remove. The third is delegation from above, which is how most managers receive their authority: the owner gives it to a general manager, the general manager gives part of it to a department head, and so on down. The fourth is custom, the habits of an organisation that people have come to accept. Each source also sets a limit. An owner may not order what the law forbids. A manager may not give a subordinate more authority than the manager holds. And custom can be changed only slowly.

Authority flows down the chain of command that Chapter 9 described, from the top box to the bottom box. This path is sometimes called the scalar chain, meaning the ladder of ranks. Orders are expected to travel along it, level by level, and answers are expected to travel back up. The path has a practical use: it tells every employee who the manager is, and where to turn when a question is too large for them. It has a cost too, because a message that passes through five levels is slow and may change on the way. Fayol (1916) accepted that a short cut may sometimes be wise, but he asked that the managers on either side agree to it, so that nobody is surprised.

Authority is never unlimited. Law limits it, since no order may require an unlawful act. Ethics limits it, since an order to deceive customers should be refused, as Chapter 5 explained. Resources limit it, since a manager may order a task only for which the money, people and time exist. The authority of the position limits it, since a counter supervisor has no right to change the price of bread. And acceptance limits it, as Barnard showed. A new manager who

Authority and responsibility must be matched.

Authority has four sources, and each sets a limit.

Authority flows down the chain of command, one level at a time.

Authority has natural limits that every manager should know.

Authority is more effective when its limits are stated.

Unity of command protects authority from confusion.

Authority differs from power, a topic that Chapter 12 treats in full.

The family and the chart show the same three ideas in different words.

A recap that fixes the three ideas.

The reader tests the idea.

A manager's right may be to command, to advise, or to rule on one subject.

Line authority is the right to direct the work of subordinates.

Line authority. The right to direct subordinates and to give them orders.

forgets these limits will soon find that an order on paper is not an order in practice.

For this reason, a good organisation states the limits of each position in writing: what the holder may decide alone, what needs approval, and what is not the holder's business. A stated limit does two things. It protects the holder, who knows how far to go. And it protects the organisation, which knows that no single person can commit it beyond a known sum. The bakery case later in this chapter shows such a limit in use.

Authority works best when each person knows one manager. Chapter 3 gave this idea the name unity of command. Suppose two managers each give the same employee an order, and the orders conflict. The employee must disobey one of them, and both managers lose authority. In the opening scene, imagine that the father and the mother had both given the son different orders about the dinner: he could not obey both. Organisations avoid this by drawing the chain of command so that each box has one line above it. Chapter 9 showed the matrix, which breaks the rule on purpose, and explained the price: confusion unless the two managers agree in advance.

Authority is not the same as power. Authority is a right that comes from the position and is accepted by those below. Power, taught in Chapter 12, means the ability to make others act, and it can come from sources other than position, such as knowledge or personal respect. A junior engineer with rare skill may have power without authority. A new manager whom nobody respects may have authority without much power. Chester Barnard (1938) argued that authority in practice depends on whether subordinates accept an order.² An order that most people refuse to obey has little authority behind it, however high the position.

Return to the dinner. The mother held authority over the kitchen and, as host, was accountable for the evening. When she told her son to cook, she should have handed over a share of authority (the key and the money) and assigned responsibility (the meal), while remaining accountable to her guests. That is the whole pattern of this chapter in a single scene.

Before moving on. Authority is the right to give orders and use resources. Responsibility is the duty to do the assigned task. Accountability is answering for the result. Authority should match responsibility, and accountability stays with the one who delegates. Power is a separate idea, taught in Chapter 12. *You should now be able to say:* "Authority is the right to direct, responsibility is the duty to do, and accountability is answering for the result."

Pause and check. (a) A clerk is told to approve refunds but has no right to use the refund account. Which rule of this section is broken? (b) Which of the three ideas looks back at a result? The answers are at the back of the book.

LINE, STAFF AND FUNCTIONAL AUTHORITY

In Chapter 9 we met line positions, which do the core work, and staff positions, which advise. The two kinds of position hold different kinds of authority. A third kind arises when a specialist has a right that reaches across departments. The three are taught together because they are easily confused.

Line authority is the right to direct subordinates and to give them orders. It runs down the chain of command, from owner to manager to employee. A line manager may assign tasks, approve leave and judge performance. Line authority

² Barnard's book of 1938 is *The Functions of the Executive*.

is complete within its area, which means that the person who holds it also decides.

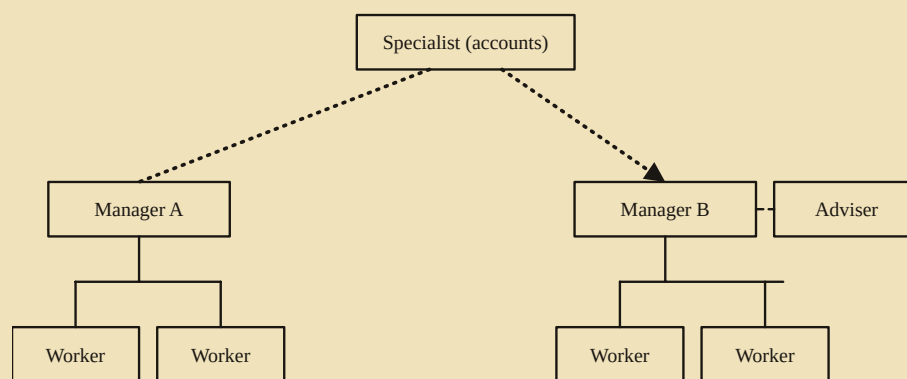
Staff authority is the right to advise and support the line, but not to command it. A legal adviser may tell a manager that a contract is unsafe. The manager may take the advice or refuse it. Staff authority usually takes one of three forms. Advisory staff recommend, as a lawyer does. Service staff supply a service, as a transport pool does. Control staff check that rules are followed, as an inspector does. In every form, the staff member does not direct line employees.

Functional authority is the right of a specialist over a process outside the specialist's own line. The specialist cannot direct a department's work in general, but may require that one process be done in one way. An accounts officer may require receipts from every department before money is repaid. A safety officer may stop a machine that is unsafe. The right is narrow: it covers one subject, in every department. Chapter 9 described the older idea of several specialist bosses; functional authority is a limited, safer version of it.

The table and the figure below compare the three kinds. Read each row as a question with three answers.

LINE, STAFF AND FUNCTIONAL AUTHORITY COMPARED

QUESTION	LINE	STAFF	FUNCTIONAL
What is the right?	to command	to advise	to rule on one subject
Over whom?	own subordinates	nobody, only advises	people in other departments
Who decides?	the line manager	the line manager	the specialist, on that subject
Example	manager assigns tasks	lawyer advises	accounts requires receipts



Solid: orders (line). Dashed: advice (staff).
Dotted: rule on the specialist subject only (functional).

FIG. 32. Three kinds of authority: orders (line), advice (staff) and a rule on one subject across departments (functional).

Friction is common. Line managers may think that advisers have never faced real pressure. Advisers may think that line managers ignore good ideas. Four habits help. State in writing what each position may and may not do. Require line managers to listen to advice before deciding. Expect advisers to explain and win support, not to order. Allow the line manager to decide, and to answer for the decision. The same care applies to functional authority: it should be kept

Staff authority is the right to advise.

Staff authority. The right to advise and support the line, but not to command it.

Functional authority is a right to rule on one subject across other departments.

Functional authority. The right of a specialist over a process outside the specialist's own line.

The three kinds of authority are compared in the table and the figure.

Line and staff often quarrel, and clear roles reduce the quarrel.

A hotel shows all three kinds of authority at work in one place.

The test of any right is to ask three questions.

Both staff and functional roles can be abused, and managers should watch for it.

The need for staff and functional roles grows with size and with complexity.

A recap of the three kinds of authority.

A manager who does everything personally limits the firm to what one person can do.

GO SLOWLY HERE.

Delegation means passing authority and duty to a subordinate.

Delegation. Passing authority and duty for a task down to a subordinate.

Delegation follows four steps, taken in order.

narrow, because every extra right reduces the unity of command that Chapter 3 recommended.

Consider a hotel with a general manager, a front-office manager, a kitchen manager and a housekeeping manager. Each of the three managers has line authority over their own team: the front-office manager directs the receptionists, the kitchen manager directs the cooks, and the housekeeping manager directs the cleaners. The hotel also employs a lawyer, who advises on guest contracts; that is staff authority. And the hotel's safety officer has the right to close any kitchen burner that fails an inspection, whichever manager protests; that is functional authority, limited to safety. If the safety officer began to tell the kitchen manager how to plan the menu, the officer would be passing the limit of functional authority.

A simple test sorts any case. Ask first whether the person may direct the daily work of others. If so, the authority is line. Ask next whether the person may only advise. If so, it is staff. Ask finally whether the person may rule on one subject across several departments. If so, it is functional. A person may hold more than one kind: the safety officer has staff authority over most matters, for advice, and functional authority over safety rules.

Misuse is possible in both directions. A staff adviser who is close to the owner may begin to give orders by hinting that the owner agrees; this is staff power grown beyond staff authority. A line manager may ignore a rule from a functional specialist by arguing that the work is urgent. Both habits damage trust. A clear written statement of each right, and a way for the owner to settle disputes, prevents most of the trouble.

Why do organisations add staff and functional roles at all? The reason is that one line manager cannot know everything. A bakery of four employees needs no lawyer; a bakery chain with forty shops does. As the firm grows, the rules of tax, safety, hiring and accounting multiply, and no line manager can master them all. A specialist who knows one subject well can serve many managers at once, which is cheaper than training every manager. The cost is the friction already described, and the risk that the specialist's rule may be applied without regard for the needs of the work. The gain and the cost must be weighed, as with every design choice in this book.

Before moving on. Line authority commands. Staff authority advises. Functional authority rules on one subject in other departments. Friction is reduced by clear roles, and the line manager keeps the final decision.

DELEGATION AND ITS OBSTACLES

Return to the mother. If she cooks every dinner herself, the family can enjoy only what one pair of hands can make. If she teaches her son to cook, the family gains a second cook, and she gains time. A manager faces the same choice. A manager who does everything personally limits the organisation to what one person can do. That choice leads to delegation.

Delegation means passing authority and duty for a task down to a subordinate. It is not the same as dumping work. When a manager gives a task without the means to do it, as the mother did first, that is not delegation. Delegation has a plan: it chooses the task, the person, the power and the way to check progress. The figure above shows the direction of travel.

Four steps make a good delegation. First, assign the duty: state the task and the result wanted, in plain words. Second, grant the authority: give the right to

act, and the resources needed, within limits that are stated. Third, create the obligation: make clear that the subordinate must answer for the result to the manager. Fourth, follow up: agree when and how progress will be checked, and then check, without taking the task back. Each step answers a question: what, with what power, answerable for what, and checked how.

Delegation has three kinds of gain. The manager gains time for planning and for work only a manager can do. The subordinate gains skill, trust and a reason to care about the result. The firm gains speed, since decisions are made close to the work, and gains managers for the future, because people learn by deciding. These gains explain why successful managers delegate more than they like to.

If delegation brings so many gains, why do managers hold back? The obstacles fall into three groups. Some lie with the manager. A manager may fear losing control, may believe that "I can do it faster myself", may fear that a subordinate will do the job better, or may not know how to explain the task. Some lie with the subordinate. A subordinate may fear blame for mistakes, may lack confidence, may lack information or tools, or may see no extra reward for extra duty. Some lie with the organisation. Duties may be poorly defined, a culture of blame may punish every error, and there may be no simple way to check results.

Each obstacle has a cure. A manager who fears losing control can set limits and agree dates for review, so that control is kept without interference. A manager who thinks it faster to do the work alone should count the cost of repeating that choice every week. A subordinate who fears blame is helped by a rule that honest mistakes are treated as lessons. A subordinate who lacks tools is given them. And the organisation can define duties in writing and reward those who take on more. Trust is built in small steps: start with a small task, check it, and then give a larger one.

Delegation is not an all-or-nothing act, and managers can choose a degree. In the cautious degree, the subordinate studies the matter and recommends, and the manager decides. In a middle degree, the subordinate decides and acts but tells the manager at once. In a free degree, the subordinate decides and acts, and reports only at the regular review. A manager who knows a subordinate well may move from the first degree toward the third as trust grows. A manager who is new to the task, or whose subordinate is untried, should begin with the first degree.

Some tasks belong to the manager and should not be passed down. They include setting the goals of the whole unit, because no one else sees the whole picture; deciding disputes between subordinates; matters of discipline and dismissal; matters that are confidential; and crises that threaten the whole firm. A manager who delegates these tasks has not delegated but abandoned them. The test is to ask whether the task needs the authority of the manager's own position, and not merely the skill of whoever does it.

Take the hotel again. The front-office manager wishes to delegate the handling of guest complaints to a supervisor. *Assign*: the manager states the duty, which is to settle complaints about rooms on the same day, and gives the result wanted, which is that no guest leaves unhappy. *Grant*: the manager gives authority to offer a free night's breakfast, or a room change, up to a stated value, without asking. *Create the obligation*: the supervisor agrees to answer to the manager for the number and the outcome of complaints each week. *Follow up*: the manager reads a short weekly note, and meets the supervisor once a month. The delegation fits the four steps, and the manager has not taken the task back.

Delegation brings gains to the manager, the subordinate and the firm.

Obstacles to delegation come from managers, subordinates and the organisation.

Each obstacle has a matching cure.

Delegation may be given in different degrees, from cautious to free.

Some tasks should not be delegated.

The hotel supervisor example applies the four steps in order.

Control and trust must be balanced in delegation.

Four rules guide delegation.

A short recap of delegation.

A manager can direct only so many people before quality falls.

Span of management. The number of subordinates who report directly to one manager.

The span fixes the number of levels in the organisation.

Tall structure. A structure with many levels and small spans.

Flat structure. A structure with few levels and large spans.

A tall structure has many levels; a flat structure has few.

Every delegation sets control against trust. If the manager keeps too much control, by checking every step, the subordinate has no real authority, and the mother's second failure is repeated. If the manager gives too much trust, with no review, errors may grow unseen. The balance is reached by agreeing in advance on the result wanted, the limits of authority and the dates of review. Within those limits, the subordinate chooses the method. Chapter 13 will show that this is control by results, in which the manager watches the outcome and not every movement.

Four rules summarise good practice. Authority should match responsibility, as the first section showed. Each subordinate should have one manager, which protects unity of command. Delegation should pass down the chain, not skip a level. And accountability cannot be delegated: the manager who passes a task down still answers for its result. The last rule is the one most often forgotten, and the final question of this chapter returns to it.

Before moving on. Delegation means passing authority and duty for a task down to a subordinate. Its four steps are assign, grant, create the obligation and follow up. Obstacles come from managers, subordinates and the organisation, and each has a cure. The manager stays accountable. *You should now be able to say:* "Delegation passes authority and duty down, but accountability stays with the manager."

SPAN OF MANAGEMENT

A manager with one subordinate has time for every detail. A manager with fifty has time for none. Between those extremes lies a question that every organisation must answer: how many people should report to one manager? The number is called the **span of management**. Span of management means the number of subordinates who report directly to one manager. A manager with three direct reports has a span of three.

The span decides the shape of the structure. If the span is small, many managers are needed, and the structure has many levels. If the span is large, fewer managers are needed, and the structure has few levels. The next paragraph names the two shapes, and a worked calculation then shows the difference in numbers.

A **tall structure** has many levels and small spans. A **flat structure** has few levels and large spans. The figure below shows both, using the same eight operatives, so that the effect of span is the only change.

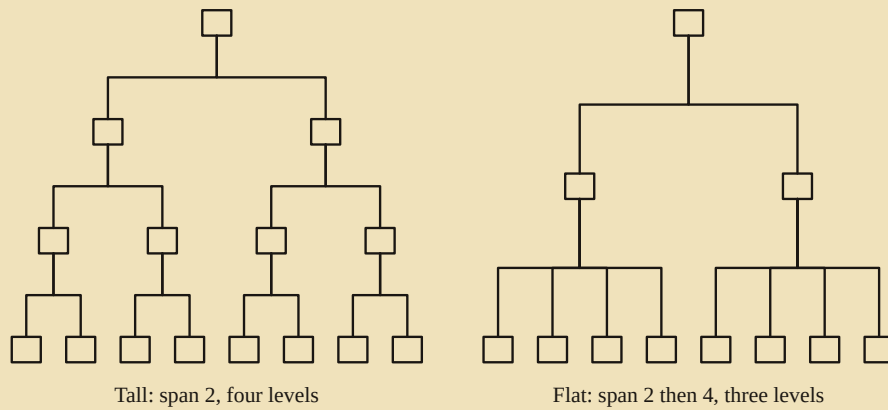


FIG. 33. The same eight operatives in a tall structure (span 2) and a flat structure (span 4 below the top).

Suppose a workshop has 64 operatives. *Given:* 64 operatives. *Find:* the number of managers needed if each manager has a span of 8, of 4, and of 2. *Method:* divide the number of people at one level by the span to find the managers needed to direct them, and repeat upward until one person is left. *Steps for span 8:* $64 \div 8 = 8$ managers; $8 \div 8 = 1$ manager. Total managers $8 + 1 = 9$, in 2 levels. *Steps for span 4:* $64 \div 4 = 16$; $16 \div 4 = 4$; $4 \div 4 = 1$. Total $16 + 4 + 1 = 21$, in 3 levels. *Steps for span 2:* $64 \div 2 = 32$; $32 \div 2 = 16$; $16 \div 2 = 8$; $8 \div 2 = 4$; $4 \div 2 = 2$; $2 \div 2 = 1$. Total $32 + 16 + 8 + 4 + 2 + 1 = 63$, in 6 levels. *Answer:* 9, 21 and 63 managers. *Check:* each answer is the sum of a chain that ends at one person.

The numbers show a trade-off. Span 2 needs 63 managers, far more than the 9 needed at span 8. More managers cost more, and orders must pass through six levels, so decisions are slow and messages may change on the way. Span 8 is cheap and quick, but each manager has less time for each person. Neither extreme is right. The question is which span suits the work.

Six factors help to decide the right span. A manager can direct more people when the work is routine, because less explanation is needed; and fewer people when the work is complex. A larger span is possible when subordinates are skilled and trained, because they need less guidance. A larger span is possible when duties are clearly defined, and when plans and standards are in writing. A manager whose subordinates work close together can direct more than one whose subordinates are spread across the city. A manager with helpers, such as an assistant, can direct more. And a manager who must spend much time on other duties, such as planning, can direct fewer.

Writers have offered numbers, but none fits all cases. Some suggested that about five or six subordinates suit a top manager, and that larger spans suit supervisors of routine work, who may direct a dozen or more. These figures are guides, not laws. The factors above matter more than any number.

The calculation may be turned round. *Given:* a manager has a span of 5, and each of those five subordinates also has a span of 5. *Find:* how many operatives are reached at the second level. *Steps:* $5 \times 5 = 25$. A third level reaches $5 \times 5 \times 5 = 125$. *Answer:* two levels below the top reach 25 people, and three levels reach 125. *Check:* $125 \div 5 = 25$ and $25 \div 5 = 5$, which returns to the manager's span. The result shows how quickly a structure widens: each new level multiplies the number of people by the span.

Span affects the speed and accuracy of messages. In a tall structure, a message from the top passes through many managers, and each manager may

A calculation shows how span changes the number of managers.

The numbers show a trade-off between cost and control.

Span depends on six situational factors.

No single number is correct, but beginners often prefer a range.

A second calculation shows how span links to the number of people reached.

Span also affects how messages travel.

shorten, soften or alter it. In a flat structure, the message passes through few hands and arrives nearly as sent, but each manager has less time to explain it. The right balance depends on how complex the message is. A simple order, such as the day's production target, travels well through many levels. A complex change, such as a new quality method, is better explained face to face by few managers.

A clinic shows how the span factors work in practice.

Take a clinic with a manager and twelve employees: two doctors, three nurses, a pharmacist, two receptionists, a laboratory technician, a cleaner and a part-time accountant. If the manager directs all twelve, the span is twelve, which is large for work that varies from patient to patient. If the manager directs only the doctors, the head nurse and the head receptionist, the span is four, and the head nurse and head receptionist direct the rest. The second arrangement suits skilled but varied work. The first might suit a clinic in which the tasks were routine and the employees experienced.

Tall and flat structures each carry their own risks.

Each shape carries risks. A tall structure slows decisions, raises cost, and lets messages change as they pass many levels. It also gives close supervision, which suits new or unskilled employees. A flat structure speeds decisions and gives subordinates room to act, but it demands skilled employees and strong delegation, since the manager has little time to watch. Many firms have made their structures flatter in recent decades, partly because communication is faster, but they have not removed the trade-off.

The span also shapes the manager's style and the subordinate's freedom.

The span also shapes how people work. A manager with a span of two can watch both subordinates closely, and subordinates may feel supervised. A manager with a span of twelve cannot watch everyone, and subordinates must decide more for themselves. This explains why a firm that wants initiative from its employees often widens spans, and why a firm with new or untrained employees often narrows them. The span is therefore not only a count of people but also a choice about how much freedom to give.

A recap of span.

Before moving on. Span of management means the number of subordinates who report directly to one manager. Small spans make tall structures, and large spans make flat ones. The right span depends on the work, the people, the clarity of duties, the closeness of the group, the help available and the manager's other duties. *You should now be able to say:* "A smaller span needs more managers and more levels; a larger span needs fewer of both."

The reader tests the idea.

Pause and check. (a) A firm has 36 operatives and a span of 6. How many managers does it need in all? (b) Name two factors that allow a larger span. The answers are at the back of the book.

CENTRALISATION AND DECENTRALISATION

Every decision has a level at which it is made, and the level can be moved.

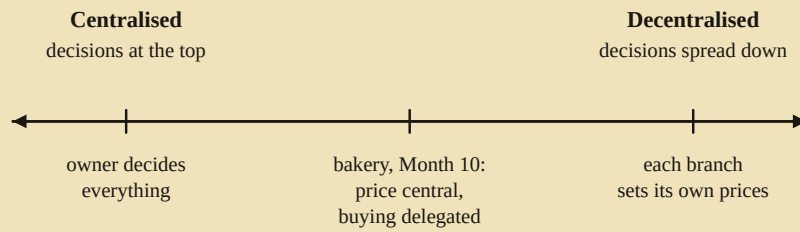
Chapter 8 taught how to make a decision. This section asks a different question: who should make it? In every organisation some decisions are taken by the owner, and others by people lower down. How far decisions are gathered at the top is a design choice, and it has a name.

Centralisation. Keeping decision authority at the top of the organisation.

Centralisation means keeping decision authority at the top of the organisation. **Decentralisation** means spreading decision authority down to lower levels. They are not two boxes but the ends of a scale, like the two design models in Chapter 9. An organisation is more or less centralised, and the degree may differ from one kind of decision to another. The figure below shows the scale.

Decentralisation. Spreading decision authority down to lower levels.

Centralisation holds authority at the top; decentralisation spreads it down.



Illustrative positions only; the choice is made decision by decision.

FIG. 34. *The centralisation scale; positions are illustrative.*

Take care to separate decentralisation from delegation. Delegation is what one manager does with one task and one subordinate, and the manager can take the task back. Decentralisation is a policy for the whole organisation: it spreads decision authority across many positions as a matter of design, and it is made permanent in the structure. Decentralisation is delegation carried widely, but delegation can happen in a very centralised firm.

Centralisation has strong arguments. Decisions are consistent across the firm. Top managers see the whole picture. Costs fall when buying or hiring is done once for all. Control is easier, and crises can be met with one voice. It also has costs. Top managers become overloaded. Decisions wait for those far from the facts. Lower managers have no chance to learn by deciding, and their interest fades.

Decentralisation has opposite strengths. Decisions are quick, because they are taken where the facts are known. Lower managers grow by deciding, which prepares them for senior posts. Top managers are free for the long view. Its risks are inconsistent decisions, duplicated effort, and a loss of control if lower managers pursue their own aims. The history at the end of this chapter shows an army that chose carefully between the two.

Seven factors guide the choice. Size: large firms tend to decentralise, because the top cannot decide everything. Importance and cost of a decision: costly decisions tend to stay at the top. Need for uniformity: a brand that must look the same everywhere is held centrally. Skill of lower managers: skilled managers can be trusted with more. Speed of communication: fast systems let the top keep more. The environment: a changing or varied market favours local decisions. And history: a firm that grew from one owner often keeps the habit of deciding at the top.

How can we tell how centralised an organisation is? Three questions help. How many decisions are taken at the top, compared with lower down? How important are the decisions that lower managers may take, measured by the money or the risk involved? And how much checking must lower managers go through before they act: do they act freely, report afterward, or ask first? A firm in which lower managers may spend only small sums, and must ask first, is more centralised than one in which they may spend large sums and report at the end of the month.

Decentralisation is wider than delegation and differs from it.

Centralisation brings uniformity and control, but also delay and distance from the facts.

Decentralisation brings speed and learning, but also risk of inconsistency.

Seven factors help decide the degree of centralisation.

Three questions help measure how centralised a decision is.

CENTRALISATION AND DECENTRALISATION COMPARED

POINT	CENTRALISED	DECENTRALISED
Speed of decisions	slower, they travel up	faster, taken near the work
Consistency	high	may differ between units
Use of local knowledge	low	high
Load on top managers	heavy	lighter
Training of managers	few chances to decide	many chances to decide
Risk	delay and distance	loss of control

The table gives the points side by side.

A chain of shops shows why the choice differs by decision.

Departmentalisation affects the degree of centralisation.

Two firms with the same size may choose different degrees of centralisation.

Moving the level of decisions should be done in steps.

A recap of centralisation.

The table above gives the points side by side. Read each row as one aspect of a decision and the two ways it can be handled. No row favours one design in every case, which is why the choice must be made with the situation in view.

Consider a chain of four bakeries. The chain wants every shop to sell loaves at the same price, so price is set at the head office; this is centralisation, for the sake of uniformity. Each shop manager, however, chooses the number of loaves to bake each morning, because only the manager knows the weather and the local events; this is decentralisation, for the sake of local knowledge. The chain is therefore both centralised and decentralised, according to the decision. This is the usual case, and it explains why the question should always be asked about a particular decision.

The basis of grouping from Chapter 9 also matters. A firm grouped by product or territory usually gives each group its own manager and so tends to be more decentralised. A firm grouped by function tends to be more centralised, because the functions must be joined at the top. A firm may centralise some decisions, such as pricing or accounts, and decentralise others, such as local buying. The question is asked decision by decision.

Size alone does not decide the degree. Consider two firms of equal size. One sells a standard product through shops that must look alike, and so holds most decisions at the centre. The other sells custom work, in which each customer's needs differ, and so gives each team wide freedom. Their sizes are the same, yet their degrees differ, because the need for uniformity and the variety of the work differ. This is another reason to treat every factor listed above as one among several, and not to expect any single factor to settle the matter.

A firm that wishes to change the degree of centralisation should do so in steps. A firm that decentralises all at once may find that lower managers lack the skill or the information, and may then pull everything back to the top, which damages trust. A better path is to begin with decisions that are small and frequent, such as ordering routine supplies, and to widen the limit as lower managers prove their judgement. The bakery case follows this path: it begins with buying up to a stated sum, not with buying without limit.

Before moving on. Centralisation keeps decision authority at the top, and decentralisation spreads it down. Delegation is one manager's act; decentralisation is a policy. Size, cost of the decision, need for uniformity, skill, communication, environment and history guide the degree.



WORKED CASE: THE BAKERY AT MONTH TEN

Case file, illustrative. Month 10. *Situation.* Morning Loaf Bakery has eight employees in three departments, and the owner is the only person who buys ingredients and sets prices. The owner spends the early morning on small orders. The head baker, who knows the stock best, waits for approval each day. The owner also needs advice on records, because receipts are missing for some purchases.

Question. What authority should the bakery give the head baker, what should it do about the records, and how many people should each manager direct?

Reasoning. First, delegation. The owner assigns the head baker the duty of keeping the kitchen supplied. The owner grants authority to buy ingredients up to Tk 5,000 per order. At Month 4 prices, a day's flour is $60 \text{ kg} \times \text{Tk } 69 = \text{Tk } 4,140$, which is within the limit; two days' flour is Tk 8,280, which needs the owner's approval. The head baker answers to the owner for supply, and the owner follows up weekly. The owner stays accountable to customers and lenders for the whole business. Second, the three authorities. The owner engages a part-time bookkeeper at Tk 8,000 a month. The bookkeeper is a staff role: an adviser who does not direct the bakers. The owner also gives the bookkeeper functional authority to require receipts from every employee who spends money. The bookkeeper may not tell the bakers how to bake.

Reasoning, continued. Third, span. The owner directs the head baker, the counter lead and the rider, a span of 3. The head baker directs two bakers and a helper, a span of 3. The counter lead directs two counter employees, a span of 2. The structure is flat, with three levels, and the spans are small enough for close help. Fourth, centralisation. Price setting stays with the owner, because prices must be the same at the counter and in delivery, and because they decide profit. Buying within the limit moves to the head baker, who knows the facts. The bakery is therefore partly decentralised.

Conclusion. The head baker holds delegated authority to buy up to Tk 5,000 per order; the bookkeeper advises and holds functional authority over receipts; spans are 3, 3 and 2; price setting is central and routine buying is delegated.

Check. The head baker's authority matches her duty to keep the kitchen supplied. The owner stays accountable. The spans cover $3 + 3 + 2 = 8$ employees, which matches the employees counted in Chapter 9, and the part-time bookkeeper is outside the chain of command. Each employee still has one manager.



SUMMARY

Authority is the right to give orders and use resources; responsibility is the duty to do an assigned task; accountability is answering for the result, and it stays with the one who delegates. Authority should match responsibility. Line authority commands, staff authority advises, and functional authority rules on one subject across departments. Delegation passes authority and duty down in four steps, and its obstacles lie with managers, subordinates and the organisation. Span of management sets the number of levels, and the right span depends on the work, the people and the situation. Centralisation and decentralisation are the ends of a scale, and the right degree is chosen decision by decision.

The case applies the chapter's ideas to one growing business.

The question asks what authority to give, to whom, and where decisions should rest.

The reasoning applies delegation, the three authorities and span.

The reasoning continues with span and centralisation.

The conclusion and the check complete the case.

The check counts the people and tests each rule.

A summary of the chapter.

Responsibility cannot be passed away completely, and the reason matters.

Mistake: authority and power are the same.

Mistake: delegating a task also delegates accountability.

Mistake: staff may give orders to the line.

Mistake: a larger span is always better.

Mistake: decentralisation means no control.

Mistake: delegation and decentralisation are the same.

A famous army found a way to centralise the aim and decentralise the means.

The method gave subordinates freedom within a clear aim.

The method had a risk, and the army took steps to control it.

THINK IT THROUGH

A restaurant owner tells the head cook to buy fish each morning, with a limit of Tk 10,000. One day the cook buys fish that is not fresh, and several customers fall ill. The owner says, "I delegated it, so it is the cook's problem." Is the owner right? Consider who customers will blame, what the owner can still do, and what the owner must have done before the day. Why can responsibility never be fully delegated, and what part of it can be?

WHERE PEOPLE GO WRONG

"Authority and power are the same." Authority is a right that comes from a position. Power is the ability to make others act, and it may come from knowledge or respect. A person can have either without the other.

"If I delegate, I am no longer answerable." The subordinate becomes responsible for the task, but the manager remains accountable for the result.

"Advisers may command." Staff authority is the right to advise only. Functional authority gives a specialist a rule on one subject, not general command.

"A wider span is always more efficient." A wide span saves managers but can leave subordinates without help. The right span depends on the factors above.

"Decentralisation means giving up control." Decentralised firms usually keep control of results, standards and key decisions, and give freedom over the means.

"Delegation and decentralisation are the same thing." Delegation is one manager's act with one subordinate; decentralisation is a lasting policy of the whole structure.

IN HISTORY: MISSION ORDERS IN THE PRUSSIAN ARMY

In the nineteenth century the Prussian army faced a problem that every large organisation knows. Armies had grown too large, and battles too fast, for one commander to direct each unit. Helmuth von Moltke the Elder, chief of the Prussian General Staff from 1857 to 1888, taught his officers a different method.³ A commander stated the aim of a task and its limits, and left the choice of means to the officer on the spot.

The method kept the aim central and made the means local. The officer who knew the ground decided how to reach the aim, and was expected to act without waiting for orders when conditions changed. It required officers who were well trained, trusted and answerable for results. Prussian forces won decisive wars in 1866 and in 1870 to 1871, and later armies copied the idea of mission orders.

The method carried a risk, which the army knew. Officers who act on their own may pull in different directions. The army met the risk in three ways. It trained its officers in the same ideas, so that independent decisions were likely to agree. It required every order to state the aim, so that an officer who acted differently still served the aim. And it judged officers by the result against the aim, not by the exact steps they took. In the language of this chapter, the aim was centralised, the means were decentralised, and training and a shared aim served as the means of coordination from Chapter 9.

³ Moltke (1800 to 1891) led the Prussian General Staff, the body that planned the army's operations.

The story shows several of this chapter's ideas at once. The commander delegated authority and duty but stayed accountable for the campaign. Authority matched responsibility. Training and trust removed the obstacles to delegation. And decisions were decentralised where the facts were known, while the aim stayed central.

The story illustrates delegation, accountability and decentralisation together.

SOURCES

Fayol, Henri (1916), *Administration industrielle et générale*, which stated that authority and responsibility belong together and that each person should have one manager. It is the origin of the pair as management teaching.

Fayol, 1916, first set out authority and responsibility as a pair.

Barnard, Chester I. (1938), *The Functions of the Executive*, which argued that authority rests on the subordinate's acceptance of the order. It changed how writers think about the source of authority.

Barnard, 1938, treated authority as dependent on acceptance.

Weber, Max, for authority as a right attached to a position, in the work cited in Chapter 9. For the Prussian method, histories of the General Staff and of Moltke's writings.

Weber and the Prussian staff.

REVIEW QUESTIONS

1. Define authority, responsibility and accountability, and give one example of each from a shop or a school.

Use the three ideas in order.

2. A new supervisor is told to keep the stockroom tidy but cannot move any shelves or order boxes. Say which rule of this chapter is broken and how to correct it.

Think of a clerk and a refund account.

3. Compare line, staff and functional authority in a table, with one example each.

Compare in a table.

4. Explain the four steps of delegation, using a manager who wants an assistant to prepare a weekly sales report.

Count the four steps.

5. List three obstacles to delegation and a cure for each, and say whether each obstacle lies with the manager, the subordinate or the organisation.

Group the obstacles by source.

6. A factory has 81 operatives. Find the number of managers needed if each manager has a span of 9, and again if the span is 3. State which structure is tall and which is flat, and name one risk of each.

Use the method from this chapter.

7. Name four factors that allow a larger span, and apply them to a team of experienced accountants doing routine work.

Name factors, then apply them.

8. Distinguish delegation from decentralisation, and say why a very centralised firm can still use delegation.

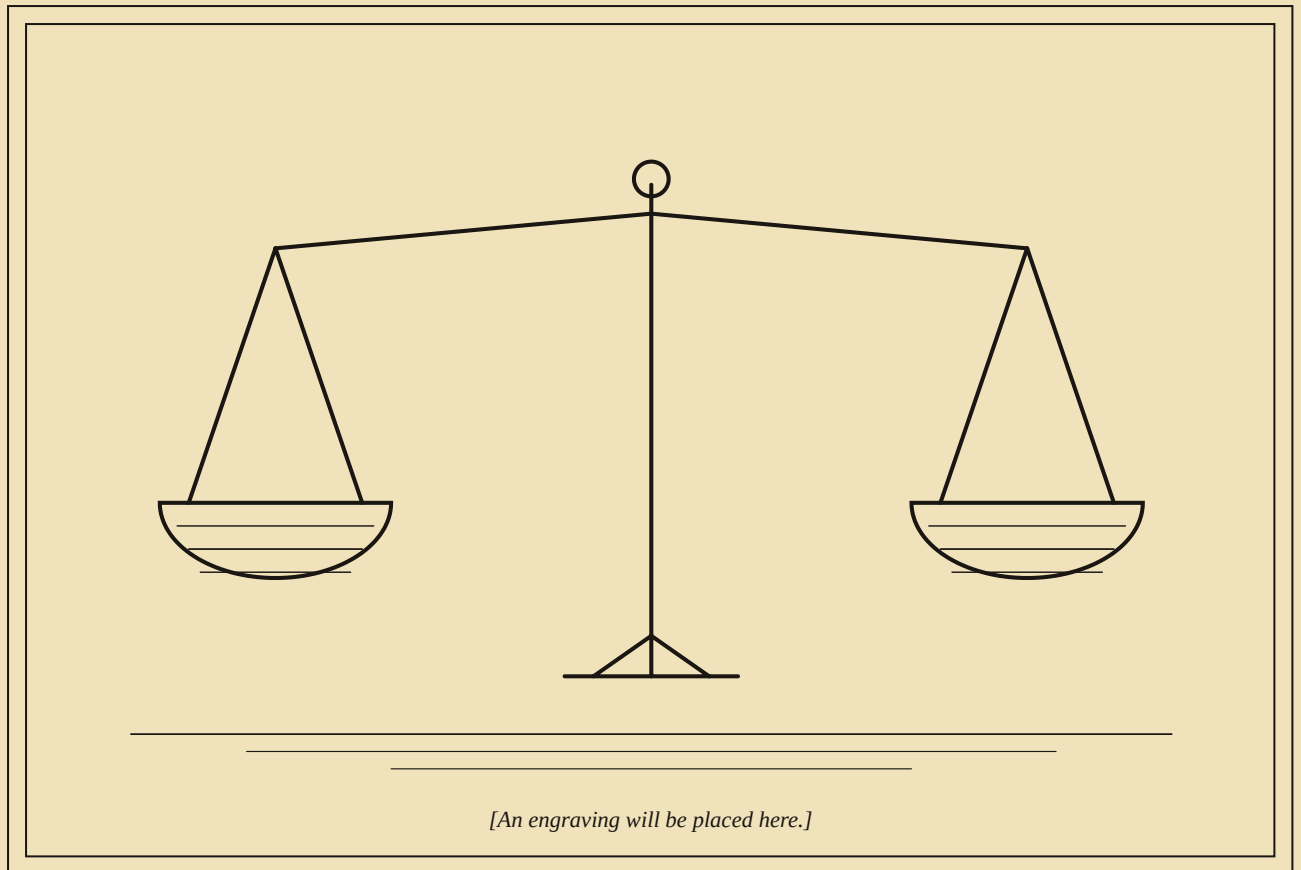
Keep the ideas apart.

9. A chain of five clothing shops sets its own prices in each shop. Give two reasons for and two against this, and say which decisions you would keep at the head office.

Use the factors.

10. A hotel manager gives the front-desk supervisor authority to give guests a room upgrade, worth up to Tk 3,000, without asking. Identify the duty, the authority, the accountability, and one way to follow up.

Think of a hotel.



CHAPTER XI

People at Work

*Why one person works with care, another with distance,
and what a manager can do about it.*



*Chapter 9 showed how work is arranged, and Chapter 10 how
authority is passed down. This chapter asks what moves people to
use that authority and their own skill well. It studies what
managers assume about workers, what starts and keeps effort
going, how needs shape behaviour, how groups form and act, and
where new ideas come from.*

Two people left a busy bakery, and the owner wants to know why.

A manager who guesses the cause of behaviour will often act on the wrong cause.

Tools for the work help only if people use them willingly and well.

This chapter studies what drives effort, in individuals and in groups.

WHY DID THEY LEAVE?

Case file, illustrative. Morning Loaf Bakery, Month 11. In the last six months, 2 of the 8 staff have left, a loss of 25%. During the month the owner starts a team bonus of Tk 1,000 for waste of 5% or less, stops checking every tray, and accepts a helper's idea for a milk bread: 40 loaves a day at Tk 80.

Suppose you were the owner. Two people you trained have gone. The shop is busy, the oven runs all day, and customers return. Your first thought may be the wage. You could raise it for everyone. But you would be guessing, and a wrong guess costs money and changes nothing. Perhaps the two wanted something the wage cannot buy: to be trusted, to learn a new skill, to belong to a team that valued them. Perhaps they did not want anything different at all, and left for a shorter journey to work.

Every manager meets this puzzle sooner or later. Chapters 6 to 10 gave you tools for the work itself: plans, objectives, choices, structure and authority. Those tools are useful only if people use them willingly and well. The word *people* then raises a harder question than any chart can answer. What makes a person put effort into a task, keep it up, and aim it at the goals of the organisation?

The chapter answers in six steps. Section 11.1 looks at what managers assume about the people they lead. Section 11.2 explains what starts effort. Section 11.3 describes the best-known theories of human need. Section 11.4 adds three theories about how people choose where to put their effort. Section 11.5 turns from the individual to the group, and section 11.6 shows where new ideas come from and how they reach use.

YOU WILL NEED

From Chapter 9, section 9.3: the difference between formal and informal organisation. From Chapter 7: objectives and the MBO cycle. From Chapter 10, section 10.3: delegation. Section 3.4 of Chapter 3 introduced the human relations approach, which this chapter builds on.



11.1 HUMAN FACTORS AND ASSUMPTIONS ABOUT PEOPLE

A worker is a whole person, not only a pair of hands.

Human factors. The personal qualities, needs and beliefs that shape how a person behaves at work.

People differ in ability, in what they want, and in how they see the same event.

When a bakery hires a baker, it hires more than the baker's hands. It also takes in the person's skill, health, hopes, worries, family duties and opinions of the owner. These are the **human factors** of work: the personal qualities, needs and beliefs that shape how a person behaves at work. Two bakers with equal skill and equal pay can behave very differently, because their human factors differ. One may value a steady wage most. The other may value praise and the chance to learn.

Three facts about human factors matter to a manager. First, people differ. The same instruction, wage or word of thanks does not reach two people in the same way. Second, a person is also a social being, who cares about friends, status and respect at work. Third, behaviour usually has more than one cause. When a worker arrives late, the cause may be distance, health, boredom or a quarrel, and the manager must find which before acting.

Consider one event. The owner moves a baker from the early shift to the late shift. One baker sees a sign of trust, because the late shift pays a little more and the closer holds the keys. Another baker sees a punishment, because the family dinner is now lost. The event is the same, and the meaning is different. To predict how a person will respond, a manager must ask how the person sees the event, and not only how the manager sees it.

Ideas about workers have changed over time. In the early years of scientific management, described in Chapter 3, the worker was seen mainly as a source of labour who answered to pay and clear methods. Later research, including the studies described at the end of this chapter, showed that workers also respond to attention, friendship and a sense of being valued. The human relations approach grew from these findings. Its lesson does not cancel the earlier one. Pay and method still matter, and people matter as well. A modern manager uses both lessons at once.

Behind every managerial action lies a belief about what people are like. A manager who locks the flour store and counts every sack believes something about the staff. A manager who leaves the key with the head baker believes something different. In 1960 the American writer Douglas McGregor, in his book *The Human Side of Enterprise*, gave names to two such sets of belief.

Theory X is a set of assumptions that holds that the average person dislikes work, avoids responsibility when possible, and works hard only when watched, directed or threatened. A manager who holds Theory X sets tight rules, checks work often, and uses penalties and fixed rewards. The manager is not cruel. The manager believes that without such pressure little would be done.

Theory Y is the opposite set of assumptions. It holds that effort at work is as natural as rest or play, that people direct themselves toward goals they accept, and that most people can learn to seek responsibility. It also holds that ideas and imagination are found widely, not only among managers. A manager who holds Theory Y agrees goals with the staff, passes down authority, and judges results rather than watching every movement.

The two theories connect to earlier chapters. A Theory X manager is drawn to the one best method and the close supervision of scientific management in Chapter 3, and to authority held at the top, as in Chapter 10, section 10.5. A Theory Y manager is drawn to the agreed objectives of Chapter 7 and to the delegation of Chapter 10, section 10.3. The chapters on structure and authority describe tools, and this chapter describes the belief that decides how the tools are used.

The two sets of assumptions lead to different habits. The table that follows sets them side by side. Read each row from left to right: the first column names a feature of management, and the other two show what a manager holding each theory tends to believe or do.

TWO SETS OF ASSUMPTIONS ABOUT PEOPLE AT WORK

FEATURE	THEORY X	THEORY Y
View of work	A burden to be avoided	As natural as rest or play
Source of effort	Pressure, reward and punishment	Goals the person accepts
Attitude to duty	Avoided if possible	Accepted and often sought
Use of ideas	Few people have useful ideas	Many people have useful ideas
Typical style	Close checking, tight rules	Agreed goals, delegation

Two people can see the same event and judge it very differently.

Earlier writers saw workers mainly as labour, and later writers saw them as people.

Before you manage people, you hold some beliefs about them, whether or not you notice.

Theory X sees workers as unwilling and so leads to close control.

Theory X. The belief that people dislike work, avoid duty and must be directed and controlled.

Theory Y sees workers as willing and so leads to trust and shared goals.

Theory Y. The belief that people can enjoy work, accept duty and direct themselves toward goals they share.

Theory X links to close control and Theory Y to agreed goals and delegation.

The table sets the two sets of assumptions side by side.

The theories describe beliefs held by managers, not two kinds of worker, and a belief can bring about the behaviour it expects.

The same manager looks different to the worker, the customer and the owner.

Theory Y does not mean freedom for every person at every time.

The bakery owner moves from checking trays to agreed targets.

A short recap of section 11.1.

Here lies the first exception, and many students miss it. McGregor did not say that workers are of two kinds. He said that managers hold one of two sets of belief, and that the belief shapes what the manager does. A further point follows. Beliefs can make themselves true. A manager who treats staff as unwilling, with close checking and no trust, may find that staff do the minimum, which seems to confirm the belief. A manager who trusts may find that staff respond with care.

The two sets of belief also look different from other seats. A worker under Theory X sees a manager who does not trust, and may answer with careful but minimal effort. A worker under Theory Y sees a manager who trusts, and may feel a duty to deserve that trust. A customer sees the effect at the counter: a team told to follow a rule exactly may refuse to correct a small error on the spot, while a team trusted to use judgement may correct it. The owner sees the result in the monthly figures for waste and sales.

Theory Y is not a gift of freedom to everyone at all times. A new helper who has never baked may need clear and close instruction at first, because freedom without skill produces waste. The choice of how much to direct depends on the person and the task, a point Chapter 12 develops. McGregor's lasting lesson is smaller and more useful: examine your assumptions, because your methods follow from them.

Case file, illustrative. Until Month 11 the owner checks every tray before it goes to the counter. In Month 11 the owner replaces this check with agreed targets: waste of 5% or less and on-time orders of 95% or more, set with each member of staff as in Chapter 7. The change moves from a Theory X habit toward a Theory Y habit. The targets remain in place, so control has not been given up; it has changed its form.

Before moving on. Human factors differ from person to person. Theory X assumes people avoid work, and Theory Y assumes they can direct themselves. A manager's assumptions shape the methods the manager chooses.



11.2 MOTIVATION: MEANING AND THE NEEDS-GOAL MODEL

GO SLOWLY HERE. ONE IDEA AT A TIME.

The hunger story has five parts that will return in a model.

Motivation is an inner push, visible only through what people do.

Motivation. The push that starts and keeps a person's effort going toward a goal.

It is eleven in the morning and you are studying. You have eaten nothing since the evening before. At first you notice little. Then your stomach feels empty and your thoughts drift. The feeling grows until reading is hard. You stand up, walk to the canteen, and eat. The feeling fades, your thoughts return to the page, and you read on. Nobody gave you an order. Something inside you started a chain of events and ended it. Keep this small story in mind, because the whole of this section is built on it.

In management this chain has a name. **Motivation** means the push that starts a person's effort and keeps it going toward a goal. In the story, hunger was the push, and the goal was a meal. At work, the push might be the wish to earn a wage, to be praised, or to finish a hard task well. A manager cannot see motivation. A manager sees only its results: a person who works with care, or one who does not.

The push begins with a need. In daily life, a *need* is something you must have in order to live. In management, a **need** is a gap that a person feels between what the person has and what the person wants. Hunger is a need for food. The wish to be noticed is a need for recognition. In both cases the person feels the gap, and the gap is uncomfortable.

The **needs-goal model** shows how a need leads to effort and then to rest. It has five steps. First, a person feels a *need*. Second, the need creates a **drive**, which means an inner tension that pushes the person to act. Third, the person takes *action*. Fourth, the action reaches a *goal*, which means the result that the person wanted. Fifth, the goal brings *relief*, which means the tension fades. The figure shows the five steps in order.

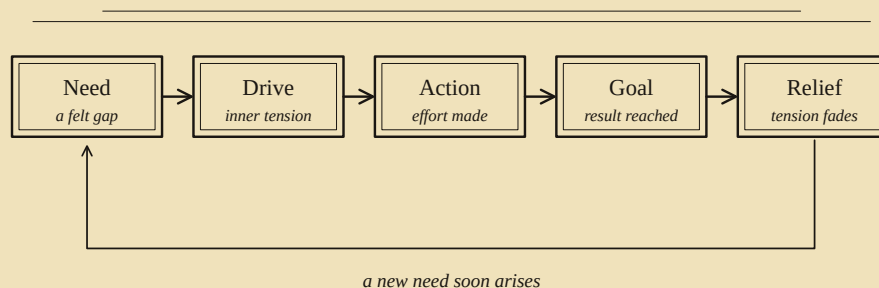


FIG. 35. *The needs-goal model: a need creates a drive, the drive leads to action, and action reaches a goal that brings relief*

The figure also shows an arrow from relief back to the start. It is there because relief does not last. Hunger returns after some hours, and the wish for praise returns after a few weeks. A satisfied need stops pushing, but a new need soon takes its place. This is why motivation is never finished. A manager who solved one need today will meet the next tomorrow.

Now place the model in a bakery. A young baker wishes to be seen as skilled. That is the need. The wish makes the baker restless, which is the drive. The baker takes action: shapes each loaf with extra care for the festival order. The goal is reached when the owner notices and says so. Relief follows, and for some weeks the baker is content. Then a new need arises, perhaps to learn a harder recipe, and the chain starts again.

What if the goal cannot be reached? Suppose the owner never notices the baker's care. The drive remains, with nowhere to go. The person feels **frustration**, which means the discomfort of a blocked goal. Some people respond well: they look for a different route, such as showing the work to a customer. Others respond badly. They may become angry, withdraw from the team, or give up. A manager who sees these signs should ask what goal has been blocked.¹

Pause and check 1. A counter worker wants to be trusted with the cash till. The owner never allows it. Which step of the model is blocked, and what may the worker do next?

A need is felt as a gap, and the gap is uncomfortable.

Need. Word in use. In daily life, something you must have to live. In management, a gap a person feels between what they have and what they want.

The model has five steps: need, drive, action, goal and relief.

Needs-goal model. A chain of five steps: need, drive, action, goal, relief.

Drive. The inner tension that a need creates and that pushes a person to act.

Relief is brief, and so the chain begins again with a new need.

The same five steps explain a baker's wish for recognition.

A blocked goal causes frustration, which can lead to anger or withdrawal.

Frustration. The discomfort a person feels when a goal is blocked.

Test your use of the model on a new case before moving on.

¹ Psychologists use other terms for these responses. The simple division into constructive and defensive responses is enough for this book.

Rewards from the work itself and rewards from outside meet different needs.

Intrinsic reward. A reward that comes from the work itself, such as pride.

Extrinsic reward. A reward that comes from outside the work, such as pay.

A manager cannot see a need, but can learn it by asking, watching and listening.

A met need no longer pushes, so the same reward can fail the second time.

Results depend on motivation, ability and means together.

The same chain looks different from the seat of the employee, owner and customer.

A one-line test of the motivation passage.

A short recap of section 11.2.

Maslow's 1943 paper is the starting point for most later need theories.

Two more ideas complete the picture. A reward that comes from the work itself, such as pride in a well-shaped loaf, is an **intrinsic reward**. A reward that comes from outside the work, such as pay or a bonus, is an **extrinsic reward**. Both can relieve a need, but they relieve different needs. Pay meets the need for food and shelter. Pride meets the need to feel skilled. This difference will matter when you ask, at the end of the chapter, whether money alone can motivate.

Since a need cannot be seen, how does a manager find it? Three habits help. The manager can ask, in a private and unhurried talk, what the person values in the work. The manager can watch, and notice what makes a person brighter or more silent. And the manager can listen to what the person says about colleagues and tasks, because complaints often point to a blocked goal. None of these gives certainty, but together they give a better guess than one made at a desk. A person may also hide a need, such as the wish for status, so the manager should be patient.

Consider what happens to a reward over time. The owner gives a baker a rise, and for some weeks the baker is pleased. Then the baker takes the new wage for granted, and its power to push fades. This is the last step of the model at work, since relief ends the drive. A manager who relies on one reward must keep raising it, and the cost grows. A manager who learns the person's open needs can offer different rewards over time, such as a new task, a new skill or a new title, and gain more for less.

Finally, keep motivation apart from two things that resemble it. Motivation is not the same as ability: a highly motivated person who has never baked still cannot make bread of high quality. Nor is it the same as means: an eager baker cannot work if the oven is broken. The result of work depends on all three, and a manager who blames low motivation for every poor result will often miss the true cause.

The same chain can be seen from three seats. For the employee, it is a felt gap and a wish to close it. For the owner, it is a question of whether the closing of that gap also serves the bakery. For the customer, the result appears as the care in the loaf and the manner at the counter. A manager succeeds when the employee's way of closing the gap, and the bakery's aims, point the same way. The rest of the chapter returns to this idea again and again.

You should now be able to say: "Motivation is the push that starts and keeps effort. It follows five steps: need, drive, action, goal and relief, and then a new need begins."

Before moving on. A need is a felt gap. A drive is the tension it creates. Action aims at a goal, relief follows, and a new need soon arises. A blocked goal causes frustration.



11.3 NEED THEORIES: MASLOW AND OTHERS

The needs-goal model shows how any need works. It does not say which needs people have. In 1943 the American psychologist Abraham Maslow published a paper called "A Theory of Human Motivation" that gave an answer. He was studying human nature in general, not factories, but managers soon found his ideas useful. The theory remains one of the best-known accounts of human need, and many later theories are compared with it.

Maslow's idea is called the **hierarchy of needs**: five levels of need arranged one above another, from the most basic to the most advanced. The levels are physiological, safety, social, esteem and self-actualisation. The figure shows them as a stepped pyramid, with the most basic level at the base. Beside each level the figure gives an example from a bakery.

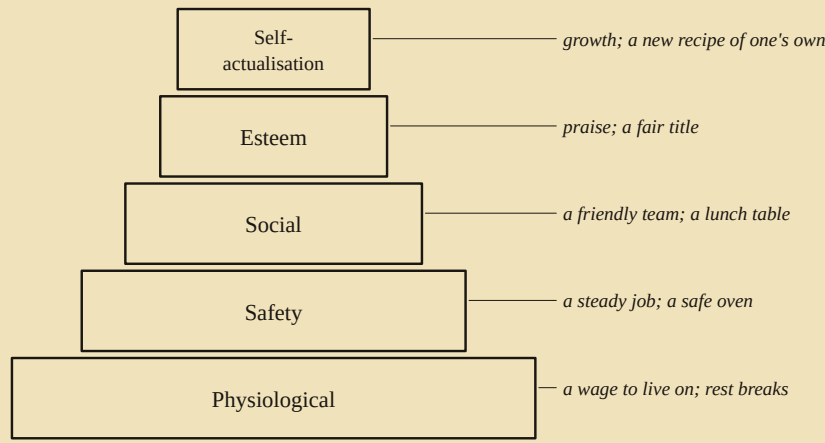


FIG. 36. Maslow's hierarchy of needs, with a bakery example beside each level

The first level holds the *physiological* needs: food, water, rest and shelter. At work these are met by a wage large enough to live on, and by breaks. The second holds the *safety* needs: freedom from danger and from sudden loss. At work they are met by a steady job, safe equipment and clear rules. A baker who fears a burn from a faulty oven, or fears dismissal without notice, is thinking about this level.

The third level holds the *social* needs: to belong, to have friends, and to be accepted. At work these are met by a friendly team and a place at the table. The fourth holds the *esteem* needs: to be respected by others and to respect oneself. They are met by praise, a fair title and real responsibility. The fifth holds the need for **self-actualisation**, which means the wish to use and grow one's abilities as far as one can. A baker who invents a new recipe may be at this level.

Maslow's theory rests on three rules. The first rule is that an unmet need motivates, and a met need does not. A baker with a good wage is not moved by the offer of more wage. The second rule is that lower needs come first: a person who is hungry or afraid has little interest in praise. The third rule is that the levels are not fully separate. A person can be partly satisfied at one level and already be pushed by the next.

The theory has limits, and a good manager knows them. Research has not found that people always climb the levels in the stated order. Some people accept hunger for a cause they value, and an artist may create while poor. Cultures also differ in what they rank first; in some, belonging to family and community can outrank personal respect. Maslow himself described the order as a general tendency, not a fixed law. Treat the hierarchy as a useful map, not as a rule for every person.²

The hierarchy can be turned into a short list of managerial actions. For the physiological level, pay a fair wage and allow rest breaks. For safety, give clear

² Studies of Maslow's theory in workplaces have given mixed results. Verify the current state of the research before relying on a stronger claim.

GO SLOWLY HERE. FIVE LEVELS, ONE AT A TIME.

The hierarchy has five levels arranged from basic to advanced.

Hierarchy of needs. Maslow's five levels of need, from basic survival at the base to personal growth at the top.

The two lowest levels are survival and safety.

The upper levels are belonging, respect and personal growth.

Self-actualisation. The wish to use and grow one's abilities as far as one can.

Unmet needs push, lower needs come first, and the levels overlap.

The order of the levels is a general tendency, not a law for each person.

Each level calls for a different kind of action from the manager.

The lower levels may stay open longer where one wage supports many people.

The practical lesson is to match the reward to the level now open.

Apply the levels to a baker before moving on.

A one-line test of the Maslow passage.

Herzberg separates items that prevent discontent from items that build effort.

Hygiene factor. An item whose absence causes discontent but whose presence does not drive effort.

Motivator. An item that raises effort and satisfaction, such as recognition or interesting work.

A bakery shows the two kinds of factor in action.

Alderfer and McClelland changed the number and kind of needs.

terms of work, keep equipment in good repair and avoid sudden changes. For belonging, build teams and allow time for people to know one another. For esteem, give honest praise, a fair title and real responsibility. For self-actualisation, offer new tasks that stretch a person, and training that opens a path. The list is a guide to what to try, not a promise that each action will work for each person.

Context changes which levels are open. In a country where many families depend on a single wage, a manager should expect the first two levels to matter strongly, and for longer, than in a place where each household has several earners. This does not mean that higher needs are absent. A baker who worries about next month's rent may still care deeply about respect. It means that a manager should not offer praise as a substitute for a fair and steady wage.

The use of the theory for a manager is practical: find the level at which a person is now pushed, and offer a reward that fits it. Consider a call centre where agents take forty calls a day from a fixed script. Pay is fair and the building is safe. Yet several agents leave each month. The theory suggests that the lower levels are met, and that the missing items are belonging and esteem. Rotating agents into small teams, and letting them handle hard calls in their own words, would aim at the levels that are open. A pay rise would aim at a level already met.

Pause and check 2. A baker has a steady job, a good wage and many friends in the team. At which level is the baker most likely to be pushed now, and what could the owner offer?

You should now be able to say: "There are five levels of need, from survival to self-actualisation. An unmet need pushes a person, a met need does not, and a manager should offer the reward that fits the level now open."

Other writers changed or added to Maslow's picture. Three are worth knowing. Frederick Herzberg and his colleagues, in a book of 1959, reported that the things which prevent unhappiness at work are not the same as the things which create real effort. He called pay, working conditions, supervision and company rules **hygiene factors**: items whose absence causes discontent, but whose presence does not by itself drive effort. He called achievement, recognition, interesting work, responsibility and growth **motivators**: items that raise effort and satisfaction. Herzberg's lesson is that a manager must supply both kinds. A good wage removes a complaint, while meaningful work builds the push.

A bakery shows the difference. A broken oven, a late wage or a rude supervisor causes complaints. When these are repaired, the complaints stop, but the staff are not yet eager. Eagerness grows from other sources: a baker who is trusted with a new recipe, a counter worker who is thanked by a regular customer, a helper whose suggestion is used. Herzberg's critics note that his method relied on what people said about past events, and that some people place pay among the motivators. Treat the two lists as a useful guide, not a final law.

Clayton Alderfer, writing in the early 1970s, reduced the five levels to three: needs of existence, of relatedness and of growth. Existence covers the first two of Maslow's levels, relatedness covers belonging and respect from others, and growth covers self-respect and self-actualisation. Alderfer differed from Maslow in one important way. He said that a person blocked at a higher level may return to a lower one. A baker refused promotion may start to care more about pay. David McClelland, in work from the 1950s and 1960s, argued that people learn three strong needs: for *achievement*, for *affiliation* and for *power*. Chapter 12 returns to power. The table compares the four theories.

FOUR NEED THEORIES COMPARED

WRITER	MAIN IDEA	KINDS OF NEED	LESSON FOR A MANAGER
Maslow	Needs rise through five levels	Five levels	Offer the reward that fits the open level
Herzberg	Two sets of factors	Hygiene and motivators	Remove complaints, then build real interest
Alderfer	Three groups; people may move down	Existence, relatedness, growth	Expect a return to lower needs when blocked
McClelland	Three learned needs	Achievement, affiliation, power	Match the job to the person's strongest need

Both writers give a practical guide. If a baker's need for growth is blocked, Alderfer would expect the baker to turn back to pay and to friends until growth is possible again, so a manager who sees a sudden interest in pay should ask whether growth has been blocked. McClelland would expect a person with a strong need for achievement to like tasks with a clear target and quick feedback, such as a daily count of waste. A person with a strong need for affiliation may prefer work in a pair or a small team, and may dislike tasks that set the person apart from colleagues.

The four theories do not agree on the number of needs or on whether they follow an order. They do agree on a central point: a manager who offers the same reward to everyone will motivate few people. Chapter 7 gave one practical use of this point. When targets are agreed one by one with each person, as in management by objectives, the manager has a chance to learn which need each person is trying to meet.³

Now combine the theories in one plan. Suppose the owner wishes to design a plan of pay and recognition for the bakery's 8 staff. The plan should reach every level of Maslow's hierarchy, and should supply both hygiene factors and motivators. The wage and written terms of work serve the first two levels, and they are hygiene factors. A monthly team lunch serves the social level. A monthly title, such as baker of the month, serves esteem. One hour each Friday, in which a baker may try a new recipe, serves self-actualisation. The last three are motivators.

An owner who offered only a higher wage would reach two levels and ignore three. The plan also has costs, in the price of the lunch and in the hour of lost baking time, and the owner must weigh them against the gain. Before adopting it, the owner should ask each person which of the rewards the person values, because the theories describe tendencies, and people differ.

Before moving on. Maslow named five levels of need, and unmet needs push people. Herzberg separated hygiene factors from motivators. Alderfer allowed a return to lower needs, and McClelland named three learned needs.

Alderfer and McClelland suggest how to fit jobs to people.

The four theories agree that people differ, and that a manager must find the open need.

A plan of pay and recognition should reach several levels at once.

A single reward reaches few levels, and each person should be asked what is valued.

A short recap of section 11.3.



³ The cross-reference is to the MBO cycle in Chapter 7, section 7.4.

11.4 PROCESS THEORIES IN BRIEF

Process theories study how people choose, not only what they want.

Process theory. A theory that explains how a person decides where to put effort.

Effort needs three beliefs: I can do it, it will be rewarded, and I want the reward.

A small example shows how one weak belief can lower effort.

People judge fairness by comparing their own give-and-get with another person's.

Clear, hard and accepted goals raise effort, but too many goals scatter it.

Behaviour that brings a pleasant result tends to be repeated.

Reinforcement. Following a behaviour with a result that makes the behaviour more likely to be repeated.

Need theories say what to offer, and process theories say how to offer it.

Section 11.3 asked what people want. A second group of theories asks a different question: how does a person decide where to put effort? These are the **process theories** of motivation, which explain the thinking by which a person chooses to work hard, or not. Three are brief enough to take together. Each gives the manager a question to ask.

The first theory is *expectancy*, set out by the American writer Victor Vroom in 1964. It says that a person works hard only when three beliefs hold together. The person believes that effort will lead to good performance. The person believes that good performance will lead to a reward. And the person values the reward. If any one belief is weak, effort falls. A baker may know that a bonus is offered for low waste, yet doubt that the oven is reliable enough to reach the target. The first belief fails, and the bonus changes nothing.

Take a small example, with each belief rated from 0 (none) to 1 (full). A baker rates the belief that effort will reach the waste target at 0.9, the belief that reaching it will bring the bonus at 0.9, and the value of the bonus at 0.2, because Tk 125 is a small sum to this baker. Multiplying the three gives $0.9 \times 0.9 \times 0.2 = 0.162$, or about 0.16. If the bonus mattered much more, and the baker rated its value at 0.9, the product would be $0.9 \times 0.9 \times 0.9 = 0.729$, or about 0.73. The numbers are illustrations, because nobody can measure belief so exactly. They show why one weak link lowers the whole.

The second theory is *equity*, described by J. Stacy Adams in 1963. People compare what they give to the job, such as effort, skill and time, with what they receive, such as pay and praise. Then they compare this ratio with the ratio of a colleague. If the two seem equal, the person is content. If the person seems to give more for less, the person feels unfairly treated and may reduce effort, ask for more, or leave. Note that the comparison is made by the person, and a pay scale that the manager thinks fair may look unfair from another seat.

The third theory is *goal-setting*, developed by Edwin Locke from 1968. It says that a clear and difficult goal, which the person accepts, produces more effort than a loose request such as "do your best". Feedback on progress adds to the effect. This links directly to Chapter 7: the agreed objectives of management by objectives are an application of the theory. The exceptions are plain. A goal that is too hard is dropped, a goal imposed without consent is resisted, and too many goals at once scatter effort.

A fourth view is simpler still, and is linked with the American psychologist B. F. Skinner. **Reinforcement** means following a behaviour with a result that makes the person likely to repeat it. A baker who is thanked after cleaning the bench early is more likely to do so again. A result that the person finds unpleasant tends to reduce the behaviour. This view does not ask what a person feels. It asks what follows the behaviour. Its use for a manager is plain: reward the behaviour that you want, and do so soon after it happens. Its limit is also plain, because it pays little attention to how the person judges the reward or its fairness.

The need theories and the process theories answer different questions, and a manager need not choose between them. Need theories help to decide what to offer. Process theories help to decide how to offer it. An owner who offers the right reward, but in a way that the staff find unfair or out of reach, will still fail. An owner who offers a fair and reachable reward that nobody wants will fail as well.

The three theories add to the need theories, and do not replace them. A manager can use them as three questions. Does the person believe that effort will pay? Does the person think the reward is fair? Is the goal clear and accepted? In the bakery, the team bonus of Tk 1,000 a month, if shared equally among 8 staff, gives each person Tk 125. Whether this changes behaviour depends on whether each person thinks the target is reachable, the share fair, and the sum worth having.

Before moving on. Expectancy asks whether effort will pay. Equity asks whether the reward is fair compared with others. Goal-setting asks whether the goal is clear, hard and accepted.



11.5 GROUPS AND TEAMS; INFORMAL GROUPS

Case file, illustrative. Month 9: when the staff grew from 4 to 8, a lunch group formed among 4 of the staff. No order created it, and the chart of the bakery does not show it.

Chapter 9 introduced the informal organisation: the friendships and habits that exist outside the chart. This section looks at the same facts from the side of the person. Why do people join groups? The reasons are old and strong. People need company, which is the social level of Maslow's hierarchy. They gain safety in numbers, because a group can protect its members. They gain standing from belonging to a respected group. And some tasks cannot be done alone.

A **group** means two or more people who meet, influence one another and see themselves as belonging together. Four friends who eat lunch together form a group. A **team** is a special kind of group: people who work together toward a joint goal, with skills that fit one another and a shared duty for the result. The bakery's production staff are a team, because the loaf cannot be made unless each does a part, and all answer for the quality of the bread.

The bakery's production staff show these features. Their joint goal is a batch of good loaves on time. Their skills fit one another: the head baker mixes and judges, the two bakers shape and bake, and the helper cleans and carries. They share a duty, since a spoiled batch costs the whole shop. The 7 am briefing of Month 9 gives them a daily chance to confirm the goal. The lunch group has none of these features by design, yet it supports the team, because friends who trust one another at lunch tend to speak plainly at the bench.

The difference matters to a manager. A group may have no purpose beyond itself, and may pull in any direction. A team has one purpose, and the manager's task is to keep that purpose clear. Not every group is a team, and not every team works well. A team needs a clear goal, a mix of skills that fit the task, and members who trust one another enough to say when something is going wrong.

Groups are also divided by how they begin. A **formal group** is created by the organisation to do a task, such as the production staff or a committee. An **informal group** forms by choice, not by order, from friendship, common interest or closeness at work. The lunch group is informal. Members of an informal group often belong also to a formal one, and the two overlap in every organisation.

Ask three questions: does effort pay, is it fair, and is the goal clear?

A short recap of section 11.4.

Nobody told four bakery staff to eat lunch together, yet they do.

People form groups to gain company, safety, standing and help with tasks.

A team is a group with a joint goal and a shared duty for results.

Group. Two or more people who meet, influence one another and see themselves as belonging together.

Team. A group with joint goals, fitting skills and shared duty for the result.

The bakery's production staff show the features of a team.

A group may have no shared purpose, but a team always has one.

Formal groups are created by order, and informal groups by choice.

Formal group. A group created by the organisation to do a task.

Informal group. A group that forms by the members' own choice, not by order.

Norms are unwritten rules, and cohesion is the strength of the bond.

Norm. An unwritten rule about how members of a group should behave.

Cohesion. How strongly the members of a group feel bound to one another.

Members often follow group norms even against their own judgement.

Informal groups help with belonging and news, but can resist change and limit output.

News spreads through informal groups faster than through notices.

Grapevine. The informal flow of news among people, outside the official channels.

Teams fail through unclear goals, poor fit of skills, loose contact and no leader.

The same group looks different to a member, the owner and the customer.

A manager should work with an informal group, not order it to end.

Teams change over time, and early disagreement is normal.

Informal groups develop two features that a manager should understand. The first is a set of **norms**, which means unwritten rules about how members should behave. A norm in the lunch group might be that nobody speaks of problems at the counter during the meal. The second is **cohesion**, which means how strongly members feel bound to the group and to one another. A group with strong cohesion pressures its members to follow its norms, and a member who breaks them may be left out.

Members of a cohesive group often follow its norms even when their own judgement differs. A new baker may slow down to match the pace of the others, because working faster would set the baker apart. A person who wishes to belong accepts the group's rules, and the price may be lower output. This is why targets agreed only with individuals may meet a quiet limit set by the group. A manager who talks with the group as well as with each person makes it more likely that the target is accepted by both.

Informal groups bring benefits to the organisation. They meet social needs, they carry news faster than the chart can, and members help one another with work in ways no rule requires. They bring risks as well. A group may resist a change that the owner wants. Rumours can spread quickly. And if the norms of a cohesive group are against high output, members may hold one another back. The first Hawthorne researchers noticed this last effect among workers in a factory in the United States.⁴

Informal groups carry news by the **grapevine**, which means the informal flow of news among people outside the official channels. If the owner plans a change in shifts, the grapevine may carry a distorted version to every counter worker by lunch. What it carries is often accurate, though it is also often incomplete. A manager cannot close it, but can use it: tell the group the facts early and plainly, and the grapevine carries the facts.

Teams fail for reasons that a manager can learn to see. The goal may be unclear, so each member works toward a different aim. The skills may not fit the task, so some members idle while others are overloaded. Size can matter as well: in a small team members know one another well, and as the team grows, contact thins and some members do less. And a team without any leader tends to drift. Chapter 12 studies leadership, and shows how a leader can keep a team pointed at its goal.

The informal group looks different from each seat. To a member, it is a place of belonging and a shield against pressure. To the owner, it is a source of help and a possible source of resistance. To the customer, it appears in the mood of the counter: a friendly group greets a regular customer by name, and a divided one does not. A manager who wishes to understand the group should try all three views before acting.

How should a manager respond? Ordering such groups to end does not work, because the needs that created them remain. A better course has four parts. First, learn that the group exists and who its natural leader is. Second, speak openly to the group about changes before they happen. Third, align the group's goal with the goals of the organisation, for example by sharing a target. Fourth, treat the informal leader as a partner. A manager who works with the informal group gains a channel of information and, often, a source of support.

A work team also changes as members come to know one another. At first they are polite and unsure of their roles. Disagreement often follows, as

⁴ The Hawthorne studies, at the Western Electric company in the United States, began in the 1920s. Verify dates, names and the later criticisms of the findings before this note is finalised.

members test one another. Later the team settles its norms and begins to work smoothly. A manager who expects the early disagreement, and does not mistake it for failure, can guide the team through it. New members added later may restart the process.

Pause and check 3. The four staff in the lunch group disagree with the owner about a new shift plan. Name one risk and one benefit of this group for the owner.

Before moving on. A group is people who influence one another. A team has a joint goal. Informal groups form by choice, create norms and cohesion, and can help or hinder the organisation.

Test the group ideas on a new case before moving on.

A short recap of section 11.5.



11.6 CREATIVITY AND INNOVATION

Case file, illustrative. Month 11: a helper suggests a milk bread, 40 loaves a day at Tk 80. A plain loaf sells at Tk 70.

Ideas do not arrive by order, yet every organisation lives on them. The bakery of Month 1 would not survive in Month 24 if it never changed its bread, its methods or its service. Two words are often used for the same thing, though they are different. **Creativity** means making new ideas, and **innovation** means putting a new idea to use. The helper showed creativity by thinking of a milk bread. The owner will show innovation only if the milk bread is baked, priced and sold.

The difference is worth keeping. A drawer full of good ideas does not help a business. Many organisations have creative people yet little innovation, because the ideas die before they reach a customer. Others try to innovate without any fresh ideas, and copy what others have done. The best organisations manage both parts: they gather ideas, and they have a way of testing and using them.

Many people think that creativity belongs to artists or to rare geniuses. The helper who thought of a milk bread was not a genius. The helper knew the work well, saw a gap in what the customers could buy, and spoke up. Creativity grows from knowledge of the work and freedom to say what one sees. Most people have it to some degree, and organisations can raise it or lower it by how they treat people who speak.

Writers on creativity have long noticed a pattern in how new ideas come. The English writer Graham Wallas described four stages in 1926. In *preparation*, the person studies the problem and gathers facts. In *incubation*, the person leaves the problem aside, and the mind works on it unseen. In *illumination*, the new idea suddenly appears. In *verification*, the person tests the idea and finds out whether it works. Innovation then adds a fifth step: the verified idea is put to use. The figure shows the whole line.

A helper had an idea, and the owner had to decide what to do with it.

Creativity makes an idea, and innovation puts it to use.

Creativity. Making new ideas.

Innovation. Putting a new idea to use.

Ideas without use change nothing, and use without ideas only copies.

Creativity is a skill of many people, not only of artists.

New ideas often follow four stages, and innovation adds a fifth.

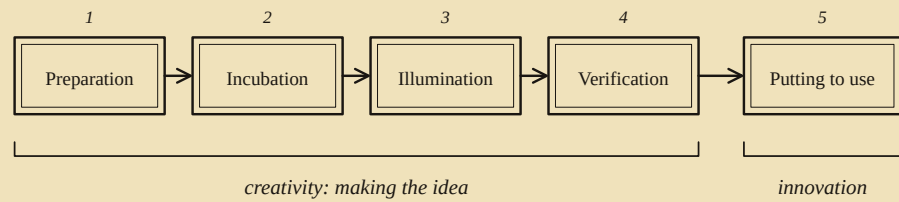


FIG. 37. From creative idea to innovation: four stages of making an idea, then putting it to use

Each stage of the idea needs a different kind of help from the manager.

Ideas come from inside the bakery and from outside it.

Blame and rigid rules block ideas, while freedom and praise encourage them.

Innovation may change the product or the process, and is often small.

Not every trial succeeds, and a manager must accept that.

A trial turns a hopeful idea into evidence.

A short recap of section 11.6.

The stages are not a strict recipe. Some ideas arrive in minutes and some take years. They are useful because they show the manager where to help. In preparation, give people time and information. In incubation, do not demand an answer at once. In verification, offer a safe way to test: a small trial, not a full launch. At the last step, innovation needs money, space in the plan, and someone to answer for the result.

Where do useful ideas come from? Inside the bakery they come from staff who handle the work every day, like the helper who suggested the milk bread. Outside, they come from customers who ask for something not on the shelf, from suppliers who offer a new flour, and from rivals whose changes can be studied. A manager who listens in all these places gains more ideas than one who waits for a flash of genius. A small habit, such as noting each customer request in a book and reading it each week, can be a steady source of ideas.

Certain features of an organisation block creativity. Fear of blame for a failed idea makes people keep quiet. Rules that cannot be questioned and workloads with no free time do the same. Other features help. These include freedom to try within agreed limits, praise for a good attempt even if it fails, and a habit of listening to ideas from every level. Notice the link with section 11.1. A manager who holds Theory Y believes that ideas are found widely, and the belief shows in these habits.

Innovation also takes different forms. A *product* innovation changes what is sold, like the milk bread. A *process* innovation changes how it is made, like a new mixer or a better layout. Both can be small. In a small bakery, innovation seldom means a great invention. More often it means a small change that is tried, measured and kept or dropped. This is the form of innovation most managers will meet, and the form this chapter's case follows.

Innovation always carries risk, because a new product may not sell. Chapter 8 showed how risk can be weighed with probabilities before a choice is made. But the manager must also decide how to treat a trial that fails. If the owner blames the helper for a milk bread that sells poorly, the staff learn to stop proposing ideas. If the owner studies why it failed and thanks the helper for trying, the staff learn that effort in new directions is safe. The first habit saves a small loss and costs many future ideas.

An innovation must also be managed like any other change. A new product needs a plan, as Chapter 6 teaches, with objectives from Chapter 7 and a choice among alternatives from Chapter 8. Results of the trial are then compared with a standard, which is the work of Chapter 13. The owner might trial the milk bread for four weeks, count the loaves sold and wasted each day, and compare the waste with the 5% target. A trial of this kind turns a hopeful idea into evidence.

Before moving on. Creativity makes new ideas, and innovation puts them to use. New ideas often pass through four stages, and each needs a

different kind of help. A manager's beliefs about people shape how many ideas are heard.



WORKED CASE: WHY TWO PEOPLE LEFT

Case file, illustrative. Situation. In six months, 2 of the 8 staff left, a loss of 25%. The owner introduced a monthly team bonus of Tk 1,000 for waste of 5% or less, stopped checking every tray, and accepted a helper's milk bread: 40 loaves a day at Tk 80. The case file does not record why the two left.

Case file, illustrative. Question. Which actions suit which needs, and what should the owner do first?

Reasoning, step 1. We begin with what the file does not say. It gives no reason for leaving, so the owner cannot yet know the open need. The theories of this chapter suggest what to ask. Was it a lower level, such as pay or a safe oven? Or a higher level, such as belonging or respect? The first step is therefore to ask the staff who remain, and, if possible, the two who left. The owner gathers information before acting.

Reasoning, step 2. We now place each action at a level of Maslow's hierarchy. The bonus is extrinsic. Shared equally, it gives each of the 8 staff $Tk\ 1,000 \div 8 = Tk\ 125$ a month. At this size it is more likely to signal that low waste matters than to meet a physiological need. Stopping the tray check and agreeing targets is a Theory Y step; it speaks to esteem, because it shows trust. Adopting the helper's milk bread speaks to esteem and to self-actualisation, because the helper's own idea reaches the customer.

Reasoning, step 3. Next we use the process theories. For expectancy, the staff must believe that waste of 5% or less is reachable. The file does not yet show whether the target can be reached, so the owner should watch the first results. For equity, the bonus is shared equally, which most members may judge fair. For goal-setting, the target is clear and measurable, and was agreed one by one, as Chapter 7 teaches.

Reasoning, step 4. Last, the milk bread. At 40 loaves a day and Tk 80 each, daily sales are $40 \times Tk\ 80 = Tk\ 3,200$. Over 26 trading days the sales are $Tk\ 3,200 \times 26 = Tk\ 83,200$ a month. This is a sales figure, not profit, because the costs of the new bread are not in the file. As an innovation, it should be tried first as a small batch, in line with the verification stage.

Case file, illustrative. Conclusion. The owner should ask first, then keep the agreed targets and the bonus as signals of trust and fairness. The milk bread should be tested as a small trial. If the answers show a lower-level need, such as pay or safety, the owner should respond at that level.

Check. Does each action match a named need? Yes: the bonus (signal), the targets (esteem), the milk bread (esteem and growth). Is the arithmetic right? $Tk\ 1,000 \div 8 = Tk\ 125$. $Tk\ 80 \times 40 = Tk\ 3,200$. $Tk\ 3,200 \times 26 = Tk\ 83,200$. Does the answer claim more than the file allows? No: it states that the reason for leaving is unknown and that the sales figure is not profit.



Situation: two leavers, three new actions, and no stated reason for leaving.

Question: match each action to a need and decide the first step.

First gather facts, because the cause of leaving is not recorded.

The bonus signals what matters, and the other two actions aim at esteem and growth.

The three process theories test whether the plan can work.

The milk bread brings Tk 83,200 a month in sales, not profit.

Conclusion: ask first, keep the targets, and test the new bread on a small scale.

Check: each action is matched to a need, and the figures are right.

SUMMARY

The chapter's main ideas, in short.

Managers hold beliefs about people. Theory X assumes people avoid work and need control; Theory Y assumes they can direct themselves. Motivation is the push that starts and keeps effort. It follows the needs-goal model: need, drive, action, goal and relief, and then a new need. Maslow's hierarchy gives five levels of need: physiological, safety, social, esteem and self-actualisation. Herzberg, Alderfer and McClelland added other views. Process theories ask how people choose: expectancy, equity and goal-setting. Groups form because people need company, safety, standing and help. Informal groups create norms and cohesion and may help or hinder. Creativity makes ideas, and innovation puts them to use.

IN SHORT

Effort begins in a need. A manager who finds the open need, offers a fitting reward, treats people as willing, and works with their groups gets better results than one who guesses.

THINK IT THROUGH

A case to reason through before turning the page.

A call centre pays its agents above the local average and has a safe, clean building. Agents take calls from a fixed script and are timed on every call. Many leave within a year. The manager proposes a pay rise of 10% for everyone. Using this chapter, say which levels of need are probably met and which are probably open. Say which of Theory X or Theory Y the timing and the script suggest. Then judge whether the pay rise is likely to work. Finally, ask whether money alone can motivate, and give one reason for your answer drawn from the hygiene factors and motivators.

WHERE PEOPLE GO WRONG

Mistake: money is all that motivates.

"Motivation means money." Money meets some needs, mainly at the lower levels, and signals worth. It cannot give belonging, trust or pride in the work.

Mistake: the levels are a fixed ladder for everyone.

"Maslow's levels are a strict ladder." Maslow described a general tendency. Levels overlap, and people differ.

Mistake: one theory is always right.

"Theory X is bad and Theory Y is good." Both are sets of belief. A new worker may need close direction, and an experienced one may need freedom. The right choice depends on the person and the task.

Mistake: informal groups only cause trouble.

"Informal groups are a threat." They meet real needs and carry useful news. The risk lies in their norms, which a manager should learn.

Mistake: the two words mean the same.

"Creativity and innovation are the same." Creativity makes an idea. Innovation puts it to use.

IN HISTORY

A true account of how one study changed ideas about workers.

From the middle of the 1920s, researchers studied workers at the Hawthorne Works of the Western Electric company, near Chicago, in the United States. The first tests looked at the effect of lighting on output. Output rose when light was increased, but it also rose in other conditions, which puzzled the researchers. Later work, joined by Elton Mayo and colleagues from Harvard, pointed to the attention paid to the workers and to the pull of the group. Output depended on

feelings, friendships and group norms as well as on conditions of work. This helped give rise to the human relations approach of Chapter 3. Later critics argued that the conclusions went beyond the data, and the studies are still debated. This chapter's lesson is smaller and still safe: people are social beings, and their groups matter.

SOURCES

McGregor, *The Human Side of Enterprise* (1960), named Theory X and Theory Y, and made managers examine their own assumptions. Maslow, "A Theory of Human Motivation" (1943), set out the hierarchy of needs. Herzberg, Mausner and Snyderman, *The Motivation to Work* (1959), separated hygiene factors from motivators. McClelland, *The Achieving Society* (1961), described the needs for achievement, affiliation and power. Vroom, *Work and Motivation* (1964), set out expectancy theory. Wallas, *The Art of Thought* (1926), described the stages of creative thought.

Where the main ideas of this chapter come from.

REVIEW QUESTIONS

1. Define motivation in your own words, and give one example from a shop or a school.

Question 1 tests the meaning of motivation.

2. Draw the needs-goal model, and use it to explain what happens when a delivery rider wants to be thanked for a hard day's work.

Question 2 asks for the model and a new case.

3. List Maslow's five levels in order. Then place each of these actions at the level it best serves: a safe oven, a fair title, a lunch table, a monthly wage, a chance to design a new loaf.

Question 3 asks for the levels and a matching task.

4. Compare Theory X and Theory Y in a table with three rows. State which one a manager holds who checks every sheet of paper each day, and give one cost of that habit.

Question 4 asks for a table of the two sets of assumptions.

5. State two differences between hygiene factors and motivators, with one example of each.

Question 5 tests Herzberg's two kinds of factor.

6. Define an informal group. Give two benefits and two risks for the organisation.

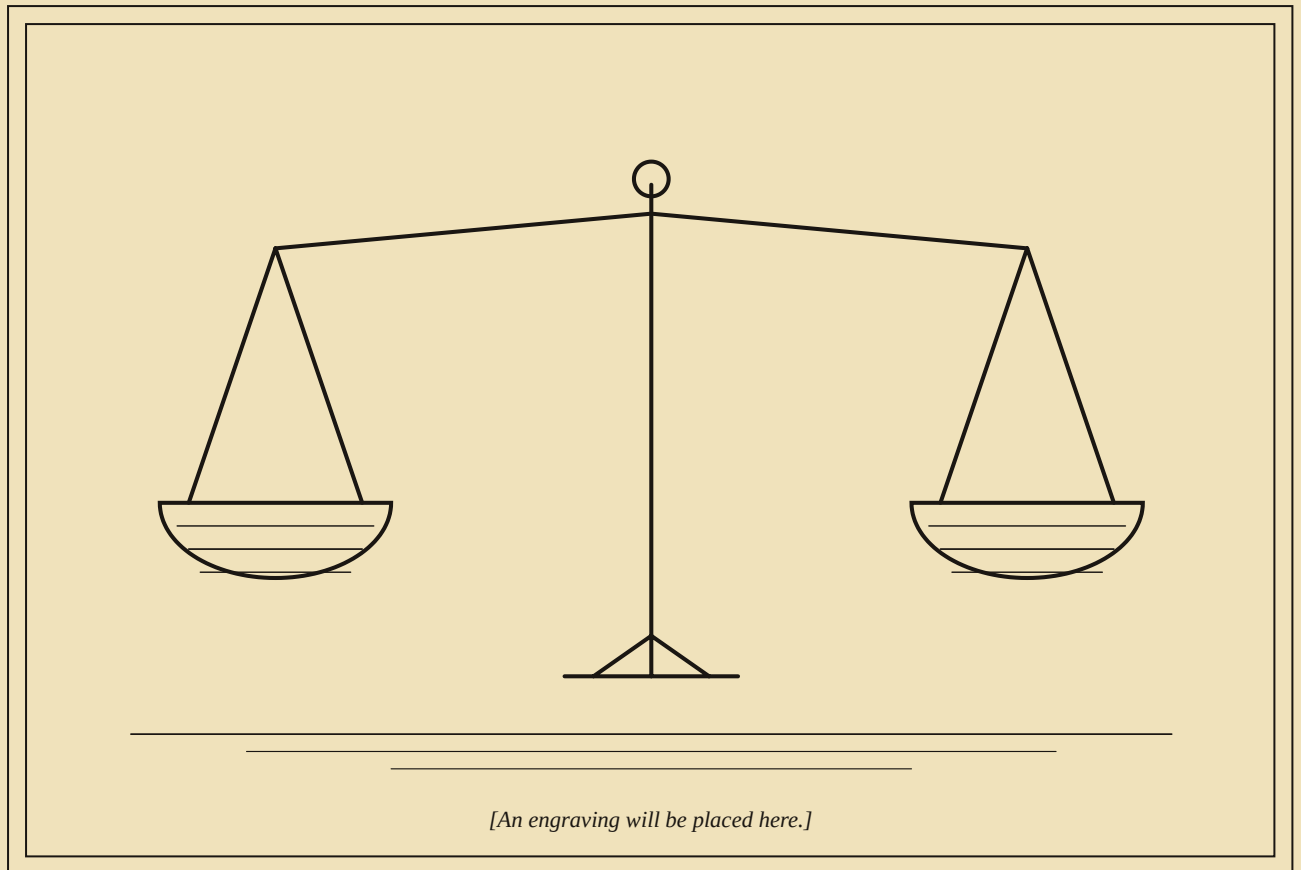
Question 6 asks for a definition with benefits and risks.

7. Distinguish creativity from innovation, using the example of a tailor who thinks of a new kind of pocket.

Question 7 asks for the two terms apart on a new case.

8. Say why a team differs from a group, and name one thing a manager can do to turn a group into a team.

Question 8 tests the difference between group and team.



CHAPTER XII

Leadership

Why people follow one person and ignore another who holds the same position.



Chapter 11 showed what moves a person to work and how groups shape that effort. This chapter asks who guides the effort, and by what means. It separates leading from managing, studies the sources of power, compares styles of leading, and shows how a manager can choose a style to suit the people and the task. It ends with newer ideas: emotional intelligence, transformational leadership, coaching and super leadership.

WHO DO THEY FOLLOW?

Two bakers hear the same instruction, and only one acts on it.

Position alone does not explain why people act on an instruction.

A right to give orders does not guarantee that people follow.

This chapter asks what makes people follow, and how a manager can use it well.

Case file, illustrative. Morning Loaf Bakery, Month 12. Until now the owner has decided most questions of work method alone. In Month 12 the owner begins to ask the experienced bakers for their views before deciding. With the new helper the owner stays directive and gives clear orders. The owner also meets the head baker each week to coach.

Imagine that the owner tells two people, on the same morning, to change the order in which the dough is mixed. The new helper makes the change at once and asks whether it is right. The senior baker, who knows the oven well, frowns, says that the old order gave a better crust, and goes on as before. The owner holds the same position in both cases. The owner has the same authority, the same wage list and the same right to give orders. Yet the two people respond in different ways.

This small scene raises a question that Chapter 10 could not answer. Authority is the right to give orders, and the owner has it. But a right to give orders does not guarantee that people will follow them, still less that they will follow willingly. Something else decides whether a person's words move others. The helper may follow because the helper is new and wants guidance. The baker may resist because the baker has skill that the owner lacks. To understand the difference, we must study how people come to follow another person.

The chapter proceeds in six steps. Section 12.1 defines leadership and separates it from management. Section 12.2 examines the five sources of power. Section 12.3 compares the early approaches: traits, and then styles of behaviour. Section 12.4 shows why the best style depends on the situation. Section 12.5 studies emotional intelligence, and section 12.6 describes three newer kinds of leadership. A worked case then follows the owner through Month 12.

YOU WILL NEED

From Chapter 10, section 10.1: authority and responsibility. From Chapter 11: the needs-goal model, Theory X and Theory Y, and informal groups. From Chapter 2: the four functions of management, including leading.



12.1 WHAT LEADERSHIP IS; LEADER AND MANAGER

GO SLOWLY HERE. NEW IDEA, FAMILIAR EXAMPLE FIRST.

A captain without any formal right can still lead the players.

Influence is the central word in the definition of leadership.

Influence. Changing what another person thinks, feels or does, without using force.

Think of a school football team. The coach decides who plays, and the teachers decide who may leave early for practice. The captain has none of these rights. The captain cannot choose the team, pay anyone or send anyone home. Yet when the team trails by two goals, the players look to the captain. They follow the captain's calm voice and the captain's way of passing the ball. Nobody ordered them to do so. The captain leads, and the coach, who holds the position, may or may not.

The captain's power rests on **influence**. Influence means changing what another person thinks, feels or does, without using force. A player who copies the captain's way of defending has been influenced. The word is the key to the definition that follows.

Leadership means influencing people toward a goal. A **leader** is a person who does so. In the bakery, a baker who persuades the others to start the morning batch early, so that the festival order is ready, is leading. Every case of leadership has three parts: a leader, followers and a goal. Without followers, nobody is led. Without a goal, people may follow a popular person, but they are not led toward anything.

The next question is how a leader differs from a **manager**, a word met in Chapter 1: a person responsible for the work of others. The two words overlap, but they are not the same. A manager holds a position, and the position gives the right to direct resources and people. A leader need not hold any position. The captain in our example leads without one. In an organisation, the same is true of an **informal leader**, which means a member who has influence over a group without holding any official post, as met in Chapter 11, section 11.5.

Chapter 2 listed four functions of management, and leading is one of them. A manager therefore must lead, as well as plan, organise and control. The reverse does not hold. A person can lead without managing anything. We may then state the relation in one sentence: leading is part of a manager's work, and leadership is wider than management. The table below sets the two side by side. Read each row from left to right: the first column names a feature, and the other two describe a typical manager and a typical leader.

MANAGER AND LEADER COMPARED

FEATURE	MANAGER	LEADER
Source of the right to direct	Position in the organisation	Influence over followers
Main concern	Order: plans, structure, control	Direction: where people are going and why
Main tools	Authority, rules, rewards	Example, trust, ideas
Followers	Subordinates who must obey	Followers who choose to follow
Typical risk	Compliance without commitment	Enthusiasm without order

The last row of the table needs care. A manager who cannot lead gets people to obey but not to care, so work is done at the minimum level. A leader who cannot manage may inspire people but fail to give them plans, structure or checks, so effort scatters. The strongest results come when one person does both, and a small business such as the bakery cannot afford to separate them. The owner is a manager and a leader at once.

There are exceptions to the table. Some writers, such as the American scholar John Kotter in 1990, stress the difference sharply. Others reply that in real work the two cannot be pulled apart.¹ Treat the table as a guide to emphasis and not as a division between two kinds of person.

Now view the same point from three seats. To the worker, a manager is the person who assigns tasks and checks them, and a leader is the person who makes the tasks seem worth doing. To the owner, a manager keeps the shop in order, and a leader makes the shop grow. To the customer, none of this is visible, but the result is: a counter staff who care about the customer, and a shop that runs on time.

Leadership has three parts: a leader, followers and a goal.

Leadership. Influencing people to work toward a goal.

A manager holds a position, while a leader needs only influence.

Manager. A person responsible for the work of others, by virtue of a position.

Informal leader. A member who influences a group without holding any official post.

Leading is one function of management, and leadership reaches beyond management.

Obedience without care and care without order both give weak results.

The table shows differences of emphasis, not two separate kinds of person.

The difference between manager and leader looks different from each seat.

¹ The distinction is discussed in several textbooks and articles. Verify the exact source and date of Kotter's argument before this note is finalised.

Test the difference on a new case before moving on.

A one-line test of the leadership definition.

A short recap of section 12.1.

Pause and check 1. In a small shop, the owner holds the keys but the staff follow a senior assistant. Who holds the position, and who leads?

You should now be able to say: "Leadership is influencing people toward a goal. A manager holds a position, and a leader needs only influence. Leading is one of the manager's four functions."

Before moving on. Influence means changing others without force. A leader has followers and a goal. A manager must lead, but a leader need not manage.



12.2 POWER AND ITS BASES

Consider the head baker of the bakery. Since Month 10 the head baker may buy ingredients up to Tk 5,000 an order, and the owner stays accountable. That is a right written into the job. But when the head baker asks the two bakers to change the order of work, the response depends on much more than the right to buy flour. It depends on whether they respect the head baker's skill, whether they like the head baker, and whether they fear or want what the head baker can give. These sources of influence have a name.

In daily life, *power* often means physical strength, or the force of an engine, or the rule of a government. In management, **power** means the ability to make others act. A baker who can persuade a colleague to stay late has power over that colleague. The word does not say whether the power is good or bad, or whether the organisation has granted it.

Chapter 10, section 10.1, gave a short note on this point, and we can now complete it. Authority is a right given by the organisation. Power is an ability, and it may come from many sources. The two can be found apart. A new manager may have full authority and little power, if the staff ignore the orders. An informal leader may have much power and no authority. The strongest position is to hold both, with authority backing power and power backing authority.

In 1959 two American psychologists, John French and Bertram Raven, set out five bases of power, which are the five sources from which a person's ability to influence others comes. Their list remains the most common one in textbooks.² We take each base in turn, with a bakery example.

Legitimate power means influence that comes from holding a position that others accept as having the right to direct. The head baker's right to order ingredients up to Tk 5,000 is legitimate power. It depends on the staff accepting that the role carries that right. It ends when the position ends, and it does not reach beyond the limits of the role.

Reward power means influence that comes from the ability to give things that others value. The owner's monthly team bonus of Tk 1,000 for waste of 5% or less is reward power. It also includes shifts, praise and training. Chapter 11 showed that a reward works only if it meets a need that is open. A reward that people do not want gives no power, and a reward that the giver cannot deliver gives none either.

Coercive power means influence that comes from the ability to punish or to threaten loss. Dismissal, fines and the least liked shifts are all forms of it. Coercive power is real, and it can work for a short time. But it brings fear. People who fear punishment may hide errors, avoid responsibility and leave when they can. Chapter 11 described such a manager as holding Theory X.

² Raven added a sixth base, informational power, in later work. This book follows the original five. Verify the date and details of the original paper.

Expert power means influence that comes from special knowledge or skill that others need. A baker who knows how to repair the mixer has expert power, whatever the baker's position. It persists as long as the skill is needed and others lack it. It fades if the knowledge becomes old or if others learn it.

Referent power means influence that comes from the admiration and liking that others feel for a person, and their wish to be like that person. A baker whom the whole team trusts and copies has referent power. It cannot be ordered or bought, and it takes time to build. It can be lost quickly through one act of unfairness.

The table that follows sets the five side by side. Read each row from left to right: the first column names the base, and the others give its source, a bakery example and its main risk.

THE FIVE BASES OF POWER

BASE	SOURCE	BAKERY EXAMPLE	MAIN RISK
Legitimate	A position others accept	The head baker's order limit of Tk 5,000	Ends outside the role
Reward	Ability to give valued things	The Tk 1,000 team bonus	Fails if the reward is not wanted
Coercive	Ability to punish	Threat of dismissal	Fear, hidden errors, departures
Expert	Special knowledge or skill	The baker who repairs the mixer	Fades as others learn
Referent	Admiration and liking	The baker the team copies	Lost by one unfair act

Writers often divide the five into two groups. Legitimate, reward and coercive power belong to the position, because the organisation grants them and can take them back. Expert and referent power belong to the person, because they stay with the individual who holds them. A manager moved to another post keeps the second group and loses the first. This division explains why some leaders are followed after they leave the post, and some managers are forgotten the day they retire.

The bases also lead to different responses from followers. Where power rests on expert or referent bases, followers tend to show **commitment**, which means willing effort that goes beyond the minimum. Where it rests on legitimate or reward power, they tend to show **compliance**, which means doing what is asked, but no more. Where it rests on coercive power, compliance is the best result, and resistance is common. A leader who wants commitment should therefore build personal power, and not rely on position alone.

No leader uses one base only. The owner has legitimate power as the owner, reward power through pay and bonus, coercive power through the right to dismiss, expert power in baking, and, with luck, referent power. The skill lies in the balance. A leader who relies on coercion alone makes a weak team. One who relies on liking alone cannot say no. Return now to the scene at the start of the chapter. The new helper follows the owner at once, because the helper has little skill and values guidance, so the owner's expert and legitimate power are strong. The senior baker has expert power of the baker's own, and for this reason the owner does better to ask than to tell.

Power must also serve the goal and not the private gain of the leader. A manager who uses reward power to favour friends, or coercive power to silence

honest criticism, harms the team and the organisation. Chapter 5 studied the ethical tests that apply, and Chapter 13 shows how control systems can guard against misuse.

Pause and check 2. Since Month 10 the part-time bookkeeper may require receipts from all staff. Which base of power does this right rest on, and why would it be weak if the owner withdrew support?

Before moving on. Power is the ability to make others act. French and Raven named five bases: legitimate, reward, coercive, expert and referent. The first three belong to the position and the last two to the person.



12.3 TRAIT AND BEHAVIOURAL APPROACHES; STYLES OF LEADING

If you could list the qualities of every great leader, could you hire a leader by checking the list? Early researchers believed so. Their search for the list is the oldest approach to leadership, and its failure taught more than its success.

The **trait approach** means the view that leaders are born with special qualities that other people lack. A *trait* is a lasting personal quality, such as height, energy, intelligence or confidence. From the early years of the twentieth century, researchers measured the traits of people who led and compared them with those who did not.³ Some traits appeared often: intelligence, drive, self-confidence and honesty among them.

But the approach met two difficulties. First, no list of traits worked in every setting. A person who led well in a sports club might fail in a hospital, and the traits did not predict which. Second, traits are hard to change. If leaders are born, then training can do little, and experience contradicts this view. Today managers use traits as one piece of information, for example when choosing among candidates, and they do not treat any list as a guarantee.

Researchers then turned from what leaders *are* to what leaders *do*. This is the behavioural approach. It studies a leader's **leadership style**, which means the usual pattern of behaviour that a leader shows toward followers. The best-known early study was made in the United States in the late 1930s by Kurt Lewin with two colleagues, Ronald Lippitt and Ralph White. The details are given at the end of the chapter.

The study named three styles. An **autocratic leader** is one who decides alone and tells the group what to do. A **democratic leader** is one who shares decisions with the group, invites discussion and then guides the result. A **laissez-faire leader** is one who leaves the group to decide and act with little guidance. *Laissez-faire* is a French phrase, now used in English for leaving things alone. The table below compares the three. Read each row from left to right.

³ Verify the main studies and dates summarised by later reviews of trait research before this note is finalised.

THREE STYLES OF LEADING

FEATURE	AUTOCRATIC	DEMOCRATIC	LAISSEZ-FAIRE
Who decides	The leader alone	The leader with the group	The group alone
Direction of talk	Mostly from leader to group	In both directions	Little from the leader
Speed of decision	Fast	Slower	Varies widely
Good when	Time is short; followers are new	Followers know the work and want a voice	Followers are skilled and self-directed
Main risk	Resentment; dependence on the leader	Delay; loss of direction	Drift; no one answers for results

The styles link to the assumptions of Chapter 11. An autocratic leader often holds Theory X, and a democratic leader holds Theory Y. A laissez-faire leader assumes a group that is already skilled and self-directed, and so it fits only a narrow set of cases. None of the three is good or bad in itself. A bakery fire demands an autocratic style, because there is no time to discuss. A discussion of a new product demands a democratic style, because the staff know the customers. A team of very experienced bakers may do well with a light touch.

A second line of research, begun at Ohio State University and the University of Michigan in the 1940s and 1950s, looked at the aim of the leader's actions. It found two themes that appear again and again.⁴ The first is concern for task: setting goals, planning schedules and checking results. The second is concern for people: showing trust, respect and warmth. A leader may show much or little of each.

In 1964 two American writers, Robert Blake and Jane Mouton, drew the two themes as a grid.⁵ Concern for people is shown upward, and concern for task is shown to the right, each on a scale from 1 (low) to 9 (high). The figure shows the five main positions on the grid. A manager at the lower left shows little concern for either and does the minimum. A manager at the lower right cares for output only, and treats people as tools. A manager at the upper left cares for people but neglects results. A manager in the middle balances the two at a modest level. A manager at the upper right is a *team* leader, who shows high concern for both.

⁴ Verify the dates, the names of the main studies and the terms each used before this note is finalised.

⁵ Verify the date and the terms Blake and Mouton used for each position.

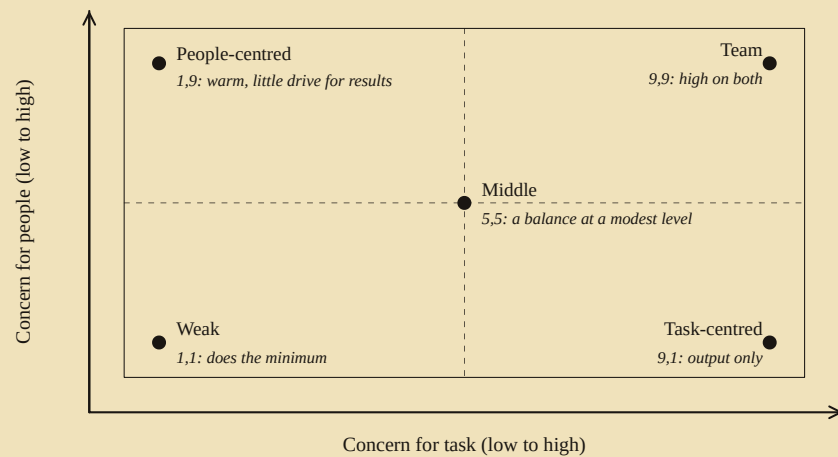


FIG. 38. *The two concerns of a leader: five positions on a grid of concern for people and concern for task*

Blake and Mouton held that the team position is the best in all cases. Many writers disagree. They point out that a firm facing an emergency may need a high concern for task and little time for discussion, and that a skilled and willing team may need little push from its leader. The behavioural approach, like the trait approach, therefore leaves a question open: what works depends on whom the leader leads and in what setting. That is the starting point for the next section.

The behavioural studies have a further limit. Lewin's groups were boys' clubs, and findings from them may not carry over to adults at work. Later research gave mixed results on whether democratic leaders always get better output. What most studies suggest is a modest point: people usually prefer to be consulted on matters that affect them, and leaders who never consult pay a price in loyalty.

Before moving on. The trait approach looks at what leaders are and finds no list that always works. The behavioural approach looks at what leaders do. Three styles are autocratic, democratic and laissez-faire, and two concerns, for task and for people, describe a leader's aims.



12.4 SITUATIONAL APPROACHES

In 1958 two American scholars, Robert Tannenbaum and Warren Schmidt, published an article titled "How to Choose a Leadership Pattern" in the *Harvard Business Review*.⁶ Managers of the time asked a question that seemed sensible: should I be firm, or should I be democratic? The two writers answered that the question is wrongly put. A manager should not choose a style once for all, but should ask which style suits this task, these people and this moment.

The **situational approach** means the view that the best way to lead depends on the situation, so that no single style is best in every case. It is the form that the contingency approach of Chapter 3, section 3.4, takes in the study of leadership. We begin with the Tannenbaum–Schmidt model, which is the clearest, and then add three others.

⁶ Verify the title, journal, year and authors of this article before this note is finalised.

THE LEADERSHIP CONTINUUM

Tannenbaum and Schmidt did not describe two or three separate styles. They drew a line. The **leadership continuum** means the range of leader behaviour, from the boss-centred end, where the leader decides, to the team-centred end, where the group decides within limits. *Continuum* means a line along which there is no sharp break between one point and the next. The figure shows seven points on the line.

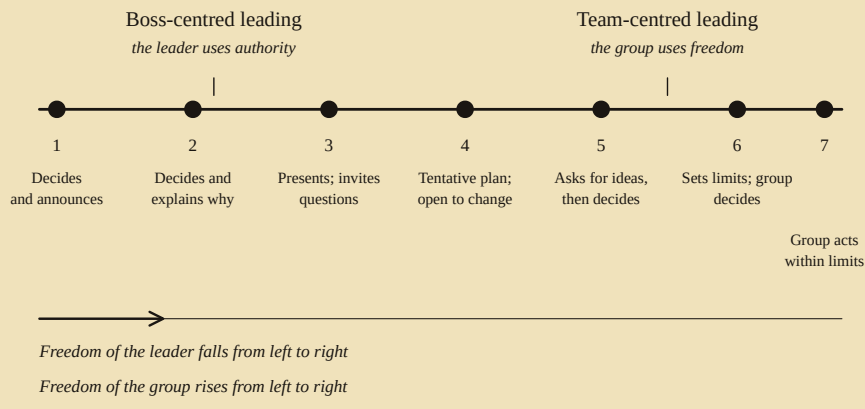


FIG. 39. The leadership continuum: seven points from boss-centred to team-centred leading

We describe the points from left to right, using the bakery's choice of a new mixing order as the example. At the first point, the leader decides and announces: "From Monday the dough is mixed in this order." At the second, the leader decides but explains the reasons, to win acceptance. At the third, the leader presents the decision and invites questions before it is final. These three points keep the decision in the leader's hands.

At the fourth point, the leader presents a tentative decision and is ready to change it after hearing the group. At the fifth, the leader presents the problem, asks the group for suggestions, and then decides. The leader has moved from announcing to asking, but the leader still decides. This is what many writers call a *consultative* style, which means asking for the views of others before deciding.

At the sixth point, the leader states the limits, for example the cost and the date, and asks the group to decide within them. At the seventh, the group acts freely within limits that the leader's own superior has set. The decision now rests with the group. The figure shows that the amount of freedom for the leader falls from left to right, and the amount of freedom for the group rises.

The two writers then asked what should decide the point on the line. They named three groups of forces. The first lies in the **leader**: the leader's beliefs about people, confidence in the group, own habits, and tolerance of risk. A leader who holds Theory X will drift to the left, and one who holds Theory Y to the right. The second lies in the **followers**: their wish for independence, readiness to take responsibility, interest in the problem, understanding of the goal, knowledge and experience, and their past expectations of how the leader behaves. The third lies in the **situation**: the size and form of the organisation, the skill of the group in working together, the kind of problem, and the time available.

The lesson is that the successful leader is not the one who is always firm or always gentle. It is the one who reads the forces correctly and acts on them, and who can move to the left when time is short and to the right when the group is ready. For the bakery, this explains the owner's choices in Month 12. With the

new helper, who knows little, the owner keeps to the left of the line. With experienced bakers who know the oven well, the owner moves to the fifth point.

THREE OTHER SITUATIONAL VIEWS

Other writers have built situational theories on different foundations. We give three, each in brief.

The American psychologist Fred Fiedler, in 1967, proposed the *contingency model of leadership*.⁷ He held that a leader's style is hard to change, and that it is better to fit the leader to the situation than to ask the leader to change. He described the situation by three features: the quality of the leader's relations with followers, how clearly the task is set out, and how much formal power the leader holds. A task-oriented leader, he suggested, does best where the situation is either very favourable or very unfavourable, and a relationship-oriented leader does best in between.

In 1969 Paul Hersey and Kenneth Blanchard proposed the *situational leadership model*.⁸ It places the fit between style and the *readiness* of followers, which means their combined ability and willingness to do a task. When readiness is low, the leader should tell. As readiness rises, the leader should sell, then take part, and finally delegate. The figure shows the four steps. A new helper who cannot yet bake needs *telling*. A helper who has learned the basics but lacks confidence needs *selling*, which means explaining and persuading. An experienced baker who is able but unsure may need *participating*, which means sharing decisions. A head baker who is able and willing may be given *delegating*, which means handing over decisions and duty.

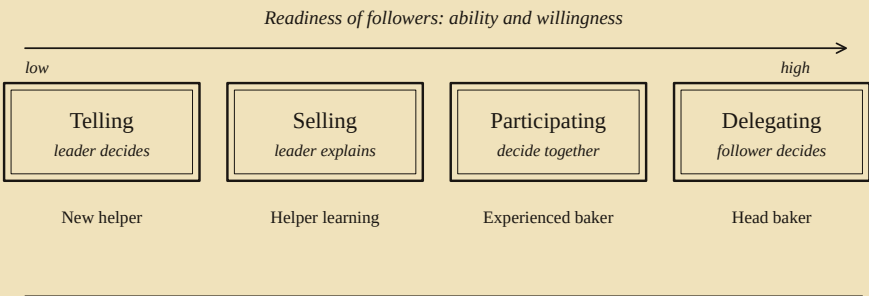


FIG. 40. Four steps of readiness and the style that fits each

The third view, set out by the American scholar Robert House in 1971, is called *path-goal theory*.⁹ It says that the leader's task is to help followers see the path to their goals and to clear obstacles from it. It links directly to the expectancy theory of Chapter 11: if followers believe that effort leads to reward, they work harder, and the leader's job is to make that belief true. House named four styles, which are directive, supportive, participative and achievement-oriented, and the leader picks among them according to the task and the followers.

The table that follows compares the four situational views. Read each row from left to right: the first column names the writers, and the others give the main idea, what the leader adjusts and what the situation is judged by.

⁷ Verify the date, the name of the model and the details of its findings before this note is finalised.
⁸ Verify the date and names used by the authors for the four styles before this note is finalised.
⁹ Verify the date, the author and the names of the four styles before this note is finalised.

FOUR SITUATIONAL VIEWS OF LEADING

WRITERS	MAIN IDEA	THE LEADER ADJUSTS	THE SITUATION IS JUDGED BY
Tannenbaum and Schmidt	A continuum of styles	Place on the line	Forces in leader, followers and situation
Fiedler	Fit the leader to the situation	Little: the situation changes	Relations, task structure, position power
Hersey and Blanchard	Style follows readiness	Telling to delegating	Ability and willingness of followers
House	Show the path to goals	Four styles of support	Task and followers' needs

The views differ on how far a leader can change style. Tannenbaum and Schmidt, and Hersey and Blanchard, expect a skilled leader to move along the line. Fiedler doubts that a person can change style easily. In daily life, most managers can adjust somewhat, but not without limit, and a manager who is naturally warm will find strict orders hard to give.

Situational leading also has a risk that connects to Chapter 11. If the owner is directive with one person and consultative with another, the two may compare their treatment, as the equity theory predicts. If the owner cannot explain why the styles differ, the difference may look like favouritism. A sound leader therefore says why: "You will have a say in this, because you know the oven. For this task, I need you to follow the steps exactly while you are learning."

The common message of the four views is plain. There is no single best style. Before acting, the leader should ask three questions. What is the task, and how urgent? Who are the followers, and how ready are they? What is the setting, and what does it allow? The answers decide the point on the continuum, and they may change next month.

Pause and check 3. On the leadership continuum, which point does each of these sentences show? "From Monday the shop opens at six." And: "Here is the plan. Tell me what you think, and I may change it."

Before moving on. The situational approach holds that the best style depends on the situation. The leadership continuum runs from boss-centred to team-centred. Fiedler, Hersey and Blanchard, and House offer three other situational views.



12.5 EMOTIONAL INTELLIGENCE

Two managers have equal skill. In a busy week, one notices that the head baker has grown silent and short with the staff, and asks quietly what is wrong. The other sees only that the work is behind, and gives more orders. After a month, the first manager has kept the head baker, and the second has lost one. The difference lies in a quality that the earlier approaches did not name.

Emotional intelligence means the ability to read and manage feelings, both one's own and those of other people. Two American psychologists, Peter Salovey and John Mayer, used the term in 1990, and the writer Daniel Goleman made it widely known in 1995.¹⁰

¹⁰ Verify the names, dates and the exact list of components in the sources before this note is finalised.

Goleman described five parts. We state each in plain words, with a bakery example.

Self-awareness means knowing one's own feelings and how they affect one's actions. An owner who knows that anger makes the owner unfair will wait before speaking. *Self-regulation* means controlling one's own reactions, and acting in a calm and fair way. *Motivation*, in this list, means an inner drive to reach goals for their own sake and not only for pay, as in the needs-goal model of Chapter 11. **Empathy** means understanding how another person feels, and showing that you do. *Social skill* means handling relations with others, so that people cooperate and trust is built.

These five parts link directly to leadership. The needs in Chapter 11 are felt, and a leader who cannot read feelings will not see an open need. Referent power, from section 12.2, rests on liking and trust, which a leader earns through empathy and fairness. Even the choice of a point on the continuum rests on reading the followers' readiness correctly, which takes empathy as well as facts.

Seen from the follower's seat, emotional intelligence feels like being noticed and treated with respect, even when a hard decision must be made. Seen from the owner's seat, it appears as lower loss of staff and fewer quarrels. The customer sees it only as a shop in which people are at ease.

There are limits. Researchers still dispute how to measure emotional intelligence, and the claims for it are sometimes larger than the evidence. Skill in feelings cannot replace skill in the work itself. A person who reads feelings well may also use the skill to manipulate others, so it must be joined to honesty, a point that Chapter 5 supports. Finally, people in different places show feelings in different ways, and a leader must learn the customs of those who are led.

Before moving on. Emotional intelligence is the ability to read and manage feelings, one's own and others'. Its five parts are self-awareness, self-regulation, motivation, empathy and social skill. It helps a leader see needs and build trust, but it does not replace skill, and it must be used honestly.



12.6 TRANSFORMATIONAL, COACHING AND SUPER LEADERSHIP

Suppose one owner asks each morning only whether the batch is finished. Another owner asks the staff what would make this the best bakery in the street, and then helps them move toward it. Both get loaves baked. But the effect on the people, and on the shop in three years, will differ. The second owner is closer to the newer kinds of leadership described in this section.

TRANSFORMATIONAL LEADERSHIP

The American writer James MacGregor Burns, in 1978, separated two kinds of leader.¹¹ A **transactional leader** means one who gets work done by an exchange: clear tasks and fair rewards in return for effort. The owner who sets targets and pays the team bonus for reaching them is acting in this way. The exchange is useful, and every organisation needs some of it, but it reaches only as far as the rewards reach.

A **transformational leader** means one who inspires followers to look beyond their own interest toward a shared purpose, and to change the way they

¹¹ Verify the date and wording of Burns' distinction, and Bass' later extension of it, before this note is finalised.

work. The writer Bernard Bass developed the idea in 1985. Four features mark such a leader. The leader holds a clear picture of a better future and shares it. The leader inspires pride and hope. The leader challenges old habits of thought and invites new ideas, as in the creative process of Chapter 11. And the leader gives each person individual attention.

The link to Chapter 11 is close. A transactional leader works mainly at the lower levels of need, with pay and safety. A transformational leader works at the upper levels, with belonging, esteem and growth. The two kinds do not compete. Most leaders need both, since a vision cannot be reached unless the daily tasks are done and paid for.

The danger is plain. A leader who inspires followers can lead them wrongly as well as rightly. Charm without honesty has done harm in business and in politics. A transformational leader must therefore be judged by the aims set, as well as by the following won. Chapter 5 gave the tests that apply.

COACHING

Coaching means developing a person's skill and confidence through regular one-to-one guidance. The aim is for the person to be able to do more, and not for the coach to do the task. A coach follows four steps. First, agree a goal with the person. Second, watch the person at work. Third, give specific feedback, saying what went well and what to change. Fourth, agree the next step and meet again.

The skill of a coach lies partly in asking. A coach who asks "What do you think went wrong with the order?" helps a person to think, while a coach who says "You did it wrong" ends the thinking. Coaching serves needs at the esteem and growth levels of Chapter 11, since it shows that the leader values the person's future. It also makes delegation, from Chapter 10, safer. A manager who is afraid to delegate may find that coaching builds the skill that makes it possible.

In the bakery, the owner coaches the head baker each week. A session might look at how a purchase was made within the Tk 5,000 limit, or how the morning briefing was run. Coaching takes time, and it needs a person who wants to learn. Where the person does not want to learn, coaching becomes nagging.

SUPER LEADERSHIP

Super leadership means leading others so that they learn to lead themselves. Two American writers, Charles Manz and Henry Sims, set out the idea in 1989.¹² The word *super* does not refer to a leader of great power. It refers to a leader who makes the followers strong enough that they need less leading.

The idea rests on *self-leadership*, which means a person's ability to set goals, check progress and reward oneself without being told. A super leader shows self-leadership by example, guides followers to set their own goals, praises independent action, and builds the team's habit of managing its own work. It links to Theory Y of Chapter 11, and to the delegating step in the readiness model.

Super leadership has limits. It needs followers who are skilled and willing, so it does not fit the new helper at the start. A manager must also accept less control, and Chapter 10 listed the fears that make this hard. And a team without a leader's guidance may drift if goals are unclear. The table compares the three kinds of leadership.

¹² Verify the date, the authors and the steps they proposed before this note is finalised.

THREE NEWER KINDS OF LEADING

KIND	MAIN AIM	THE LEADER'S METHOD	FITS BEST
Transformational	Change and a shared purpose	Vision, inspiration, challenge, attention	Times of change; followers ready for more
Coaching	Growth of one person	Goals, watching, feedback, next step	A person who wants to learn
Super leadership	Self-led followers	Example; guiding goals; praising independence	Skilled and willing followers

Before moving on. A transactional leader exchanges rewards for effort, while a transformational leader inspires a shared purpose. Coaching develops one person at a time. Super leadership helps followers lead themselves.



WORKED CASE: TWO STYLES IN ONE BAKERY

Case file, illustrative. Situation. In Month 12 the owner changes from deciding most questions of work method alone to asking the experienced bakers for their views first. With the new helper the owner stays directive. The owner also coaches the head baker each week. Since Month 10 the head baker may buy ingredients up to Tk 5,000 an order, and the owner stays accountable.

Case file, illustrative. Question. Where does each approach lie on the leadership continuum, which bases of power does the owner use, and is the choice sound?

Reasoning, step 1. We place each approach on the continuum of section 12.4. With the experienced bakers, the owner presents the problem, hears suggestions and then decides. That is the fifth point, the consultative style. With the new helper, the owner decides and gives clear orders. That is the first or second point, depending on whether reasons are given.

Reasoning, step 2. We name the bases of power from section 12.2. The owner holds legitimate power as the owner, reward power through the Tk 1,000 team bonus, and coercive power through the right to dismiss. The owner also has expert power in baking, which is strong with the helper and weaker with the experienced bakers, who know the oven well. The head baker holds legitimate power up to Tk 5,000 an order, and expert power as the most skilled baker. Asking the bakers first respects their expert power.

Reasoning, step 3. We test the choice against the forces of section 12.4. The experienced bakers are able and willing, so by the readiness model they suit the participating style. The helper has little skill, so the helper suits telling. The head baker, who is both able and willing, may be given more freedom, and the weekly coaching builds the skill that makes delegation safe.

Reasoning, step 4. We check the links to Chapter 11. Asking first is a step toward Theory Y, and it speaks to the esteem need of the bakers. The directive style with the helper does not mean Theory X. It reflects the helper's present readiness, and the owner should move to the right on the line as the helper learns. The owner should also explain why the two styles differ, so that nobody sees it as favouritism.

Case file, illustrative. Conclusion. The choice is sound. The owner uses a consultative style where readiness is high and a directive style where it is

low, uses several bases of power, and builds the head baker's skill through coaching.

Check. Is each style tied to a named point on the line? Yes: point five for the bakers, point one or two for the helper. Is the choice tied to readiness and not to liking? Yes. Does the answer use only facts in the case file? Yes: the Tk 5,000 limit, the Tk 1,000 bonus, and the weekly coaching. No figures were added.



SUMMARY

Leadership is influencing people toward a goal. A manager holds a position and must lead, but a leader needs only influence. Power is the ability to make others act, and it rests on five bases: legitimate, reward, coercive, expert and referent. The trait approach found no list of qualities that works in every case. The behavioural approach named three styles, autocratic, democratic and laissez-faire, and two concerns, for task and for people. The situational approach holds that the best style depends on the task, the followers and the setting, and the leadership continuum shows a range from boss-centred to team-centred. Emotional intelligence helps a leader read feelings. Transformational leadership inspires a shared purpose, coaching develops one person at a time, and super leadership helps followers lead themselves.

IN SHORT

No single style of leading is best. A leader reads the task, the followers and the setting, uses the bases of power with care, and moves along the continuum as people grow.

THINK IT THROUGH

A hospital ward has a new team of junior nurses and two senior nurses who have worked there for many years. The ward manager has used one style, firm and detailed instruction, with everyone since the start. The senior nurses have begun to complain, and the junior nurses are learning fast. Using this chapter, say where on the continuum the manager now stands and where the manager should stand with each group. Name the bases of power the manager might use with each. Then answer this question: should the manager's style change as the team matures? Give one reason in favour and one risk of changing style.

WHERE PEOPLE GO WRONG

"A leader is the same as a manager." A manager holds a position. A leader needs only influence, and may hold no position at all.

"Power is always bad." Power is an ability. It is used well or badly, depending on the aims and the means.

"Leaders are born." The trait approach found no list of qualities that works in every setting, and skills such as coaching and listening can be learned.

"Democratic leading is always best." Short time, new followers or a crisis may call for a more directive style, and very skilled followers may need only light guidance.

"Changing style means having no principles." A sound leader keeps the same values and changes the method to suit the people and the task.

IN HISTORY

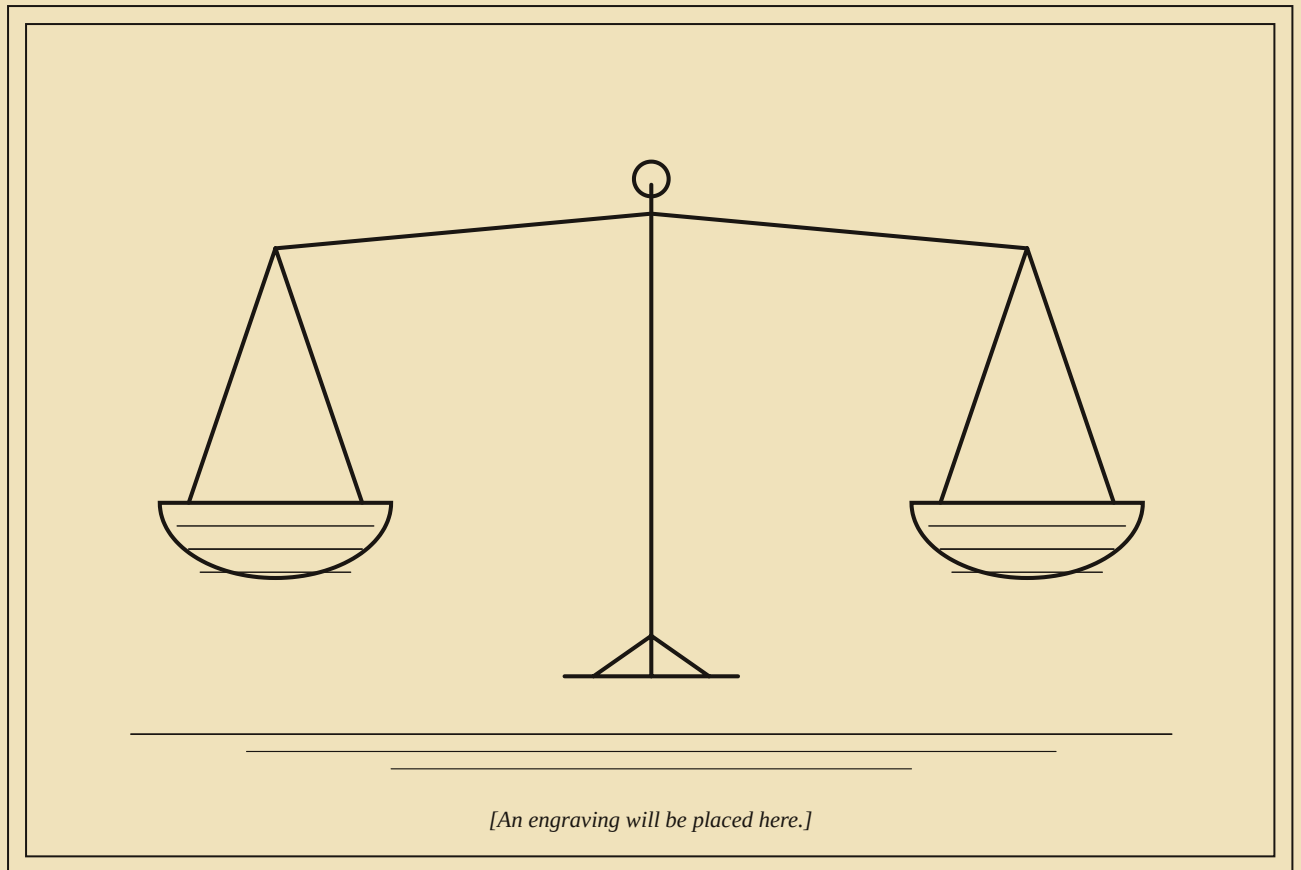
In the late 1930s Kurt Lewin, a German-born psychologist working in the United States, studied boys' clubs with two colleagues, Ronald Lippitt and Ralph White. Adult leaders were trained to act in turn in an autocratic, a democratic or a laissez-faire way toward groups of boys, while observers recorded what happened. The reports said that the autocratic groups worked hard while the leader was present, but showed more hostility, and work fell when the leader left the room. The democratic groups worked a little less in some respects, but with greater originality, and work continued when the leader was absent. The laissez-faire groups did the least. The study was small and the subjects were children, so it cannot settle how adults should be led. It did show that style affects both output and the feelings of the group, which opened the study of leadership by behaviour.

SOURCES

French and Raven, "The Bases of Social Power" (1959), named the five bases of power. Tannenbaum and Schmidt, "How to Choose a Leadership Pattern" (*Harvard Business Review*, 1958), set out the continuum. Lewin, Lippitt and White (1939) reported the early style experiments. Blake and Mouton (1964) drew the managerial grid. Fiedler (1967), Hersey and Blanchard (1969) and House (1971) developed the situational views. Burns (1978) and Bass (1985) described transformational leadership. Goleman (1995) made emotional intelligence widely known. Manz and Sims (1989) described super leadership.

REVIEW QUESTIONS

1. Define leadership. State two ways in which a leader differs from a manager, in a table with three rows.
2. List the five bases of power. Give one example of each from a shop or a school, and say which two belong to the person and not to the position.
3. Compare the autocratic, democratic and laissez-faire styles. Say in which situation each would suit best.
4. Draw the leadership continuum with its seven points. Place on it the three cases your teacher gives you, and give a reason for each placement.
5. Explain why the question "Which leadership style is best?" has no single answer. Use two situational views in your reply.
6. Distinguish a transactional leader from a transformational leader, with one example of each.
7. Define emotional intelligence and name its five parts. Say why a leader who lacks empathy may fail to see a worker's needs.
8. Describe the four steps of coaching, and explain how coaching and delegation can work together.



CHAPTER XIII

Controlling

A plan is only a wish until someone checks what happened.



Every plan can drift from its goal, a little each day. This chapter shows how a manager sets a standard, measures the result, compares the two, and corrects the gap. It describes three moments for control, the barriers that weaken it, and the way control joins planning, efficiency and power.

A PLAN IS NOT ENOUGH

A bus driver watches the road, the clock and the fuel all the way.

A manager is in the driver's position: plans meet a changing world.

At Month 12 the bakery has made 280 loaves a day, not 300.

This chapter gives the method, called control, and returns to the bakery at the end.

Imagine a bus that leaves Dhaka for Rajshahi at six in the morning. The driver knows the plan: arrive at noon, by the main road, with one stop for rest. Does the driver hold the wheel straight and stop watching? No. The driver looks at the road, the clock, the fuel gauge and the speed. If a road is closed, the driver changes the route. Without these checks the plan would exist, but the bus might not arrive.

A manager is in the same position as the driver. In earlier chapters we learned to make plans and to set objectives. But a plan is only a statement about the future, and the future then arrives. Prices rise, workers fall ill, machines stop, and customers choose another shop. How does a manager learn that the plan is slipping? How soon must the manager learn it? What should be done next?

Case file, illustrative. At the end of Month 12, the owner of Morning Loaf Bakery looks at the plan made in Months 6 and 7. The plan asked for 300 loaves a day. The shop now makes 280. Is that a small miss or a large one? Why did it happen? Should the owner change the work, or change the plan? The owner cannot answer these questions by feeling alone. The owner needs a method.

This chapter teaches that method. It is called control, and it is the last of the four functions that we met in Chapter 2. We shall learn what control is, how the process works step by step, when to apply it, and what weakens it. We shall then see how it fits with planning, efficiency and power. At the end we return to Morning Loaf and answer the owner's questions.

WHAT CONTROL IS

THE MEANING OF CONTROL

In daily English, to control a thing means to hold power over it. A driver controls a bus, and a parent controls a child's spending. In management the word has a narrower meaning. **Control** is the work of checking results against plans and correcting any gap between them. The manager does not try to hold power for its own sake. The manager tries to keep the work on the road that the plan has chosen.¹

Let us see the idea in a family. A family plans to spend Tk 12,000 on food in one month. On the fifteenth day the mother counts the money spent, and the total is Tk 9,000. That is more than half of the plan after only half of the month. The mother does not ignore this fact. She buys less fish for the second half of the month and cooks more rice at home. In this small story four things happened. A plan existed. A result was counted. The result was compared with the plan. An action followed. These four things are the heart of control.

Fayol, whom we met in Chapter 3, listed control among the main elements of management in 1916. For him, control meant seeing that everything happens in agreement with the plan and the orders given. Later writers made control one of four functions, with planning, organising and leading. In this book we use the four functions of Chapter 2, so control is the fourth. It is the function that closes the circle, because it sends information back to planning.

The everyday word control means holding power; in management it means checking and correcting.

WORD IN USE. CONTROL. IN DAILY LIFE, TO HOLD POWER OVER A PERSON OR A THING. IN MANAGEMENT, TO CHECK RESULTS AGAINST PLANS AND CORRECT THEM.

Control. Checking results against plans and correcting any gap.

A family budget shows the whole idea in four small actions.

GO SLOWLY HERE.

Early writers on management already placed control beside planning and organising.

¹ Fayol wrote in French. The French word for control, *contrôle*, carries the sense of checking and inspecting, which is close to the meaning used in this book.

WHY MANAGERS CONTROL

Managers control for five reasons. The first is to keep the plan on course, so that small differences do not grow into large ones. The second is to find mistakes early, when they cost little to correct. The third is to protect resources from waste, whether the resource is money, material, time or effort. The fourth is to give information to the next plan, since good planning needs facts about what happened before. The fifth is to show who is accountable, because a result can only be questioned when someone can be named as answerable for it.

Control has five purposes, each protecting a different part of the plan.

THE NATURE OF CONTROL

Control has six features. First, it is a function of every manager, and not only of senior managers. A first-line manager controls a shift, and a general manager controls a whole unit. Second, it is continuous. It is a cycle that repeats, and not an event that happens once at the end. Third, it looks forward and backward. It uses past results to correct present work, and it also tries to prevent future error. Fourth, it depends on planning, because a result can only be judged against something planned. Fifth, it depends on information, since a manager cannot correct what the manager cannot see. Sixth, it involves people, because people do the work, and people accept or reject the checking.

Control has six features that distinguish it from a single act of checking.

The sixth feature deserves special care. A thermostat in a room has no feelings about the temperature that it reads. A worker does. If a worker believes that control exists only to find fault, the worker may hide errors or change the numbers. If the worker believes that control helps the work, the worker will often report problems early. Later in this chapter we shall see that many failures of control are failures of trust, and not failures of arithmetic.

The sixth feature, people, decides whether control is accepted or resisted.

Control is needed in a small firm as much as in a large one. The owner of a one-shop bakery may check everything personally, and may keep no written plans or reports. Even so, the owner compares results with a plan, in the head if not on paper. The difference between a small and a large firm is not whether control exists, but how it is organised. In a small firm one person does all four steps. In a large firm the steps are shared among many people, and so the system must be written down.

Control is needed in a small firm as much as in a large one.

THE CONTROL SYSTEM AND THE CONTROLLER

A single check is not enough for a large organisation. A bakery with one shop can rely on the owner's eyes. A business with many shops needs a **control system**. This is the whole set of standards, measuring methods, reports and responsible people that together keep the plans on course. The system should work whether or not any one manager is present.

A control system is the whole set of parts that keep a plan on course.

Control system. The standards, measures, reports and people that keep a plan on course.

Someone must design the system and keep it running. We call this person the **controller**. In a small firm, the owner is the controller. In a large firm, the controller is often a specialist who collects the figures, prepares the reports and tells managers where results differ from plans. The controller does not replace the other managers. Each manager still controls his or her own area, in the way that the head baker controls the baking. The controller supplies the system and the facts.

The controller runs the system; every manager still controls his or her own area.

Controller. The person who runs the control system.

Case file, illustrative. In Month 10 the owner of Morning Loaf Bakery engaged a part-time bookkeeper at Tk 8,000 a month. The bookkeeper may require receipts from all staff. This is a form of functional authority, which we studied in Chapter 10. It is the right to give instructions about one process, even to people outside one's own line. In this way the bookkeeper helps to run the

In the bakery the part-time bookkeeper acts as a controller of records.

The controller should be independent of the work that is checked.

Six principles guide the design of any control, large or small.

First principle: control follows the plan.

Second principle: attention goes to the critical points and to large differences.

Critical point. A place in the work where a failure would harm the goal most.

Third principle: a check must lead to action.

Fourth principle: prevent where possible.

Fifth principle: every difference must be traceable to a person who can act.

Sixth principle: the checker is independent of the work checked.

Monitoring watches; control watches and then acts.

Monitoring. Watching results without yet acting on them.

The table sets the two side by side on five points.

control of money records, while the owner remains answerable for the shop as a whole.

An important rule follows from the meaning of the word. A person who checks should not be the same person who did the work. If a baker counts his own waste, he may be honest, but the owner cannot be sure. When the checker is independent, the owner can trust the report more. We shall see in the history section of this chapter what happens when this rule is broken in a large organisation.

PRINCIPLES OF CONTROL

Principles are general rules that hold in many cases. Six principles guide the design of control. They do not give exact figures. They tell a manager what to keep in mind when building a way of checking.

The first principle is that control follows the plan. A manager cannot check what was never planned. If the plan says 300 loaves a day, control can compare the daily count with 300. If the plan says only "make more bread", control has nothing to measure. A clearer plan makes a clearer control. For this reason Chapters 6 and 7 come before this chapter.

The second principle is that attention goes to the critical points. A **critical point** is a place in the work where a failure would harm the goal most. A manager who watches every detail will be tired and will miss the important facts. In a bakery the critical points include the quality of flour at delivery, the heat of the oven and the count of loaves packed. A manager should also give most attention to large differences from the plan, and less to small ones.

The third principle is that a check must lead to action. A report that nobody reads, or a difference that nobody corrects, wastes the effort of the people who prepared it. Before adding a new check, a manager should ask: who will receive this information, and what will they do if the result is poor?

The fourth principle is that control should prevent error where possible. It is better to stop poor flour entering the bakery than to throw away poor bread afterward. Control that looks forward saves more than control that only looks backward. In the third section of this chapter we shall see how managers do this.

The fifth principle is that every difference from the plan should be traceable to a person who can act. If a result is poor and nobody is responsible, nobody will correct it. This links control to the authority and delegation of Chapter 10. A manager who is made answerable for a result must also have the authority to change the work that produces it.

The sixth principle is the one that we met above: the checker should be independent of the work that is checked. Where this is not possible, as in a very small firm, the manager should at least make the records open to another person.

CONTROL AND MONITORING

Two words are often mixed. **Monitoring** means watching results without yet acting on them. A baker who reads the oven thermometer is monitoring. Control includes monitoring, and it adds two further steps: comparing the reading with a standard (the planned level, defined in the next section), and correcting if the reading is wrong. Monitoring is therefore a part of control, and not a rival to it. A report that is only read, and never compared or acted upon, stays at the level of monitoring.

The next table compares the two ideas on five points. Read each row from left to right.

CONTROL AND MONITORING COMPARED

POINT	MONITORING	CONTROL
Main question	What is happening?	What is happening, and what must change?
Action	Watches and records	Compares with a standard and corrects
Standard needed	Not always	Always
Result	A report	A report and a change in the work
Bakery example	The baker reads the oven thermometer	The baker reads it, finds it above the standard, and lowers the heat

IN SHORT

Control is the work of checking results against plans and correcting any gap. It has four parts: a plan, a measured result, a comparison, and an action. Monitoring is the watching part only.

Monitoring becomes control only when comparison and correction are added.

THE CONTROL PROCESS

THE FOUR STEPS

Every act of control, in a bakery or a bank, follows the same four steps. The manager sets a standard. The manager measures the actual result. The manager compares the result with the standard. The manager then acts: this means correcting the work, or accepting the result, or changing the standard. After the action the cycle starts again. The next figure shows these steps as a circle.

The control process has four steps, and the fourth sends the manager back to the first.

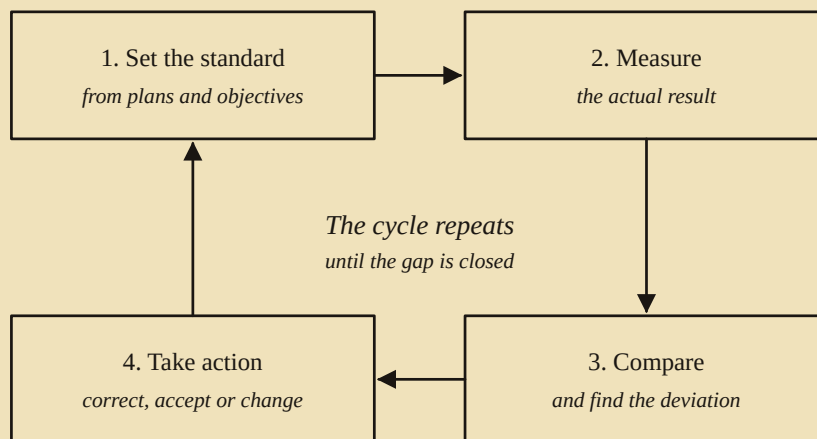


FIG. 41. *The control cycle: four steps, each leading to the next*

Some books name only three steps and join measuring and comparing into one. We keep four, for a practical reason. Measuring and comparing are different tasks, often done by different people, and they fail in different ways. A shop can measure well and compare badly, or the reverse. When each task has its own name, a manager can find the weak point more easily.

Some books name three steps; this book keeps four so that each task has its own name.

A standard is the exact level of result that is to be reached.

WORD IN USE. STANDARD. IN DAILY LIFE, HOW GOOD A THING IS, AS IN A HIGH STANDARD OF WORK. IN MANAGEMENT, THE EXACT LEVEL OF RESULT TO BE REACHED.

Standard. The level of result that is to be reached.

Standards come from the objectives and plans that the manager has already made.

Standards are of five kinds, which answer five different questions.

Standards are found from past results, from other firms, from careful study, or from judgment.

Benchmark. A result reached by a good firm, used as a guide by others.

A good standard is exact, measurable, reachable, accepted and tied to a time.

A tolerance allows a small difference without action.

Tolerance. The range around a standard inside which a result is accepted.

STEP ONE: SET STANDARDS

The first step is to decide what result is wanted. The **standard** is the level of result that is to be reached. In daily English, people say a school has a high standard, which means good quality in a general way. In control, the word is exact. A standard says what level, of what thing, by what time. "Good service" is a hope. "Ninety-five orders in every hundred delivered on time" is a standard.

Standards do not appear from nowhere. They come from the objectives of Chapter 7 and the plans of Chapter 6. When an owner decides that waste must be at most 5% by Month 12, the objective becomes the standard. This is why a manager who has written clear objectives has already done much of the first step. It also explains why unclear objectives lead to weak control.

Standards are of five kinds. Quantity standards say how much. Quality standards say how good. Time standards say how fast or by when. Cost standards say how much money may be used. Behaviour standards say how people must act. The next table gives an example of each from Morning Loaf Bakery.

FIVE KINDS OF STANDARD, WITH EXAMPLES FROM THE BAKERY

KIND	IT ASKS	BAKERY EXAMPLE
Quantity	How much?	300 loaves a day
Quality	How good?	Waste of 5% or less
Time	How fast?	95% of orders or more on time
Cost	How costly?	Ingredients of about Tk 24 a loaf
Behaviour	How must people act?	No claim on a label that is not checked

Managers find standards in four ways. The first is past results: what the shop achieved last year. The second is a **benchmark**, which is a result reached by a good firm and used as a guide by others. The third is careful study of the work, such as the time study of Taylor that we met in Chapter 3. The fourth is the judgment of experienced managers, which is useful when no other source exists. Each source has a weakness. Past results may repeat past errors. A benchmark may come from a firm unlike our own. Study takes time and money. Judgment may be wrong. A manager should use at least two sources when it is possible.

A good standard has five qualities. It is exact, so that two people read it in the same way. It is measurable, which means it can be counted or checked, as Chapter 7 explained. It is reachable, because a standard that nobody can meet teaches people to ignore standards. It is accepted by the people who must meet it, and this is easier when they helped to set it. And it is tied to a time, so that the manager knows when to check.

No result matches a standard exactly. A baker cannot make exactly 300 loaves every day, and the oven cannot hold exactly the same heat all hours. For this reason managers often set a **tolerance**. A tolerance is the range around a standard inside which a result is accepted.² If the standard for output is 300 loaves and the tolerance is 5%, then any day from 285 to 315 loaves needs no action. A tolerance saves a manager from reacting to small and natural changes. But it must be chosen with thought. If it is too wide, real problems pass

² In this chapter, percentages are rounded to one decimal place.

unseen. If it is too narrow, the manager is busy with differences that mean nothing.

STEP TWO: MEASURE ACTUAL RESULTS

The second step is to **measure**, which means to find the actual result. The word actual is important. A standard states what should happen, and the measure states what did happen. A measure may be a count, such as loaves baked. It may be a ratio, such as loaves per working hour, which we met in Chapter 2. It may be a rate, such as complaints per week, or a time, such as minutes of waiting.

Managers have four common ways to find actual results. They may observe the work themselves, as an owner does when walking through the kitchen. They may ask for oral or written reports from the people who do the work. They may read records, such as daily count sheets and sales slips. Or they may take a sample, which means checking a small part to judge the whole. A sample of ten trays from one hundred is quicker and cheaper than counting every loaf. But a sample must be chosen fairly, or it will mislead.

A measure should meet three tests. It must be accurate, since a wrong number leads to a wrong action. It must be timely, since a number that arrives after the problem has grown is of little use. And it must be cheap enough to repeat, because control is a cycle and not a single act. A measure that costs more than the problem it detects is not worth keeping.

The most serious mistake in measuring is to count what is easy and ignore what matters. A clinic finds it easy to count the patients seen in an hour. It finds it harder to count how many patients were helped. A manager who measures only the easy figure may push the staff in the wrong direction. We return to this danger in the section on barriers. For now, we keep one rule: ask first what the plan is trying to achieve, and then find a measure that truly reflects it.

STEP THREE: COMPARE RESULTS WITH STANDARDS

The third step is to compare. The manager finds the **deviation**, which is the gap between the actual result and the standard. We write it in words and then in symbols:

Deviation equals actual result minus standard. In symbols, $D = A - S$. If the result is above the standard, D is positive. If the result is below, D is negative. A positive deviation is not always good. For output, a positive deviation is good. For waste, it is bad. The sign shows direction only. The manager must always ask whether the direction helps or harms the plan.

The size of a deviation is easier to judge when it is shown against the standard. For this we use the percentage deviation. In words, it is the deviation divided by the standard, multiplied by 100. In symbols, $\text{percentage deviation} = D \div S \times 100$. A deviation of 20 loaves means more when the standard is 100 than when it is 1,000.

One more idea needs care. Many standards are themselves percentages, such as waste of 5%. If the actual waste is 8%, the gap is not 3%. It is 3 **percentage points**. A percentage point is one unit on a percentage scale, so the step from 5% to 8% is three of them. But the gap is also 60% of the standard, because 3 is 60% of 5. Both statements are true, and they say different things. To avoid confusion, a manager should state which one is meant.

To measure is to find the actual result, using a method the manager can trust.

Measure. To find the actual result.

Managers measure through observation, reports, records and samples.

The measure must be accurate, timely and cheap enough to repeat.

Measure what matters, not what is easy to count.

Comparing finds the deviation, which is the gap between actual result and standard.

Deviation. The gap between the actual result and the standard.

The deviation is found by subtracting the standard from the actual result.

Percentage deviation shows the size of the gap relative to the standard.

A percentage point is a step on a percentage scale, which must not be confused with a percentage of a percentage.

Percentage point. One unit on a percentage scale, such as the step from 5% to 6%.

GO SLOWLY HERE.

The manager then asks whether the deviation lies inside or outside the tolerance.

Next the manager asks whether the deviation is inside or outside the tolerance. A deviation inside the tolerance usually needs no action. A deviation outside it needs attention. A deviation at the edge needs a closer watch, because the next result may fall outside. The next figure shows weekly results against a standard and a tolerance. It is a simple form of a control chart, which we describe in the next section.

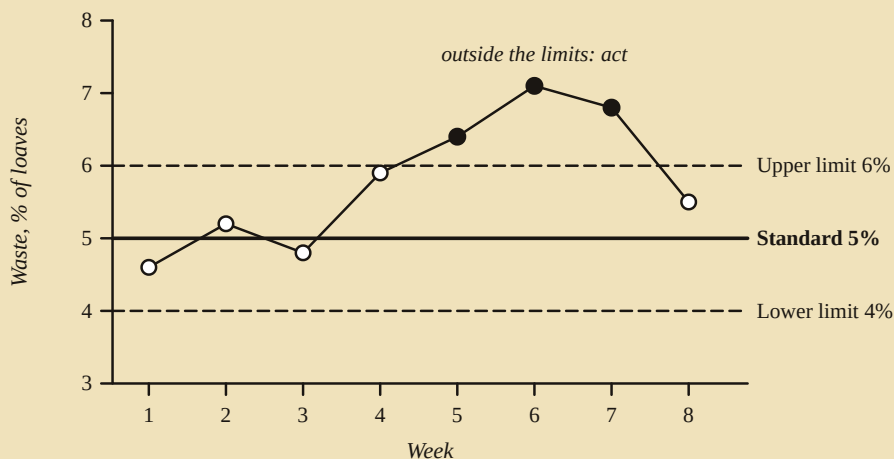


FIG. 42. Weekly results against a standard, with a tolerance band (illustrative figures)

A deviation is a symptom; the cause must be found before acting.

A deviation tells the manager that something is wrong. It does not say what. A doctor treats a fever by finding its cause, and a manager must do the same. Five kinds of cause appear again and again. The work may have been done badly. The standard may have been set too high or too low. The plan may have rested on a wrong premise, such as a price that later changed. An outside force of the kind we studied in Chapter 4 may have changed the situation. Or the measure itself may be wrong. If the manager does not find which cause is at work, the action may fix the wrong thing.

STEP FOUR: TAKE CORRECTIVE ACTION

Corrective action is the step that closes the gap, or closes the cycle if no gap exists.

Corrective action. A step taken to close the gap between result and standard.

The fourth step is **corrective action**, which means a step taken to close the gap between result and standard. It is the step that makes control different from mere watching. A manager has three choices. The first is to accept the result, when the deviation is inside the tolerance. The second is to correct the performance, by changing the way the work is done, the people who do it, or the resources they use. The third is to change the standard or the plan, when the cause is a standard that was never reachable or a world that has changed.

A good corrective action treats the cause, not the symptom, and is checked afterward.

Two rules improve corrective action. The first is to act on the cause and not on the symptom. If waste is high because trays wait too long, a warning to the bakers does not help; a change in the loading of the ovens might. The second is to check afterward. The manager returns to step two and measures again. If the gap has closed, the action worked. If not, the cause was misjudged, and the cycle continues. In this way control is never finished. It is a loop.

TWO WORKED EXAMPLES

The first example uses clean numbers so that the method is the only thing to learn.

The first example is not from the bakery but from a small snack stall, so that the numbers stay clean. *Given.* The stall prepares 100 portions a day. The standard for waste is 5% of the portions. The actual waste in one week is 8%. *Find.* The deviation in percentage points, in portions, and as a percentage of the standard. *Formula.*

Deviation equals actual minus

standard; percentage deviation equals deviation divided by standard, multiplied by 100.

Steps. In percentage points: $8\% - 5\% = +3$ percentage points. In portions: $8 - 5 = +3$ portions a day. As a percentage of the standard: $3 \div 5 \times 100 = +60\%$. *Answer.* Waste is 3 percentage points above the standard. That equals 3 more portions wasted each day, or 60% more waste than the standard allows. *Check.* The standard plus the deviation must give the actual result: $5 + 3 = 8$. And 60% of 5 is 3, which agrees with the gap.

The second example is messier, as real work often is. A school canteen has three standards, and each has its own unit and its own tolerance. The manager must decide which deviations need action. The next table gives the data. Read it first, and then follow the steps below it.

CANTEEN STANDARDS, RESULTS AND TOLERANCES (ILLUSTRATIVE FIGURES)

MEASURE	STANDARD	ACTUAL	TOLERANCE
Meals served a day	400	372	Within 5% of standard
Cost of one meal	Tk 38	Tk 41	Within 10% of standard
Complaints a week	5 or fewer	9	Up to 6 accepted

Steps. Meals: $372 - 400 = -28$, and $-28 \div 400 \times 100 = -7.0\%$. This lies outside the tolerance of 5%. Cost: $41 - 38 = +\text{Tk } 3$, and $3 \div 38 \times 100 = +7.9\%$. This lies inside the tolerance of 10%. Complaints: $9 - 5 = +4$, and $4 \div 5 \times 100 = +80\%$. Nine complaints are above the accepted level of six, so this lies outside the tolerance.

Answer. Two deviations need action: meals served and complaints. The cost deviation is inside its tolerance, but it is close to the edge. The three may be connected. When fewer meals are served, the fixed costs of the kitchen are shared among fewer meals, so the cost of one meal rises. A manager who finds the cause of the low meal count may therefore improve two measures with one action. *Check.* $400 - 28 = 372$, $38 + 3 = 41$, and $5 + 4 = 9$, so each actual result equals standard plus deviation. Each percentage was divided by its own standard, and so the units agree.

The same cycle works in any organisation. Consider a small clinic. Its plan says that 90% of patients should be seen within 30 minutes of their appointment time. This is the standard. The manager measures it from the appointment book, which records both times. In one month, 78% of patients were seen on time. The deviation is $78 - 90 = -12$ percentage points, and $-12 \div 90 \times 100 = -13.3\%$ of the standard. The tolerance is 5 percentage points, so the result is outside it.

The manager then tests the five kinds of cause. The standard seems fair, because other clinics of the same size reach it. The measure is sound, since the front desk fills the book at the time of each visit. No outside force has changed, and the plan assumed the same number of patients as before. The remaining cause is the work. The records show that the delays occur on mornings when only one nurse is present. The corrective action is to arrange the nurses' leave so that two nurses work on the busiest mornings. The manager will measure again in the next month. *Check.* Standard plus deviation gives the actual result: $90 - 12 = 78$.

Three slips are common. The first is a wrong sign: subtract the standard from the actual result, not the reverse. The second is dividing by the actual result instead of the standard. The third is mixing percentage points with percentages of the standard, as we saw above. If the

The steps show each line of arithmetic with its unit.

The second example is messier: three standards, each with its own tolerance.

Each measure gets its own deviation and its own test against tolerance.

The answer names the two measures that need action, and the link between them.

A clinic shows the same cycle in a new setting, with no baking at all.

The manager tests the five causes in turn and finds that the cause lies in the work.

The common slips in these calculations are few and easy to avoid.

manager checks that standard plus deviation equals actual, most slips appear at once.

TYPES AND TOOLS OF CONTROL

THREE MOMENTS FOR CONTROL

Every piece of work has three parts. Inputs enter, such as flour, workers and money. A process changes the inputs, such as mixing and baking. Outputs leave, such as loaves for sale. Control can be applied at any of these moments. The moment gives its name to the type of control, and each type has its own strengths and weaknesses. The next figure shows the three moments on one line.

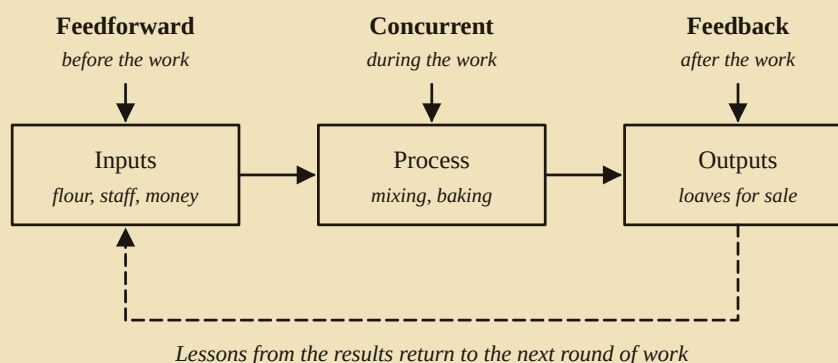


FIG. 43. Three moments for control, drawn along the path of the work

Feedforward control acts before the work begins, to prevent error.

Feedforward control. Control that acts before the work starts.

Concurrent control acts during the work, so that correction is immediate.

Concurrent control. Control that acts while the work is being done.

Feedback control acts after the work is done and teaches the next plan.

Feedback control. Control that checks results after the work is done.

The table compares the three types on timing, strength, weakness and example.

The first type acts before the work begins. We call it **feedforward control**, which means control that acts before the work starts. Its aim is to prevent error, by making sure that the inputs and the plans are right before the process begins. A bakery that checks the flour at delivery uses feedforward control. A hospital that tests the sterility of instruments before an operation does the same. So does a firm that tests new staff before hiring them. Its great strength is that it stops a problem at the cheapest point. Its weakness is that it needs good knowledge of what could go wrong, and it cannot catch errors that start during the work.

The second type acts during the work. **Concurrent control** means control that acts while the work is being done. A baker who watches the oven thermometer and lowers the heat at once uses concurrent control. So does a supervisor who stands at a packing bench and corrects a worker on the spot. Its strength is speed: the error is corrected before it grows. Its weaknesses are cost and pressure. It needs a person or an instrument at the place of work, and workers who are watched closely may feel distrusted.

The third type acts after the work. **Feedback control** means control that checks results after the work is done. A monthly profit report is feedback control, as is a customer survey, or a test of finished loaves. Its strength is that it gives the full picture of the result and shows what to change in the next plan. Its weakness is delay. By the time the report arrives, the loaves are sold and the money is spent, and the error cannot be undone.³

The next table puts the three types side by side.

³ In daily English, feedback often means a person's comment on someone's work. In this chapter it has the narrower meaning of control that acts after the work is done.

THREE TYPES OF CONTROL COMPARED

POINT	FEEDFORWARD	CONCURRENT	FEEDBACK
Moment	Before the work	During the work	After the work
Aim	Prevent error	Correct at once	Learn and improve
Strength	Cheapest point to act	Fast correction	Full picture of results
Weakness	Needs good forecasts	Needs presence; may feel like pressure	Late; cannot undo
Bakery example	Flour checked at delivery	Oven temperature watched	Monthly profit report

No type is best. A sound control system uses all three, in a balance that suits the work. The more costly an error is, the earlier the control should come. A mistake in a medicine is so serious that feedback alone would not be enough, and checks must come before and during the work. A mistake in the colour of a poster is less serious, and a check at the end may be enough. The balance also depends on cost, since feedforward and concurrent control often need more people or instruments.

Case file, illustrative. At Month 12 Morning Loaf Bakery has three controls. The ingredient check at delivery is feedforward control. The oven temperature check is concurrent control. The monthly profit report is feedback control. The first two work every day. The third arrives once a month. This difference in timing will matter when we return to the bakery's problem later in this chapter.

A sound system uses all three types, in balance.

In Month 12 the bakery already used all three types, but its feedback was slow.

COMMON TOOLS OF CONTROL

Managers use many tools to carry out the steps of control. A tool collects facts or makes them easy to read. It does not decide what to do. That decision belongs to the manager. We describe seven common tools. The next table lists them with what each checks and a bakery use.

Tools are the instruments of control; none of them is control itself.

SEVEN COMMON TOOLS OF CONTROL

TOOL	WHAT IT CHECKS	BAKERY USE
Budget	Money used against money planned	Monthly flour cost against the budget
Records and reports	Daily and monthly facts	Count sheet of loaves baked
Personal observation	Work as it happens	Owner walks through the kitchen
Ratio	One figure divided by another	Waste divided by loaves baked
Audit	Honesty and accuracy of records	Check of the sales book by an outsider
Control chart	Results over time against limits	Weekly waste plotted against 5% limits
Schedule chart	Tasks against planned dates	Festival-week order timetable

The first tool is the budget. In Chapter 6 we defined a budget as a plan in numbers. When the period ends, or during it, the manager compares actual money used with the budget. This is called budgetary control. It works well because money is a common unit: flour, wages and rent can all be compared in taka.

A budget is a plan in numbers, so it becomes a standard for money.

Records and reports carry the facts from the work to the manager.

Personal observation gives the manager facts that numbers do not show.

A ratio compares two numbers by division and so allows fair comparison.

An audit is an independent check of records, by a person outside the work.

Audit. An independent check of records and accounts.

A control chart shows results over time with limits drawn on the graph.

Control chart. A graph of results over time, with the standard and limits drawn on it.

A schedule chart compares planned and actual dates for the tasks of a project.

Records and reports carry the facts from the place of work to the manager. A good report is short, it arrives in time, and it shows the difference from the standard. A report that is long and late is rarely read. Management should ask of every report: who needs it, and what will they do with it?

Personal observation means the manager looks at the work. It gives facts that numbers cannot show, such as the mood of the workers or the state of the machines. It also costs the manager's time, and workers may see it as pressure. Many managers combine it with reports: the report points to the problem, and the visit explains it.

A ratio compares one number with another by division. Productivity, which we studied in Chapter 2, is a ratio of output to input. Waste divided by loaves baked is also a ratio. Ratios are useful because they allow fair comparison between weeks, between shops, or between firms of different size.

An **audit** is an independent check of records and accounts. An internal audit is done by staff of the organisation who are not part of the work checked. An external audit is done by outsiders. Audit supports the sixth principle: the checker should be independent of the work. It checks that the records are honest and accurate before managers rely on them.

A **control chart** is a graph of results over time, with the standard and its limits drawn on it. The weekly chart shown earlier in this chapter is a simple example. A point outside the limits calls for action. A run of several points on one side of the standard also calls for action, even if each point is inside the limits, because it suggests a steady drift.⁴

The last tool is the schedule chart, which lays out the tasks of a project against time. The planned dates are drawn first, and the actual dates are added as the work proceeds. Where the actual line falls behind the plan, the manager sees at once which task is late. Henry Gantt, a colleague of Taylor, made such charts popular in the early twentieth century. They suit the single-use plans of Chapter 6, such as a festival-week order schedule.

BARRIERS AND REQUIREMENTS

BARRIERS TO CONTROL

Control often fails in practice, even when the four steps are known. A **barrier to control** is anything that stops control from working as intended. Some barriers are in the system, some are in the people, and some are in the world outside. The next table lists six common barriers, how each shows itself, and what a manager may do about it. Six short paragraphs then add detail.

A barrier to control is anything that stops control from working as intended.

Barrier to control. Anything that stops control from working as intended.

⁴ Control charts were developed for quality control in factories in the 1920s by Walter Shewhart, an engineer in the United States.

SIX BARRIERS TO CONTROL AND REMEDIES

BARRIER	HOW IT SHOWS	REMEDY
Resistance by people	Hidden errors; changed numbers	Involve staff; explain the purpose
Too much control	Many reports; little time to work	Keep checks at critical points
Too little control	Problems found late or never	Add checks; shorten reports
Wrong measure	Numbers improve; goal does not	Test the measure against the goal
Poor information	Action on old or wrong figures	Check the source; speed the report
No authority to correct	Problem seen but not solved	Give authority with duty

The first barrier is resistance by people. Workers may fear that control will be used to blame them, or that it will take away their freedom to decide. They may react in several ways. They may hide errors. They may do only what is measured. They may change the figures. A manager reduces resistance by involving staff in setting standards, by explaining why a check exists, and by using results to help the worker and not only to judge. The self-control that Theory Y assumes, which we studied in Chapter 11, is the best remedy of all.

People resist control when they fear blame or loss of freedom.

The second and third barriers are two sides of one problem. Too much control means so many checks that the workers spend more time reporting than working, and the manager cannot see what matters. Too little control means problems grow unseen. The remedy for both is the second principle: place the checks at the critical points, and do not place them everywhere.

Too much control wastes effort, and too little lets problems grow.

The fourth barrier is the wrong measure, which we mentioned earlier. People tend to do what is measured. If a clinic rewards the number of patients seen, the doctors may hurry. If a shop rewards the number of calls made, the staff may make short and useless calls. The measure then improves while the goal suffers. A manager should ask often: if people do exactly what is measured, will the goal be reached?

A wrong measure sends people in the wrong direction.

The fifth barrier is poor information. If a report arrives late, the problem may have grown. If a report is wrong, the action will be wrong. Managers should know where each figure comes from, who prepared it, and how it was checked. They should also be careful with reports that are always good, since a report that never shows a problem may be hiding one.

Poor information, whether late or wrong, damages every later step.

The sixth barrier is the lack of authority to correct. A head baker may see that waste is high, but if the head baker cannot change the oven schedule, nothing will improve. This links back to the principles: those who are answerable for a result need authority over the work that produces it, as Chapter 10 explained.

A person who sees the deviation but cannot act has no real control.

Two further barriers are worth noting. Some work is hard to measure, such as creative work or the quality of care. For such work, managers may use indirect measures and judgment, and must accept that control will be less exact. Second, standards may become outdated when the world changes. If the price of flour rises sharply, a cost standard set last year no longer fits. Control must therefore include an occasional review of the standards themselves.

Some work cannot be measured easily, and standards can become outdated.

Effective control meets requirements in three groups: information, design and people.

The information must be timely, accurate and easy to understand.

The design must focus on critical points, cost little and bend when needed.

The people must accept the control, find it fair and know who is responsible.

Planning and control form a loop: plans give standards, and results improve plans.

REQUIREMENTS OF EFFECTIVE CONTROL

The barriers suggest what an effective control system must be. We group its requirements in three: the information it uses, the way it is designed, and the people it serves. If a system meets all three, the manager can have confidence in it.

The first group concerns information. The information must be timely, reaching the manager while action can still help. It must be accurate, because a wrong figure leads to wrong action. And it must be understandable, which means the reader sees quickly what the figure says and what it requires.

The second group concerns design. The system must focus on critical points and on large deviations, as the second principle said. It must be economical: the cost of the control must be less than the loss that it prevents. And it must be flexible, so that when the plan or the world changes, the standards and the checks can change with them.

The third group concerns people. The staff must accept the control, which is easier when they helped to design it. They must find it fair, which means the same rules apply to all, from the owner to the newest helper. And they must know who is responsible for each result. Finally, a system should balance numbers with judgment. Figures show what happened. Only a manager who talks with the people can learn why.

CONTROL IN THE WHOLE SYSTEM

CONTROL AND PLANNING

Control does not stand alone. It is joined to planning in a loop. Plans give the standards that control uses. Control gives the facts that improve the next plan. The next figure shows the loop. Plans are made, work is done, results are measured and compared, corrective action is taken, and the lessons return to the next plan.

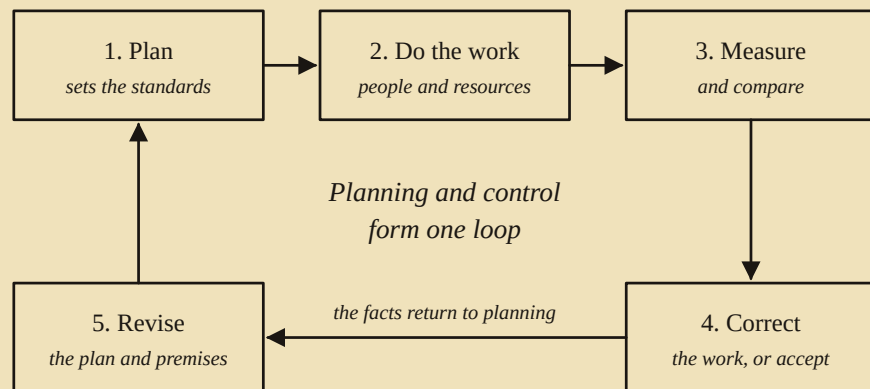


FIG. 44. *The planning and control loop*

Control also tests the premises on which a plan rests.

In Chapter 6 we learned that every plan rests on premises, which are assumptions about the future. Control tests these premises. If a deviation arises because a premise was wrong, then correcting the work will not help. The manager should change the plan. A bakery that planned for flour prices to rise by 5%, and then saw a rise of 15%, would meet deviations in cost that no baker could remove. The right action is a new plan with a new premise.

CONTROL, EFFICIENCY AND EFFECTIVENESS

In Chapter 2 we kept three ideas apart: productivity, efficiency and effectiveness. Control uses all three. Waste measures show efficiency, which is doing a task with little waste. Output and quality measures against the objective show effectiveness, which is reaching the goal set. A productivity figure may serve as a standard in its own right, as when a shop sets a standard of loaves per working hour.

There is a limit. Control checks whether the plan is carried out, and it cannot decide whether the plan is wise. A firm may have excellent control and still be ineffective, if its standards point at the wrong goal. A firm that cuts waste to very low levels while its customers leave for other shops is efficient and ineffective. For this reason managers should review the goals as well as the results.

Control checks efficiency through waste and effectiveness through goals.

Good control cannot make a wrong goal right.

CONTROL AND POWER

In Chapter 12 we studied power and its bases. Control touches power in three ways. First, the controller needs legitimate power, the right that comes from the position, in order to ask for facts and to require action. Second, results often lead to reward or to punishment, which are reward power and coercive power. If these are used unfairly, people will resist the control. Third, control itself is a use of power over the work of others. It must therefore be exercised with fairness and with open rules.

Leadership style, also studied in Chapter 12, shapes how control is felt. A leader who uses control only to punish makes people hide facts. A leader who uses it to coach, by discussing deviations and agreeing the next step, makes people report early. In this way control and leadership support one another.

Control needs authority, and control is itself a use of power.

The leader's style shapes how control is felt.

THE WORKED CASE: MORNING LOAF AT MONTH 13

The worked case follows the layout used for descriptive courses: situation, question, reasoning, conclusion and check. It runs the whole cycle on the figures of the bakery, which were built up over twelve chapters.

Case file, illustrative. *Situation.* In Month 7, the owner agreed three objectives with the staff: 300 loaves a day by Month 12, waste of 5% or less, and 95% of orders or more on time. At Month 12 the results are 280 loaves a day, waste of 8%, and 93% of orders on time. The shop works 26 trading days in a month. Ingredients cost about Tk 24 a loaf. *Question.* How large are the deviations, what is their cost and cause, and what should the owner do?

The next table gives the standards, the results and the deviations.

The case runs the whole cycle on the bakery's figures from earlier chapters.

The situation states the standards, the results and the facts so far.

The table sets out standards, results and deviations for the three measures.

MORNING LOAF BAKERY AT MONTH 12: STANDARDS, RESULTS AND DEVIATIONS

MEASURE	STANDARD	ACTUAL	DEVIATION	PERCENTAGE
Output a day	300 loaves	280 loaves	-20 loaves	-6.7%
Waste	5% or less	8%	+3 percentage points	+60.0%
Orders on time	95% or more	93%	-2 percentage points	-2.1%

Reasoning, part one: the figures. Output: $280 - 300 = -20$ loaves, and $-20 \div 300 \times 100 = -6.7\%$. Waste: $8 - 5 = +3$ percentage points, and $3 \div 5 \times 100 = +60.0\%$. Orders on time: $93 - 95 = -2$ percentage points, and $-2 \div 95 \times 100 = -2.1\%$. The waste has a money cost. Three percentage points of 280 loaves is

The calculations show the three deviations and the money cost of the waste.

The owner checks the five kinds of cause one by one and finds the cause in the work.

The corrective action treats the cause and adds a faster check.

The conclusion puts waste first because it is largest in size and has a money cost.

The check tests the arithmetic and the logic, and sets the next cycle.

The case uses ideas from every earlier part of the book.

Control checks results against plans and corrects any gap, in a repeating cycle of four steps.

The three types, the tools and the barriers help a manager to design and judge a control system.

$0.03 \times 280 = 8.4$ loaves a day. At Tk 24 of ingredients a loaf this is $8.4 \times 24 =$ Tk 201.60 a day. Over 26 trading days it is $201.60 \times 26 =$ Tk 5,241.60 a month.

Reasoning, part two: the cause. The owner tests the five kinds of cause in turn. The standards were agreed with the staff in Month 7, so they are not the problem. The day sheets agree with the sales records, so the measure is sound. No new outside force appears in the records. The plan's premises have not changed in a way that explains the gap. The cause lies in the work. The two ovens are loaded at the same hour, so some trays wait and are baked too long, and the second batch starts late. This one cause explains the low output, the high waste, and part of the late orders.

Reasoning, part three: the action. The owner changes the oven-load schedule so that the two ovens are loaded at different hours. The owner also asks the head baker to count waste every week. The existing controls remain: the delivery check is feedforward, the oven temperature check is concurrent, and the monthly profit report is feedback. The weekly count shortens the delay of the feedback from one month to one week. The owner and the head baker discuss the weekly figures at their regular coaching meeting.

Conclusion. Waste is the deviation with the largest relative size and a direct money cost, so it comes first. The oven schedule treats a single cause behind three deviations, and output and punctuality are expected to improve with it. Orders on time are only two points under standard, so they are watched and not yet treated separately.

Check. Standard plus deviation equals actual in each row: $300 - 20 = 280$, $5 + 3 = 8$, and $95 - 2 = 93$. If waste returns to 5%, it will be $0.05 \times 280 = 14$ loaves a day, against 22.4 now, and the saving of 8.4 loaves agrees with the cost found above. The next cycle begins at once: the weekly count measures waste again, and after four weeks the owner compares the result with the standard.

Notice how many earlier chapters stand behind this one case. The standards came from the objectives of Chapter 7. The plan came from Chapter 6. The two ovens were the result of the decision of Chapter 8. The head baker's duty to count waste rests on the authority and delegation of Chapter 10. The monthly team bonus for waste of 5% or less, which came from Chapter 11, gives staff a reason to care about the weekly count. And the coaching meeting comes from Chapter 12. Control does not replace these ideas. It keeps them working together.

SUMMARY

Control is the work of checking results against plans and correcting any gap. It serves five purposes: to keep plans on course, to find mistakes early, to protect resources, to inform the next plan, and to show who is accountable. The process has four steps: set a standard, measure the actual result, compare the two to find the deviation, and take corrective action. The cycle then repeats. A tolerance allows small differences, and the cause of a deviation must be found before action is taken.

Control may act before the work, during it, or after it, and a sound system uses all three. Managers use tools such as budgets, reports, ratios, audits, control charts and schedule charts. Barriers such as resistance, wrong measures and poor information weaken control. Effective control needs timely and accurate information, a design that focuses on critical points, and people who accept it. Control joins

planning in a loop, checks efficiency and effectiveness, and is itself a use of power.

You should now be able to say what control is and how it differs from monitoring. You should be able to run the four steps on a new set of figures, and to name the type of control in a given case. You should also be able to list the barriers, and to explain how control joins planning.

You should now be able to say what control is, run the cycle on new figures and name the type of control.

THINK IT THROUGH

A clinic wants its doctors to work faster. The manager sets a standard of eight patients seen in each hour, and reports each doctor's figure every week. Within a month the doctors reach nine patients an hour. But the number of patients who return for the same illness rises, and the patients say that the doctors do not listen. Was this measure better than having no measure at all? Write your answer in three or four sentences before you read the next paragraph.

The problem asks why a wrong measure may be worse than none.

A measure steers behaviour. When people know what is counted, they give it their effort. Here the count was patients seen, but the plan wanted patients helped. With no measure, the doctors would have used their own judgment about how long each patient needs, and the clinic would at least have been steering by a purpose, if not by a number. With the wrong measure, the clinic steered the doctors away from its purpose, and it also gave the manager false comfort, since the figures looked good. The lesson is to ask first what the plan is trying to achieve, and then to find a measure that reflects it. A better measure here might combine patients seen with the share who return for the same problem.

The answer: a measure steers behaviour, so the wrong one steers people the wrong way.

WHERE PEOPLE GO WRONG

The first mistake is to think that control means close supervision of people. Control checks results against plans, and supervision is only one way of doing it. Many firms use control without watching anyone, as when they read a weekly report. The best control is often self-control, where the worker knows the standard and checks the work personally.

Mistake one: control means close supervision of people.

The second mistake is to think that control happens only after the work is finished. That is feedback control, and it is one of three. A firm that waits for the end of the month to learn that its flour was poor has used the least helpful moment. Feedforward and concurrent control often act at lower cost.

Mistake two: control happens only after the work is finished.

The third mistake is to think that every deviation means that somebody failed. A deviation may come from a standard that was too high, from a plan that rested on a wrong premise, or from a change in the outside world. A manager who blames first and asks later will find that people hide the facts. The question is why the gap arose, and not who is at fault.

Mistake three: every deviation means that somebody failed.

The fourth mistake is to think that more control is always better. Each check costs time and money, and each can make workers feel distrusted. Beyond a certain point, an added check costs more than the loss it prevents. Good control is selective: it watches the critical points.

Mistake four: more control is always better.

The fifth mistake is a mistake of arithmetic. A change from 5% to 8% is three percentage points, and it is also sixty percent of the standard. These two statements measure different things. A manager who writes "waste rose by 3%" may be understood to mean either. It is safer to say which is meant.

Mistake five: a gap of three percentage points is the same as a gap of three percent.

The sixth mistake is to treat monitoring and control as one thing. Monitoring watches. Control watches, compares and acts. A firm may have excellent reports and poor control, if the reports are read and nothing is changed.

Mistake six: monitoring and control are the same thing.

IN HISTORY

Barings Bank failed in 1995 because one trader both did the work and checked it.

The losses were hidden in an account that London did not examine closely.

The case shows the cost of missing independence, missing questions and slow feedback.

The sources name the thinkers whose ideas shaped this chapter.

The first group asks for recall and explanation of the main ideas.

The second group asks for calculation on new figures.

In February 1995 Barings Bank, one of the oldest merchant banks in London, failed. The cause was the trading of one man, Nick Leeson, at the bank's office in Singapore. Leeson traded futures, which are agreements to buy or sell at a fixed price on a later date. He was also in charge of the part of the office that recorded and settled his own trades. In this way the same man did the work and checked it. The sixth principle of control, that the checker should be independent of the work checked, was broken.

Losses on his trades were placed in a little-used account, so the reports that went to London did not show them. London saw profits, and it did not ask how a small office could earn so much. Later inquiries found that the bank's own management had not acted on warnings about this arrangement. Control by feedback was late and weak, and concurrent and feedforward control were almost absent. When the hidden losses came to light, they were larger than the bank's capital, and the bank was sold to the Dutch group ING for a token price of one pound.

The case teaches three lessons. A checker must be independent of the work checked. Good news should be questioned as carefully as bad news. And a system that depends on reports from the person being checked is not a control system at all. The loss of a bank that had stood for more than two centuries came from a failure in the oldest and simplest rule of control.

SOURCES

Henri Fayol, in *Administration industrielle et générale* (1916), named control among the main elements of management, and Frederick W. Taylor, in *The Principles of Scientific Management* (1911), showed how standards can be found by careful study of the work. Norbert Wiener, in *Cybernetics* (1948), gave the idea of feedback in machines and living things, which later writers applied to organisations. Walter Shewhart, in *Economic Control of Quality of Manufactured Product* (1931), described the control chart. Peter Drucker, in *The Practice of Management* (1954), argued for management by objectives and self-control. On Barings, the inquiry report of the Board of Banking Supervision (1995) and Nick Leeson's own book, *Rogue Trader* (1996), give the main accounts.

REVIEW QUESTIONS

(1) Explain control in your own words, and state how it differs from monitoring. (2) List the four steps of the control process in order, and say what each step produces. (3) Name the five kinds of standard and give one example of each from a business that you know. (4) Describe the three moments for control, and give one example of each. (5) List six barriers to control, and give a remedy for each. (6) Explain how planning and control form a loop, and say what a manager does when a deviation comes from a wrong premise.

(7) A clinic sets a standard of 20 minutes for the waiting time of a patient. The average in one week is 31 minutes. Find the deviation in minutes and the percentage deviation. If the tolerance is 10% of the standard, does the result need action? (8) A workshop has three standards. Output is 250 units a day, and the actual output is 262. Rejected units are 2% or fewer, and the actual rate is 3.5%. Deliveries on time are 90% or more, and the actual rate is 90%. Find each deviation, state which are favourable, and decide which need action first. (9) A

bakery's waste is 6% of 400 loaves baked in a day, against a standard of 5%. Find the deviation in percentage points, the number of extra loaves wasted, and the daily cost at Tk 24 a loaf.

(10) Name the type of control in each case. (a) A hospital tests the sterility of instruments before an operation. (b) A manager watches a call-centre screen and helps a caller who is in difficulty. (c) A school studies the results at the end of a term to change its teaching in the next term. (d) A cook tastes a sauce while it cooks and adds salt. (e) A firm tests applicants for a job before it hires them.

The third group asks the reader to name the type of control in short cases.

GLOSSARY

Accommodative. A stance in which the firm meets legal duties and also meets reasonable requests from society when it is asked. (*Chapter V*)

Accountability. Being answerable for results. (*Chapter I*)

Adapt. To change the organisation so that it fits an outside force. (*Chapter IV*)

Alternative. One of the possible actions from which a choice is made. (*Chapter VIII*)

Audit. An independent check of records and accounts. (*Chapter XIII*)

Authority. The right to give orders and to use the firm's resources to get results. (*Chapter III*)

Barrier to control. Anything that stops control from working as intended. (*Chapter XIII*)

Behavioural model. A design that stresses people, teamwork and flexibility more than rules and ranks. (*Chapter IX*)

Benchmark. A result reached by a good firm, used as a guide by others. (*Chapter XIII*)

Boundary. The line between what managers can direct and what they cannot. (*Chapter IV*)

Bounded rationality. The limits of time, information and mental capacity that stop a person from weighing every alternative. (*Chapter VIII*)

Budget. A plan expressed in numbers: money, materials or time to be used over a period. (*Chapter VI*)

Bureaucratic model. A design built on written rules, fixed ranks and records. (*Chapter IX*)

Business ethics. The standards of right and wrong that guide the conduct of people in business. (*Chapter V*)

Centralisation. The degree to which authority is held at the top of the firm. (*Chapter III*)

Certainty. The condition in which the outcome of each alternative is known in advance. (*Chapter VIII*)

Chain of command. The path along which orders pass from the top of an organisation to the bottom. (*Chapter IX*)

Chance node. A point on a tree, drawn as a circle, where an outside event decides which branch follows. (*Chapter VIII*)

Code of ethics. A written statement of the standards of conduct that an organisation expects of its people. (*Chapter V*)

Cohesion. How strongly the members of a group feel bound to one another. (*Chapter XI*)

Competitive forces. Outside pressures from rivals, customers, suppliers and others who shape the market. (*Chapter IV*)

Concurrent control. Control that acts while the work is being done. (*Chapter XIII*)

Contingency approach. The view that the best way to manage depends on the situation. (*Chapter III*)

Control. Checking results against plans and correcting any gap. (*Chapter XIII*)

Control chart. A graph of results over time, with the standard and limits drawn on it. (*Chapter XIII*)

Control system. The standards, measures, reports and people that keep a plan on course. (*Chapter XIII*)

Controller. The person who runs the control system. (*Chapter XIII*)

- Controlling.** Checking results against the goals and correcting what is wrong. *(Chapter II)*
- Coordination.** Fitting the parts of an organisation together so that they work toward one goal. *(Chapter IX)*
- Corrective action.** A step taken to close the gap between result and standard. *(Chapter XIII)*
- Creativity.** Making new ideas. *(Chapter XI)*
- Criterion.** A standard used to judge an alternative; the plural is criteria. *(Chapter VIII)*
- Critical point.** A place in the work where a failure would harm the goal most. *(Chapter XIII)*
- Davis model.** A set of propositions linking the social power of a firm to its social responsibility. *(Chapter V)*
- Decentralisation.** Spreading decision authority down to lower levels. *(Chapter X)*
- Decision.** A choice made from two or more possible actions. *(Chapter VIII)*
- Decision maker.** The person or group that chooses and is accountable for the choice. *(Chapter VIII)*
- Decision node.** A point on a tree, drawn as a square, where the decision maker chooses a branch. *(Chapter VIII)*
- Decision support system.** A computer tool that helps a manager study alternatives by using data and models. *(Chapter VIII)*
- Decision tree.** A branching diagram that shows choices, chance events, outcomes and payoffs. *(Chapter VIII)*
- Defensive.** A stance in which the firm does only what the law requires and argues that more is not its duty. *(Chapter V)*
- Delegation.** Passing authority and duty for a task down to a subordinate. *(Chapter X)*
- Department.** A group of jobs placed under one manager. *(Chapter IX)*
- Departmentalisation.** Grouping jobs into departments. *(Chapter IX)*
- Design.** Choosing a structure on purpose, so that it fits the organisation's needs. *(Chapter IX)*
- Deviation.** The gap between the actual result and the standard. *(Chapter XIII)*
- Discipline.** Obedience to agreed rules and respect for the agreements between firm and staff. *(Chapter III)*
- Division of labour.** Splitting a job into parts so that each person does one part. *(Chapter IX)*
- Division of work.** Splitting a job into parts so that each person does one kind of task. *(Chapter III)*
- Drive.** The inner tension that a need creates and that pushes a person to act. *(Chapter XI)*
- Economic forces.** Outside conditions of money and trade, such as prices, incomes and interest rates. *(Chapter IV)*
- Environment** ■. Everything, outside and inside the organisation, that affects its work and results. In daily speech: nature. *(Chapter IV)*
- Environmental scanning.** Watching outside forces for early signs of change. *(Chapter IV)*
- Equity.** Fair and kind treatment of staff. *(Chapter III)*
- esprit de corps.** Team spirit: harmony and unity among the members of a firm. *(Chapter III)*

- Ethical dilemma.** A choice in which each option has something right about it and something wrong or costly. (*Chapter V*)
- Expected value.** In daily life, expected means thought likely. In decisions, an average payoff found by weighting each payoff by its probability. (*Chapter VIII*)
- External environment.** The forces outside the organisation's control. (*Chapter IV*)
- Extrinsic reward.** A reward that comes from outside the work, such as pay. (*Chapter XI*)
- Fayol's principles.** Fourteen general rules of management set out by Henri Fayol. (*Chapter III*)
- Feedback.** Information about results that returns to the firm to guide later work. (*Chapter III*)
- Feedback control.** Control that checks results after the work is done. (*Chapter XIII*)
- Feedforward control.** Control that acts before the work starts. (*Chapter XIII*)
- First-line manager.** A manager who directly guides the daily work of workers. (*Chapter I*)
- Flat structure.** A structure with few levels and large spans. (*Chapter X*)
- Formal group.** A group created by the organisation to do a task. (*Chapter XI*)
- Formal organisation.** The structure that is officially described, with its jobs, departments and reporting lines. (*Chapter IX*)
- Frustration.** The discomfort a person feels when a goal is blocked. (*Chapter XI*)
- Function.** A main kind of work that managers do. Example: checking results. (*Chapter II*)
- Functional authority.** The right of a specialist over a process outside the specialist's own line. (*Chapter X*)
- Functional foremanship.** Taylor's plan to replace one foreman with several specialist foremen, each directing one part of a worker's task. (*Chapter III*)
- Functional manager.** A manager who runs one specialist area of an organisation. (*Chapter I*)
- General manager.** A manager who runs a whole unit, with all its specialist areas. (*Chapter I*)
- Global forces.** Outside influences from other countries and from trade between countries. (*Chapter IV*)
- Goal.** A result someone wants to reach. Example: the rent paid by the first of the month. (*Chapter I*)
- Grapevine.** The informal flow of news among people, outside the official channels. (*Chapter XI*)
- Group.** Two or more people who meet, influence one another and see themselves as belonging together. (*Chapter XI*)
- Groupthink.** The habit of a group of seeking agreement so strongly that its members stop testing ideas. (*Chapter VIII*)
- Hierarchy of needs.** Maslow's five levels of need, from basic survival at the base to personal growth at the top. (*Chapter XI*)
- Human factors.** The personal qualities, needs and beliefs that shape how a person behaves at work. (*Chapter XI*)
- Human relations approach.** An approach that stresses people's feelings, attention and group membership. (*Chapter III*)
- Hygiene factor.** An item whose absence causes discontent but whose presence does not drive effort. (*Chapter XI*)
- Influence.** To try to change an outside force. (*Chapter IV*)

Informal group. A group that forms by the members' own choice, not by order. *(Chapter XI)*

Informal leader. A member who influences a group without holding any official post. *(Chapter XII)*

Informal organisation. The friendships, habits and unwritten rules among people, outside the chart. *(Chapter IX)*

Initiative. The freedom to think out a plan and to carry it through. *(Chapter III)*

Innovation. Putting a new idea to use. *(Chapter XI)*

Internal environment. The owners, the staff and the culture inside the organisation. *(Chapter IV)*

Intrinsic reward. A reward that comes from the work itself, such as pride. *(Chapter XI)*

Job. The set of tasks that one person is asked to do. *(Chapter IX)*

Key result area. A part of the firm's work where results matter so much that objectives must be set there. *(Chapter VII)*

Leadership. Influencing people to work toward a goal. *(Chapter XII)*

Leading. Guiding and influencing people so that they work toward the goals. *(Chapter II)*

Line. In daily life, a row of people or things. In an organisation, the positions that do its core work, joined in one chain of command. *(Chapter IX)*

Line authority. The right to direct subordinates and to give them orders. *(Chapter X)*

Management. Getting things done through people and resources to reach goals. *(Chapter I)*

Management by objectives. A method in which managers and staff agree objectives together and review results at fixed times (MBO). *(Chapter VII)*

Management process. The continuing cycle of the four functions. *(Chapter II)*

Management science. Using mathematical models and calculation to help managers solve problems. *(Chapter III)*

Manager. A person responsible for the work of others. *(Chapter I)*

Matrix. Two lines of command crossed, so that one person reports to two managers. *(Chapter IX)*

MBO cycle. The repeating round of agreeing objectives, planning, acting, reviewing and resetting. *(Chapter VII)*

Measurable. Able to be counted or checked by a number or a plain test. *(Chapter VII)*

Measure. To find the actual result. *(Chapter XIII)*

Middle manager. A manager who runs a department or branch and answers to top managers. *(Chapter I)*

Model. A simplified picture of a real problem, made with numbers and rules. *(Chapter III)*

Monitoring. Watching results without yet acting on them. *(Chapter XIII)*

Motivation. The push that starts and keeps a person's effort going toward a goal. *(Chapter XI)*

Motivator. An item that raises effort and satisfaction, such as recognition or interesting work. *(Chapter XI)*

Need. Word in use. In daily life, something you must have to live. In management, a gap a person feels between what they have and what they want. *(Chapter XI)*

Needs-goal model. A chain of five steps: need, drive, action, goal, relief. *(Chapter XI)*

Norm. An unwritten rule about how members of a group should behave.

(Chapter XI)

Objective. A specific result to be reached by a set time, stated so that it can be checked. *(Chapter VII)*

Obstructive. A stance in which the firm denies a problem and resists those who press it to act. *(Chapter V)*

Operational plan. A short-range plan for daily or weekly work in a single unit. *(Chapter VI)*

Opportunity. An outside change that could help the organisation if it acts. *(Chapter IV)*

Order. A proper, known place for each thing and each person. *(Chapter III)*

Organisation. Word in use. In daily life, "she is well organised" means tidy. Here, a group working for shared goals. *(Chapter I)*

Organisation chart. A diagram of the jobs in an organisation and the reporting lines between them. *(Chapter IX)*

Organising. Arranging people, tasks and resources so that the work can be done. *(Chapter II)*

Outcome. What happens after an alternative is chosen. *(Chapter VIII)*

Payback period. The time that a purchase takes to earn back its cost. *(Chapter VIII)*

Payoff. The value of an outcome to the decision maker, usually measured in taka. *(Chapter VIII)*

Percentage point. One unit on a percentage scale, such as the step from 5% to 6%. *(Chapter XIII)*

Piece rate. Pay that depends on the number of pieces a worker produces. *(Chapter III)*

Plan. A chosen route from where we are to where we want to be. *(Chapter VI)*

Planning. Choosing goals and deciding how to reach them. *(Chapter II)*

Planning premise. An assumption about the future conditions under which a plan will be carried out. *(Chapter VI)*

Policy. A general guide to thinking and decision, which leaves room for judgement. *(Chapter VI)*

Political-legal forces. Outside laws, rules, taxes and government policies that apply to the organisation. *(Chapter IV)*

Principle. A general rule that guides action in many different situations. *(Chapter III)*

Proactive. A stance in which the firm looks ahead, acts before it is pressed, and tries to lead others. *(Chapter V)*

Probability. A number from 0 to 1 that measures how likely an outcome is. *(Chapter VIII)*

Procedure. A fixed series of steps for doing a repeated task. *(Chapter VI)*

Process theory. A theory that explains how a person decides where to put effort. *(Chapter XI)*

Profession. An occupation with trained skill, a code of conduct and set standards. *(Chapter III)*

Programme. A single-use plan that groups the goals, steps, resources and timing for a large set of tasks. *(Chapter VI)*

Project. A smaller single-use plan, often one part of a programme, with its own tasks and deadline. *(Chapter VI)*

Rational decision process. A series of logical steps that leads from a problem to a chosen action. *(Chapter VIII)*

Reinforcement. Following a behaviour with a result that makes the behaviour more likely to be repeated. (*Chapter XI*)

Resource. Anything a group uses to do its work. Example: money, tools, flour, time, skill. (*Chapter I*)

Responsibility. The duty to do the task that has been assigned. (*Chapter X*)

Risk. In daily life, danger. In decisions, a condition in which the outcomes are known and their chances can be stated. (*Chapter VIII*)

Rule. A fixed instruction that allows no choice. (*Chapter VI*)

Rule of thumb. A method based on habit and guesswork, not on measurement. (*Chapter III*)

Satisficing. Choosing the first alternative that meets all the criteria, instead of searching for the best. (*Chapter VIII*)

Scalar chain. The line of authority from the top of the firm to the lowest level. (*Chapter III*)

School of thought. A group of ideas about management that share one main belief. (*Chapter III*)

Scientific management. Studying work by measurement to find the best method, then teaching it. (*Chapter III*)

Self-actualisation. The wish to use and grow one's abilities as far as one can. (*Chapter XI*)

Single-use plan. A plan made for one occasion or one project, and not used again in the same form. (*Chapter VI*)

Social forces. Outside changes in people's numbers, ages, tastes, habits and values. (*Chapter IV*)

Social responsibility. The duty of an organisation to act for the good of society as well as for its own profit. (*Chapter V*)

Soldiering. Working slower than one could, on purpose, so as not to do more than others. (*Chapter III*)

Span of management. The number of subordinates who report directly to one manager. (*Chapter X*)

Stability of tenure. Keeping staff in their posts long enough to learn and give good service. (*Chapter III*)

Staff. In daily life, all the employees of a firm. In organisation design, advisers who support the line with special knowledge but do not direct it. (*Chapter IX*)

Staff authority. The right to advise and support the line, but not to command it. (*Chapter X*)

Stakeholder. Anyone who is affected by the organisation, and may also affect it. (*Chapter IV*)

Standard. The level of result that is to be reached. (*Chapter XIII*)

Standing plan. A plan made once and used again and again for situations that repeat. (*Chapter VI*)

Strategic plan. A long-range plan for the whole firm, setting its main direction. (*Chapter VI*)

Structure. The way an organisation divides its work among people and links those people together. (*Chapter IX*)

Subordinate. A person who reports to a manager. (*Chapter X*)

Sustainability. Meeting the needs of today without harming the ability of future people to meet their own needs. (*Chapter V*)

Systems approach. Seeing an organisation as a set of linked parts that work together. (*Chapter III*)

Tactical plan. A middle-range plan that turns part of a strategic plan into work for one department. (*Chapter VI*)

Tall structure. A structure with many levels and small spans. (*Chapter X*)

Task. One piece of work, such as cutting wire or serving a customer. (*Chapter IX*)

Team. A group with joint goals, fitting skills and shared duty for the result. (*Chapter XI*)

Technological forces. Outside advances in machines, methods and knowledge that an organisation can use or must face. (*Chapter IV*)

Theory X. The belief that people dislike work, avoid duty and must be directed and controlled. (*Chapter XI*)

Theory Y. The belief that people can enjoy work, accept duty and direct themselves toward goals they share. (*Chapter XI*)

Threat. An outside change that could harm the organisation if it does not act. (*Chapter IV*)

Time-and-motion study. Timing a task and examining each movement to remove waste. (*Chapter III*)

Tolerance. The range around a standard inside which a result is accepted. (*Chapter XIII*)

Top manager. A manager who directs the whole organisation. (*Chapter I*)

Uncertainty. The condition in which the chances of outcomes cannot be stated, and some outcomes may be unknown. (*Chapter VIII*)

Unity of command. One boss for each person: orders come from one superior only. (*Chapter III*)

Unity of direction. One head and one plan for each group of activities with the same aim. (*Chapter III*)